Skill Navora

COMPLETE HANDBOOK

Data Science & AI Handbook

116 lessons across Excel, SQL, Power BI, AI for analytics, Python, statistics and machine learning, generative and agentic AI, and career preparation.

116 lessons | Version 2026.10 | Updated October 2026

Created by **Varun Kumar**
www.skillnavora.in

Contents

Every entry is a clickable link. The PDF also has bookmarks for each lesson and section.

Advanced Excel

01 Getting Started with Excel for Data Analysis

02 Data Entry, Data Validation & Data Cleaning

03 Number, Custom & Conditional Formatting

04 Formulas, Cell References & Aggregate Functions

05 Text Functions, Logical Functions & IF

06 Sorting, Filtering & Dynamic Arrays

07 Lookup Functions: VLOOKUP, XLOOKUP and INDEX-MATCH

08 Pivot Tables from Basic to Advanced

09 Date, Time & Number Functions and Named Ranges

10 Charts and Advanced Charts

11 Interactive Dashboard: Superstore Case Study

12 Power Query, Power Pivot, Analysis ToolPak & What-If Analysis

SQL for Data Analysis

01 Databases, RDBMS & SQL Basics

02 ER Diagrams, Relationships, Keys & Normalisation

03 Data Types, Operators, NULLs, Variables & Comments

04 DDL, DML & Constraints: CREATE, ALTER, INSERT, UPDATE, DELETE

05 SELECT, WHERE, ORDER BY & Filtering Operators

06 SQL Functions: Aggregate, Text, Date & Number

07 GROUP BY, HAVING, CASE WHEN, ROLLUP, CUBE & COALESCE

08 Joins and a Retail Case Study

09 Set Operators: UNION, UNION ALL, INTERSECT & EXCEPT

10 Subqueries, Derived Tables & EXISTS

11 OFFSET-FETCH and Window Functions I: OVER, PARTITION BY & Ranking

12 Window Functions II: Running Totals, LEAD, LAG & Frames

13 CTEs, Views, Temporary Tables, Stored Procedures & Functions

14 Transactions, Indexes, Performance & Security

15 End-to-End SQL Case Study & Interview Practice

Power BI

01 Getting Started with Power BI & Your First HR Report

02 Power Query, Star Schema & Data Modelling (Adventure Cycles)

03 DAX Fundamentals: Measures, Filter Context & CALCULATE

04 DAX Time Intelligence, Ranking, Parameters & Dynamic Measures

05 Visuals, Filters, Drill-through, Tooltips, Buttons & Report Design

06 Power BI Service, Connectivity Modes, Security & Capstone

Tableau

01 Getting Started with Tableau: Interface, Data Connections and Your First Views

02 Charts in Tableau: From Bar and Line to Maps, Dual Axis, Histograms and Pareto

03 Calculations in Tableau: Calculated Fields, Table Calculations and LOD Expressions

04 Filters, Parameters, Sets, Groups and Hierarchies in Tableau

05 Data Modelling in Tableau: Relationships, Joins, Unions, Blending, Extracts and Tableau Prep

06 Dashboards, Actions, Stories and Publishing in Tableau (Capstone)

AI for Data Analysis

01 Generative AI for Analysts: How It Works, Prompting & AI vs GenAI vs Agentic AI

02 AI with SQL: Text-to-SQL, Docker, Metabase & MinusX

03 AI in Excel & Office: Copilot, Gemini, Claude, Python in Excel & Agents

04 The AI-Assisted Analyst Workflow, Agents & Governance (Capstone)

Python for Data Science

01 Introduction to Python: Why, Install & Jupyter Notebooks

02 Variables, Data Types, Type Conversion, print(), f-strings & input()

03 Operators, Strings, Indexing, Slicing & String Methods
04 Control Flow: if, elif, else, Nested Conditions & match
05 Loops: while, for, range, break, continue & Patterns
06 Lists: Creating, Slicing, Methods, Sorting & Comprehensions
07 Tuples & Sets: Immutable Records, Unpacking and Set Algebra
08 Dictionaries: Key-Value Data, Counting and Grouping
09 Functions: Reusable Code, Arguments, Scope and Docstrings
10 Comprehensions, Lambda, map/filter and Generators
11 Object-Oriented Python: Classes, Objects, Inheritance and Dataclasses
12 Files, Exceptions, JSON and CSV
13 Modules, Packages, Virtual Environments and Git
14 Debugging, Testing, Regular Expressions and Code Quality
15 NumPy: Fast Arrays, Vectorisation and Broadcasting
16 pandas I: Series, DataFrames, Reading Data, Selecting and Filtering
17 pandas II: Data Cleaning: Missing Values, Types, Duplicates, Text and Outliers
18 pandas III: GroupBy, Pivot Tables, Merge/Join and Reshape (melt/unpivot)
19 pandas IV: Dates, Time Series, Resampling and Rolling Windows
20 Data Visualisation I: Matplotlib and Choosing the Right Chart
21 Data Visualisation II: Seaborn Statistical Charts and Interactive Plotly
22 Exploratory Data Analysis (EDA) Case Study: Telecom Customer Churn
23 Python and SQL: sqlite3, DuckDB, SQLAlchemy and pandas.read_sql
24 APIs, JSON and Web Scraping with requests and BeautifulSoup
25 Performance and Bigger Data: Vectorisation, Memory, Parquet, Polars and DuckDB
26 DSA I: Big-O, Arrays, Strings, Hashing, Two Pointers and Sliding Window
27 DSA II: Stacks, Queues, Linked Lists, Recursion, Sorting and Binary Search
28 DSA III: Trees, Heaps, Graphs (BFS/DFS) and Dynamic Programming
29 Python Capstone: End-to-End Retail Sales Analytics Project

Statistics & Machine Learning

01 Descriptive Statistics: Centre, Spread, Shape and Relationships
02 Probability and Distributions: Rules, Bayes, Binomial, Poisson and Normal
03 Sampling, the Central Limit Theorem and Confidence Intervals
04 Hypothesis Testing I: The Framework, p-values, z-tests and t-tests
05 Hypothesis Testing II: Chi-Square, ANOVA, Non-Parametric Tests and Correlation Tests
06 A/B Testing and Experimentation: Design, Power, Analysis and Pitfalls
07 Introduction to Machine Learning: Types, Business Framing and the ML Lifecycle
08 The scikit-learn Workflow: Train/Test Split, Baselines, Pipelines and Saving Models
09 Data Preprocessing and Feature Engineering for ML
10 Linear Regression: Fitting, Interpreting, Assumptions and Diagnostics
11 Gradient Descent: How Models Learn (Built from Scratch in NumPy)
12 Overfitting, Polynomial Regression and Regularisation (Ridge, Lasso, Elastic Net)
13 Logistic Regression: Probabilities, Odds Ratios and Credit-Risk Scoring
14 Classification Metrics: Confusion Matrix, Precision, Recall, ROC, PR and Business Value
15 Cross-Validation and Hyperparameter Tuning: K-Fold, GridSearchCV, RandomizedSearchCV and Optuna
16 K-Nearest Neighbours: Classifier and Regressor (Bank Loans Case Study)
17 Decision Trees: Splits, Impurity, Pruning and Interpretable Rules
18 Ensembles I: Bagging and Random Forests
19 Ensembles II: Gradient Boosting, XGBoost, LightGBM and CatBoost
20 Support Vector Machines and Naive Bayes
21 Imbalanced Classification: Fraud Detection with Class Weights, Resampling and Thresholds
22 Clustering I: K-Means and RFM Customer Segmentation

23 Clustering II and Dimensionality Reduction: Hierarchical, DBSCAN, Gaussian Mixtures, PCA, t-SNE and UMAP

24 Time-Series Forecasting: Decomposition, Baselines, ETS, ARIMA and ML with Lag Features

25 NLP and Text Analytics: Cleaning, TF-IDF, Sentiment Classification, Topics and Embeddings

26 Neural Networks and Deep Learning: From a Neuron to Transformers

27 Explainable AI and MLOps: SHAP, Fairness, Deployment, Monitoring and Responsible ML

28 ML Capstone: End-to-End Churn Prediction and Retention Strategy

Generative AI & Agentic AI

01 How Large Language Models Work: Tokens, Embeddings, Transformers and Training

02 Prompt Engineering in Depth: Patterns, Templates, Few-Shot, Reasoning and Structured Output

03 Calling LLM APIs from Python: Messages, Streaming, Structured Output, Costs and Reliability

04 Embeddings and Vector Search: Meaning as Numbers, Similarity and Vector Databases

05 Retrieval-Augmented Generation (RAG): Build, Evaluate and Improve a Q&A Assistant

06 Tool Use and Function Calling: Letting LLMs Query Data, Call APIs and Take Actions

07 AI Agents and Agentic AI: Loops, Planning, Memory, Multi-Agent Systems and When to Use Them

08 Model Context Protocol (MCP): Connecting AI Assistants to Your Data and Tools

09 Evaluating LLM Applications and Building Guardrails: Test Sets, LLM-as-Judge, Injection, PII and Safety

10 Customising Models: Prompting vs RAG vs Fine-Tuning, LoRA, Distillation and Open-Weight Models

11 GenAI for Data Work: Text-to-SQL, AI Coding Assistants, Data Cleaning and Automated Insights

12 Building and Shipping a GenAI App: Architecture, Streamlit/FastAPI, Cost, Observability and Responsible AI (Capstone)

Capstone, Portfolio & Interviews

01 Portfolio, GitHub and Resume for Data Roles

02 Data Analyst Interview Preparation: SQL, Excel, Power BI, Statistics and Case Questions

03 Data Scientist, ML and GenAI Interview Preparation: Questions and Model Answers

04 Your 90-Day Plan: Learning Roadmap, Job Search Strategy and Growing in Your First Role

Statistics & Machine Learning

Probability, CLT, hypothesis tests, the ML lifecycle, regression, classification, trees, ensembles, clustering, PCA, time series, NLP, deep learning basics, evaluation, explainability and deployment.

1. Descriptive Statistics: Centre, Spread, Shape and Relationships
2. Probability and Distributions: Rules, Bayes, Binomial, Poisson and Normal
3. Sampling, the Central Limit Theorem and Confidence Intervals
4. Hypothesis Testing I: The Framework, p-values, z-tests and t-tests
5. Hypothesis Testing II: Chi-Square, ANOVA, Non-Parametric Tests and Correlation Tests
6. A/B Testing and Experimentation: Design, Power, Analysis and Pitfalls
7. Introduction to Machine Learning: Types, Business Framing and the ML Lifecycle
8. The scikit-learn Workflow: Train/Test Split, Baselines, Pipelines and Saving Models
9. Data Preprocessing and Feature Engineering for ML
10. Linear Regression: Fitting, Interpreting, Assumptions and Diagnostics
11. Gradient Descent: How Models Learn (Built from Scratch in NumPy)
12. Overfitting, Polynomial Regression and Regularisation (Ridge, Lasso, Elastic Net)
13. Logistic Regression: Probabilities, Odds Ratios and Credit-Risk Scoring
14. Classification Metrics: Confusion Matrix, Precision, Recall, ROC, PR and Business Value
15. Cross-Validation and Hyperparameter Tuning: K-Fold, GridSearchCV, RandomizedSearchCV and Optuna
16. K-Nearest Neighbours: Classifier and Regressor (Bank Loans Case Study)
17. Decision Trees: Splits, Impurity, Pruning and Interpretable Rules
18. Ensembles I: Bagging and Random Forests
19. Ensembles II: Gradient Boosting, XGBoost, LightGBM and CatBoost
20. Support Vector Machines and Naive Bayes
21. Imbalanced Classification: Fraud Detection with Class Weights, Resampling and Thresholds
22. Clustering I: K-Means and RFM Customer Segmentation
23. Clustering II and Dimensionality Reduction: Hierarchical, DBSCAN, Gaussian Mixtures, PCA, t-SNE and UMAP
24. Time-Series Forecasting: Decomposition, Baselines, ETS, ARIMA and ML with Lag Features
25. NLP and Text Analytics: Cleaning, TF-IDF, Sentiment Classification, Topics and Embeddings
26. Neural Networks and Deep Learning: From a Neuron to Transformers
27. Explainable AI and MLOps: SHAP, Fairness, Deployment, Monitoring and Responsible ML
28. ML Capstone: End-to-End Churn Prediction and Retention Strategy

STATISTICS & MACHINE LEARNING · LESSON 1

Descriptive Statistics: Centre, Spread, Shape and Relationships

Describe data with numbers before modelling it. Types of data and measurement scales, mean, median and mode, range, variance, standard deviation, IQR and coefficient of variation, percentiles and z-scores, skewness and kurtosis, covariance and correlation (Pearson and Spearman), all on the house-price and HR datasets.

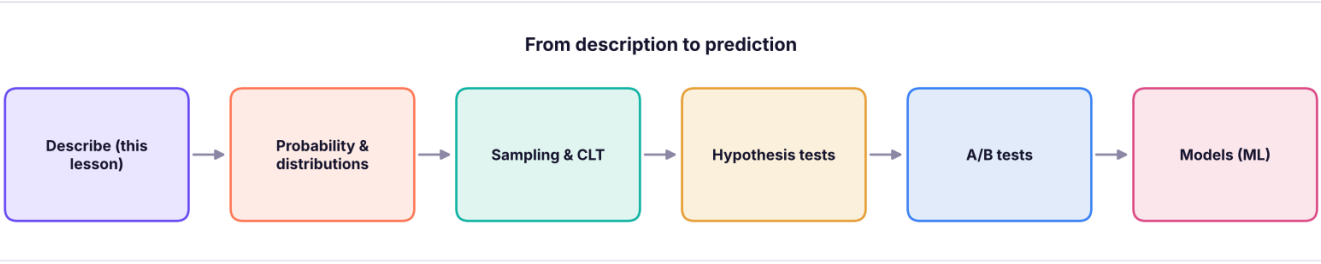
Python class 14 · pandas 3 · SciPy 1.1x · Beginner · 100 min

By the end of this lesson you can

- Classify variables as categorical (nominal, ordinal) or numeric (discrete, continuous; interval, ratio)
- Compute and interpret mean, median, mode, weighted mean and trimmed mean
- Compute and interpret range, IQR, variance, standard deviation and coefficient of variation
- Use percentiles and z-scores and describe skewness and kurtosis
- Measure association with covariance, Pearson and Spearman correlation, and explain why correlation is not causation

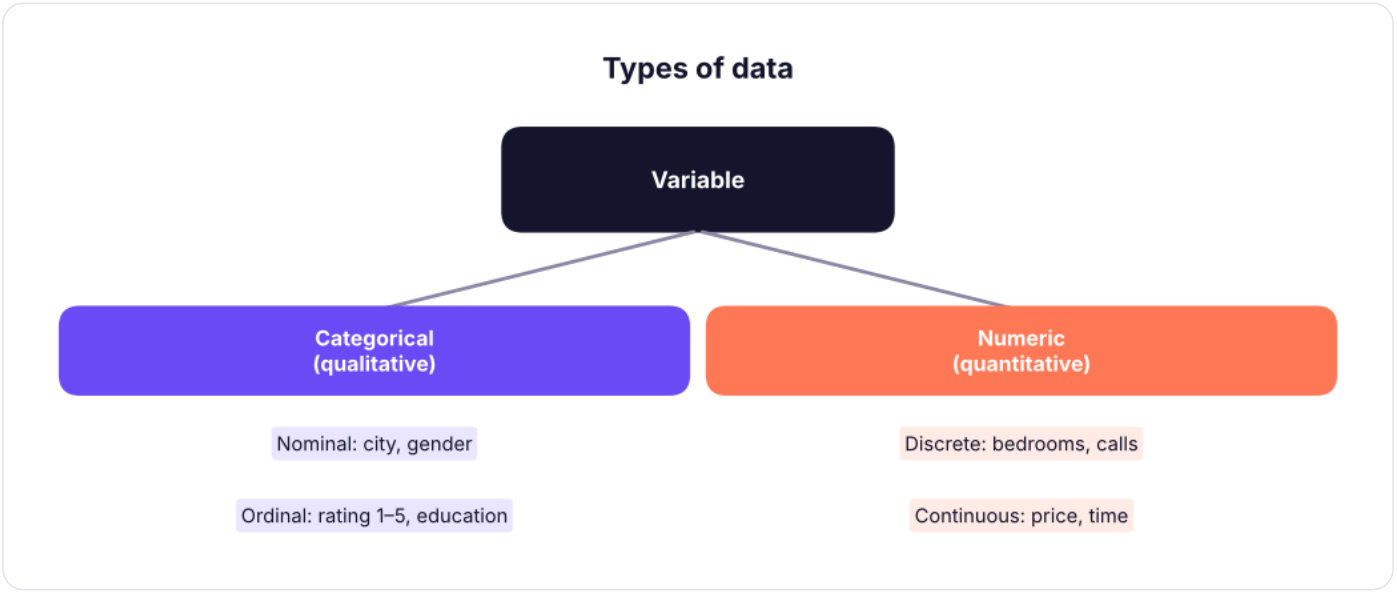
Statistics in data science

Descriptive statistics summarise the data you have. **Inferential statistics** use a sample to draw conclusions about a larger population (next lessons). Machine learning builds on both.



The statistics roadmap for this track.

Types of data



Variable types decide which statistics and charts make sense.

Scale	Has order?	Equal intervals?	True zero?	Example	Allowed statistics
Nominal	No	–	–	City	Mode, counts
Ordinal	Yes	No	–	Satisfaction 1–4	Median, percentiles
Interval	Yes	Yes	No	Temperature °C	Mean, sd (no ratios)
Ratio	Yes	Yes	Yes	Income, price	Everything, including ratios

Measures of centre

Python

```python
import pandas as pd, numpy as np
from scipy import stats
h = pd.read_csv("house_prices.csv")
p = h["price_lakh"]
print(f"mean {p.mean():.1f} | median {p.median():.1f} | mode (rounded to 10) {(p.round(-1)).mode()
[0]:.0f} | trimmed mean 10% {stats.trim_mean(p, 0.1):.1f}")
print("bedrooms mode:", h["bedrooms"].mode().tolist(), "| city mode:", h["city"].mode()[0])
w = pd.DataFrame({"city_avg_price": [62, 88, 75], "houses": [100, 20, 50]})
print("simple mean of city averages:", w.city_avg_price.mean().round(1), "| weighted mean:",
np.average(w.city_avg_price, weights=w.houses).round(1))
```

OUTPUT

```
mean 80.4 | median 78.9 | mode (rounded to 10) 80 | trimmed mean 10% 78.9
bedrooms mode: [2] | city mode: Bengaluru
simple mean of city averages: 75.0 | weighted mean: 68.9
```

Measure	Use when	Sensitive to outliers?
Mean	Symmetric numeric data; totals matter	Yes
Median	Skewed data (income, prices), ordinal data	No
Mode	Categories, most common value	No
Trimmed mean	Want a mean but robust to extremes	Less
Weighted mean	Groups of different sizes	Depends

Python

```
salaries = pd.Series([40, 42, 45, 47, 50, 52, 55, 60, 65, 900])      # one CEO in thousands
print(f"mean {salaries.mean():.1f}K vs median {salaries.median():.1f}K: one extreme value moves the
mean, not the median")
```

OUTPUT

```
mean 135.6K vs median 51.0K: one extreme value moves the mean, not the median
```

Measures of spread

Python

```
def spread(x):
    q1, q3 = x.quantile([.25, .75])
    return pd.Series({"range": x.max() - x.min(), "IQR": q3 - q1, "variance": x.var(), "std": x.std(),
"CV_%": 100 * x.std() / x.mean(), "MAD": (x - x.median()).abs().median()})
print(pd.DataFrame({"price_lakh": spread(p), "area_sqft": spread(h["area_sqft"].dropna()), "age_years":
spread(h["age_years"])}).round(1))
```

OUTPUT

```
          price_lakh  area_sqft  age_years
range          518.0     2130.0       29.0
IQR             42.2      580.0       15.0
variance      1261.9   158572.6       74.2
std             35.5      398.2        8.6
CV_%            44.2       32.1       59.8
MAD             21.0      290.0        7.0
```

- **Variance** = average squared distance from the mean; **standard deviation** (sd) = its square root, in the original units.
- pandas `.std()` uses the **sample** formula (divide by *n* − *1*, Bessel's correction); NumPy `np.std` defaults to population (*n*).
- **Coefficient of variation** (CV = sd ÷ mean) compares spread across variables with different units.
- **IQR** (Q3 − Q1) is the spread of the middle 50%, robust to outliers.

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Percentiles, z-scores and the five-number summary

Python

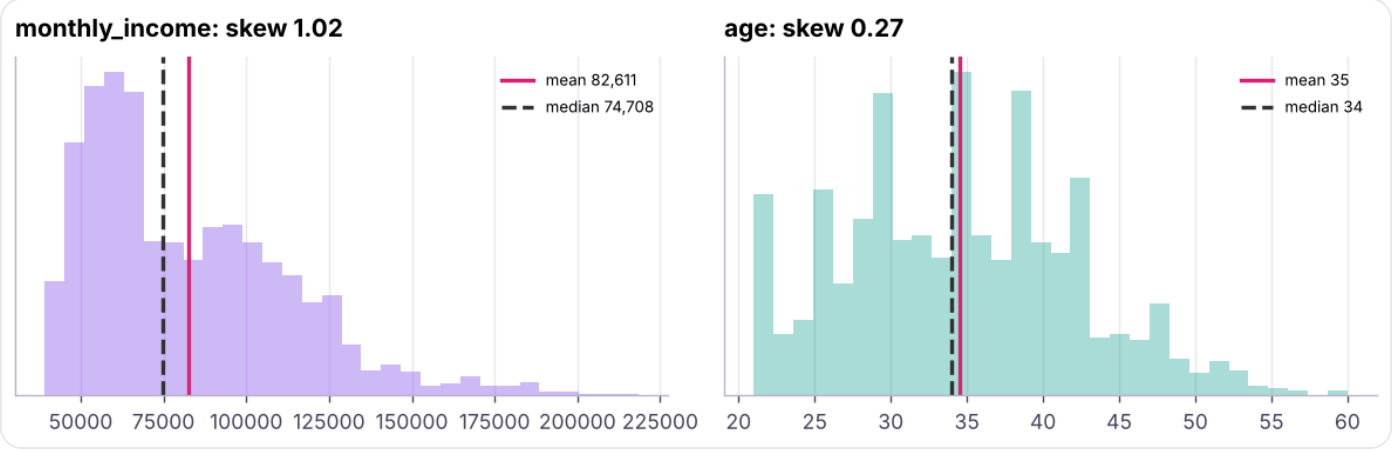
```
print(p.quantile([0, .1, .25, .5, .75, .9, 1]).round(1).to_dict())
z = (p - p.mean()) / p.std()
print(f"a ₹150 lakh house has z = {(150 - p.mean()) / p.std():.2f} | {(z.abs() > 2).mean():.1%} of
houses are beyond ±2 sd")
print(f"percentile rank of ₹150 lakh: {stats.percentileofscore(p, 150):.1f}")
```

OUTPUT

```
{0.0: 2.0, 0.1: 40.5, 0.25: 57.3, 0.5: 78.9, 0.75: 99.4, 0.9: 120.1, 1.0: 520.0}
a ₹150 lakh house has z = 1.96 | 1.7% of houses are beyond ±2 sd
percentile rank of ₹150 lakh: 98.2
```

A **z-score** says how many standard deviations a value is from the mean: it puts different variables on one scale (used in standardisation for ML).

Shape: skewness and kurtosis



Right-skewed (income), roughly symmetric (age) and the mean/median positions.

Python

```
for col in ["monthly_income", "age", "years_at_company", "distance_from_home_km"]:
    x = hr[col]; print(f"{col:24s} skew {x.skew():6.2f} | excess kurtosis {x.kurt():6.2f} | mean
{x.mean():10,.1f} | median {x.median():10,.1f}")
print("log(income) skew:", round(np.log(hr.monthly_income).skew(), 2))
```

OUTPUT

```
monthly_income           skew   1.02 | excess kurtosis   0.95 | mean   82,610.7 | median
74,708.5
age                      skew   0.27 | excess kurtosis  -0.39 | mean       34.5 | median
34.0
years_at_company         skew   1.55 | excess kurtosis   3.00 | mean        3.8 | median
3.0
distance_from_home_km    skew   0.00 | excess kurtosis  -1.26 | mean       15.2 | median
15.0
log(income) skew: 0.28
```

Skewness	Shape	Mean vs median
≈ 0	Symmetric	Mean ≈ median
> 0	Right (positive) skew, long right tail	Mean > median
< 0	Left skew	Mean < median

Kurtosis describes tail heaviness (excess kurtosis 0 = normal; positive = more extreme values). A log transform often makes right-skewed money variables more symmetric.

Relationships: covariance and correlation

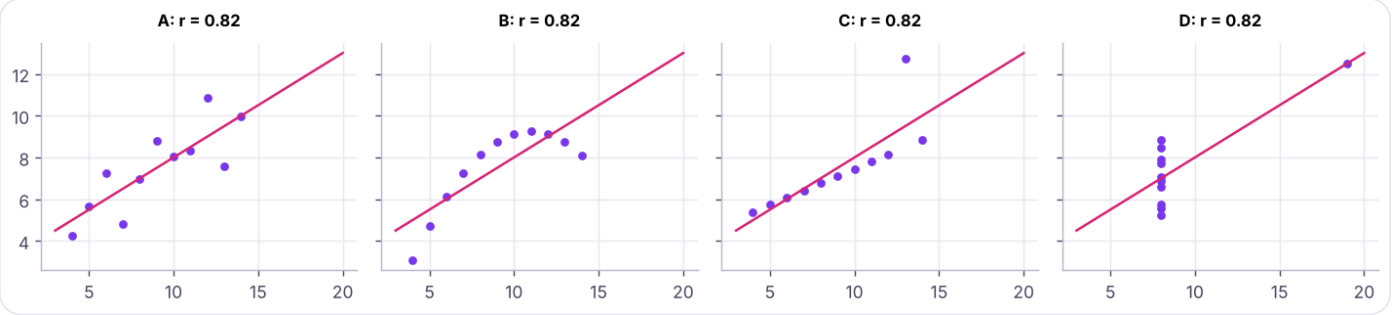
Python

```python
num = h[["price_lakh", "area_sqft", "bedrooms", "age_years", "distance_to_center_km"]].dropna()
print("covariance price-area:", round(num.price_lakh.cov(num.area_sqft), 1), "(units: lakh × sq ft,
hard to interpret)")
print(num.corr().round(2)["price_lakh"].sort_values(ascending=False))
print("Spearman (rank):", num.corr(method="spearman").round(2)
["price_lakh"].drop("price_lakh").to_dict())
r, pval = stats.pearsonr(num.area_sqft, num.price_lakh); print(f"Pearson r = {r:.3f}, p-value =
{pval:.1e}")
```

OUTPUT

```
covariance price-area: 11182.8 (units: lakh × sq ft, hard to interpret)
price_lakh               1.00
area_sqft                0.79
bedrooms                 0.58
distance_to_center_km   -0.07
age_years               -0.11
Name: price_lakh, dtype: float64
Spearman (rank): {'area_sqft': 0.9, 'bedrooms': 0.68, 'age_years': -0.13,
'distance_to_center_km': -0.06}
Pearson r = 0.787, p-value = 3.3e-248
```

- **Pearson r** measures linear association (−1 to +1). Rough guide: |r| < 0.3 weak, 0.3–0.7 moderate, > 0.7 strong.
- **Spearman ρ** correlates ranks: robust to outliers and captures monotonic non-linear relations.
- **Cramér's V** (categorical × categorical) and point-biserial (binary × numeric) cover other pairs.



Anscombe's quartet: four datasets with the same mean, variance and correlation (r ≈ 0.82) but completely different shapes. Always plot your data.

WATCH OUT **Correlation is not causation**

Ice-cream sales and drowning both rise in summer (a **confounder**: temperature). Correlation can also come from reverse causation, selection bias or chance. Only experiments or careful causal methods show cause.

Summary

KEY TAKEAWAYS

- Know the variable type and scale first: it decides valid statistics.
- Centre: mean (sensitive), median (robust), mode (categories); use weighted means for groups of different sizes.
- Spread: range, IQR (robust), variance and sd (sample uses n − 1), CV to compare variables.
- Shape: skewness (mean vs median), kurtosis (tails); log transforms reduce right skew.
- z-scores standardise; percentiles rank. Correlation (Pearson linear, Spearman rank) measures association, not causation; always plot.

Quick quiz

6 questions · tap an answer to check it

1. **Which is the best measure of centre for household income?**

- **A** Mean
- **B** Median
- **C** Mode
- **D** Range

Answer: B. Income is right-skewed; the median resists extremes.

2. **In a right-skewed distribution, which is usually true?**

- **A** Mean > median
- **B** Mean < median
- **C** Mean = median
- **D** Mode > mean

Answer: A. The long right tail pulls the mean up.

3. **Why does the sample standard deviation divide by n − 1?**

- **A** To make it larger for fun
- **B** To correct the bias from estimating the mean from the same sample (Bessel's correction)
- **C** Because n is unknown
- **D** It is a convention with no reason

Answer: B. It gives an unbiased variance estimate.

4. A z-score of −1.5 means...

A The value is 1.5 standard deviations below the mean

B The value is negative

C The value is in the 15th percentile exactly

D An error

Answer: A. z = (x − mean) / sd.

5. Which correlation suits a monotonic but non-linear relationship?

A Pearson

B Spearman

C Covariance

D None

Answer: B. Spearman uses ranks.

6. What does Anscombe's quartet teach?

A Correlation proves causation

B Identical summary statistics can hide very different patterns, so plot the data

C Means are useless

D Outliers never matter

Answer: B. Visualise before you summarise.

EXERCISE **Practice (hr_employees.csv)**

(1) Mean, median and sd of `monthly_income` by department; which department is most variable (CV)? (2) The 90th percentile of income. (3) The share of employees beyond ±3 sd of income. (4) Pearson and Spearman correlation between age and income. (5) Classify each column by type and scale.

Show solution (worked answer)

Python

```
g = hr.groupby("department")["monthly_income"].agg(["mean", "median", "std"]); g["cv"] = g["std"]
/ g["mean"]; print(g.sort_values("cv"))
print(hr.monthly_income.quantile(.9))
zi = (hr.monthly_income - hr.monthly_income.mean()) / hr.monthly_income.std(); print((zi.abs() >
3).mean())
print(hr[["age", "monthly_income"]].corr().iloc[0, 1], hr[["age",
"monthly_income"]].corr("spearman").iloc[0, 1])
# gender, department, city: nominal; education, job_satisfaction, performance_rating: ordinal;
age, income: ratio
```

INTERVIEW CORNER **Questions you may be asked**

Q: Mean vs median: when do you report each? Median for skewed data or outliers (income, house prices, delivery times); mean for symmetric data or when totals matter.

Q: What does a standard deviation of 0 mean? All values are identical.

Q: Correlation is 0. Are the variables independent? Not necessarily: Pearson only captures linear relationships; a perfect U-shape can have r = 0.

STATISTICS & MACHINE LEARNING · LESSON 2

Probability and Distributions: Rules, Bayes, Binomial, Poisson and Normal

The language of uncertainty. Probability rules, conditional probability and Bayes' theorem (with a fraud-alert example), random variables and expected value, and the distributions analysts use most (Bernoulli, binomial, Poisson, uniform, normal, exponential), with scipy.stats and simulation.

Python class 14 · SciPy 1.1x · NumPy 2 · Beginner · 110 min

By the end of this lesson you can

- Apply the addition, multiplication and complement rules and conditional probability
- Use Bayes' theorem to update probabilities and explain base-rate effects
- Compute expected value and variance of a random variable
- Model counts and measurements with binomial, Poisson, normal and exponential distributions using scipy.stats
- Use the empirical (68-95-99.7) rule and check normality with a Q-Q plot

Probability basics

Probability is a number from 0 (impossible) to 1 (certain). For equally likely outcomes, P(event) = favourable outcomes ÷ all outcomes. In data, it is a long-run frequency.

Python

```
import numpy as np, pandas as pd
from scipy import stats
rng = np.random.default_rng(7)
rolls = rng.integers(1, 7, 100_000)
print("P(six) theory 0.1667 | simulated", round((rolls == 6).mean(), 4))
print("P(even or > 4) theory", round(4 / 6, 4), "| simulated", round(((rolls % 2 == 0) | (rolls >
4)).mean(), 4))
```

OUTPUT

```
P(six) theory 0.1667 | simulated 0.1682
P(even or > 4) theory 0.6667 | simulated 0.6679
```

Rule	Formula	Example
Complement	P(not A) = 1 − P(A)	P(no conversion) = 1 − 0.10
Addition	P(A or B) = P(A) + P(B) − P(A and B)	Even or > 4: 3/6 + 2/6 − 1/6
Multiplication (independent)	P(A and B) = P(A) × P(B)	Two sixes: 1/36
Conditional	P(A | B) = P(A and B) ÷ P(B)	P(churn | month-to-month)

Python

```
tc = pd.DataFrame({"contract": ["M2M"] * 1690 + ["1yr"] * 739 + ["2yr"] * 571,
                   "churn": [1] * 607 + [0] * 1083 + [1] * 125 + [0] * 614 + [1] * 40 + [0] * 531})   # counts from the churn EDA
joint = pd.crosstab(tc.contract, tc.churn, normalize="all").round(3)
print(joint)
print("P(churn) =", round(tc.churn.mean(), 3), "| P(churn | M2M) =", round(tc[tc.contract == "M2M"].churn.mean(), 3), "| P(M2M | churn) =", round((tc[tc.churn == 1].contract == "M2M").mean(), 3))
```

OUTPUT

```
churn         0      1
contract
1yr       0.205  0.042
2yr       0.177  0.013
M2M       0.361  0.202
P(churn) = 0.257 | P(churn | M2M) = 0.359 | P(M2M | churn) = 0.786
```

P(churn | M2M) and P(M2M | churn) are **different** questions. Mixing them up is one of the most common reasoning errors.

Bayes' theorem

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

A fraud model flags 99% of fraudulent payments and wrongly flags 2% of genuine ones. Only 0.5% of payments are fraud. If a payment is flagged, how likely is it to be fraud?

Python

```
p_fraud, sens, fpr = 0.005, 0.99, 0.02
p_flag = sens * p_fraud + fpr * (1 - p_fraud)                # law of total probability
p_fraud_given_flag = sens * p_fraud / p_flag
print(f"P(flag) = {p_flag:.4f} | P(fraud | flag) = {p_fraud_given_flag:.1%}")
n = 100_000; fraud = rng.random(n) < p_fraud
flag = np.where(fraud, rng.random(n) < sens, rng.random(n) < fpr)
print(f"simulated: {fraud[flag].mean():.1%} of flagged payments are fraud ({flag.sum():,} flags, {(fraud & flag).sum()} real fraud)")
```

OUTPUT

```
P(flag) = 0.0249 | P(fraud | flag) = 19.9%
simulated: 19.9% of flagged payments are fraud (2,485 flags, 495 real fraud)
```

Only about one flag in five is real fraud, because genuine payments vastly outnumber fraud (the **base rate**). This is why precision matters in imbalanced ML problems (lesson 21) and why naive Bayes classifiers work (lesson 20).

Random variables and expected value

A **random variable** assigns a number to each outcome. Its **expected value** is the probability-weighted average: the long-run mean.

Python

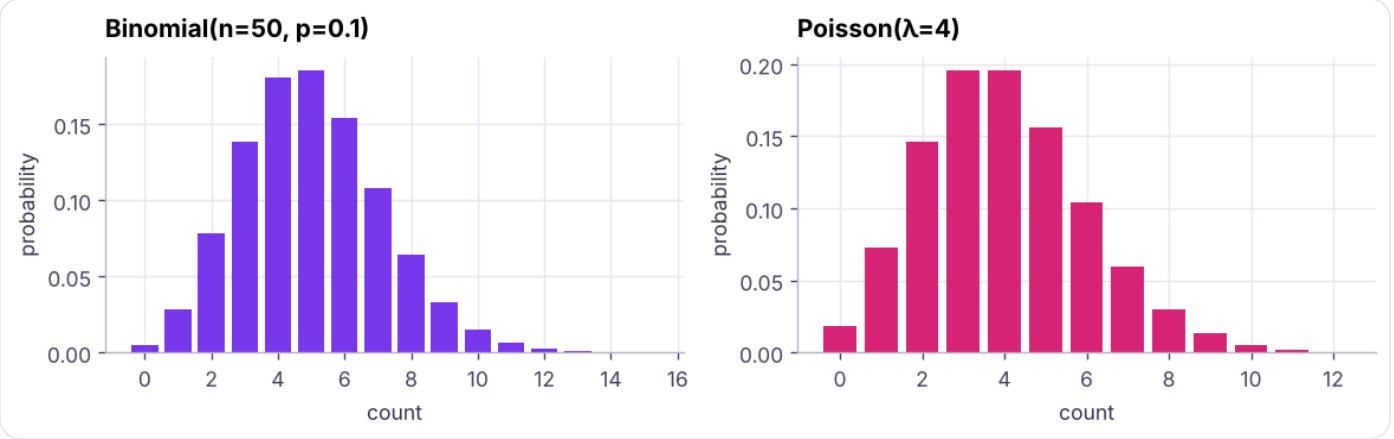
```python
values = np.array([0, 500, 2000]); probs = np.array([0.80, 0.15, 0.05])      # customer spend next month
ev = (values * probs).sum(); var = ((values - ev) ** 2 * probs).sum()
print(f"E[spend] = ₹{ev:.0f} | sd = ₹{var ** 0.5:.0f}")
cost, uplift = 60, 0.25                                                      # campaign: costs ₹60, raises spend by 25%
print(f"expected profit per customer from the campaign: ₹{ev * uplift - cost:.0f} → {'run it' if ev * uplift > cost else 'do not run it'}")
```

OUTPUT

```
E[spend] = ₹175 | sd = ₹455
expected profit per customer from the campaign: ₹-16 → do not run it
```

Expected value drives decisions: pricing, insurance, campaign ROI, and the "expected loss" in credit risk (PD × LGD × EAD).

Discrete distributions



Binomial (number of conversions out of 50 visitors, p = 0.1) and Poisson (support tickets per hour, mean 4).

Distribution	Models	Parameters	Mean	Variance
Bernoulli	One yes/no trial (did this user convert?)	p	p	p(1 − p)
Binomial	Number of successes in n trials	n, p	np	np(1 − p)
Poisson	Count of events in an interval (calls/hour, defects/batch)	λ	λ	λ
Geometric	Trials until first success	p	1/p	(1 − p)/p²

Python

```
b = stats.binom(n=50, p=0.10)
print(f"P(exactly 5) = {b.pmf(5):.3f} | P(at most 2) = {b.cdf(2):.3f} | P(10 or more) = {b.sf(9):.4f} |
mean {b.mean()} sd {b.std():.2f}")
po = stats.poisson(mu=4)
print(f"tickets/hour: P(0) = {po.pmf(0):.3f} | P(more than 8) = {po.sf(8):.4f} | 95th percentile =
{po.ppf(0.95):.0f}")
ab = pd.read_csv("ab_test.csv"); ctrl = ab[ab.group == "A_control"]
print(f"A/B control: {ctrl.converted.sum()} conversions from {len(ctrl)} users, p̂ =
{ctrl.converted.mean():.3f}")
```

OUTPUT

```
P(exactly 5) = 0.185 | P(at most 2) = 0.112 | P(10 or more) = 0.0245 | mean 5.0 sd 2.12
tickets/hour: P(0) = 0.018 | P(more than 8) = 0.0214 | 95th percentile = 8
A/B control: 198 conversions from 2000 users, p̂ = 0.099
```

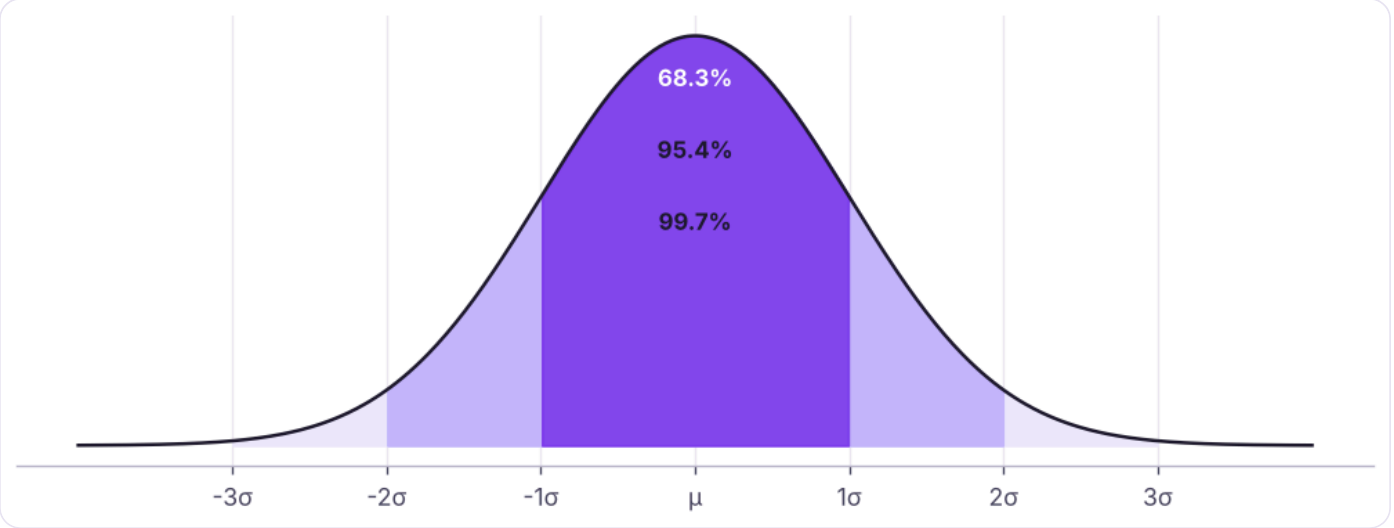
scipy.stats methods: `pmf` / `pdf` (probability or density at a point), `cdf` (P(X ≤ x)), `sf` (P(X > x) = 1 − cdf), `ppf` (inverse cdf: percentiles), `rvs` (random samples).

Continuous distributions

For continuous variables, probability is the **area under the density curve**; P(X = exact value) is 0.

The normal distribution

Bell-shaped, symmetric, described by mean μ and standard deviation σ. Heights, measurement errors and, crucially, **sample means** (next lesson) are approximately normal.



The empirical rule: about 68%, 95% and 99.7% of a normal distribution lie within 1, 2 and 3 standard deviations of the mean.

Python

```
d = pd.read_csv("delivery_times.csv"); pune = d[d.city == "Pune"].delivery_minutes
mu, sd = pune.mean(), pune.std()
nd = stats.norm(mu, sd)
print(f"Pune delivery: mean {mu:.1f} min, sd {sd:.1f}")
print(f"P(late, > 40 min) model {nd.sf(40):.1%} vs data {(pune > 40).mean():.1%}")
print(f"promise time covering 95% of orders: {nd.ppf(0.95):.1f} min | within ±1 sd: {((pune - mu).abs()
⩽ sd).mean():.0%} of orders")
print("68-95-99.7 check:", [round(float(stats.norm.cdf(k) - stats.norm.cdf(-k)), 4) for k in (1, 2,
3)], "| z for 95% two-sided:", round(stats.norm.ppf(0.975), 3))
```

OUTPUT

```
Pune delivery: mean 31.7 min, sd 6.4
P(late, > 40 min) model 9.7% vs data 13.3%
promise time covering 95% of orders: 42.2 min | within ±1 sd: 68% of orders
68-95-99.7 check: [0.6827, 0.9545, 0.9973] | z for 95% two-sided: 1.96
```

The **standard normal** has μ = 0, σ = 1; any normal value converts with z = (x − μ)/σ. The number **1.96** (95% two-sided) appears in confidence intervals and tests.

Other continuous distributions

Distribution	Models	Example
Uniform(a, b)	Every value equally likely	Random number generators, A/B bucket assignment
Exponential(λ)	Waiting time between Poisson events	Time between support calls
Log-normal	Positive, right-skewed values	Income, house prices, order values
t, χ^2, F	Sampling distributions for tests	t-tests, chi-square tests, ANOVA

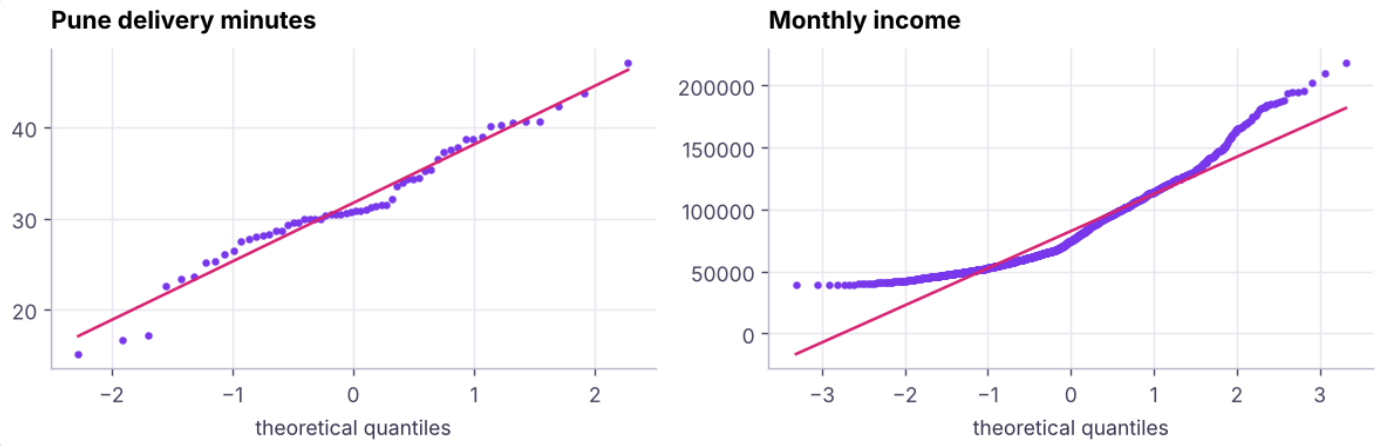
Python

```
wait = stats.expon(scale=1 / 4)                    # 4 calls per hour → mean wait 15 minutes
print(f"P(no call for 30+ minutes) = {wait.sf(0.5):.3f} | memoryless: P(wait > 45 | already 15) =
{wait.sf(0.75) / wait.sf(0.25):.3f}")
```

OUTPUT

```
P(no call for 30+ minutes) = 0.135 | memoryless: P(wait > 45 | already 15) = 0.135
```

Is it normal? Q-Q plots and tests



Q-Q plots compare data quantiles with normal quantiles: points on the line mean "approximately normal". Pune delivery times look normal; HR income is right-skewed.

Python

```
for name, data in [("delivery (Pune)", pune), ("income", hr.monthly_income), ("log income",
np.log(hr.monthly_income))]:
    print(f"{name:16s} Shapiro-Wilk p = {stats.shapiro(data.sample(min(len(data), 500),
random_state=0)).pvalue:.4f} | skew {stats.skew(data):.2f}")
```

OUTPUT

```
delivery (Pune)  Shapiro-Wilk p = 0.1629 | skew -0.15
income           Shapiro-Wilk p = 0.0000 | skew 1.02
log income       Shapiro-Wilk p = 0.0000 | skew 0.28
```

A small p-value (< 0.05) suggests the data are **not** normal. With large samples, tiny departures become "significant", so rely on plots and skewness too.

Summary

KEY TAKEAWAYS

- Probability rules: complement, addition, multiplication (independence), conditional P(A | B) ≠ P(B | A).
- Bayes' theorem updates beliefs; low base rates make most alerts false positives.
- Expected value = probability-weighted average, the basis of ROI and risk decisions.
- Binomial (successes in n trials), Poisson (counts per interval), normal (measurements, sample means), exponential (waiting times); use `scipy.stats` `pmf/pdf`, `cdf`, `sf`, `ppf`, `rvs`.
- Empirical rule 68-95-99.7; z = (x − μ)/σ; check normality with Q-Q plots and skewness.

Quick quiz

6 questions · tap an answer to check it

1. A fraud test is 99% sensitive with a 2% false-positive rate, and fraud is 0.5% of payments. Roughly what share of flagged payments are fraud?

A 99%

B 50%

C About 20%

D 2%

Answer: C. The low base rate means most flags are genuine payments.

2. Which distribution models the number of support calls per hour?

A Normal

B Poisson

C Uniform

D Binomial with n = 1

Answer: B. Counts of events in a fixed interval.

3. For a binomial with n = 50 and p = 0.1, what is the mean?

A 0.1

B 5

C 50

D 4.5

Answer: B. Mean = np.

4. About what share of a normal distribution lies within ±2 standard deviations?

A 68%

B 95%

C 99.7%

D 50%

Answer: B. The empirical rule.

5. Which scipy method gives P(X > x)?

A pdf

B cdf

C sf

D ppf

Answer: C. sf = survival function = 1 − cdf.

6. P(churn | month-to-month) equals P(month-to-month | churn).

A True

B False

Answer: B. They condition on different groups.

EXERCISE **Practice**

(1) Probability of at least one conversion from 20 visitors if p = 0.08. (2) A call centre averages 6 calls per 10 minutes; probability of more than 10 calls in 10 minutes. (3) Hyderabad delivery times: estimate P(> 45 min) with a normal model and compare with the data. (4) Simulate the Bayes fraud example with fraud at 2% instead of 0.5%. What changes?

Show solution (worked answer)

Python

```
print(1 - 0.92 ** 20, stats.binom.sf(0, 20, 0.08))
print(stats.poisson.sf(10, 6))
hyd = d[d.city == "Hyderabad"].delivery_minutes; print(stats.norm(hyd.mean(), hyd.std()).sf(45),
(hyd > 45).mean())
p_fraud = 0.02; print(sens * p_fraud / (sens * p_fraud + fpr * (1 - p_fraud)))     # about 50%:
the base rate matters
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain Bayes' theorem with an example. It updates a prior probability with new evidence. For a rare disease with an accurate test, a positive result can still be more likely a false positive because healthy people vastly outnumber sick ones.

Q: When would you use a Poisson distribution? For counts of independent events in a fixed time or space with a constant average rate, such as orders per minute, defects per batch or calls per hour.

Q: What is the difference between pdf and cdf? The pdf gives density at a point (probability for discrete pmf); the cdf gives the cumulative probability P(X ≤ x).

STATISTICS & MACHINE LEARNING · LESSON 3

Sampling, the Central Limit Theorem and Confidence Intervals

Learn how a sample tells us about a population. Population vs sample, sampling methods and biases, sampling distributions and standard error, the Central Limit Theorem by simulation, confidence intervals for means and proportions (z, t and bootstrap), and how to choose a sample size.

● Python classes 14–15 · ML class 1 ✦ SciPy 1.1x · statsmodels 0.1x Intermediate ⏱ 110 min

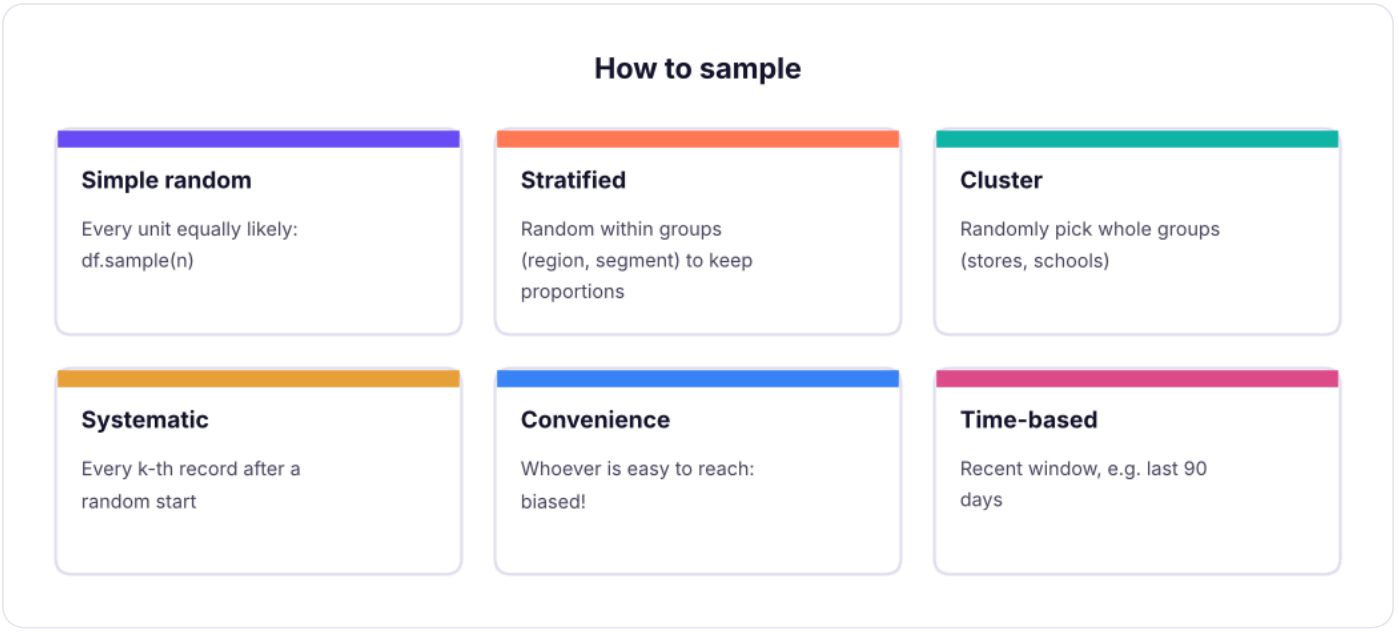
By the end of this lesson you can

- Distinguish population and sample, parameter and statistic, and name common sampling methods and biases
- Explain sampling distributions and standard error
- Demonstrate the Central Limit Theorem by simulation and state when it applies
- Compute and interpret confidence intervals for a mean and a proportion, including the bootstrap
- Estimate the sample size needed for a target margin of error

Population and sample

Term	Population	Sample
What	Everyone/everything of interest	The part you observe
Summary	**Parameter** (μ, σ, p)	**Statistic** ($\bar{x}$, s, $\hat{p}$)
Example	All 1,500 employees	100 surveyed employees

We almost never see the population (all customers, all future orders), so we estimate parameters from samples and state our **uncertainty**.



Sampling methods.

Python

```
import pandas as pd, numpy as np
from scipy import stats
hr = pd.read_csv("hr_employees.csv")
pop_mean, pop_rate = hr.monthly_income.mean(), (hr.attrition == "Yes").mean()
print(f"population: mean income {pop_mean:,.0f}, attrition {pop_rate:.1%}, N = {len(hr)}")
srs = hr.sample(150, random_state=1)
strat = hr.groupby("department", group_keys=False).sample(frac=0.1, random_state=1)     # stratified
10%
print(f"simple random n=150: mean {srs.monthly_income.mean():,.0f}, attrition {(srs.attrition ==
'Yes').mean():.1%}")
print(f"stratified n={len(strat)}: mean {strat.monthly_income.mean():,.0f}; department shares kept:",
      (strat.department.value_counts(normalize=True) -
hr.department.value_counts(normalize=True)).abs().max().round(3) < 0.01)
```

OUTPUT

```
population: mean income 82,611, attrition 25.9%, N = 1500
simple random n=150: mean 85,587, attrition 21.3%
stratified n=149: mean 81,315; department shares kept: True
```

WATCH OUT **Sampling bias beats sample size**

A huge biased sample is worse than a small random one. Watch for **selection bias** (only app users surveyed), **non-response bias** (unhappy customers ignore surveys), **survivorship bias** (studying only companies that still exist) and **time bias** (a festive week is not a normal week).

Sampling distributions and standard error

Take many samples and compute the mean of each: those means form the **sampling distribution**. Its standard deviation is the **standard error (SE)**.

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} \qquad SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Python

```
rng = np.random.default_rng(0)
inc = hr.monthly_income.to_numpy()
for n in (10, 40, 160, 640):
    means = [rng.choice(inc, n).mean() for _ in range(3000)]
    print(f"n={n:4d}: sd of sample means {np.std(means):7,.0f} | formula σ/√n {inc.std() /
np.sqrt(n):7,.0f}")
```

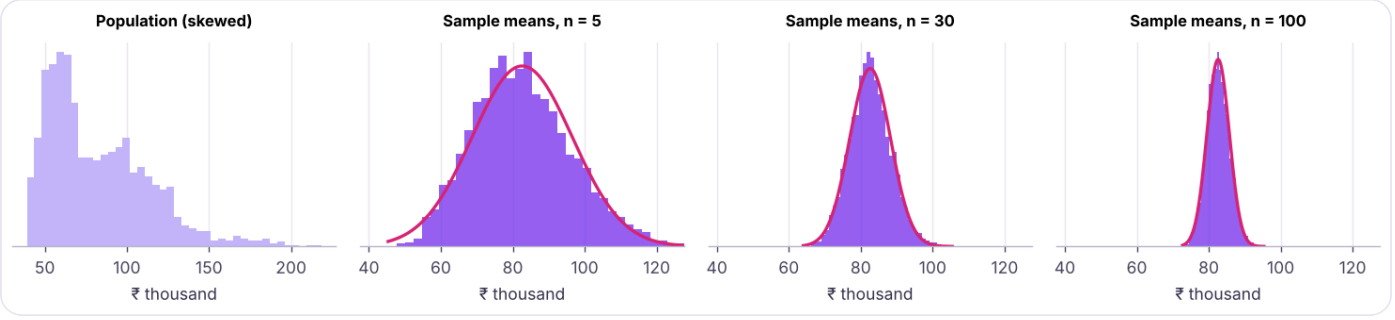
OUTPUT

```
n=  10: sd of sample means   9,854 | formula σ/√n   9,846
n=  40: sd of sample means   4,970 | formula σ/√n   4,923
n= 160: sd of sample means   2,492 | formula σ/√n   2,461
n= 640: sd of sample means   1,232 | formula σ/√n   1,231
```

Quadrupling the sample size halves the standard error: precision grows with √n, so doubling precision costs four times the data.

The Central Limit Theorem

CLT: for a large enough sample (rule of thumb n ≥ 30, more for very skewed data), the sampling distribution of the mean is approximately **normal**, whatever the shape of the population, with mean μ and SE σ/√n.



Income is right-skewed (left), yet the means of samples of 5, 30 and 100 employees become ever more normal and narrow.

The CLT is why z- and t-tests, confidence intervals and A/B tests work on non-normal business data such as revenue.

Confidence intervals

A **95% confidence interval** is a range built by a method that captures the true parameter in 95% of repeated samples.

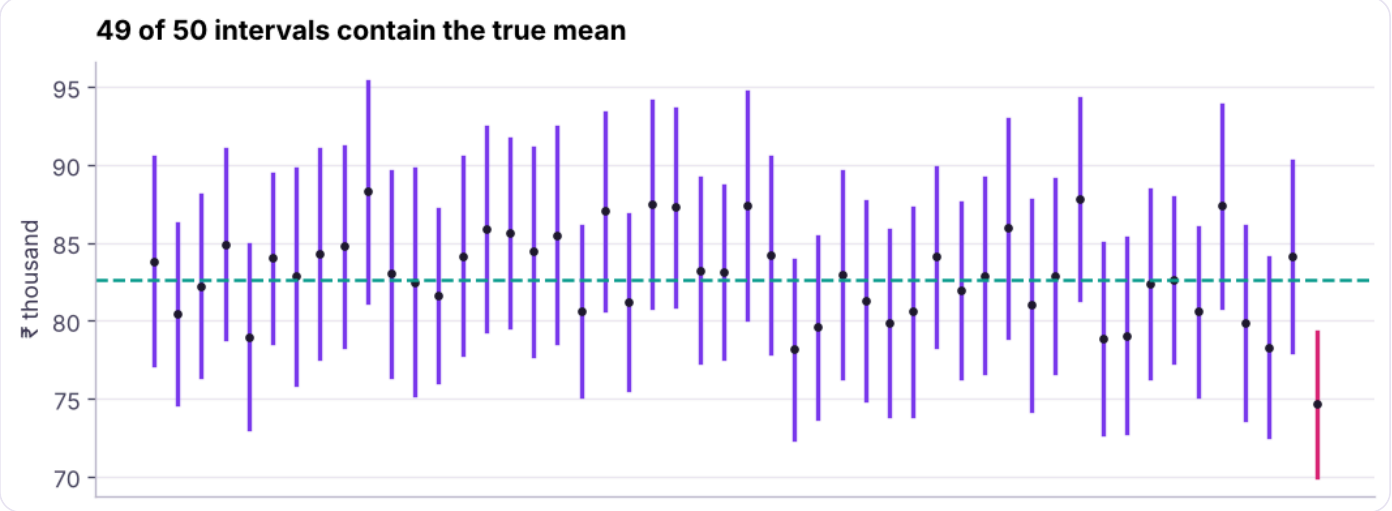
$$\bar{x} \pm t_{\alpha/2,\, n-1} \frac{s}{\sqrt{n}} \qquad \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Python

```
sample = hr.sample(100, random_state=3)
x = sample.monthly_income
se = x.std() / np.sqrt(len(x)); tcrit = stats.t.ppf(0.975, df=len(x) - 1)
lo, hi = x.mean() - tcrit * se, x.mean() + tcrit * se
print(f"mean {x.mean():,.0f} | 95% CI {lo:,.0f} to {hi:,.0f} | t* = {tcrit:.3f} | true mean
{pop_mean:,.0f} inside? {lo ⩽ pop_mean ⩽ hi}")
print("scipy:", [round(v) for v in stats.t.interval(0.95, len(x) - 1, loc=x.mean(), scale=se)])
k = (sample.attrition == "Yes").sum(); n = len(sample); ph = k / n
z = stats.norm.ppf(0.975)
print(f"attrition p̂ = {ph:.2f}, 95% CI (Wald) {ph - z * np.sqrt(ph * (1 - ph) / n):.3f} to {ph + z *
np.sqrt(ph * (1 - ph) / n):.3f}")
from statsmodels.stats.proportion import proportion_confint
print("Wilson CI (better for small n or extreme p):", [round(v, 3) for v in proportion_confint(k, n,
method="wilson")])
```

OUTPUT

```
mean 82,653 | 95% CI 77,094 to 88,213 | t* = 1.984 | true mean 82,611 inside? True
scipy: [77094, 88213]
attrition p̂ = 0.32, 95% CI (Wald) 0.229 to 0.411
Wilson CI (better for small n or extreme p): [0.237, 0.417]
```



50 samples of 100 employees, each with its 95% CI. Most intervals (purple) contain the true mean; about 1 in 20 (pink) misses.

NOTE **What a 95% CI does not mean**

It does **not** mean "there is a 95% probability that the true value is in this particular interval": the true value is fixed. It means the *procedure* captures the true value 95% of the time. In practice, say "we are 95% confident the average income is between … and …".

Wider interval when…	Narrower when…
Higher confidence (99% vs 95%)	Larger sample size
More variable data	Less variable data
Smaller sample	Lower confidence (90%)

Bootstrap confidence intervals

When there is no simple formula (median, ratios, percentiles), **resample the sample with replacement** many times and take percentiles of the statistic.

Python

```
boot = np.array([rng.choice(x.to_numpy(), len(x), replace=True) for _ in range(5000)])
med = np.median(boot, axis=1)
print(f"sample median {x.median():,.0f} | 95% bootstrap CI {np.percentile(med, 2.5):,.0f} to
{np.percentile(med, 97.5):,.0f}")
res = stats.bootstrap((x.to_numpy(),), np.median, confidence_level=0.95, n_resamples=5000,
random_state=0, method="percentile")
print("scipy.stats.bootstrap:", [round(v) for v in (res.confidence_interval.low,
res.confidence_interval.high)])
```

OUTPUT

```
sample median 82,666 | 95% bootstrap CI 68,750 to 90,542
scipy.stats.bootstrap: [68746, 89750]
```

Sample size planning

Required n for a margin of error E (half-width of the CI):

$$n = \left(\frac{z\sigma}{E}\right)^2 \qquad n = \frac{z^2\, p(1-p)}{E^2}$$

Python

```
def n_for_mean(sigma, E, conf=0.95): return int(np.ceil((stats.norm.ppf(1 - (1 - conf) / 2) * sigma / E) ** 2))
def n_for_prop(E, p=0.5, conf=0.95): return int(np.ceil(stats.norm.ppf(1 - (1 - conf) / 2) ** 2 * p * (1 - p) / E ** 2))
print("income within ±₹5,000:", n_for_mean(inc.std(), 5000), "employees | ±₹2,000:", n_for_mean(inc.std(), 2000))
print("survey proportion within ±5 points:", n_for_prop(0.05), "| ±3 points:", n_for_prop(0.03), "| ±1 point:", n_for_prop(0.01))
```

OUTPUT

```
income within ±₹5,000: 149 employees | ±₹2,000: 931
survey proportion within ±5 points: 385 | ±3 points: 1068 | ±1 point: 9604
```

This is why opinion polls survey about 1,000 people (±3 points) and why A/B tests on small effects need tens of thousands of users (lesson 6).

Summary

KEY TAKEAWAYS

- Parameters describe populations; statistics estimate them from samples. Random (and stratified) sampling avoids bias; size cannot fix bias.
- The standard error σ/√n measures how much a statistic varies from sample to sample.
- The CLT makes sample means approximately normal for large n, whatever the population shape.
- CI for a mean: $\bar{x} \pm t \cdot s/\sqrt{n}$; *for a proportion:* $\hat{p} \pm z \cdot \sqrt{\hat{p}(1-\hat{p})/n}$ (Wilson for small samples); bootstrap for anything else.
- Plan sample size from the margin of error: halving the error needs four times the data.

Quick quiz

6 questions · tap an answer to check it

1. What happens to the standard error when the sample size quadruples?

A It quadruples

B It doubles

C It halves

D It stays the same

Answer: C. SE ∝ 1/√n.

2. What does the Central Limit Theorem say?

A All data are normal

B Sample means are approximately normal for large samples, whatever the population shape

C Samples must be normal

D Means equal medians

Answer: B. It underpins most tests on business data.

3. A 95% CI for mean spend is ₹820–₹910. Best interpretation?

A We are 95% confident the true mean spend lies between ₹820 and ₹910 (the method captures the truth 95% of the time)

B 95% of customers spend between ₹820 and ₹910

C The mean is 865 with certainty

D There is a 5% chance the sample is wrong

Answer: A. A CI is about the parameter, not individual values.

4. Which makes a confidence interval narrower?

A Higher confidence level

B Larger sample

C More variable data

D Smaller sample

Answer: B. More data, more precision.

5. How do you get a CI for a median without a formula?

- **A** Use the z-table
- **B** Bootstrap: resample with replacement and take percentiles
- **C** Use the mean instead
- **D** It is impossible

Answer: B. Works for almost any statistic.

6. Which bias occurs when only active app users answer a satisfaction survey?

- **A** Selection bias
- **B** Rounding bias
- **C** Recall bias only
- **D** No bias

Answer: A. The sample does not represent all customers.

EXERCISE **Practice**

(1) From `ab_test.csv`, compute a 95% CI for the conversion rate of each group. Do they overlap? (2) Bootstrap a 95% CI for the difference in mean order value between groups. (3) How many users per group would you need to estimate a conversion rate within ±1 point?

Show solution (worked answer)

Python

```
ab = pd.read_csv("ab_test.csv")
for g, d in ab.groupby("group"): print(g, [round(v, 4) for v in
proportion_confint(d.converted.sum(), len(d), method="wilson")])
a, b = ab[ab.group == "A_control"].order_value.to_numpy(), ab[ab.group ==
"B_new_checkout"].order_value.to_numpy()
diffs = [rng.choice(b, len(b)).mean() - rng.choice(a, len(a)).mean() for _ in range(3000)]
print(np.percentile(diffs, [2.5, 97.5]))
print(n_for_prop(0.01, p=0.12))
```

INTERVIEW CORNER

Questions you may be asked

Q: Explain the Central Limit Theorem to a non-technical manager. If we average enough randomly chosen customers, those averages behave predictably (a bell curve) even when individual spending is wildly uneven, so we can say how confident we are in an average.

Q: What is the difference between standard deviation and standard error? Standard deviation measures spread of individual values; standard error measures spread of a statistic (like the mean) across samples: SE = sd/√n.

Q: How would you choose a sample size for a survey? Decide the margin of error and confidence level, estimate variability (use p = 0.5 for proportions if unknown), compute n, then inflate for expected non-response.

STATISTICS & MACHINE LEARNING · LESSON 4

Hypothesis Testing I: The Framework, p-values, z-tests and t-tests

Turn business questions into testable hypotheses. Null and alternative hypotheses, significance level, test statistics, p-values and critical values, one- vs two-tailed tests, Type I and II errors and power, then one-sample, two-sample (Welch) and paired t-tests and z-tests for proportions, on a delivery-time case study.

● ML classes 1–2 ✦ SciPy 1.1x · statsmodels 0.1x Intermediate ⏱ 120 min

By the end of this lesson you can

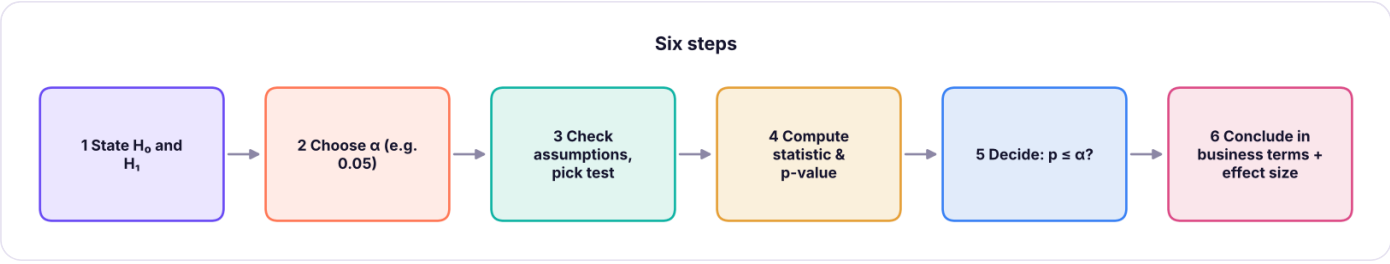
- Translate a business question into null and alternative hypotheses
- Explain significance level, test statistic, p-value and critical value, and decide with both
- Choose one- or two-tailed tests and describe Type I/II errors and power
- Run and interpret one-sample, independent (Welch) and paired t-tests and a z-test for proportions
- Report effect sizes and confidence intervals, not just p-values

From business question to hypothesis

Case. A quick-commerce company promises "delivery in 30 minutes". Operations claim Pune meets the promise on average. Is the average delivery time in Pune really 30 minutes, or is it slower? Is Hyderabad different from Pune?

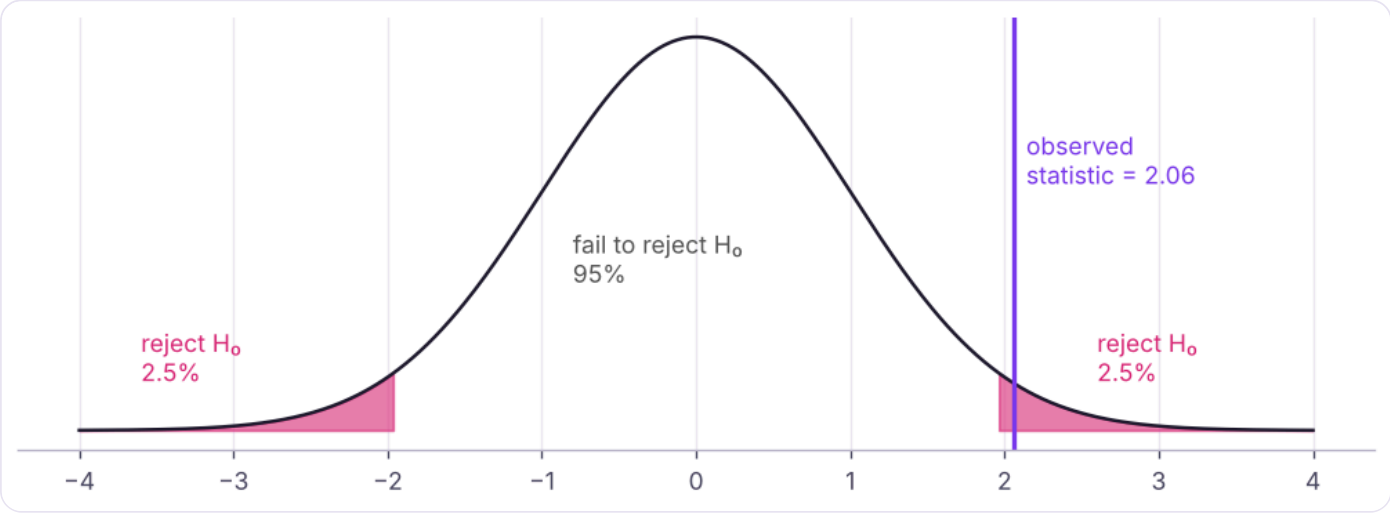
A **hypothesis test** checks whether data are consistent with a default claim.

Term	Meaning	Delivery case
H_0 (null)	No effect / status quo; assumed true until evidence	μ_Pune = 30 min
H_1 (alternative)	What we want evidence for	μ_Pune > 30 min (one-tailed) or ≠ 30 (two-tailed)
α (significance level)	Accepted risk of a false alarm	0.05
Test statistic	Distance from H_0 in standard-error units	$t = (\bar{x} - 30)/(s/\sqrt{n})$
p-value	P(data at least this extreme | H_0 true)	from the t distribution
Decision	$p \le \alpha$ → reject H_0; $p > \alpha$ → fail to reject H_0	



The hypothesis-testing procedure.

p-values and critical values



Two-tailed test at α = 0.05: the rejection regions are the 2.5% tails beyond ±1.96 (z). The p-value is the tail area beyond the observed statistic.

Python

```
print("critical values: two-tailed z ±", round(stats.norm.ppf(0.975), 3), "| one-tailed z",
round(stats.norm.ppf(0.95), 3), "| two-tailed t(df=59) ±", round(stats.t.ppf(0.975, 59), 3))
for z in (1.2, 1.7, 2.06, 3.1): print(f"z = {z}: two-tailed p = {2 * stats.norm.sf(abs(z)):.4f} | one-
tailed p = {stats.norm.sf(z):.4f}")
```

OUTPUT

```
critical values: two-tailed z ± 1.96 | one-tailed z 1.645 | two-tailed t(df=59) ± 2.001
z = 1.2: two-tailed p = 0.2301 | one-tailed p = 0.1151
z = 1.7: two-tailed p = 0.0891 | one-tailed p = 0.0446
z = 2.06: two-tailed p = 0.0394 | one-tailed p = 0.0197
z = 3.1: two-tailed p = 0.0019 | one-tailed p = 0.0010
```

WATCH OUT **What a p-value is not**

It is **not** the probability that H_0 is true, and 1 − p is not the probability that H_1 is true. A large p-value does not prove H_0 ("absence of evidence is not evidence of absence"). A tiny p-value does not mean a large or important effect.

One- vs two-tailed

	Two-tailed	One-tailed
H_1	$\mu \neq 30$ (any difference)	$\mu > 30$ (only "slower")
α split	2.5% in each tail	5% in one tail
When	Default; direction unknown	Direction fixed **before** looking at data, and the other direction is irrelevant

Errors and power

	H_o true	H_o false
Reject H_o	**Type I error** (false positive), probability α	Correct: **power** = $1 - \beta$
Fail to reject	Correct	**Type II error** (false negative), probability β

Fire alarm analogy: a Type I error rings without a fire; a Type II error stays silent during a fire. Lower α means fewer false alarms but more misses unless you collect more data. Power (commonly 80%) grows with sample size, effect size and α.

One-sample t-test: does Pune meet 30 minutes?

Python

```
import pandas as pd
d = pd.read_csv("delivery_times.csv")
pune, hyd = d[d.city == "Pune"].delivery_minutes, d[d.city == "Hyderabad"].delivery_minutes
print(f"Pune: n={len(pune)}, mean {pune.mean():.2f}, sd {pune.std():.2f}")
t_two = stats.ttest_1samp(pune, 30)
t_one = stats.ttest_1samp(pune, 30, alternative="greater")
se = pune.std() / np.sqrt(len(pune))
print(f"t = {t_two.statistic:.3f}, df = {len(pune) - 1} | two-tailed p = {t_two.pvalue:.4f} | one-tailed (greater) p = {t_one.pvalue:.4f}")
print(f"by hand: t = ({pune.mean():.2f} - 30) / {se:.3f} = {(pune.mean() - 30) / se:.3f}")
ci = stats.t.interval(0.95, len(pune) - 1, loc=pune.mean(), scale=se)
print(f"95% CI for mean: {ci[0]:.2f} to {ci[1]:.2f} minutes (30 is {'outside' if not ci[0] ⩽ 30 ⩽ ci[1] else 'inside'})")
print(f"effect size (Cohen's d) = {(pune.mean() - 30) / pune.std():.2f}")
```

OUTPUT

```
Pune: n=60, mean 31.70, sd 6.39
t = 2.060, df = 59 | two-tailed p = 0.0439 | one-tailed (greater) p = 0.0219
by hand: t = (31.70 - 30) / 0.825 = 2.060
95% CI for mean: 30.05 to 33.35 minutes (30 is outside)
effect size (Cohen's d) = 0.27
```

Conclusion. At $\alpha = 0.05$ we reject H_o: Pune's average delivery time (about 31.7 minutes) is statistically higher than the 30-minute promise ($p \approx 0.02$ one-tailed). The difference is small (under 2 minutes, $d \approx 0.27$), so the

business question is whether 1–2 minutes matter to customers, not just whether p < 0.05.

Two-sample t-test: Pune vs Hyderabad

Python

```
print(f"Hyderabad: n={len(hyd)}, mean {hyd.mean():.2f}, sd {hyd.std():.2f}")
lev = stats.levene(pune, hyd); print(f"Levene test for equal variances p = {lev.pvalue:.3f} →
{'unequal' if lev.pvalue < 0.05 else 'equal'} variances")
w = stats.ttest_ind(hyd, pune, equal_var=False)             # Welch's t-test: safe default
print(f"Welch t = {w.statistic:.3f}, p = {w.pvalue:.4f}, difference = {hyd.mean() - pune.mean():.2f}
min")
pooled = np.sqrt(((len(pune) - 1) * pune.var() + (len(hyd) - 1) * hyd.var()) / (len(pune) + len(hyd) -
2))
print(f"Cohen's d = {(hyd.mean() - pune.mean()) / pooled:.2f}")
```

OUTPUT

```
Hyderabad: n=55, mean 33.81, sd 7.60
Levene test for equal variances p = 0.047 → unequal variances
Welch t = 1.601, p = 0.1123, difference = 2.11 min
Cohen's d = 0.30
```

Conclusion. p ≈ 0.11 > 0.05: we **fail to reject** H_o. The data do not show a significant difference between the cities (Hyderabad is about 2 minutes slower in this sample, but that could be chance). Use **Welch's test** (`equal_var=False`) by default: it does not assume equal variances and loses almost nothing when they are equal.

Paired t-test: before and after

Use a **paired** test when the same units are measured twice (same riders before/after training, same stores before/after a layout change).

Python

```
rng = np.random.default_rng(5)
before = rng.normal(34, 5, 30).round(1)
after = (before - rng.normal(1.5, 1.5, 30)).round(1)          # simulated route-training effect
pt = stats.ttest_rel(before, after)
it = stats.ttest_ind(before, after, equal_var=False)
print(f"mean improvement {np.mean(before - after):.2f} min | paired p = {pt.pvalue:.4f} | (wrong)
independent p = {it.pvalue:.4f}")
```

OUTPUT

```
mean improvement 1.17 min | paired p = 0.0003 | (wrong) independent p = 0.3342
```

Pairing removes rider-to-rider variation, so the same improvement is detected far more clearly.

z-test for proportions

Python

```
from statsmodels.stats.proportion import proportions_ztest
hr = pd.read_csv("hr_employees.csv")
ot = hr[hr.overtime == "Yes"].attrition.eq("Yes"); no = hr[hr.overtime == "No"].attrition.eq("Yes")
stat, p = proportions_ztest([ot.sum(), no.sum()], [len(ot), len(no)])
print(f"attrition: overtime {ot.mean():.1%} (n={len(ot)}) vs no overtime {no.mean():.1%} (n={len(no)})
| z = {stat:.2f}, p = {p:.2e}")
stat1, p1 = proportions_ztest(hr.attrition.eq("Yes").sum(), len(hr), value=0.20, alternative="larger")
print(f"is company attrition above the 20% industry benchmark? z = {stat1:.2f}, p = {p1:.2e}")
```

OUTPUT

```
attrition: overtime 36.7% (n=433) vs no overtime 21.6% (n=1067) | z = 6.07, p = 1.26e-09
is company attrition above the 20% industry benchmark? z = 5.24, p = 7.89e-08
```

Choosing a test (means)

Situation	Test	Python
One mean vs a target, σ unknown	One-sample t	`stats.ttest_1samp`
Two independent groups	Welch's t	`stats.ttest_ind(..., equal_var=False)`
Same units twice	Paired t	`stats.ttest_rel`
One or two proportions	z-test	`statsmodels proportions_ztest`
Non-normal small samples	Mann-Whitney U / Wilcoxon (next lesson)	`stats.mannwhitneyu`, `stats.wilcoxon`
3+ groups	ANOVA (next lesson)	`stats.f_oneway`

Assumptions for t-tests: independent observations, roughly normal data **or** large samples (CLT), and for the pooled version equal variances.

Reporting

> Pune deliveries averaged **31.7 minutes** (95% CI 30.1–33.4, n = 60), significantly above the 30-minute promise (one-sample t = 2.06, one-tailed p = 0.022). The gap is small (d = 0.27). **Recommendation:** monitor the share of orders above 40 minutes (13%) rather than the average.

Report the **estimate, CI, test, p-value, effect size and business meaning**.

Summary

KEY TAKEAWAYS

- H_o is the default (no effect); H_1 is what you want evidence for; fix α (usually 0.05) and the tails before looking.
- p-value = probability of data this extreme if H_o were true; reject H_o when $p \le \alpha$ (equivalently when the statistic passes the critical value).
- Type I (false positive, α) vs Type II (false negative, β); power = 1 − β grows with n and effect size.
- One-sample, Welch two-sample and paired t-tests compare means; z-tests compare proportions.
- Always report effect size and a confidence interval: statistical significance is not business significance.

Quick quiz

6 questions · tap an answer to check it

1. The p-value is 0.03 and α = 0.05. What do you conclude?

A Reject H_o: the result is statistically significant

B Accept H_1 with 97% probability

C Fail to reject H_o

D H_o is true

Answer: A. $p \le \alpha$.

2. Which is a Type I error?

A Concluding a new checkout works when it actually has no effect

B Missing a real improvement

C Using the wrong chart

D A small sample

Answer: A. A false positive.

3. Which test compares the same riders before and after training?

A Welch's t-test

B Paired t-test

C Chi-square test

D One-sample z-test

Answer: B. Measurements are paired by rider.

4. Why prefer Welch's t-test for two groups?

A It does not assume equal variances and works well either way

B It is always more significant

C It needs no data

D It is for proportions

Answer: A. `equal_var=False`.

5. p = 0.40. What can you say?

A H_o is proven true

B The data do not provide enough evidence against H_o

C The effect is 40%

D The test failed

Answer: B. Absence of evidence is not evidence of absence.

6. What increases the power of a test?

A Smaller sample

B Larger sample size or larger true effect

C Lower α only

D Using a two-tailed test always

Answer: B. Power = probability of detecting a real effect.

EXERCISE **Practice**

(1) Test whether Hyderabad's mean delivery time exceeds 30 minutes. (2) Test whether employees who left had lower monthly income than those who stayed (one-tailed Welch). Report the difference and d. (3) Test whether the share of overtime workers differs between Sales and IT.

Show solution (worked answer)

Python

```
print(stats.ttest_1samp(hyd, 30, alternative="greater"))
left, stay = hr[hr.attrition == "Yes"].monthly_income, hr[hr.attrition == "No"].monthly_income
r = stats.ttest_ind(left, stay, equal_var=False, alternative="less"); print(r, left.mean() -
stay.mean())
print((left.mean() - stay.mean()) / np.sqrt((left.var() + stay.var()) / 2))
s_, i_ = hr[hr.department == "Sales"].overtime.eq("Yes"), hr[hr.department ==
"IT"].overtime.eq("Yes")
print(proportions_ztest([s_.sum(), i_.sum()], [len(s_), len(i_)]))
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain a p-value to a business stakeholder. If there were truly no difference, how surprising would our data be? A p-value of 0.01 means a result this extreme would happen only 1% of the time by chance alone.

Q: What is the difference between statistical and practical significance? Statistical significance says an effect is unlikely to be zero; practical significance asks whether it is large enough to matter (effect size, cost, revenue).

Q: How do you choose α? By the cost of errors: 0.05 by convention; lower (0.01) when false positives are expensive (medical, financial launches); sometimes higher for early exploration.

STATISTICS & MACHINE LEARNING · LESSON 5

Hypothesis Testing II: Chi-Square, ANOVA, Non-Parametric Tests and Correlation Tests

Test relationships between categories with chi-square tests, compare three or more groups with one-way ANOVA and Tukey post-hoc tests, use non-parametric alternatives (Mann-Whitney, Wilcoxon, Kruskal-Wallis) when assumptions fail, test correlations, and control false discoveries when running many tests.

ML classes 1 and 3 · Python class 15 | ✦ SciPy 1.1x · statsmodels 0.1x | Intermediate | ⏱ 110 min

By the end of this lesson you can

- Run chi-square tests of independence and goodness of fit and read expected counts
- Compare several group means with one-way ANOVA and locate differences with Tukey HSD
- Choose parametric vs non-parametric tests and run Mann-Whitney, Wilcoxon and Kruskal-Wallis
- Test correlations with Pearson and Spearman
- Adjust for multiple comparisons (Bonferroni, Benjamini-Hochberg)

Which test? A decision map



Choosing a test from the type of variables and the number of groups.

Parametric (assumes ~normal)	Non-parametric (ranks; fewer assumptions)
One-sample / paired t	Wilcoxon signed-rank
Independent t (Welch)	Mann-Whitney U
One-way ANOVA	Kruskal-Wallis
Pearson correlation	Spearman correlation

Parametric tests are more powerful when their assumptions hold; non-parametric tests are safer for small, skewed samples, ordinal data and outliers.

Chi-square test of independence

> **Case.** A retailer surveyed 600 customers: is the **preferred shopping channel** (App, Store, Website) related to **age group**?

Python

```
import pandas as pd, numpy as np
from scipy import stats
c = pd.read_csv("channel_survey.csv")
obs = pd.crosstab(c.age_group, c.preferred_channel)
print(obs, "\n")
print((pd.crosstab(c.age_group, c.preferred_channel, normalize="index") * 100).round(1))
res = stats.chi2_contingency(obs)
print(f"\nx² = {res.statistic:.2f}, df = {res.dof}, p = {res.pvalue:.2e}")
print("expected counts if independent:\n", pd.DataFrame(res.expected_freq, index=obs.index,
columns=obs.columns).round(1))
n = obs.values.sum(); cramers_v = np.sqrt(res.statistic / (n * (min(obs.shape) - 1)))
print(f"Cramér's V = {cramers_v:.2f} (effect size: 0.1 small, 0.3 medium, 0.5 large)")
```

OUTPUT

```
preferred_channel  App  Store  Website
age_group
18-25              106     47       65
26-40               72     70       90
41+                 31     56       63

preferred_channel   App  Store  Website
age_group
18-25              48.6   21.6     29.8
26-40              31.0   30.2     38.8
41+                20.7   37.3     42.0

x² = 33.67, df = 4, p = 8.71e-07
expected counts if independent:
 preferred_channel   App  Store  Website
age_group
18-25              75.9   62.9     79.2
```

```
26-40              80.8   66.9     84.3
41+                52.2   43.2     54.5
Cramér's V = 0.17 (effect size: 0.1 small, 0.3 medium, 0.5 large)
```

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The test compares **observed** counts with the counts **expected** if age and channel were independent. p < 0.001, so channel preference depends on age: 18–25-year-olds favour the app (49%), customers over 41 favour stores and the website. Cramér's V ≈ 0.17 shows a modest association.



Standardised residuals show which cells drive the result: positive = more than expected.

Assumptions: independent observations and **expected** counts of at least 5 in (almost) every cell; otherwise combine categories or use Fisher's exact test (`stats.fisher_exact` for 2 × 2).

Goodness of fit

Does an observed distribution match an expected one?

Python

```python
shares_expected = np.array([0.35, 0.30, 0.35])                # last year's channel mix
observed = obs.sum().reindex(["App", "Store", "Website"]).to_numpy()
gof = stats.chisquare(observed, f_exp=shares_expected * observed.sum())
print(observed, (shares_expected * observed.sum()).round(0), f"x² = {gof.statistic:.2f}, p =
{gof.pvalue:.3f}")
```

OUTPUT

```
[209 173 218] [210. 180. 210.] x² = 0.58, p = 0.748
```

One-way ANOVA

> **Case.** 120 volunteers were randomly assigned to Keto, Low-fat or Mediterranean diets for 8 weeks. Do average weight losses differ?

Running three t-tests inflates the false-positive rate ($1 - 0.95^3 \approx 14\%$). **ANOVA** tests all means at once: H_0: $\mu_1 = \mu_2 = \mu_3$.

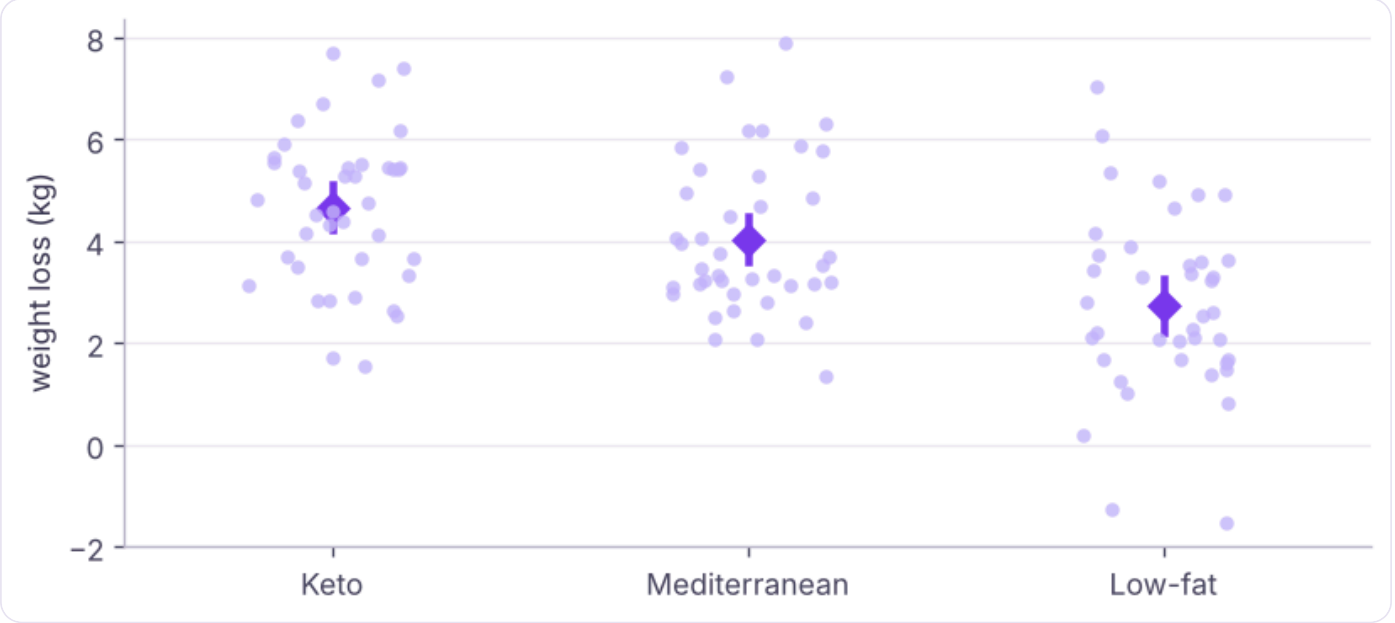
Python

```
d = pd.read_csv("diet_study.csv")
print(d.groupby("diet").weight_loss_kg.agg(["count", "mean", "std"]).round(2))
groups = [g.weight_loss_kg for _, g in d.groupby("diet")]
print("Levene equal variances p =", round(stats.levene(*groups).pvalue, 3), "| Shapiro p per group:",
[round(float(stats.shapiro(g).pvalue), 3) for g in groups])
f = stats.f_oneway(*groups)
print(f"F = {f.statistic:.2f}, p = {f.pvalue:.2e}")
grand = d.weight_loss_kg.mean()
ss_between = sum(len(g) * (g.mean() - grand) ** 2 for g in groups); ss_total = ((d.weight_loss_kg -
grand) ** 2).sum()
print(f"n² (share of variance explained by diet) = {ss_between / ss_total:.2f}")
```

OUTPUT

```
               count  mean   std
diet
Keto              40  4.65  1.49
Low-fat           40  2.75  1.78
Mediterranean     40  4.03  1.50
Levene equal variances p = 0.615 | Shapiro p per group: [0.655, 0.698, 0.036]
F = 14.74, p = 1.96e-06
n² (share of variance explained by diet) = 0.20
```

ANOVA splits total variation into **between-group** and **within-group** parts; F = (between variance) ÷ (within variance). A large F means group means differ more than random noise would suggest.



Weight loss by diet with group means: Keto and Mediterranean beat Low-fat.

Post-hoc: which groups differ?

Python

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd
print(pairwise_tukeyhsd(d.weight_loss_kg, d.diet, alpha=0.05))
```

OUTPUT

```
    Multiple Comparison of Means - Tukey HSD, FWER=0.05
=============================================================
 group1     group2    meandiff p-adj   lower   upper  reject
-------------------------------------------------------------
   Keto       Low-fat  -1.9012    0.0 -2.7496 -1.0529   True
   Keto Mediterranean  -0.6152 0.2015 -1.4636  0.2331  False
Low-fat Mediterranean    1.286 0.0014  0.4376  2.1344   True
-------------------------------------------------------------
```

Tukey's HSD compares every pair while keeping the overall error at 5%. Keto and Mediterranean both beat Low-fat; Keto vs Mediterranean is not significant.

Two-way ANOVA (two factors and their interaction) uses `statsmodels.formula.api.ols("y ~ C(a) * C(b)")` with `anova_lm`.

Non-parametric alternatives

Python

```
kw = stats.kruskal(*groups); print(f"Kruskal-Wallis (diets): H = {kw.statistic:.2f}, p =
{kw.pvalue:.2e}")
hr = pd.read_csv("hr_employees.csv")
left, stay = hr[hr.attrition == "Yes"].monthly_income, hr[hr.attrition == "No"].monthly_income
print(f"income medians: left {left.median():,.0f} vs stayed {stay.median():,.0f} | Mann-Whitney p =
{stats.mannwhitneyu(left, stay).pvalue:.2e}")
sat_ot = hr[hr.overtime == "Yes"].job_satisfaction; sat_no = hr[hr.overtime == "No"].job_satisfaction
print(f"job satisfaction (ordinal 1-4), overtime vs not: Mann-Whitney p = {stats.mannwhitneyu(sat_ot,
sat_no).pvalue:.3f}")
rng = np.random.default_rng(1); before = rng.lognormal(3, .5, 20); after = before * rng.uniform(.75,
1.05, 20)
print("Wilcoxon signed-rank (paired, skewed):", round(stats.wilcoxon(before, after).pvalue, 4))
```

OUTPUT

```
Kruskal-Wallis (diets): H = 23.29, p = 8.78e-06
income medians: left 67,917 vs stayed 77,750 | Mann-Whitney p = 3.81e-06
job satisfaction (ordinal 1-4), overtime vs not: Mann-Whitney p = 0.191
Wilcoxon signed-rank (paired, skewed): 0.001
```

Mann-Whitney compares whether values in one group tend to be larger than in the other (often described as comparing medians). It suits skewed income data and ordinal ratings.

Testing correlations

Python

```
for a, b in [("age", "monthly_income"), ("distance_from_home_km", "job_satisfaction"),
("years_at_company", "monthly_income")]:
    pr, sp = stats.pearsonr(hr[a], hr[b]), stats.spearmanr(hr[a], hr[b])
    print(f"{a:22s} vs {b:16s} Pearson r {pr.statistic:+.3f} (p {pr.pvalue:.3g}) | Spearman ρ
{sp.statistic:+.3f} (p {sp.pvalue:.3g})")
print("Department vs attrition: x² p =", round(stats.chi2_contingency(pd.crosstab(hr.department,
hr.attrition)).pvalue, 3))
```

OUTPUT

```
age                    vs monthly_income   Pearson r +0.135 (p 1.52e-07) | Spearman ρ +0.108 (p
2.95e-05)
distance_from_home_km  vs job_satisfaction Pearson r +0.008 (p 0.764) | Spearman ρ +0.007 (p
0.776)
years_at_company       vs monthly_income   Pearson r +0.679 (p 1.63e-203) | Spearman ρ +0.577 (p
7.98e-134)
Department vs attrition: x² p = 0.196
```

With 1,500 rows, even r = 0.1 can be "significant": look at the size of r, not only the p-value.

Many tests: the multiple-comparison problem

Test 100 segments where nothing really changed, at α = 0.05, and about five will look "significant" by chance.

Python

```python
from statsmodels.stats.multitest import multipletests
rng = np.random.default_rng(42)
null_p = [stats.ttest_ind(rng.normal(0, 1, 50), rng.normal(0, 1, 50)).pvalue for _ in range(100)]
# 100 segments with NO real effect
real_p = [stats.ttest_ind(rng.normal(0, 1, 50), rng.normal(0.8, 1, 50)).pvalue for _ in range(5)]
# 5 segments with a real effect
pvals = np.array(null_p + real_p); truth = np.array([False] * 100 + [True] * 5)
for name, sig in [("raw p < 0.05", pvals < 0.05), ("Bonferroni", multipletests(pvals, 0.05,
"bonferroni")[0]), ("Benjamini-Hochberg", multipletests(pvals, 0.05, "fdr_bh")[0])]:
    print(f"{name:20s} significant {sig.sum():3d} | true effects found {(sig & truth).sum()} of 5 |
false positives {(sig & ~truth).sum()}")
```

OUTPUT

```
raw p < 0.05         significant  12 | true effects found 5 of 5 | false positives 7
Bonferroni           significant   3 | true effects found 3 of 5 | false positives 0
Benjamini-Hochberg   significant   6 | true effects found 5 of 5 | false positives 1
```

Bonferroni (α ÷ number of tests) controls the chance of any false positive but is strict; **Benjamini-Hochberg** controls the false discovery rate and keeps more power. Use them for dashboards that test many segments, A/B tests with many metrics, and feature screening.

Summary

KEY TAKEAWAYS

- Chi-square tests whether two categorical variables are independent (expected counts ≥ 5); report Cramér's V and residuals.
- ANOVA compares 3+ means in one test (F = between/within variance); follow with Tukey HSD; report η^2.
- Non-parametric tests (Mann-Whitney, Wilcoxon, Kruskal-Wallis, Spearman) use ranks for skewed, ordinal or small data.
- Test correlations with Pearson (linear) or Spearman (monotonic); large n makes tiny r significant.
- Many tests → adjust p-values (Bonferroni, Benjamini-Hochberg).

Quick quiz

6 questions · tap an answer to check it

1. Which test checks whether preferred channel depends on age group?

A ANOVA

B Chi-square test of independence

C Paired t-test

D Pearson correlation

Answer: B. Both variables are categorical.

2. Why use ANOVA instead of several t-tests for 3 groups?

A Multiple t-tests inflate the chance of a false positive

B t-tests cannot handle numbers

C ANOVA always gives smaller p-values

D It is required by law

Answer: A. ANOVA tests all means together at α.

3. ANOVA gives $p < 0.001$ for three diets. What next?

A Stop: all diets differ

B Run a post-hoc test (Tukey HSD) to see which pairs differ

C Run a chi-square test

D Delete outliers

Answer: B. ANOVA only says at least one mean differs.

4. Which test compares two independent groups of skewed incomes?

A Paired t-test

B Mann-Whitney U

C Chi-square goodness of fit

D One-sample t-test

Answer: B. A rank-based alternative to the t-test.

5. What does the Benjamini-Hochberg method control?

A Type II errors

B The false discovery rate among significant results

C The sample size

D Normality

Answer: B. It is less strict than Bonferroni.

6. What is a chi-square assumption?

A Expected counts of at least about 5 per cell

B Normal data

C Equal variances

D Paired observations

Answer: A. Use Fisher's exact test for small tables.

EXERCISE **Practice (hr_employees.csv)**

(1) Is overtime related to department? (chi-square + Cramér's V). (2) Does mean monthly income differ across departments (ANOVA + Tukey)? (3) Repeat (2) with Kruskal-Wallis. (4) Is education level related to attrition?

Show solution (worked answer)

Python

```
t = pd.crosstab(hr.department, hr.overtime); r = stats.chi2_contingency(t); print(r.pvalue,
np.sqrt(r.statistic / (t.values.sum() * (min(t.shape) - 1))))
gs = [g.monthly_income for _, g in hr.groupby("department")]; print(stats.f_oneway(*gs))
print(pairwise_tukeyhsd(hr.monthly_income, hr.department))
print(stats.kruskal(*gs))
print(stats.chi2_contingency(pd.crosstab(hr.education, hr.attrition)).pvalue)
```

INTERVIEW CORNER **Questions you may be asked**

Q: When would you use a non-parametric test? Small samples that are clearly non-normal, ordinal data (ratings), or heavy outliers; for large samples the CLT usually makes parametric tests fine.

Q: What does a chi-square test tell you and what does it not? Whether two categorical variables are associated; not the direction, strength (use Cramér's V) or causation.

Q: Explain ANOVA's F-statistic. The ratio of variation between group means to variation within groups; large values mean group differences exceed what noise explains.

STATISTICS & MACHINE LEARNING · LESSON 6

A/B Testing and Experimentation: Design, Power, Analysis and Pitfalls

Run product experiments the way data teams at e-commerce and fintech companies do. Choose a metric and guardrails, size the test with power analysis, randomise, analyse conversion and revenue with z-tests, t-tests and bootstrap intervals, check sample-ratio mismatch, avoid peeking and novelty effects, and write the decision memo, using a new-checkout experiment.

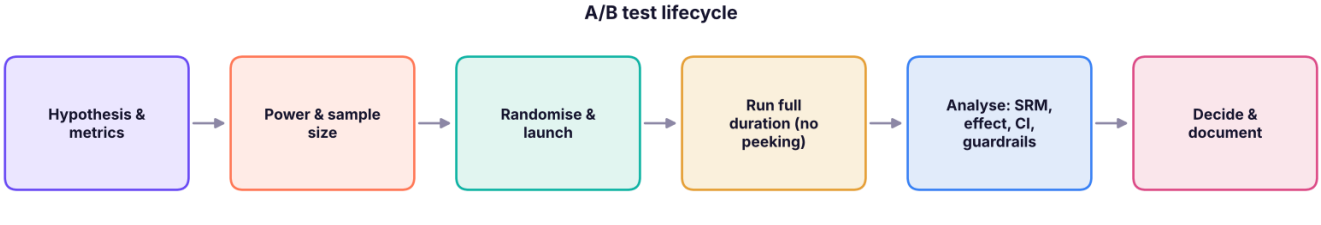
ML class 2 (case study) · ✦ statsmodels 0.1x · SciPy 1.1x · Intermediate · ⏱ 120 min

By the end of this lesson you can

- Design an experiment with a hypothesis, primary metric, guardrail metrics and randomisation unit
- Compute the required sample size with power analysis and a minimum detectable effect
- Analyse conversion (two-proportion z-test) and revenue per user (Welch t-test and bootstrap)
- Check sample-ratio mismatch and recognise peeking, novelty and network effects
- Write a launch/no-launch recommendation with uncertainty and business impact

What is an A/B test?

A **randomised controlled experiment**: users are randomly split into **control (A)** and **treatment (B)**; the only systematic difference is the change you test, so differences in outcomes are **caused** by the change (up to chance, which statistics quantifies).



The experiment lifecycle.

Case. An online store tests a **new one-page checkout (B)** against the current checkout (A). 4,000 users were split 50/50. **Hypothesis:** the new checkout increases the purchase conversion rate. **Primary metric:** conversion = purchasers ÷ users. **Secondary:** revenue per user, average order value. **Guardrails:** refund rate, page load time, support tickets.

Design choice	Good practice
Randomisation unit	User (not session or page view), so one person sees one version
Split	50/50 maximises power
Duration	At least one full business cycle (1–2 weeks, including weekends); fixed in advance
Metric	Defined before launch; one primary metric
Population	Who is eligible (new users only? all platforms?)

Sample size and power

Before launching: how many users do we need to detect the smallest effect worth acting on (the **minimum detectable effect**, MDE)?

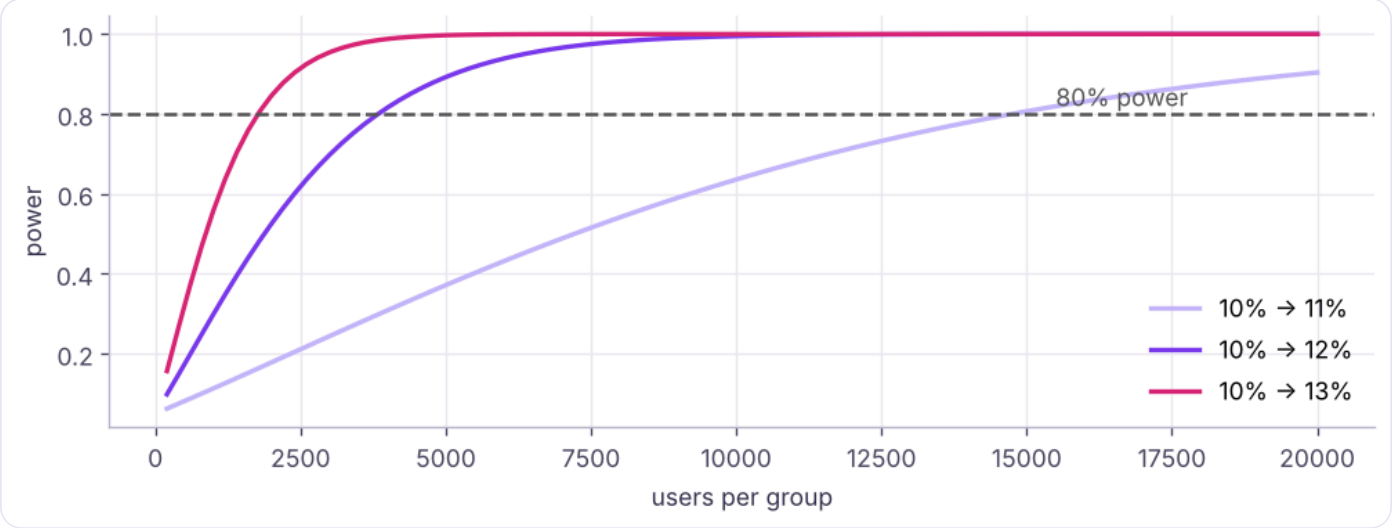
Python

```
import numpy as np, pandas as pd
from scipy import stats
from statsmodels.stats.power import NormalIndPower
from statsmodels.stats.proportion import proportion_effectsize
base = 0.10
for target in (0.11, 0.12, 0.13):
    h = proportion_effectsize(target, base)                    # Cohen's h
    n = NormalIndPower().solve_power(effect_size=h, alpha=0.05, power=0.8, ratio=1, alternative="two-
sided")
    print(f"detect {base:.0%} → {target:.0%} (+{(target - base) / base:.0%} relative):
{int(np.ceil(n)):,} users per group")
```

OUTPUT

```
detect 10% → 11% (+10% relative): 14,745 users per group
detect 10% → 12% (+20% relative): 3,835 users per group
detect 10% → 13% (+30% relative): 1,769 users per group
```

Small lifts need very large samples. With 2,000 users per group, this test can reliably detect a lift from about 10% to 13%, not smaller ones.



Power curves: the chance of detecting a true lift grows with sample size and effect size.

Analysis step 1: sanity checks

Python

```
ab = pd.read_csv("ab_test.csv")
counts = ab.group.value_counts().sort_index(); print(counts.to_dict())
srm = stats.chisquare(counts.values)                                    # expected 50/50
print(f"sample-ratio mismatch test p = {srm.pvalue:.3f} → {'OK' if srm.pvalue > 0.01 else 'INVESTIGATE:
randomisation or logging problem'}")
print("duplicate users:", ab.user_id.duplicated().sum(), "| missing:", ab.isna().sum().sum())
```

OUTPUT

```
{'A_control': 2000, 'B_new_checkout': 2000}
sample-ratio mismatch test p = 1.000 → OK
duplicate users: 0 | missing: 0
```

A **sample-ratio mismatch (SRM)** (for example 52/48 when you planned 50/50, with p < 0.01) signals a bug in assignment or tracking; do not trust results until it is explained.

Analysis step 2: conversion rate

Python

```
from statsmodels.stats.proportion import proportions_ztest, confint_proportions_2indep
g = ab.groupby("group").agg(users=("user_id", "size"), buyers=("converted", "sum"))
g["conversion"] = g.buyers / g.users
print(g)
z, p = proportions_ztest(g.buyers.values[::-1], g.users.values[::-1])          # B vs A
lo, hi = confint_proportions_2indep(g.buyers.iloc[1], g.users.iloc[1], g.buyers.iloc[0],
g.users.iloc[0], compare="diff")
pa, pb = g.conversion.iloc[0], g.conversion.iloc[1]
print(f"absolute lift {100 * (pb - pa):.2f} pts (95% CI {100 * lo:.2f} to {100 * hi:.2f}) | relative
lift {(pb - pa) / pa:.1%} | z = {z:.2f}, p = {p:.5f}")
```

OUTPUT

```
                users  buyers  conversion
group
A_control        2000     198      0.0990
B_new_checkout   2000     273      0.1365
absolute lift 3.75 pts (95% CI 1.75 to 5.75) | relative lift 37.9% | z = 3.68, p = 0.00023
```

Analysis step 3: revenue per user and order value

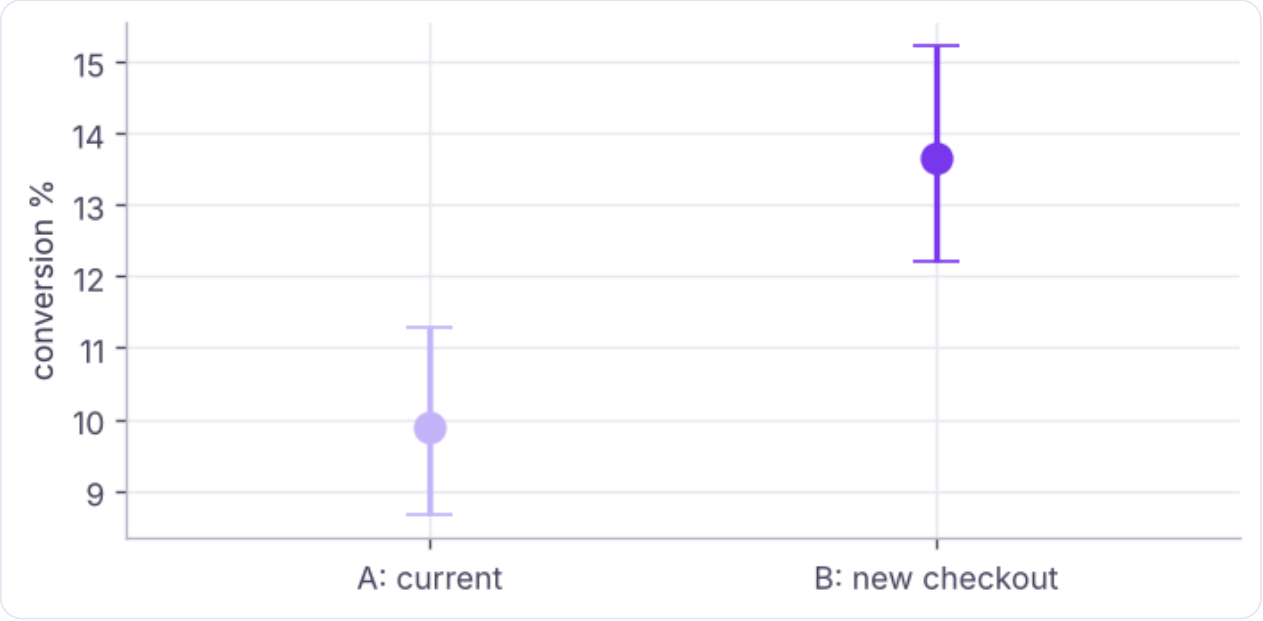
Python

```
a, b = ab[ab.group == "A_control"], ab[ab.group == "B_new_checkout"]
for name, col, fa, fb in [("revenue per user", "order_value", a, b), ("average order value (buyers)",
"order_value", a[a.converted == 1], b[b.converted == 1])]:
    t = stats.ttest_ind(fb[col], fa[col], equal_var=False)
    print(f"{name:30s} A {fa[col].mean():8.1f} | B {fb[col].mean():8.1f} | diff {fb[col].mean() -
fa[col].mean():7.1f} | Welch p = {t.pvalue:.4f}")
rng = np.random.default_rng(0)
boot = [rng.choice(b.order_value.values, len(b)).mean() - rng.choice(a.order_value.values,
len(a)).mean() for _ in range(5000)]
print(f"bootstrap 95% CI for revenue-per-user difference: {np.percentile(boot, 2.5):.1f} to
{np.percentile(boot, 97.5):.1f}")
```

OUTPUT

```
revenue per user               A    137.5 | B    202.8 | diff    65.3 | Welch p = 0.0000
average order value (buyers)   A   1389.3 | B   1485.6 | diff    96.4 | Welch p = 0.0082
bootstrap 95% CI for revenue-per-user difference: 35.5 to 94.9
```

Revenue per user mixes conversion and order size and has many zeros and a long tail; Welch's t-test (thanks to the CLT) and the bootstrap both work at this sample size.



Conversion rate with 95% confidence intervals for each variant.

Analysis step 4: business impact

Python

```
monthly_users = 300_000
extra_buyers = monthly_users * (pb - pa)
extra_rev = monthly_users * (b.order_value.mean() - a.order_value.mean())
print(f"at {monthly_users:,} users/month: about {extra_buyers:,.0f} extra orders and ₹{extra_rev / 1e5:,.1f} lakh extra revenue per month")
print(f"conservative (lower CI bound of conversion lift): {monthly_users * lo:,.0f} extra orders")
```

OUTPUT

```
at 300,000 users/month: about 11,250 extra orders and ₹195.8 lakh extra revenue per month
conservative (lower CI bound of conversion lift): 5,263 extra orders
```

Pitfalls

Pitfall	What goes wrong	Fix
Peeking	Checking daily and stopping at the first p < 0.05 inflates false positives to 20–30%	Fix the sample size; or use sequential tests / always-valid p-values
Too many metrics	Some will be "significant" by chance	One primary metric; adjust others (lesson 5)
Novelty / primacy effects	Users react to change itself, then revert	Run longer; analyse by week
SRM	Broken randomisation or logging	Chi-square check before analysis
Network effects	Treated users affect control users (marketplaces, social)	Cluster or geo randomisation
Simpson's paradox	Mix of segments differs between periods	Randomise; compare within segments
Underpowered tests	"No effect" just means "too small a test"	Power analysis first; report CIs
HARKing	Choosing the hypothesis after seeing results	Pre-register the plan

Python

```
rng = np.random.default_rng(11); false_wins = 0
for sim in range(400):                                          # A/A tests: there is NO real difference
    xa, xb = rng.random(4000) < 0.10, rng.random(4000) < 0.10
    for day in range(1, 15):                                    # peek every "day" (≈285 users per group per day)
        n = day * 285
        if proportions_ztest([xa[:n].sum(), xb[:n].sum()], [n, n])[1] < 0.05: false_wins += 1; break
print(f"stopping at the first significant daily peek: {false_wins / 400:.0%} false positives (should be 5%)")
```

OUTPUT

```
stopping at the first significant daily peek: 20% false positives (should be 5%)
```

Beyond the classic test

- **CUPED / regression adjustment:** use pre-experiment behaviour as a covariate to cut variance (and required sample size) by 20–50%.
- **Bayesian A/B testing:** reports "probability that B beats A" and expected loss; intuitive for stakeholders.
- **Multi-armed bandits:** shift traffic to the winning variant during the test (good for short campaigns, worse for clean inference).
- **Quasi-experiments** when you cannot randomise: difference-in-differences, synthetic control, regression discontinuity, propensity matching.
- Platforms: Optimizely, Statsig, GrowthBook, Eppo, LaunchDarkly, Firebase A/B Testing.

The decision memo

KEY TAKEAWAYS

Experiment: new one-page checkout, 4,000 users, 50/50, no sample-ratio mismatch. **Result:** conversion rose from 9.9% to 13.7% (+3.75 points, 95% CI about +1.7 to +5.8; $p < 0.001$). Revenue per user rose from about ₹138 to ₹203, and buyers' average order value also rose. **Guardrails:** check refunds, load time and support tickets before rollout. **Decision:** launch B to all users, with a 5% holdout for four weeks to confirm the effect lasts.

Summary

KEY TAKEAWAYS

- Randomisation makes A/B tests causal; define the hypothesis, one primary metric, guardrails and the duration before launch.
- Power analysis sets the sample size for a minimum detectable effect; small lifts need large samples.
- Check sample-ratio mismatch first; analyse conversion with a two-proportion z-test and revenue with Welch's t-test or the bootstrap; report CIs and business impact.
- Avoid peeking, metric fishing and underpowered tests; consider CUPED, Bayesian methods and quasi-experiments where appropriate.

Quick quiz

6 questions · tap an answer to check it

1. Why randomise users into A and B?

- **A** So the only systematic difference between groups is the change being tested
- **B** To make the test faster
- **C** To increase conversion
- **D** To avoid statistics

Answer: A. Randomisation balances all other factors on average.

2. What does a power analysis give you before the test?

- **A** The p-value
- **B** The sample size needed to detect a given minimum effect with a chosen probability
- **C** The winner
- **D** The conversion rate

Answer: B. Typical settings: $\alpha = 0.05$, power = 0.8.

3. The planned split was 50/50 but you observe 53/47 with SRM $p < 0.001$. What do you do?

- **A** Report the result
- **B** Investigate the assignment or logging bug before trusting the results
- **C** Increase α
- **D** Remove users until it is 50/50

Answer: B. SRM invalidates naive analysis.

4. Why is stopping as soon as p < 0.05 ("peeking") a problem?

- **A** It inflates the false-positive rate well above 5%
- **B** It wastes traffic
- **C** It makes p-values larger
- **D** It is fine

Answer: A. Use fixed horizons or sequential methods.

5. Which test suits comparing conversion rates of A and B?

- **A** Paired t-test
- **B** Two-proportion z-test (or chi-square)
- **C** ANOVA
- **D** Pearson correlation

Answer: B. Conversion is a proportion.

6. A test with 300 users per group shows no significant difference. Best conclusion?

- **A** The change has no effect
- **B** The test may be underpowered; look at the confidence interval for the plausible range of effects
- **C** Launch anyway
- **D** The metric is wrong

Answer: B. Absence of evidence is not evidence of absence.

EXERCISE **Practice**

(1) Compute the users per group needed to detect a lift from 13.7% to 15% conversion. (2) Analyse average order value among buyers with a Mann-Whitney test as well. (3) Simulate an A/A test 1,000 times with a fixed horizon and confirm about 5% false positives. (4) Draft a one-paragraph memo for the case where the result had been p = 0.20.

Show solution (worked answer)

Python

```
print(NormalIndPower().solve_power(proportion_effectsize(0.15, 0.137), alpha=0.05, power=0.8))
print(stats.mannwhitneyu(b[b.converted == 1].order_value, a[a.converted == 1].order_value))
fp = np.mean([proportions_ztest([(rng.random(4000) < .1).sum(), (rng.random(4000) < .1).sum()],
[4000, 4000])[1] < .05 for _ in range(1000)]); print(fp)
# Memo: "No significant difference (p = 0.20); CI ranges from -x to +y points, so we cannot rule
out a meaningful lift. Extend the test to N users or test a bolder change."
```

INTERVIEW CORNER **Questions you may be asked**

Q: How would you design an A/B test for a new recommendation widget? State the hypothesis and primary metric (e.g. revenue per user), guardrails (latency, returns), randomise by user, size it with power analysis, run full weeks, check SRM, analyse with CIs, and decide with business impact.

Q: Your test is significant on day 3. Do you stop? No: stopping early inflates false positives and novelty effects can fade; finish the planned duration or use a sequential design.

Q: What if you cannot randomise? Use quasi-experimental methods: difference-in-differences with a comparison group, synthetic control, regression discontinuity or matching, and state the assumptions.

STATISTICS & MACHINE LEARNING · LESSON 7

Introduction to Machine Learning: Types, Business Framing and the ML Lifecycle

What machine learning is and is not, how it differs from rules and statistics, supervised vs unsupervised vs reinforcement learning, regression vs classification vs clustering, turning a business problem into an ML problem, the end-to-end ML lifecycle (CRISP-DM to MLOps), linear vs non-linear models and link functions, and where GenAI fits.

ML classes 3–4 | ✦ scikit-learn 1.7+ · 2026 practice | Beginner | ⏱ 100 min

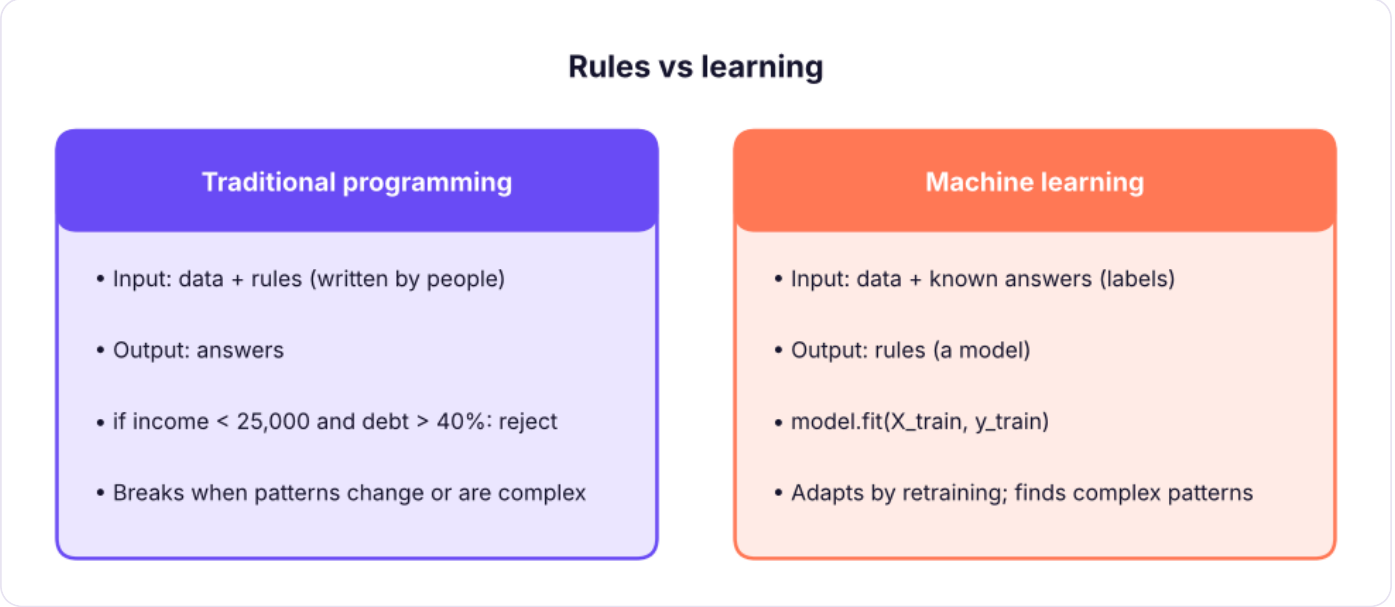
By the end of this lesson you can

- Define machine learning and distinguish it from rule-based systems and classical statistics
- Classify problems as supervised (regression, classification), unsupervised (clustering, dimensionality reduction) or reinforcement learning
- Translate a business question into target, features, unit of prediction, metric and decision
- Describe each stage of the ML lifecycle and common failure points
- Explain linear vs non-linear models, link functions, and when not to use ML

What is machine learning?

> "A computer program is said to learn from experience E with respect to some task T and performance measure P if its performance at T, as measured by P, improves with experience E." (Tom Mitchell)

Instead of writing rules by hand, we give the computer **examples** (data) and let an algorithm **learn the rules** (a model) that map inputs to outputs.

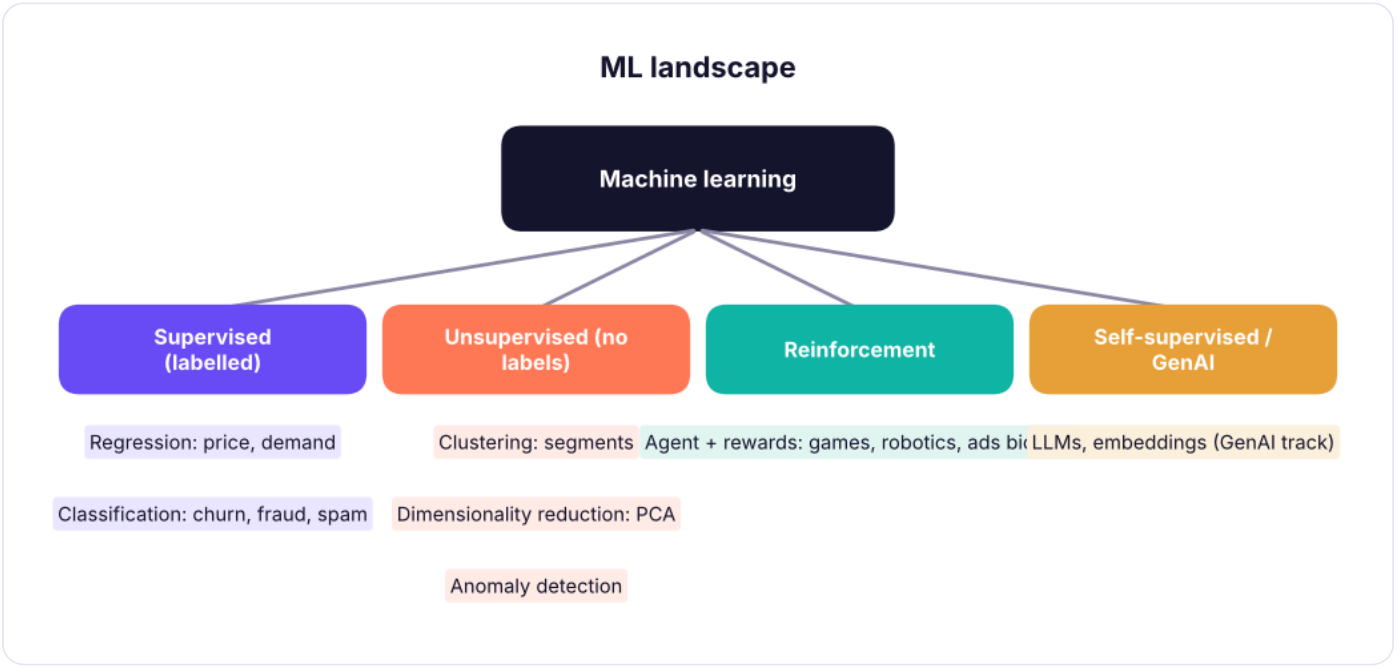


Traditional programming vs machine learning.

	Statistics	Machine learning
Main goal	Inference: understand relationships, test hypotheses	Prediction: accurate outputs on new data
Typical question	"Does overtime increase attrition, and by how much?"	"Which employees are likely to leave next quarter?"
Model choice	Interpretable, with assumptions	Whatever predicts best (with interpretability as needed)
Validation	p-values, confidence intervals	Held-out test data, cross-validation

They overlap heavily: linear and logistic regression belong to both.

Types of machine learning



The main families of machine learning.

Problem	Target type	Example	Algorithms (this track)
Regression	Number	House price, delivery time, demand	Linear, Ridge/Lasso, trees, boosting
Classification	Category	Churn yes/no, loan default, spam	Logistic, KNN, trees, RF, boosting, SVM, NB
Clustering	None	Customer segments	K-means, hierarchical, DBSCAN
Dimensionality reduction	None	Compress 50 features to 5	PCA
Time-series forecasting	Number over time	Next month's sales	Seasonal naive, ETS, ARIMA, ML with lags
Recommendation / ranking	Ranked list	Products to show	Collaborative filtering, learning to rank

Python

```
import pandas as pd
h = pd.read_csv("house_prices.csv"); tc = pd.read_csv("telecom_churn.csv")
print("Regression target:", "price_lakh", "| examples:", h.price_lakh.head(3).tolist(), "| features:",
[c for c in h.columns if c not in ("house_id", "price_lakh")])
print("Classification target: churn", tc.churn.value_counts().to_dict(), "| features:", [c for c in
tc.columns if c not in ("customer_id", "churn")])
```

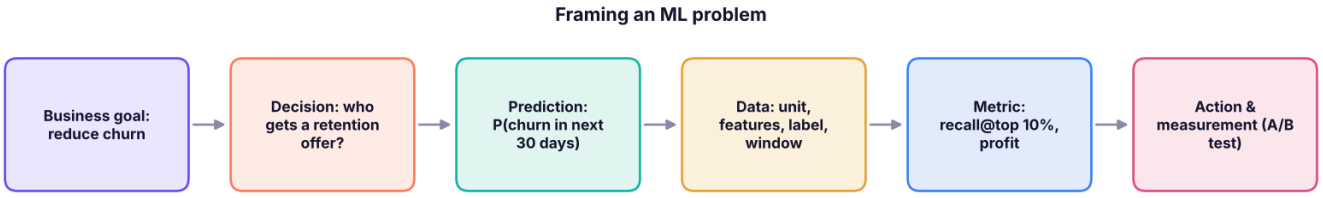
OUTPUT

```
Regression target: price_lakh | examples: [60.2, 60.9, 43.7] | features: ['city', 'area_sqft',
'bedrooms', 'bathrooms', 'age_years', 'distance_to_center_km', 'parking']
Classification target: churn {'No': 2228, 'Yes': 772} | features: ['senior_citizen',
```

```
'tenure_months', 'contract', 'internet_service', 'payment_method', 'monthly_charges',
'total_charges', 'support_calls']
```

Vocabulary: a row is an **observation/sample**; input columns are **features** (X, predictors, independent variables); the column to predict is the **target** (y, label, dependent variable); the learned function is the **model**; using it on new data is **inference/prediction**.

From business problem to ML problem



Framing comes before algorithms.

Question	Churn example
What decision will the prediction change?	Which 10% of customers receive a ₹200 retention offer
Unit of prediction	One customer, scored monthly
Target definition	Cancelled within 30 days after the scoring date
Features (only information available **at prediction time**)	Tenure, contract, charges, support calls in the previous 90 days
Success metric	Recall in the top 10%, net profit of the campaign
Baseline	Current rule: target month-to-month customers with 3+ calls
Constraints	Explainability for the retention team, privacy, weekly batch scoring

WATCH OUT **Data leakage**

Using information that would not be available at prediction time (for example "number of cancellation calls" recorded **after** the customer decided to leave) gives brilliant test scores and useless production models. Define a clear cut-off date for features and a later window for the label.

When not to use ML

Simple rules work; there is little data or no labels; mistakes are unacceptable and must be explained exactly; the pattern changes faster than you can retrain; or nobody will act on the predictions. Always compare with a simple baseline.

The ML lifecycle



CRISP-DM extended with MLOps: an iterative cycle, not a straight line.

Stage	Key work	Typical tools
Business understanding	Goal, decision, metric, baseline, constraints	Workshops, documents
Data understanding	Sources, EDA, quality, label definition	SQL, pandas, plots (Python track)
Data preparation	Cleaning, encoding, scaling, feature engineering, train/test split	pandas, scikit-learn pipelines
Modelling	Train candidate models, tune hyperparameters	scikit-learn, XGBoost, LightGBM
Evaluation	Test-set metrics, error analysis, fairness, business value	metrics, SHAP
Deployment	Batch scoring or API, integration with apps	FastAPI, Docker, cloud ML platforms
Monitoring	Data drift, performance decay, retraining	MLflow, Evidently, dashboards

In practice, most time goes to framing, data preparation and deployment, not to choosing algorithms.

Linear vs non-linear models and link functions

A **linear model** predicts with a weighted sum of features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ \ldots \ + \beta_p x_p$$

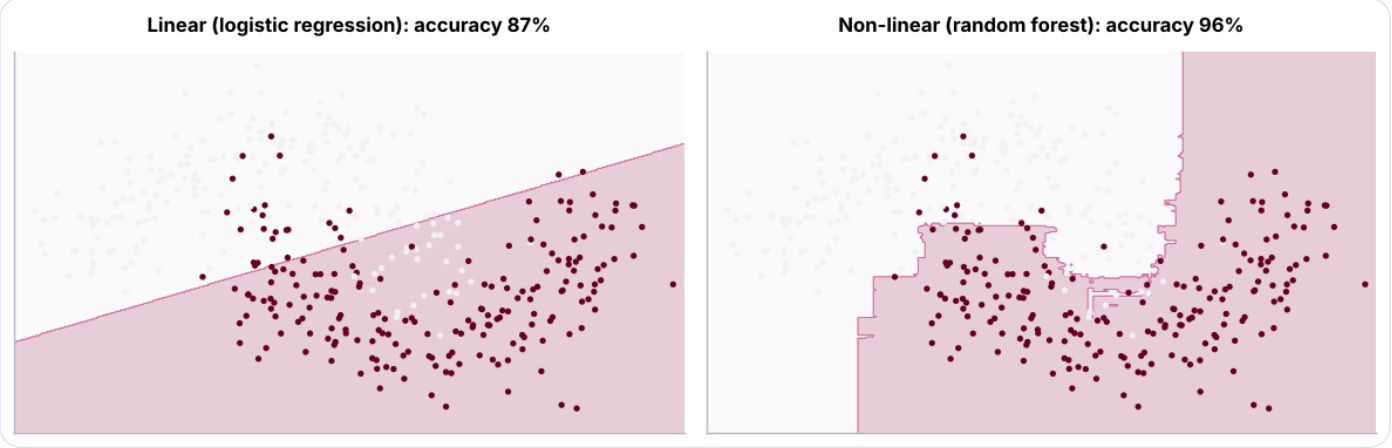
Generalised linear models pass that linear score through a **link function** so the output fits the target type:

Target	Model	Link (inverse)	Output
Any real number	Linear regression	Identity	ŷ = score
Probability (0–1)	Logistic regression	Logit ⇄ sigmoid 1/(1 + e^(−score))	P(y = 1)
Count (0, 1, 2…)	Poisson regression	Log ⇄ exp(score)	Expected count



The same linear score through three link functions: identity (regression), sigmoid (classification probability) and exp (counts).

Non-linear models (decision trees, random forests, gradient boosting, KNN, neural networks) capture thresholds and interactions automatically, usually at the cost of interpretability.



A linear model draws one straight boundary; a non-linear model can bend around the data.

Overfitting and generalisation (preview)

A model must work on **new** data. Memorising the training set (overfitting) fails in production; being too simple (underfitting) misses real patterns. We therefore always evaluate on data the model has not seen (lessons 8 and 15).

Where does GenAI fit?

Large language models are trained with self-supervised learning on huge text corpora, then adapted with instruction tuning and reinforcement learning from human feedback. In analytics they **complement** classical ML: LLMs read and write text, code and SQL; classical models remain best for structured, tabular prediction (churn, credit risk, demand), where gradient boosting usually wins. The GenAI track covers LLMs, RAG and agents.

Summary

KEY TAKEAWAYS

- ML learns rules from data; statistics focuses on inference, ML on prediction, with much overlap.
- Supervised (regression, classification), unsupervised (clustering, PCA, anomalies), reinforcement and self-supervised (LLMs) learning.
- Frame first: decision, unit, target and time windows, leak-free features, metric, baseline.
- Lifecycle: business → data → preparation → modelling → evaluation → deployment → monitoring, iteratively.
- Linear models use link functions (identity, sigmoid, exp); non-linear models capture thresholds and interactions.

Quick quiz

6 questions · tap an answer to check it

1. Predicting next month's sales in rupees is...

A Regression

B Classification

C Clustering

D Reinforcement learning

Answer: A. The target is a number.

2. Grouping customers into segments without labels is...

A Supervised learning

B Unsupervised learning (clustering)

C Regression

D A/B testing

Answer: B. No target column.

3. What is data leakage?

A Using information in training that would not be available at prediction time

B Losing data in a crash

C Sharing data publicly

D Missing values

Answer: A. It inflates test scores.

4. Which link function does logistic regression use to output a probability?

A Identity

B Sigmoid (inverse of the logit)

C Exponential

D Square root

Answer: B. It maps any score to 0–1.

5. Which stage often takes the most time in real ML projects?

- **A** Picking the algorithm
- **B** Data understanding and preparation
- **C** Writing the final slide
- **D** Installing Python

Answer: B. Data work dominates.

6. When might you NOT use ML?

- **A** When a simple rule works well or nobody will act on predictions
- **B** When you have lots of data
- **C** When the pattern is complex
- **D** When predictions save money

Answer: A. Start with a baseline.

EXERCISE **Frame these problems**

For each, state the ML type, target, unit, two leak-free features, the metric and the action: (1) predicting loan default at application; (2) forecasting daily units for each store; (3) grouping 5,000 products by sales pattern; (4) flagging suspicious card transactions in real time.

Show solution (worked answer)

(1) Classification; default within 12 months; applicant; income, debt-to-income; recall/precision at the approval threshold and expected loss; approve/decline/price. (2) Regression/forecasting; units per store-day; store-day; lagged sales, calendar/holidays; MAPE/WAPE; stock orders. (3) Clustering; none; product; weekly sales profile, price; silhouette + business sense; assortment strategy. (4) Classification (imbalanced); fraud label; transaction; amount vs customer average, merchant risk; precision/recall at alert capacity; block or step-up authentication.

INTERVIEW CORNER

Questions you may be asked

Q: Explain machine learning to a non-technical person. Instead of writing rules ourselves, we show the computer many past examples with the right answers, and it learns patterns it can apply to new cases, like an employee learning from experience.

Q: Supervised vs unsupervised learning? Supervised learning uses labelled examples to predict a target; unsupervised learning finds structure such as groups or compressed representations without labels.

Q: How do you start an ML project? Clarify the business decision and success metric, define the target and prediction time, check data availability and leakage, set a simple baseline, then iterate on models.

STATISTICS & MACHINE LEARNING · LESSON 8

The scikit-learn Workflow: Train/Test Split, Baselines, Pipelines and Saving Models

Learn the standard machine-learning workflow in scikit-learn. The estimator API (fit, predict, transform), splitting data into train, validation and test sets with stratification and time order, baselines, ColumnTransformer and Pipeline to avoid leakage, comparing models, and saving and loading a trained pipeline.

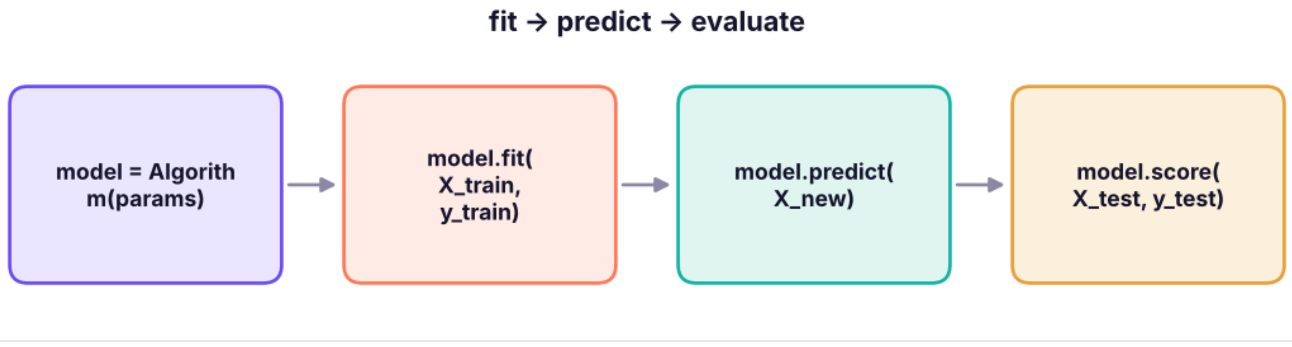
ML class 4 · scikit-learn 1.7+ · Intermediate · 110 min

By the end of this lesson you can

- Use the fit / predict / predict_proba / transform / score API consistently
- Split data into train and test (stratified, time-based) and explain why the test set is touched once
- Build dummy baselines and compare real models against them
- Combine preprocessing and a model in a Pipeline with ColumnTransformer
- Save, load and use a trained pipeline on new data

The estimator API

Every scikit-learn model follows the same pattern, which makes swapping algorithms easy.



One consistent interface for every estimator.

Method	Available on	Does
`fit(X, y)`	Every estimator	Learns parameters from training data
`predict(X)`	Models	Predicted labels or values
`predict_proba(X)`	Most classifiers	Class probabilities
`transform(X)`	Transformers (scalers, encoders, PCA)	Converts features using fitted parameters
`fit_transform(X)`	Transformers	Fit then transform (training data only!)
`score(X, y)`	Models	Default metric (accuracy or R^2)
`get_params()` / `set_params()`	All	Hyperparameters

X is a 2-D table of features (DataFrame or array, shape *n_samples* × *n_features*); **y** is a 1-D target.

Train/test split

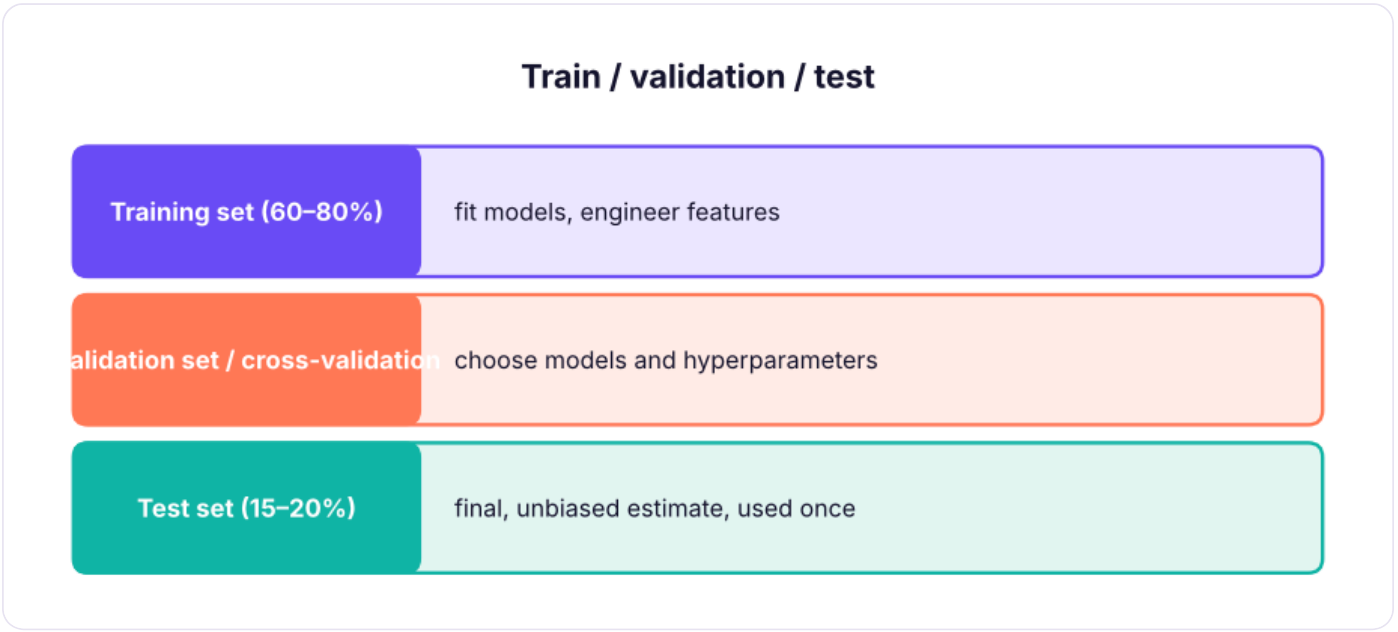
We judge a model on data it has **never seen**. Hold out a test set before doing anything else.

Python

```python
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split
tc = pd.read_csv("telecom_churn.csv")
X = tc.drop(columns=["customer_id", "churn", "total_charges"])          # total_charges ≈ tenure × monthly: redundant
y = (tc["churn"] == "Yes").astype(int)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
print(X_train.shape, X_test.shape, f"churn rate train {y_train.mean():.3f} | test {y_test.mean():.3f}")
```

OUTPUT

```
(2400, 7) (600, 7) churn rate train 0.258 | test 0.257
```



Typical data splits. The test set is a final exam: use it once, at the end.

Option	Use
`test_size=0.2`	Hold out 20%
`stratify=y`	Keep the class balance the same in both sets (classification)
`random_state=42`	Reproducible split
Time-based split	For time series and anything predicted in the future: train on the past, test on the most recent period
Group split (`GroupShuffleSplit`)	Keep all rows of a customer/store in one set

WATCH OUT **Never tune on the test set**

If you try many models and keep the one with the best **test** score, the test score is no longer an unbiased estimate. Choose with cross-validation on the training set (lesson 15); look at the test set once.

Baselines

A model is only useful if it beats a simple baseline.

Python

```
from sklearn.dummy import DummyClassifier, DummyRegressor
from sklearn.metrics import accuracy_score, recall_score
base = DummyClassifier(strategy="most_frequent").fit(X_train, y_train)
pred = base.predict(X_test)
print(f"always 'no churn': accuracy {accuracy_score(y_test, pred):.3f} | recall of churners
{recall_score(y_test, pred):.3f}")
rule = ((X_test.contract == "Month-to-month") & (X_test.tenure_months <= 12)).astype(int)       #
business rule from the EDA
print(f"rule 'new month-to-month': accuracy {accuracy_score(y_test, rule):.3f} | recall
{recall_score(y_test, rule):.3f}")
```

OUTPUT

```
always 'no churn': accuracy 0.743 | recall of churners 0.000
rule 'new month-to-month': accuracy 0.777 | recall 0.240
```

74% accuracy by predicting "no churn" for everyone shows why **accuracy alone is misleading** for imbalanced problems (lesson 14).

Preprocessing with ColumnTransformer

Models need numbers on comparable scales. Numeric and categorical columns need different treatment.

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline, make_pipeline
num_cols = ["senior_citizen", "tenure_months", "monthly_charges", "support_calls"]
cat_cols = ["contract", "internet_service", "payment_method"]
prep = ColumnTransformer([
    ("num", make_pipeline(SimpleImputer(strategy="median"), StandardScaler()), num_cols),
    ("cat", make_pipeline(SimpleImputer(strategy="most_frequent"),
OneHotEncoder(handle_unknown="ignore")), cat_cols),
])
Xt = prep.fit_transform(X_train)
print(Xt.shape, list(prep.get_feature_names_out())[:7], "...")
```

OUTPUT

```
(2400, 14) ['num__senior_citizen', 'num__tenure_months', 'num__monthly_charges',
'num__support_calls', 'cat__contract_Month-to-month', 'cat__contract_One year',
'cat__contract_Two year'] ...
```

Pipelines: preprocessing + model as one object

Python

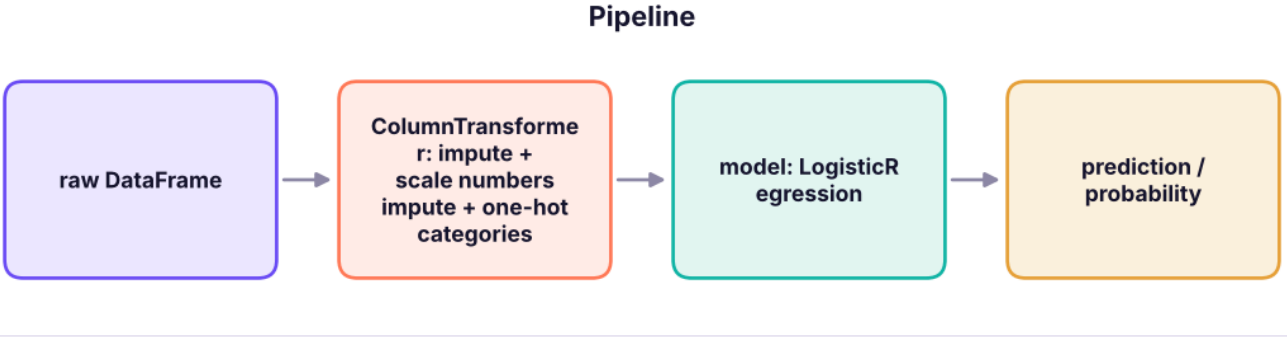
```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import roc_auc_score
models = {"logistic regression": LogisticRegression(max_iter=1000), "k-nearest neighbours":
KNeighborsClassifier(25),
          "random forest": RandomForestClassifier(300, min_samples_leaf=5, random_state=0), "gradient
boosting": HistGradientBoostingClassifier(random_state=0)}
for name, model in models.items():
    pipe = Pipeline([("prep", prep), ("model", model)]).fit(X_train, y_train)
    proba = pipe.predict_proba(X_test)[:, 1]
    print(f"{name:22s} accuracy {pipe.score(X_test, y_test):.3f} | ROC-AUC {roc_auc_score(y_test,
proba):.3f} | recall {recall_score(y_test, pipe.predict(X_test)):.3f}")
```

OUTPUT

```
logistic regression    accuracy 0.790 | ROC-AUC 0.803 | recall 0.396
k-nearest neighbours   accuracy 0.785 | ROC-AUC 0.759 | recall 0.344
random forest          accuracy 0.798 | ROC-AUC 0.793 | recall 0.403
gradient boosting      accuracy 0.760 | ROC-AUC 0.755 | recall 0.455
```

Why pipelines?

1. **No leakage:** scalers and imputers learn their statistics from the training data only, inside each fit (and inside every cross-validation fold).
2. **One object** to tune, save and deploy: raw data in, predictions out.
3. **Less code** and fewer mistakes when switching models.



A pipeline chains steps; fit runs fit_transform on each step, predict runs transform then predict.

A regression example

Python

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
h = pd.read_csv("house_prices.csv")
Xh, yh = h.drop(columns=["house_id", "price_lakh"]), h["price_lakh"]
Xh_tr, Xh_te, yh_tr, yh_te = train_test_split(Xh, yh, test_size=0.2, random_state=1)
prep_h = ColumnTransformer([("num", make_pipeline(SimpleImputer(strategy="median"), StandardScaler()),
["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km", "parking"]),
                            ("cat", OneHotEncoder(handle_unknown="ignore"), ["city"])])
for name, m in [("mean baseline", DummyRegressor()), ("linear regression", LinearRegression())]:
    p = Pipeline([("prep", prep_h), ("m", m)]).fit(Xh_tr, yh_tr).predict(Xh_te)
    print(f"{name:18s} MAE {mean_absolute_error(yh_te, p):6.2f} lakh | R² {r2_score(yh_te, p):.3f}")
```

OUTPUT

```
mean baseline      MAE  23.53 lakh | R² -0.005
linear regression  MAE   7.18 lakh | R² 0.900
```

Saving and loading a model

Python

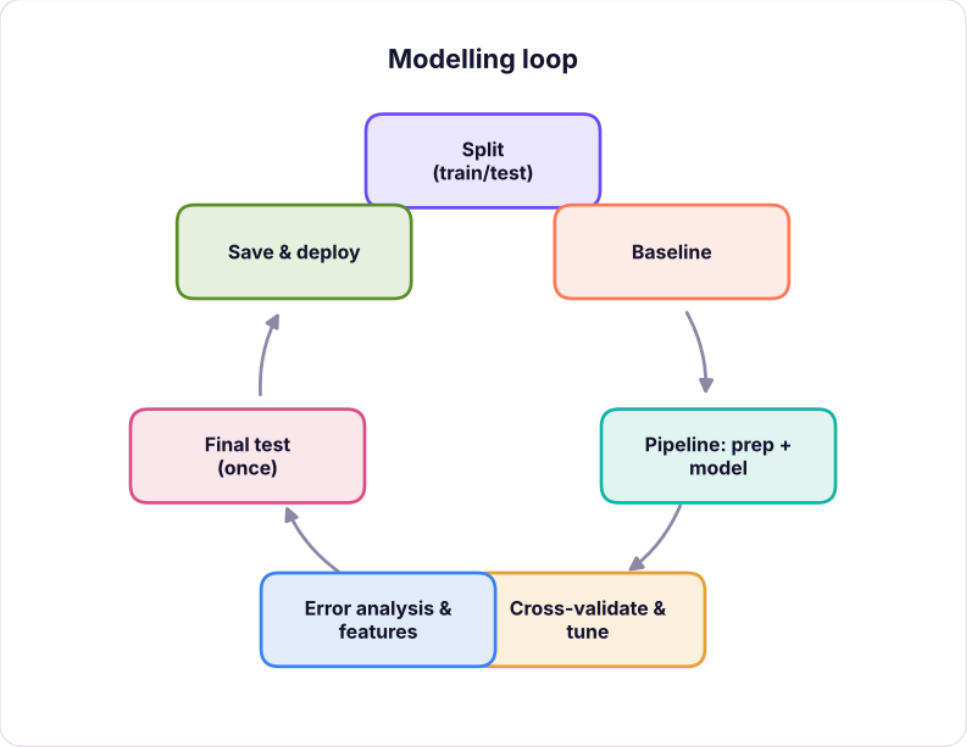
```
import joblib, os
best = Pipeline([("prep", prep), ("model", LogisticRegression(max_iter=1000))]).fit(X_train, y_train)
joblib.dump(best, "churn_model.joblib")
loaded = joblib.load("churn_model.joblib")
new_customers = pd.DataFrame([{"senior_citizen": 0, "tenure_months": 3, "contract": "Month-to-month",
"internet_service": "Fiber", "payment_method": "UPI", "monthly_charges": 95.0, "support_calls": 3},
                              {"senior_citizen": 1, "tenure_months": 60, "contract": "Two year",
"internet_service": "DSL", "payment_method": "Auto-debit", "monthly_charges": 55.0, "support_calls":
0}])
print(loaded.predict_proba(new_customers)[:, 1].round(3), f"| file size
{os.path.getsize('churn_model.joblib') / 1024:.0f} KB")
os.remove("churn_model.joblib")
```

OUTPUT

```
[0.864 0.008] | file size 5 KB
```

Save the **whole pipeline**, note the library versions (`sklearn.__version__`), and keep a model card (data, date, metrics, intended use). Load models only from sources you trust: joblib/pickle files can run code.

The workflow, end to end



The modelling loop you will repeat in every lesson of this track.

Summary

KEY TAKEAWAYS

- Every estimator uses `fit`, then `predict` / `predict_proba` or `transform`; X is 2-D, y is 1-D.
- Split first (stratified for classification, by time for forecasting); the test set is used once.
- Always compare with a baseline (`DummyClassifier`, `DummyRegressor`, or a business rule).
- `ColumnTransformer + Pipeline` apply the right preprocessing to each column type and prevent leakage.
- Save the fitted pipeline with joblib and document versions and metrics.

Quick quiz

6 questions · tap an answer to check it

1. Why use `stratify=y` in `train_test_split` ?

- **A** To keep the same class proportions in train and test sets
- **B** To shuffle the data
- **C** To scale features
- **D** To remove outliers

Answer: A. Important for imbalanced classes.

2. Where should a StandardScaler learn its mean and standard deviation?

- **A** On the full dataset
- **B** On the training data only (inside the pipeline)
- **C** On the test data
- **D** It does not matter

Answer: B. Fitting on all data leaks test information.

3. A churn model has 74% accuracy and churn is 26%. What can you say?

- **A** It is good
- **B** It may be no better than predicting "no churn" for everyone; check recall and AUC
- **C** It is perfect
- **D** It is overfitting

Answer: B. Compare with a baseline.

4. What does `ColumnTransformer` do?

- **A** Applies different preprocessing to different columns and combines the results
- **B** Trains several models
- **C** Splits the data
- **D** Removes columns with missing values

Answer: A. e.g. scale numbers, one-hot encode categories.

5. Which split suits forecasting next month's demand?

- **A** Random split
- **B** Time-based: train on earlier periods, test on the latest period
- **C** Stratified split
- **D** No split

Answer: B. Random splits leak the future into training.

6. What should you save for deployment?

- **A** Only the model coefficients
- **B** The entire fitted pipeline (preprocessing + model) with version notes
- **C** The test set
- **D** The notebook only

Answer: B. One object turns raw data into predictions.

EXERCISE **Practice**

(1) Build a pipeline for `hr_employees.csv` that predicts attrition with logistic regression; compare it with a dummy baseline using ROC-AUC. (2) Add `job_title` as a categorical feature: does AUC change? (3) Save and reload the pipeline and score two made-up employees.

Show solution (worked answer)

Python

```
hr = pd.read_csv("hr_employees.csv")
y2 = (hr.attrition == "Yes").astype(int); X2 = hr.drop(columns=["emp_id", "attrition",
"hire_date"])
nums = X2.select_dtypes("number").columns.tolist(); cats = [c for c in X2.columns if c not in
nums]
p2 = Pipeline([("prep", ColumnTransformer([("n", StandardScaler(), nums), ("c",
OneHotEncoder(handle_unknown="ignore"), cats)])), ("m", LogisticRegression(max_iter=2000))])
a, b, c, d = train_test_split(X2, y2, test_size=.2, stratify=y2, random_state=0)
print(roc_auc_score(d, p2.fit(a, c).predict_proba(b)[:, 1]), roc_auc_score(d,
DummyClassifier().fit(a, c).predict_proba(b)[:, 1]))
```

INTERVIEW CORNER **Questions you may be asked**

Q: Why do we need a train/test split? To estimate how the model performs on new, unseen data; training performance is optimistic because the model has seen those examples.

Q: What is a pipeline and why use one? A chain of preprocessing steps and a model trained and applied as one object; it prevents leakage, simplifies tuning and makes deployment reliable.

Q: What is the difference between fit_transform and transform? `fit_transform` learns parameters (e.g. mean, categories) and applies them, used on training data; `transform` applies already-learned parameters, used on validation, test and new data.

STATISTICS & MACHINE LEARNING · LESSON 9

Data Preprocessing and Feature Engineering for ML

Prepare features that models can learn from. Imputation strategies, outlier handling, encoding categorical variables (one-hot, ordinal, target encoding), scaling (standard, min-max, robust), transformations (log, Box-Cox/Yeo-Johnson), binning, interactions and polynomial features, date and text features, and feature selection, all inside leak-free pipelines.

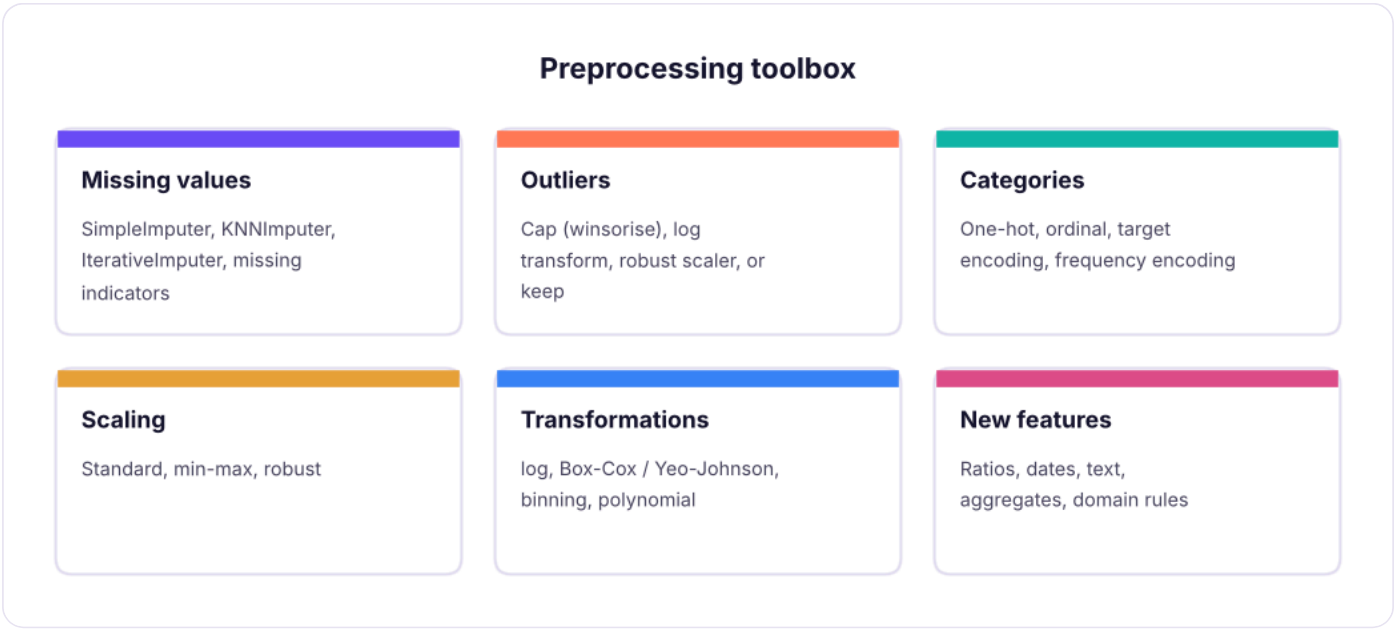
ML class 6 · scikit-learn 1.7+ · Intermediate · 120 min

By the end of this lesson you can

- Impute missing values with simple, group-aware and model-based strategies and add missing indicators
- Encode nominal and ordinal categories correctly, including high-cardinality columns
- Choose a scaler and know which algorithms need scaling
- Engineer features from numbers, dates, text and domain knowledge
- Select features with filter, wrapper and embedded methods without leaking test data

Why preprocessing matters

Algorithms only see numbers. How you represent the data often matters more than which algorithm you choose ("better data beats fancier models").



The preprocessing toolbox, applied inside a pipeline.

Missing values

Python

```
import pandas as pd, numpy as np
from sklearn.impute import SimpleImputer, KNNImputer
from sklearn.experimental import enable_iterative_imputer  # noqa: F401
from sklearn.impute import IterativeImputer
h = pd.read_csv("house_prices.csv")
num = h[["area_sqft", "bedrooms", "bathrooms", "age_years", "price_lakh"]]
mask = num.area_sqft.isna(); print("missing area:", mask.sum())
for name, imp in [("median", SimpleImputer(strategy="median")), ("KNN (5 neighbours)",
KNNImputer(n_neighbors=5)), ("iterative (regression)", IterativeImputer(random_state=0))]:
    filled = pd.DataFrame(imp.fit_transform(num), columns=num.columns)
    print(f"{name:24s} imputed areas: mean {filled.loc[mask, 'area_sqft'].mean():7.0f}, sd
{filled.loc[mask, 'area_sqft'].std():6.0f}")
print("SimpleImputer(add_indicator=True) adds a column:", SimpleImputer(strategy="median",
add_indicator=True).fit(num[["area_sqft"]]).get_feature_names_out().tolist())
```

OUTPUT

```
missing area: 25
median                   imputed areas: mean    1240, sd      0
KNN (5 neighbours)       imputed areas: mean    1396, sd    294
iterative (regression)   imputed areas: mean    1384, sd    310
SimpleImputer(add_indicator=True) adds a column: ['area_sqft', 'missingindicator_area_sqft']
```

The median fills every gap with one value (sd 0); KNN and iterative imputers use the other columns (bedrooms, price), so they produce realistic, varied values. In real projects compare strategies with cross-validation, and never use the target to impute features at prediction time.

Encoding categorical variables

Python

```
from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder, TargetEncoder
hr = pd.read_csv("hr_employees.csv")
print(hr.education.unique().tolist(), hr.department.nunique(), "departments |", hr.job_title.nunique(),
"job titles")
oh = OneHotEncoder(handle_unknown="ignore", sparse_output=False).fit(hr[["department"]])
print("one-hot:", oh.get_feature_names_out().tolist())
order = [["Diploma", "Bachelor's", "Master's", "PhD"]]
present = [e for e in order[0] if e in set(hr.education)]
oe = OrdinalEncoder(categories=[present]).fit(hr[["education"]])
print("ordinal:", dict(zip(present, oe.transform(pd.DataFrame({"education": present})).ravel())))
y = (hr.attrition == "Yes").astype(int)
te = TargetEncoder(target_type="binary", random_state=0).fit(hr[["job_title"]], y)
print("target encoding (smoothed attrition rate per job title, first 4):", dict(zip(te.categories_[0]
[:4], te.encodings_[0][:4].round(3))))
print("frequency encoding:",
hr.job_title.map(hr.job_title.value_counts(normalize=True)).round(3).head(3).tolist())
```

OUTPUT

```
["Bachelor's", "Master's", 'Diploma', 'PhD'] 6 departments | 18 job titles
one-hot: ['department_Finance', 'department_HR', 'department_IT', 'department_Operations',
'department_R&D', 'department_Sales']
ordinal: {'Diploma': np.float64(0.0), "Bachelor's": np.float64(1.0), "Master's":
np.float64(2.0), 'PhD': np.float64(3.0)}
target encoding (smoothed attrition rate per job title, first 4): {'Accountant':
np.float64(0.256), 'Analyst': np.float64(0.28), 'Data Scientist': np.float64(0.247),
'Developer': np.float64(0.295)}
frequency encoding: [0.049, 0.063, 0.063]
```

Encoding	For	Notes
One-hot	Nominal, few categories	One column per category; `handle_unknown="ignore"`; `drop="first"` for linear models if needed
Ordinal	Ordered categories (education, size S/M/L)	Give the order explicitly
Target encoding	High-cardinality nominal (job title, city, merchant)	Uses the target: must be cross-fitted (sklearn's `TargetEncoder` does this)
Frequency / count	High cardinality	Simple, leak-free
Hashing, embeddings	Very high cardinality (IDs, text)	Fixed size; embeddings in deep learning

WATCH OUT **Label encoding nominal data**

Mapping cities to 0, 1, 2... tells linear models and KNN that "Pune (2) > Delhi (1)", a false order. Use one-hot encoding for nominal variables (tree models tolerate ordinal codes better).

Scaling

Python

```
from sklearn.preprocessing import StandardScaler, MinMaxScaler, RobustScaler
inc = hr[["monthly_income"]]
out = pd.DataFrame({"raw": inc.monthly_income})
for name, sc in [("standard", StandardScaler()), ("minmax", MinMaxScaler()), ("robust",
RobustScaler())]: out[name] = sc.fit_transform(inc).ravel()
print(out.describe().loc[["mean", "std", "min", "50%", "max"]].round(2))
```

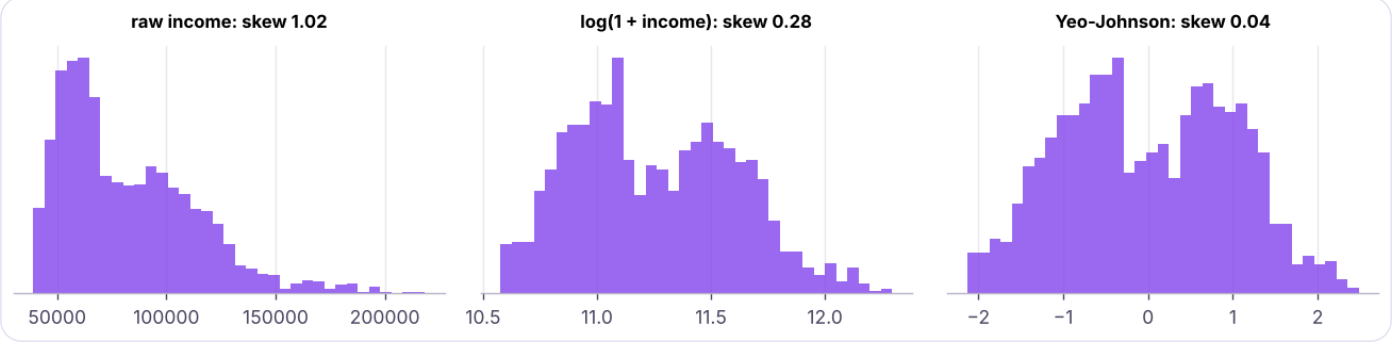
OUTPUT

```
            raw  standard  minmax  robust
mean    82610.67     -0.00    0.24    0.18
std     31145.80      1.00    0.17    0.71
min     39167.00     -1.40    0.00   -0.81
50%     74708.50     -0.25    0.20    0.00
max    218417.00      4.36    1.00    3.28
```

Scaler	Result	Use
StandardScaler	mean 0, sd 1	Default for linear models, SVM, PCA, neural networks
MinMaxScaler	range 0–1	Bounded inputs (neural networks, images)
RobustScaler	median 0, IQR 1	Data with outliers

Needs scaling: KNN, K-means, SVM, PCA, regularised linear/logistic regression, neural networks, gradient descent. **Does not:** decision trees, random forests, gradient boosting (they split on thresholds).

Transformations



Log and Yeo-Johnson transforms make right-skewed income closer to symmetric.

Log transforms help linear models with multiplicative effects (prices, incomes) and can be applied to the **target** too (`TransformedTargetRegressor`). Binning (`KBinsDiscretizer` , `pd.cut`) trades detail for robustness and interpretability.

Feature engineering

Good features encode domain knowledge the algorithm cannot easily find.

Python

```python
hr["hire_date"] = pd.to_datetime(hr.hire_date)
fe = pd.DataFrame({
    "income_per_year_tenure": hr.monthly_income / (hr.years_at_company + 1),       # ratio
    "is_long_commute": (hr.distance_from_home_km > 20).astype(int),                # threshold from
domain knowledge
    "overtime_x_low_sat": ((hr.overtime == "Yes") & (hr.job_satisfaction <= 2)).astype(int),   #
interaction
    "hire_month": hr.hire_date.dt.month, "hired_in_q4": hr.hire_date.dt.quarter.eq(4).astype(int),   #
date parts
    "income_vs_dept_median": hr.monthly_income /
hr.groupby("department").monthly_income.transform("median"),   # group-relative
    "title_is_manager": hr.job_title.str.contains("Manager").astype(int),          # text pattern
})
fe["left"] = (hr.attrition == "Yes").astype(int)
print(fe.corr()["left"].drop("left").round(3).sort_values())
print(fe.groupby("overtime_x_low_sat").left.mean().round(3).to_dict())
```

OUTPUT

```
income_vs_dept_median    -0.109
title_is_manager         -0.048
hire_month                0.038
hired_in_q4               0.043
income_per_year_tenure    0.057
is_long_commute           0.094
overtime_x_low_sat        0.165
Name: left, dtype: float64
{0: 0.229, 1: 0.43}
```

Source	Feature ideas
Numbers	Ratios, differences, logs, bins, polynomial and interaction terms
Dates	Day of week, month, holiday flags, days since last event, tenure, seasonality (sin/cos)
Transactions	Recency, frequency, monetary (RFM), averages over 7/30/90 days, trends
Text	Length, keyword flags, TF-IDF, embeddings (lesson 25 and GenAI track)
Geography	Distance to centre/store, region aggregates
Groups	Value relative to group mean/median (income vs department)

Python

```python
from sklearn.preprocessing import PolynomialFeatures
pf = PolynomialFeatures(degree=2, include_bias=False).fit(pd.DataFrame({"area": [1], "age": [1]}))
print(pf.get_feature_names_out().tolist())
```

OUTPUT

```
['area', 'age', 'area^2', 'area age', 'age^2']
```

Feature selection

More features are not always better: irrelevant ones add noise, cost and overfitting.

Python

```python
from sklearn.feature_selection import mutual_info_classif, SelectKBest, RFE
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
Xn = pd.get_dummies(hr[["age", "monthly_income", "years_at_company", "job_satisfaction",
"distance_from_home_km", "performance_rating", "overtime", "department"]],
drop_first=True).astype(float)
mi = pd.Series(mutual_info_classif(Xn, y, random_state=0),
index=Xn.columns).sort_values(ascending=False)
print("filter (mutual information):", mi.round(3).head(5).to_dict())
rfe = RFE(LogisticRegression(max_iter=2000),
n_features_to_select=4).fit(StandardScaler().fit_transform(Xn), y)
print("wrapper (RFE, logistic):", Xn.columns[rfe.support_].tolist())
rf = RandomForestClassifier(300, random_state=0, min_samples_leaf=5).fit(Xn, y)
print("embedded (random forest importance):", pd.Series(rf.feature_importances_,
Xn.columns).sort_values(ascending=False).round(3).head(5).to_dict())
```

OUTPUT

```
filter (mutual information): {'years_at_company': 0.025, 'job_satisfaction': 0.019,
'department_HR': 0.017, 'monthly_income': 0.009, 'overtime_Yes': 0.007}
wrapper (RFE, logistic): ['years_at_company', 'job_satisfaction', 'distance_from_home_km',
'overtime_Yes']
embedded (random forest importance): {'monthly_income': 0.242, 'distance_from_home_km': 0.18,
'age': 0.155, 'years_at_company': 0.117, 'job_satisfaction': 0.108}
```

The three methods disagree. That is normal: they measure different things. Random-forest (impurity) importances also favour continuous features with many distinct values, so check them with permutation importance (lesson 27).

Method	How	Pros / cons
Filter	Score each feature (correlation, chi-square, mutual information)	Fast; ignores interactions
Wrapper	Search subsets with a model (RFE, forward selection)	Strong; slow, risk of overfitting
Embedded	Selection during training (Lasso L1, tree importances)	Efficient; model-specific

Do feature selection **inside** cross-validation (in the pipeline) so the test data never influences which features are chosen.

Putting it together: a leak-free pipeline

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline, make_pipeline
from sklearn.model_selection import cross_val_score
X = hr.drop(columns=["emp_id", "attrition", "hire_date"])
pre = ColumnTransformer([
    ("num", make_pipeline(SimpleImputer(strategy="median"), StandardScaler()), ["age",
"years_at_company", "monthly_income", "job_satisfaction", "distance_from_home_km",
"performance_rating"]),
    ("ord", OrdinalEncoder(categories=[present]), ["education"]),
    ("ohe", OneHotEncoder(handle_unknown="ignore"), ["gender", "department", "overtime", "city"]),
    ("te", TargetEncoder(target_type="binary", random_state=0), ["job_title"]),
])
pipe = Pipeline([("pre", pre), ("sel", SelectKBest(mutual_info_classif, k=12)), ("model",
LogisticRegression(max_iter=2000))])
scores = cross_val_score(pipe, X, y, cv=5, scoring="roc_auc")
print(f"5-fold ROC-AUC {scores.mean():.3f} ± {scores.std():.3f}")
```

OUTPUT

```
5-fold ROC-AUC 0.661 ± 0.013
```

Summary

KEY TAKEAWAYS

- Impute with medians/modes, KNN or iterative imputers, and add missing-value indicators when missingness carries information.
- One-hot for nominal, ordinal encoding with an explicit order, target or frequency encoding for high-cardinality columns.
- Scale for distance- and gradient-based models (standard, min-max, robust); trees do not need scaling.
- Transform skewed variables (log, Yeo-Johnson); engineer ratios, interactions, date and group-relative features from domain knowledge.
- Select features with filter, wrapper or embedded methods, always inside the cross-validated pipeline.

Quick quiz

6 questions · tap an answer to check it

1. Which encoding suits "education level" (Diploma < Bachelor's < Master's < PhD)?

- **A** Ordinal encoding with the order given explicitly
- **B** One-hot only
- **C** Target encoding only
- **D** Leave as text

Answer: A. The categories have a natural order.

2. Which algorithm does NOT need feature scaling?

- **A** KNN
- **B** SVM
- **C** Random forest
- **D** K-means

Answer: C. Trees split on thresholds, unaffected by scale.

3. Why must target encoding be cross-fitted?

- **A** Using a row's own target to encode it leaks the answer and overfits
- **B** It is slow otherwise
- **C** It needs scaling
- **D** It only works on numbers

Answer: A. sklearn's TargetEncoder handles this internally.

4. What does RobustScaler use?

A Mean and standard deviation

B Median and interquartile range

C Min and max

D Logarithms

Answer: B. Robust to outliers.

5. Where should feature selection happen?

A On the full dataset before splitting

B Inside the pipeline, within cross-validation on training data

C On the test set

D After deployment

Answer: B. Otherwise test information leaks into the choice.

6. Which is an interaction feature?

A monthly_income

B overtime AND low job satisfaction

C log(income)

D hire_month

Answer: B. It combines two features.

EXERCISE **Practice (house_prices.csv)**

(1) Engineer `price_per_sqft`, `rooms = bedrooms + bathrooms`, `is_new = age_years ⩽ 5`. Which correlate most with price? (Careful: price_per_sqft uses the target, so it cannot be a feature!) (2) Build a pipeline with KNN imputation, one-hot city, standard scaling and Ridge regression; report 5-fold R^2. (3) Try a log-transformed target with `TransformedTargetRegressor`.

Show solution (worked answer)

Python

```
from sklearn.linear_model import Ridge
from sklearn.compose import TransformedTargetRegressor
h["rooms"] = h.bedrooms + h.bathrooms; h["is_new"] = (h.age_years ⩽ 5).astype(int)
Xh = h.drop(columns=["house_id", "price_lakh"]); yh = h.price_lakh
pre_h = ColumnTransformer([("num", make_pipeline(KNNImputer(), StandardScaler()), ["area_sqft",
"bedrooms", "bathrooms", "age_years", "distance_to_center_km", "parking", "rooms", "is_new"]),
("cat", OneHotEncoder(handle_unknown="ignore"), ["city"])])
print(cross_val_score(Pipeline([("p", pre_h), ("m", Ridge())]), Xh, yh, cv=5,
scoring="r2").mean())
print(cross_val_score(Pipeline([("p", pre_h), ("m", TransformedTargetRegressor(Ridge(),
func=np.log1p, inverse_func=np.expm1))]), Xh, yh, cv=5, scoring="r2").mean())
```

INTERVIEW CORNER **Questions you may be asked**

Q: How do you handle categorical variables with 1,000 levels? Target encoding (cross-fitted), frequency encoding, grouping rare levels into "Other", hashing, or learned embeddings; tree-based libraries like CatBoost handle them natively.

Q: Normalisation vs standardisation? Normalisation (min-max) rescales to 0–1; standardisation centres to mean 0 and sd 1. Use standardisation by default; min-max for bounded inputs.

Q: Give an example of a feature you engineered that helped. Use STAR: e.g. "days since last purchase" and "spend in the last 30 days vs previous 30" for churn, which raised recall in the top decile from X to Y.

STATISTICS & MACHINE LEARNING · LESSON 10

Linear Regression: Fitting, Interpreting, Assumptions and Diagnostics

Predict a number and explain it. Simple and multiple linear regression with ordinary least squares, interpreting coefficients, dummy variables and log targets, R^2, adjusted R^2, RMSE and MAE, the LINE assumptions and residual diagnostics, multicollinearity and VIF, outliers and influence, with statsmodels for inference and scikit-learn for prediction on the house-price data.

Python class 16 · ML class 4 | statsmodels 0.1x · scikit-learn 1.7+ | Intermediate | 130 min

By the end of this lesson you can

- Fit simple and multiple linear regression with statsmodels and scikit-learn
- Interpret intercepts, slopes, dummy variables and log-transformed coefficients
- Evaluate models with R^2, adjusted R^2, MAE, RMSE and MAPE on test data
- Check linearity, independence, normality and equal variance of residuals
- Detect multicollinearity with VIF and influential points with Cook's distance

Simple linear regression

Find the straight line that best predicts *y* from *x*:

$$\hat{y} = \beta_0 + \beta_1 x$$

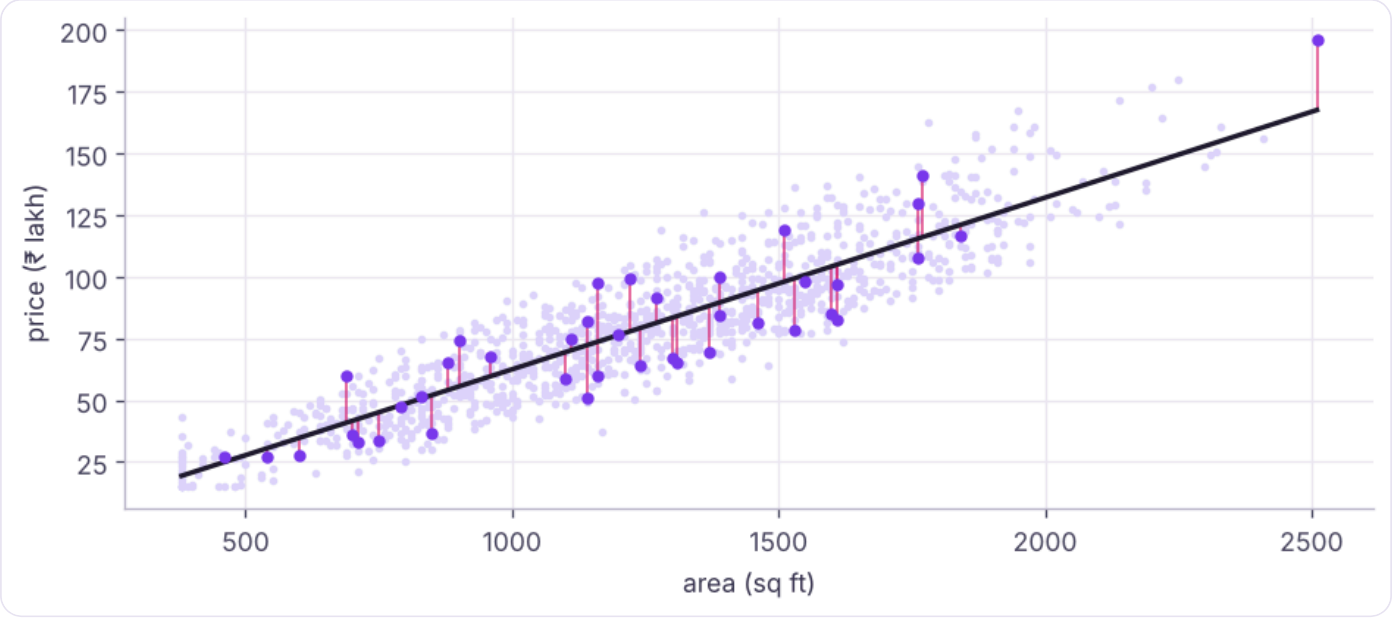
Ordinary least squares (OLS) chooses β_0 and β_1 to minimise the **sum of squared residuals**, where a residual is actual − predicted.

Python

```
import pandas as pd, numpy as np
import statsmodels.formula.api as smf
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"])
pps = h.price_lakh * 1e5 / h.area_sqft                          # price per sq ft
h = h[pps.between(pps.median() / 3, pps.median() * 3)]          # drop 4 impossible prices (data-entry
errors found in EDA)
print(len(h), "houses")
m1 = smf.ols("price_lakh ~ area_sqft", data=h).fit()
print(m1.params.round(4).to_dict(), "| R² =", round(m1.rsquared, 3))
b0, b1 = m1.params
print(f"price ≈ {b0:.2f} + {b1:.4f} × area → each extra 100 sq ft adds about ₹{100 * b1:.1f} lakh")
print("prediction for 1,500 sq ft:", round(m1.predict(pd.DataFrame({"area_sqft": [1500]}))[0], 1),
"lakh")
```

OUTPUT

```
1171 houses
{'Intercept': -7.0873, 'area_sqft': 0.0696} | R² = 0.827
price ≈ -7.09 + 0.0696 × area → each extra 100 sq ft adds about ₹7.0 lakh
prediction for 1,500 sq ft: 97.3 lakh
```



The least-squares line and the residuals (vertical distances) it minimises, for 40 sample houses.

Multiple linear regression

Python

```
m2 = smf.ols("price_lakh ~ area_sqft + bedrooms + bathrooms + age_years + distance_to_center_km +
parking + C(city)", data=h).fit()
print(m2.summary().tables[1])
print(f"R² {m2.rsquared:.3f} | adjusted R² {m2.rsquared_adj:.3f} | F-test p {m2.f_pvalue:.1e} | n =
{int(m2.nobs)}")
```

OUTPUT

```
==============================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                 15.2624      1.057     14.446      0.000      13.190      17.335
C(city)[T.Chennai]       -24.2893      0.625    -38.870      0.000     -25.515     -23.063
C(city)[T.Hyderabad]     -15.6931      0.637    -24.641      0.000     -16.943     -14.444
C(city)[T.Pune]          -20.0518      0.624    -32.122      0.000     -21.277     -18.827
area_sqft                  0.0686      0.001     81.096      0.000       0.067       0.070
bedrooms                   0.9105      0.555      1.641      0.101      -0.178       1.999
bathrooms                 -0.3698      0.472     -0.783      0.434      -1.296       0.557
age_years                 -0.4218      0.026    -16.187      0.000      -0.473      -0.371
distance_to_center_km     -0.1990      0.037     -5.407      0.000      -0.271      -0.127
parking                    0.5099      0.487      1.046      0.296      -0.446       1.466
==============================================================================================
R² 0.937 | adjusted R² 0.937 | F-test p 0.0e+00 | n = 1171
```

Reading the output

Column	Meaning
`coef`	Change in predicted price for +1 unit of the feature, **holding the others constant**
`std err`	Uncertainty of the coefficient
`t`, `P>|t|`	Test of H_o: coefficient = 0
`[0.025 0.975]`	95% confidence interval
`C(city)[T.X]`	Dummy variable: difference from the **reference city** (the first alphabetically)

Python

```python
coef = m2.params; ci = m2.conf_int()
for f in ["area_sqft", "age_years", "distance_to_center_km", "parking"]:
    print(f"{f:22s} {coef[f]:+8.3f} lakh per unit (95% CI {ci.loc[f, 0]:+.3f} to {ci.loc[f, 1]:+.3f}),
p = {m2.pvalues[f]:.3g}")
ref = sorted(h.city.unique())[0]
print(f"city effects vs {ref}:", {k.split('T.')[1][:-1]: round(v, 1) for k, v in coef.items() if
k.startswith("C(city)")})
```

OUTPUT

```
area_sqft              +0.069 lakh per unit (95% CI +0.067 to +0.070), p = 0
age_years              -0.422 lakh per unit (95% CI -0.473 to -0.371), p = 2.69e-53
distance_to_center_km  -0.199 lakh per unit (95% CI -0.271 to -0.127), p = 7.76e-08
parking                +0.510 lakh per unit (95% CI -0.446 to +1.466), p = 0.296
city effects vs Bengaluru: {'Chennai': -24.3, 'Hyderabad': -15.7, 'Pune': -20.1}
```

Coefficients are **associations conditional on the other features**, not proof that renovating (changing age) would cause that price change.

Log target: percentage effects

Prices are right-skewed and effects are often multiplicative. Modelling log(price) turns coefficients into approximate percentage effects.

Python

```
m3 = smf.ols("np.log(price_lakh) ~ np.log(area_sqft) + bedrooms + bathrooms + age_years +
distance_to_center_km + parking + C(city)", data=h).fit()
print(f"elasticity: +1% area → +{m3.params['np.log(area_sqft)']:.2f}% price | each year of age → {100 *
(np.exp(m3.params['age_years']) - 1):+.2f}% | R² (log scale) {m3.rsquared:.3f}")
```

OUTPUT

```
elasticity: +1% area → +1.12% price | each year of age → -0.66% | R² (log scale) 0.927
```

Evaluating predictions

Python

```
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score,
mean_absolute_percentage_error
train, test = train_test_split(h, test_size=0.2, random_state=0)
fit = smf.ols("price_lakh ~ area_sqft + bedrooms + bathrooms + age_years + distance_to_center_km +
parking + C(city)", data=train).fit()
p_train, p_test = fit.predict(train), fit.predict(test)
for name, yt, yp in [("train", train.price_lakh, p_train), ("test", test.price_lakh, p_test)]:
    print(f"{name:5s} R² {r2_score(yt, yp):.3f} | MAE {mean_absolute_error(yt, yp):.2f} | RMSE
{mean_squared_error(yt, yp) ** 0.5:.2f} | MAPE {100 * mean_absolute_percentage_error(yt, yp):.1f}%")
```

OUTPUT

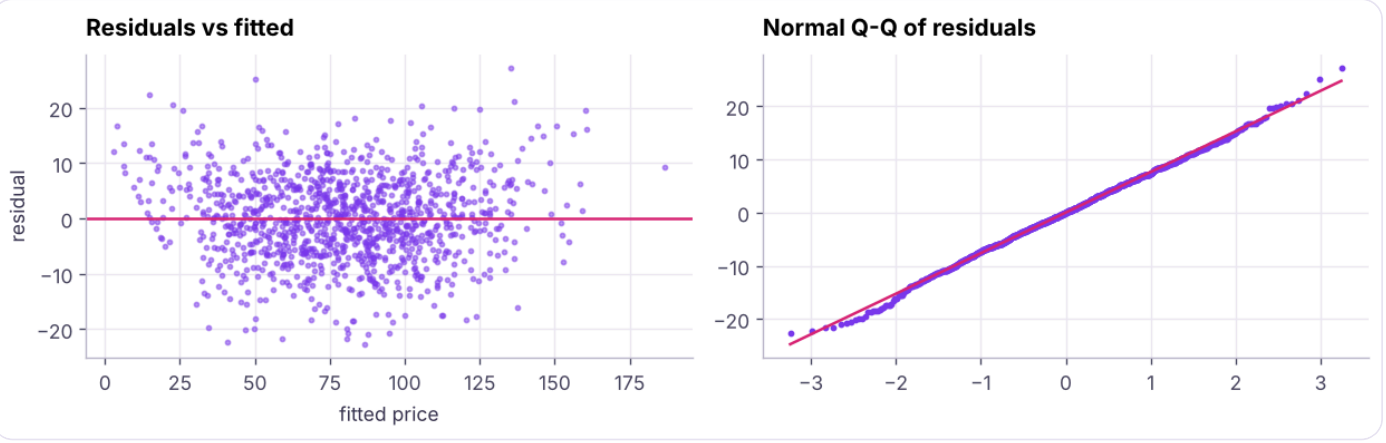
```
train R² 0.938 | MAE 5.95 | RMSE 7.51 | MAPE 9.4%
test  R² 0.936 | MAE 6.47 | RMSE 8.06 | MAPE 10.7%
```

Metric	Formula idea	Read as
MAE	mean |error|	Typical error in target units; robust
RMSE	√mean(error²)	Penalises large errors more
MAPE	mean |error| / actual	Percentage error; unstable near 0
R^2	1 − SSE/SST	Share of variance explained (0–1 on test; can be negative)
Adjusted R^2	R^2 penalised for number of features	Compare models with different feature counts

Similar train and test scores mean the model generalises; a large gap signals overfitting.

Assumptions (LINE) and diagnostics

Assumption	Meaning	Check	If violated
Linearity	Mean of y is linear in features	Residuals vs fitted: no curve	Transform, add polynomial terms, use trees
Independence	Errors not related to each other	Durbin-Watson ≈ 2; study design	Time-series models, clustered errors
Normality	Residuals ~ normal	Q-Q plot, histogram	Matters for small-sample inference; transform y
Equal variance	Constant spread of residuals	Residuals vs fitted: no funnel; Breusch-Pagan	log(y), robust (HC3) standard errors



Residual diagnostics for the multiple regression. Left: residuals vs fitted values. Right: Q-Q plot of residuals.

Python

```
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.stattools import durbin_watson
bp = het_breuschpagan(m2.resid, m2.model.exog)
print(f"Breusch-Pagan p = {bp[1]:.2e} ({'unequal variance (heteroscedastic)' if bp[1] < 0.05 else
'equal variance OK'}) | Durbin-Watson {durbin_watson(m2.resid):.2f} | residual skew
{stats.skew(m2.resid):.2f}")
robust = smf.ols(m2.model.formula, data=h).fit(cov_type="HC3")
print("area_sqft p-value: ordinary", f"{m2.pvalues['area_sqft']:.2e}", "| robust HC3", f"
{robust.pvalues['area_sqft']:.2e}")
```

OUTPUT

```
Breusch-Pagan p = 1.05e-01 (equal variance OK) | Durbin-Watson 1.99 | residual skew -0.05
area_sqft p-value: ordinary 0.00e+00 | robust HC3 0.00e+00
```

Multicollinearity

When features are strongly correlated with each other, coefficients become unstable and hard to interpret (predictions can still be fine).

Python

```
from statsmodels.stats.outliers_influence import variance_inflation_factor
Xv = h[["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km",
"parking"]].assign(const=1.0)
vif = pd.Series([variance_inflation_factor(Xv.values, i) for i in range(Xv.shape[1] - 1)],
index=Xv.columns[:-1]).round(2)
print(vif.sort_values(ascending=False).to_dict())
print(h[["area_sqft", "bedrooms", "bathrooms"]].corr().round(2))
```

OUTPUT

```
{'bedrooms': 5.53, 'bathrooms': 4.09, 'area_sqft': 2.26, 'age_years': 1.01,
'distance_to_center_km': 1.01, 'parking': 1.0}
             area_sqft  bedrooms  bathrooms
area_sqft         1.00      0.74       0.63
bedrooms          0.74      1.00       0.87
bathrooms         0.63      0.87       1.00
```

VIF above about 5–10 signals a problem. Fixes: drop or combine correlated features, use regularisation (Ridge, lesson 12), or PCA.

Outliers and influential points

Python

```
infl = m2.get_influence(); cooks = infl.cooks_distance[0]
top = np.argsort(cooks)[-3:][::-1]
print("largest Cook's distances:", cooks[top].round(3), "| threshold 4/n =", round(4 / len(h), 4))
print(h.iloc[top][["city", "area_sqft", "bedrooms", "price_lakh"]])
```

OUTPUT

```
largest Cook's distances: [0.011 0.009 0.009] | threshold 4/n = 0.0034
          city  area_sqft  bedrooms  price_lakh
464  Bengaluru     1780.0         3       162.4
113  Hyderabad     1430.0         2        66.8
863       Pune      470.0         1        37.3
```

Investigate influential points: data errors should be fixed; genuine extremes may need robust methods (Huber regression) or separate treatment.

scikit-learn version

Python

```
from sklearn.linear_model import LinearRegression
from sklearn.compose import make_column_transformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import make_pipeline
feats = ["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km", "parking", "city"]
pipe = make_pipeline(make_column_transformer((OneHotEncoder(drop="first"), ["city"]),
remainder="passthrough"), LinearRegression())
pipe.fit(train[feats], train.price_lakh)
print(f"sklearn test R² {pipe.score(test[feats], test.price_lakh):.3f} (same model as statsmodels:
{np.isclose(pipe.score(test[feats], test.price_lakh), r2_score(test.price_lakh, p_test))})")
```

OUTPUT

```
sklearn test R² 0.936 (same model as statsmodels: True)
```

Use **statsmodels** for inference (p-values, CIs, diagnostics) and **scikit-learn** for prediction pipelines, cross-validation and deployment.

Summary

KEY TAKEAWAYS

- OLS minimises squared residuals; coefficients are the change in y per unit of x, holding other features constant.
- Dummy variables compare categories with a reference; log targets give percentage effects.
- Evaluate on test data with MAE, RMSE, MAPE and R^2; adjusted R^2 penalises extra features.
- Check LINE assumptions with residual plots, Q-Q plots, Breusch-Pagan and Durbin-Watson; use robust standard errors or transformations when needed.
- VIF detects multicollinearity; Cook's distance finds influential points. Coefficients show association, not causation.

Quick quiz

6 questions · tap an answer to check it

1. What does OLS minimise?

- **A** The sum of absolute errors
- **B** The sum of squared residuals
- **C** The number of features
- **D** The p-values

Answer: B. Hence "least squares".

2. The coefficient of `age_years` is −0.6. Interpretation?

- **A** Each additional year of age is associated with ₹0.6 lakh lower price, holding other features constant
- **B** Older houses cost 60% less
- **C** Age causes prices to fall
- **D** Age is not significant

Answer: A. Conditional association, in target units.

3. Residuals fan out as fitted values grow. Which assumption is violated?

- **A** Linearity
- **B** Independence
- **C** Equal variance (homoscedasticity)
- **D** Normality only

Answer: C. Try log(y) or robust standard errors.

4. What does a VIF of 12 indicate?

- **A** Strong multicollinearity with other features
- **B** A great feature
- **C** Non-normal residuals
- **D** An outlier

Answer: A. Coefficients become unstable.

5. Test R^2 is 0.90 and train R^2 is 0.91. What does that suggest?

- **A** The model generalises well (no strong overfitting)
- **B** Severe overfitting
- **C** Data leakage for sure
- **D** Underfitting

Answer: A. Similar scores on unseen data are good news.

6. Why use a log target for prices?

- **A** Prices are skewed and effects are multiplicative; coefficients become percentage effects
- **B** To make R^2 larger always
- **C** Because statsmodels requires it
- **D** To remove outliers

Answer: A. It also stabilises variance.

EXERCISE **Practice**

(1) Add `area_sqft:C(city)` interaction: does the price per sq ft differ by city? (2) Compare test MAE of the plain and log-target models (back-transform with `np.exp`). (3) Remove the feature with the highest VIF and compare adjusted R^2. (4) Plot actual vs predicted on the test set.

Show solution (worked answer)

Python

```
mi = smf.ols("price_lakh ~ area_sqft * C(city) + bedrooms + bathrooms + age_years +
distance_to_center_km + parking", data=h).fit()
print(mi.params.filter(like="area_sqft:"))
lg = smf.ols("np.log(price_lakh) ~ area_sqft + bedrooms + bathrooms + age_years +
distance_to_center_km + parking + C(city)", data=train).fit()
print(mean_absolute_error(test.price_lakh, np.exp(lg.predict(test))),
mean_absolute_error(test.price_lakh, p_test))
m_novif = smf.ols("price_lakh ~ area_sqft + bathrooms + age_years + distance_to_center_km +
parking + C(city)", data=h).fit(); print(m_novif.rsquared_adj, m2.rsquared_adj)
plt.scatter(test.price_lakh, p_test, s=6); plt.plot([0, 300], [0, 300])
```

INTERVIEW CORNER **Questions you may be asked**

Q: What are the assumptions of linear regression? Linearity, independence of errors, normality of residuals (for inference), equal variance of residuals, and no perfect multicollinearity.

Q: R^2 vs adjusted R^2? R^2 never decreases when you add features; adjusted R^2 penalises the number of features, so it can drop when a feature adds little.

Q: How do you interpret a coefficient in a log-log model? As an elasticity: a 1% change in x is associated with a β% change in y.

STATISTICS & MACHINE LEARNING · LESSON 11

Gradient Descent: How Models Learn (Built from Scratch in NumPy)

Understand the optimisation engine behind linear regression, logistic regression and neural networks. Cost functions, gradients and the update rule, implementing batch gradient descent for linear regression in NumPy, the learning rate, feature scaling, stochastic and mini-batch variants, convergence checks, and comparison with the closed-form OLS solution and scikit-learn's SGDRegressor.

ML class 5 · NumPy 2 · scikit-learn 1.7+ · Intermediate · 110 min

By the end of this lesson you can

- Define a cost function (MSE) and explain what its gradient tells us
- Implement batch gradient descent for linear regression with NumPy
- Choose a learning rate and diagnose divergence and slow convergence with cost curves
- Explain why feature scaling speeds up gradient descent
- Compare batch, stochastic and mini-batch gradient descent and use SGDRegressor

Learning = minimising a cost

A model has **parameters** (weights). Learning means finding the weights that make a **cost function** as small as possible. For linear regression the usual cost is the **mean squared error**:

$$J(\mathbf{w}, b) = \frac{1}{2n}\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2, \qquad \hat{y}_i = \mathbf{w}\cdot\mathbf{x}_i + b$$

(The ½ simply makes the derivative tidier.) OLS has a formula for the minimum, but most models (logistic regression, neural networks) do not, so we **search** for it.

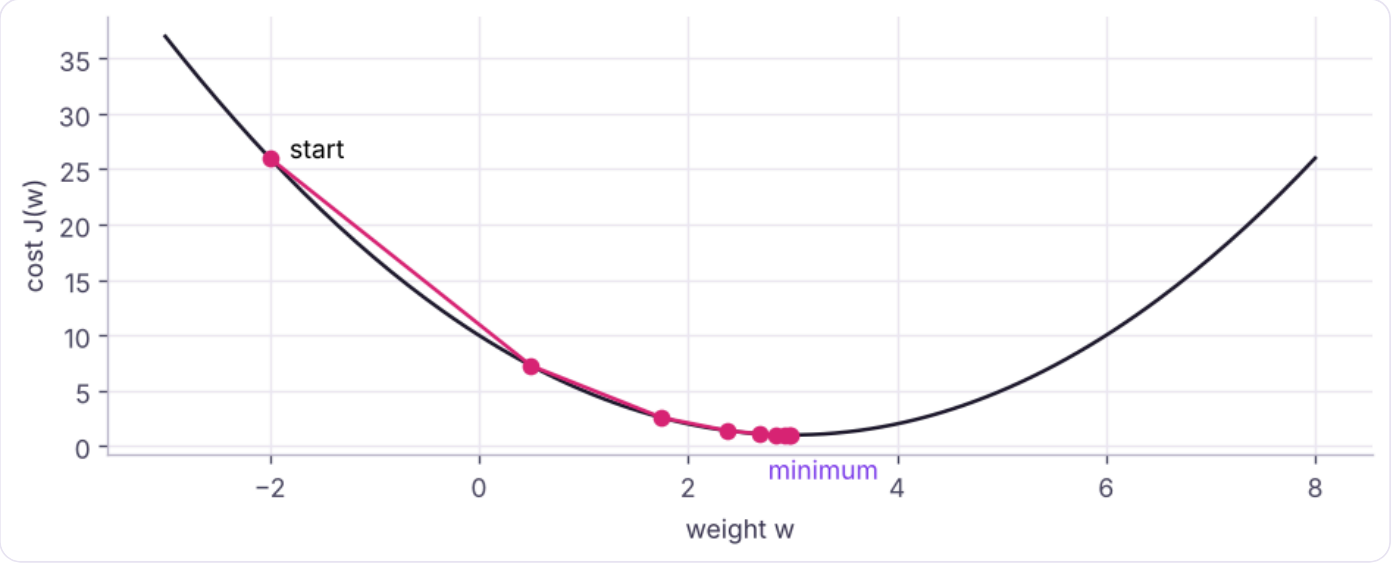
The idea: walk downhill

Imagine standing on a foggy hillside: feel the slope under your feet and step downhill; repeat until the ground is flat. The **gradient** is the slope; we step in the opposite direction.

$$w \leftarrow w - \alpha\frac{\partial J}{\partial w}, \qquad b \leftarrow b - \alpha\frac{\partial J}{\partial b}$$

α is the **learning rate** (step size). For linear regression the gradients are:

$$\frac{\partial J}{\partial \mathbf{w}} = \frac{1}{n} X^T(\hat{\mathbf{y}} - \mathbf{y}) \qquad \frac{\partial J}{\partial b} = \frac{1}{n}\sum_i (\hat{y}_i - y_i)$$



Gradient descent on a one-parameter cost curve: each step moves against the slope; steps shrink as the slope flattens near the minimum.

Implementation in NumPy

Python

```
import pandas as pd
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"])
pps = h.price_lakh * 1e5 / h.area_sqft; h = h[pps.between(pps.median() / 3, pps.median() * 3)]
feats = ["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km"]
X_raw = h[feats].to_numpy(float); y = h.price_lakh.to_numpy(float)
mu, sd = X_raw.mean(axis=0), X_raw.std(axis=0); X = (X_raw - mu) / sd          # standardise features

def gradient_descent(X, y, lr=0.1, epochs=300):
    n, p = X.shape; w, b = np.zeros(p), 0.0; history = []
    for _ in range(epochs):
        y_hat = X @ w + b
        err = y_hat - y
        history.append((err ** 2).mean() / 2)                                  # cost J
        w -= lr * (X.T @ err) / n                                              # gradient step for
weights
        b -= lr * err.mean()                                                   # gradient step for
bias
    return w, b, history

w, b, hist = gradient_descent(X, y)
print("cost: start", round(hist[0], 1), "→ after 10", round(hist[9], 1), "→ end", round(hist[-1], 2))
print("weights (per 1 sd of each feature):", dict(zip(feats, w.round(2).tolist())), "| bias", round(b,
2))
```

OUTPUT

```
cost: start 3604.0 → after 10 592.8 → end 72.59
weights (per 1 sd of each feature): {'area_sqft': 26.94, 'bedrooms': 1.29, 'bathrooms': -0.48,
'age_years': -3.81, 'distance_to_center_km': -1.12} | bias 79.25
```

Check against the exact OLS solution

Python

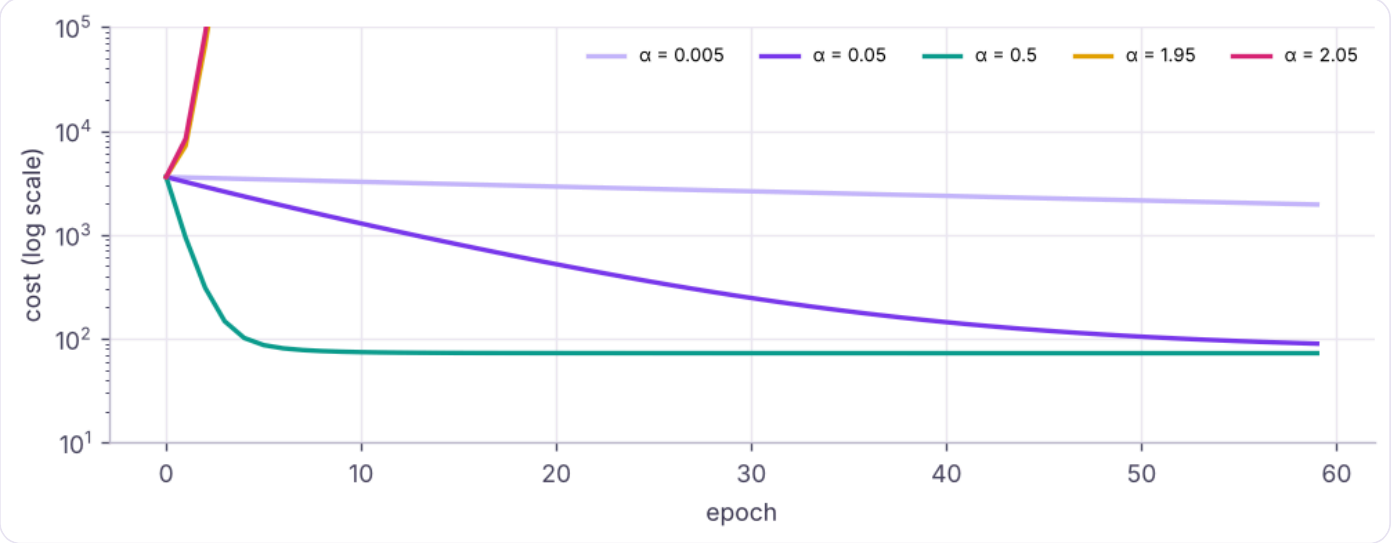
```python
X1 = np.c_[np.ones(len(X)), X]
theta = np.linalg.lstsq(X1, y, rcond=None)[0]                                    # closed form (normal
equation)
print("OLS bias", theta[0].round(2), "weights", theta[1:].round(2))
print("max difference from gradient descent:", np.abs(theta[1:] - w).max().round(4))
w_orig = w / sd; b_orig = b - (w_orig * mu).sum()                                # convert back to
original units
print(f"per sq ft: {w_orig[0]:.4f} lakh (≈ ₹{w_orig[0] * 1e5:,.0f}) | per year of age: {w_orig[3]:.3f}
lakh")
```

OUTPUT

```
OLS bias 79.25 weights [26.98  1.16 -0.38 -3.81 -1.12]
max difference from gradient descent: 0.1303
per sq ft: 0.0677 lakh (≈ ₹6,769) | per year of age: -0.442 lakh
```

Gradient descent reached the same answer as the exact formula. The **normal equation** $\theta = (X^TX)^{-1}X^Ty$ is fine for a few thousand features, but gradient descent scales to millions of rows and parameters and to models without closed-form solutions.

The learning rate



Cost curves for different learning rates: too small is slow, a good rate converges quickly, too large diverges.

Symptom	Cause	Fix
Cost falls very slowly	α too small	Increase α (×3 steps: 0.001, 0.003, 0.01…)
Cost goes up or oscillates, becomes NaN	α too large	Decrease α
Cost flattens high	Stuck / underfitting / unscaled features	Scale features, more epochs, better model

Plot the cost every time: it should go down and flatten.

Why scale features?

Python

```
best = ((X1 @ theta - y) ** 2).mean() / 2                                    # the lowest possible
cost (OLS)
with np.errstate(all="ignore"):
    for name, Xs, lr in [("raw features", X_raw, 1e-7), ("raw features", X_raw, 1e-6), ("standardised",
X, 0.1)]:
        _, _, hh = gradient_descent(Xs, y, lr=lr, epochs=300)
        gap = "diverged" if not np.isfinite(hh[-1]) or hh[-1] > 1e6 else f"{100 * (hh[-1] / best -
1):.1f}% above the optimum"
        print(f"{name:13s} lr={lr:<6} cost after 300 epochs: {gap}")
```

OUTPUT

```
raw features  lr=1e-07  cost after 300 epochs: 14.0% above the optimum
raw features  lr=1e-06  cost after 300 epochs: 13.3% above the optimum
standardised  lr=0.1    cost after 300 epochs: 0.0% above the optimum
```

With raw features, area (hundreds to thousands) and bedrooms (1–5) differ hugely in scale, so the cost surface is a long narrow valley: a learning rate small enough for area is far too small for bedrooms. Standardising makes the valley round, so one learning rate suits all weights.

Batch, stochastic and mini-batch

Variant	Gradient from	Pros	Cons
Batch GD	All n rows each step	Smooth, exact	Slow for big data
Stochastic GD (SGD)	One random row	Fast updates, escapes shallow minima	Noisy
Mini-batch GD	32–1,024 rows	Best of both; GPU friendly	Batch size to tune

Python

```
def minibatch_gd(X, y, lr=0.05, epochs=20, batch=32, seed=0):
    rng = np.random.default_rng(seed); n, p = X.shape; w, b = np.zeros(p), 0.0; costs = []
    for _ in range(epochs):
        idx = rng.permutation(n)
        for start in range(0, n, batch):
            j = idx[start:start + batch]; err = X[j] @ w + b - y[j]
            w -= lr * X[j].T @ err / len(j); b -= lr * err.mean()
        costs.append(((X @ w + b - y) ** 2).mean() / 2)
    return w, b, costs
wm, bm, cm = minibatch_gd(X, y)
print("mini-batch cost per epoch:", [round(float(c), 1) for c in cm[:5]], "...", round(float(cm[-1]),
2), "| weights close to OLS:", np.allclose(wm, theta[1:], atol=0.5))
```

OUTPUT

```
mini-batch cost per epoch: [170.3, 79.1, 74.1, 73.5, 73.0] ... 72.62 | weights close to OLS:
True
```

Deep learning uses mini-batch gradient descent with improved optimisers: **momentum** (keeps velocity), **RMSProp** and **Adam** (adaptive learning rates per parameter), plus learning-rate schedules.

scikit-learn: SGDRegressor

Python

```
from sklearn.linear_model import SGDRegressor, LinearRegression
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
Xtr, Xte, ytr, yte = train_test_split(h[feats], y, test_size=0.2, random_state=0)
sgd = make_pipeline(StandardScaler(), SGDRegressor(learning_rate="invscaling", eta0=0.01,
max_iter=2000, tol=1e-6, random_state=0)).fit(Xtr, ytr)
ols = make_pipeline(StandardScaler(), LinearRegression()).fit(Xtr, ytr)
print(f"test R²: SGDRegressor {sgd.score(Xte, yte):.4f} | LinearRegression {ols.score(Xte, yte):.4f} |
SGD epochs used {sgd[-1].n_iter_}")
```

OUTPUT

```
test R²: SGDRegressor 0.8513 | LinearRegression 0.8514 | SGD epochs used 23
```

`SGDRegressor` / `SGDClassifier` also support `partial_fit` for data that arrives in streams or does not fit in memory, and penalties (L2, L1, elastic net) for regularisation.

Where gradient descent appears

Logistic regression (minimising log loss, lesson 13), neural networks and LLMs (backpropagation computes gradients for billions of weights, lesson 26), gradient boosting (fits each new tree to the gradient of the loss, lesson 19), and matrix factorisation in recommenders.

Summary

KEY TAKEAWAYS

- Learning = minimising a cost function; gradient descent repeatedly steps against the gradient: $w \leftarrow w - \alpha \cdot \partial J/\partial w$.
- For linear regression: gradient = $X^T(\hat{y} - y)/n$; it converges to the same solution as OLS.
- The learning rate controls step size: too small is slow, too large diverges; always plot the cost.
- Scale features so one learning rate suits all weights.
- Batch, stochastic and mini-batch variants trade accuracy of each step for speed; Adam and momentum power deep learning.

Quick quiz

6 questions · tap an answer to check it

1. In the update $w \leftarrow w - \alpha \cdot \partial J/\partial w$, what is α?

A The learning rate (step size)

B The gradient

C The cost

D The bias

Answer: A. It is a hyperparameter you choose.

2. The cost increases every epoch and becomes NaN. Most likely cause?

A Learning rate too small

B Learning rate too large

C Too much data

D Features are scaled

Answer: B. Steps overshoot the minimum.

3. Why standardise features before gradient descent?

- **A** Features on very different scales create a narrow valley that slows convergence
- **B** To change the final OLS solution
- **C** To remove outliers
- **D** It is required by NumPy

Answer: A. Scaling makes one learning rate work for all weights.

4. What does mini-batch gradient descent use for each update?

- **A** All rows
- **B** One row
- **C** A small random subset of rows
- **D** Only the test set

Answer: C. Common sizes: 32–512.

5. Gradient descent and the normal equation on the same linear regression should give...

- **A** Approximately the same weights
- **B** Completely different weights
- **C** The same only without scaling
- **D** No relation

Answer: A. Both minimise the same convex cost.

6. Which algorithm fits each new tree to the gradient of the loss?

A K-means

B Gradient boosting

C KNN

D PCA

Answer: B. Hence the name.

EXERCISE **Practice**

(1) Add an early-stopping rule: stop when the cost improves by less than 0.001. How many epochs are needed at α = 0.1? (2) Implement gradient descent for a model with only `area_sqft` and plot the fitted line. (3) Try mini-batch sizes 8, 64 and 512 and compare the final cost after 10 epochs.

Show solution (worked answer)

Python

```
def gd_early(X, y, lr=0.1, tol=1e-3, max_epochs=10_000):
    n, p = X.shape; w, b = np.zeros(p), 0.0; prev = np.inf
    for e in range(max_epochs):
        err = X @ w + b - y; cost = (err ** 2).mean() / 2
        if prev - cost < tol: return w, b, e
        prev = cost; w -= lr * X.T @ err / n; b -= lr * err.mean()
    return w, b, max_epochs
print(gd_early(X, y)[2])
w1, b1, _ = gradient_descent(X[:, [0]], y)
for bs in (8, 64, 512): print(bs, minibatch_gd(X, y, batch=bs, epochs=10)[2][-1])
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain gradient descent intuitively. Start with random weights, compute how wrong the model is, find the direction that reduces the error fastest (the negative gradient), take a small step, and repeat until the error stops falling.

Q: Batch vs stochastic gradient descent? Batch uses all data per step (stable, slow); stochastic uses one example (fast, noisy); mini-batch is the practical compromise.

Q: Can gradient descent get stuck? For convex costs like linear and logistic regression there is one global minimum. For neural networks it can land in local minima or saddle points; momentum, Adam and random restarts help.

STATISTICS & MACHINE LEARNING · LESSON 12

Overfitting, Polynomial Regression and Regularisation (Ridge, Lasso, Elastic Net)

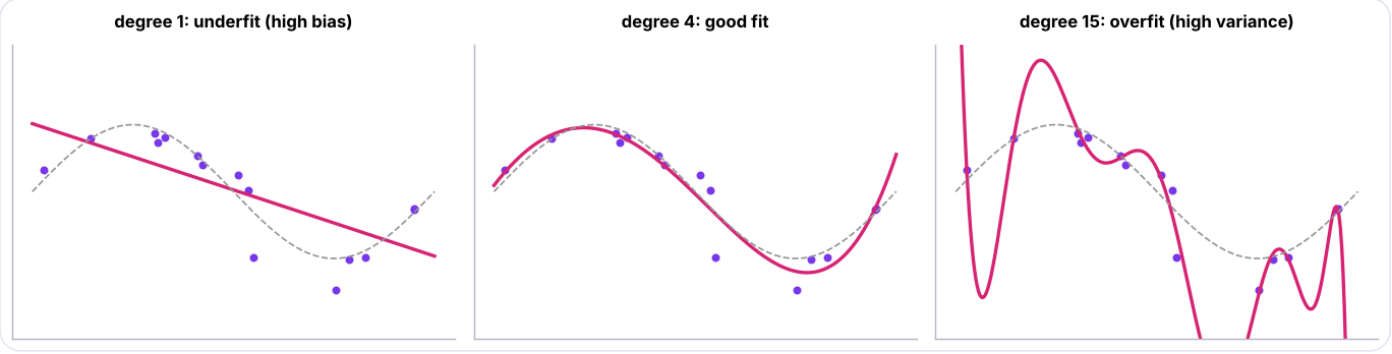
See overfitting and underfitting happen, learn the bias-variance trade-off with polynomial regression, and control model complexity with L2 (Ridge), L1 (Lasso) and Elastic Net penalties, including choosing the penalty strength with cross-validation, feature selection with Lasso, and learning curves.

ML class 6 · scikit-learn 1.7+ · Intermediate · 110 min

By the end of this lesson you can

- Recognise underfitting and overfitting from train vs test errors and plots
- Explain the bias-variance trade-off and how model complexity affects it
- Fit polynomial regression and control its degree
- Apply Ridge, Lasso and Elastic Net and tune alpha with RidgeCV/LassoCV
- Use learning and validation curves to decide between more data and a different model

Underfitting and overfitting



Polynomial models of degree 1, 4 and 15 on the same noisy data. Degree 1 underfits, degree 4 fits well, degree 15 chases noise.

	Underfitting	Good fit	Overfitting
Training error	High	Low	Very low
Test error	High	Low	High
Cause	Model too simple, missing features		Model too complex, too little data, leakage
Fix	More features, more flexible model, less regularisation		More data, simpler model, regularisation, feature selection, early stopping

The bias-variance trade-off

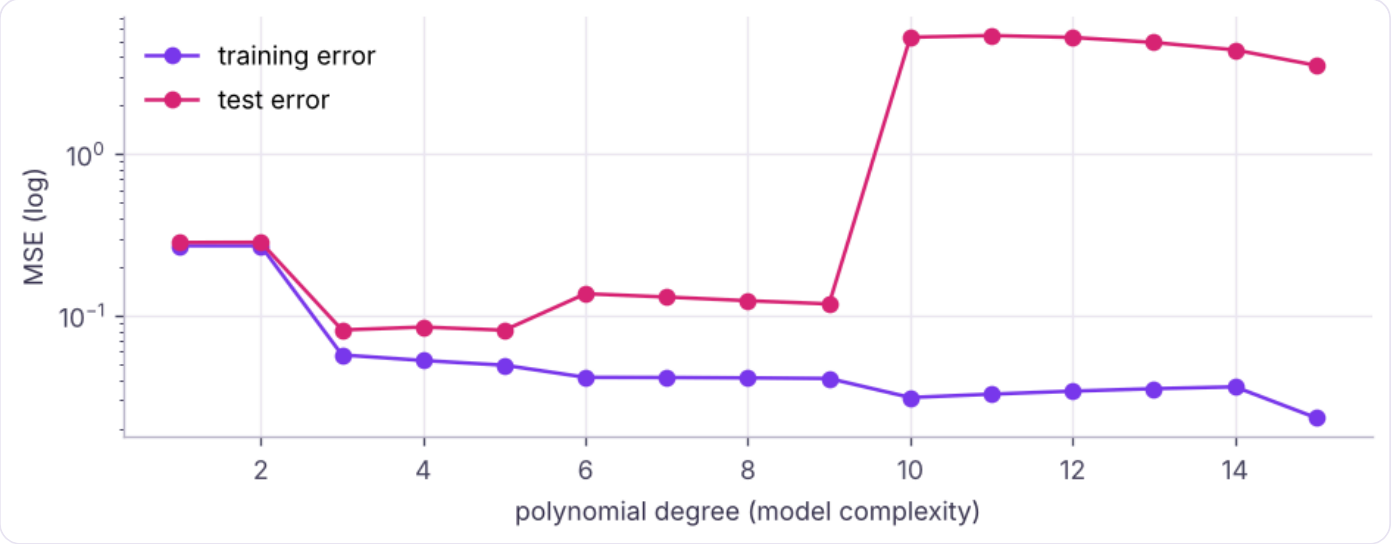
Bias is the error from wrong assumptions (a line for a curve). **Variance** is the error from sensitivity to the particular training sample. Total expected error = bias² + variance + irreducible noise.

Python

```
from sklearn.metrics import mean_squared_error
x_test = rng.uniform(0, 1, 500); y_test = np.sin(2 * np.pi * x_test) + rng.normal(0, .25, 500)
for d in (1, 2, 3, 4, 6, 9, 12, 15):
    m = make_pipeline(PolynomialFeatures(d), StandardScaler(), LinearRegression()).fit(x[:, None], y)
    tr = mean_squared_error(y, m.predict(x[:, None])); te = mean_squared_error(y_test,
m.predict(x_test[:, None]))
    print(f"degree {d:2d}: train MSE {tr:7.3f} | test MSE {te:10.3f}")
```

OUTPUT

```
degree  1: train MSE   0.271 | test MSE      0.284
degree  2: train MSE   0.271 | test MSE      0.285
degree  3: train MSE   0.057 | test MSE      0.082
degree  4: train MSE   0.053 | test MSE      0.085
degree  6: train MSE   0.042 | test MSE      0.137
degree  9: train MSE   0.041 | test MSE      0.118
degree 12: train MSE   0.034 | test MSE      5.285
degree 15: train MSE   0.023 | test MSE      3.548
```



As complexity grows, training error keeps falling while test error falls then rises: the sweet spot is in between.

Regularisation

Instead of choosing fewer features by hand, add a **penalty** on large coefficients to the cost:

$$\text{Ridge (L2): } J = \text{MSE} + \alpha \sum_j w_j^2 \qquad \text{Lasso (L1): } J = \text{MSE} + \alpha \sum_j |w_j|$$

Elastic Net mixes both (`l1_ratio`). Larger α means stronger shrinkage, simpler models, more bias and less variance. Always **scale features** first, so the penalty treats them fairly.

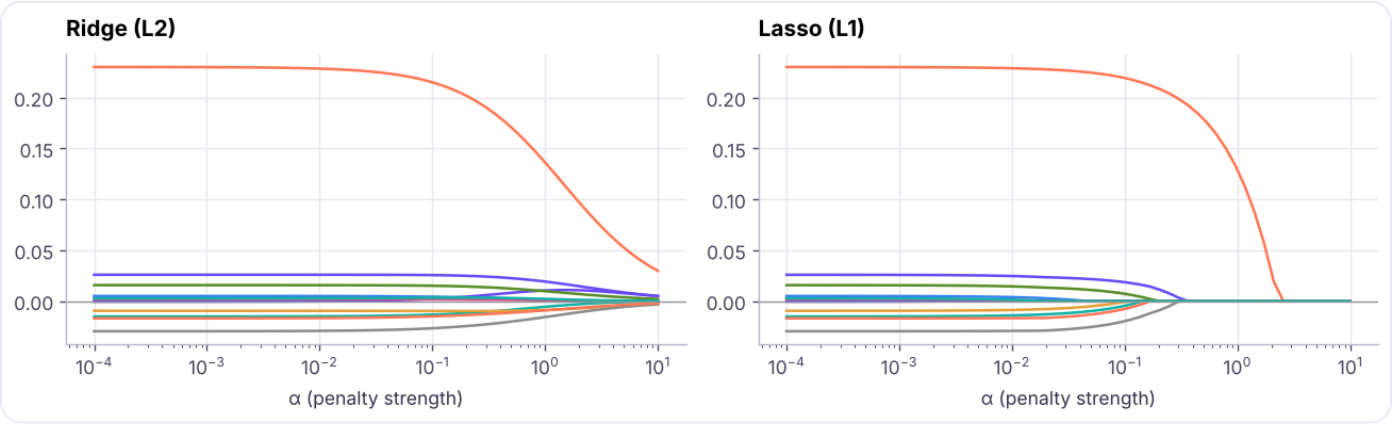
Python

```
for name, model in [("no penalty", LinearRegression()), ("Ridge α=0.01", Ridge(alpha=0.01)), ("Lasso
α=0.001", Lasso(alpha=0.001, max_iter=100000))]:
    m = make_pipeline(PolynomialFeatures(15), StandardScaler(), model).fit(x[:, None], y)
    coefs = m[-1].coef_
    print(f"degree-15 {name:12s} test MSE {mean_squared_error(y_test, m.predict(x_test[:, None])):9.3f}
| largest |coef| {np.abs(coefs).max():12.1f} | zero coefs {(np.abs(coefs) < 1e-8).sum():2d} of
{len(coefs)}")
```

OUTPUT

```
degree-15 no penalty   test MSE     3.548 | largest |coef|       9560.9 | zero coefs  0 of 16
degree-15 Ridge α=0.01 test MSE     0.113 | largest |coef|          2.2 | zero coefs  1 of 16
degree-15 Lasso α=0.001 test MSE     0.089 | largest |coef|          4.1 | zero coefs 12 of 16
```

The same degree-15 model becomes well behaved once its coefficients are penalised. **Lasso** sets some coefficients exactly to zero (automatic feature selection); **Ridge** shrinks all of them smoothly and handles correlated features well.



Coefficient paths: as α grows, Ridge shrinks every coefficient gradually while Lasso drops features to exactly zero one by one.

Choosing α with cross-validation

Python

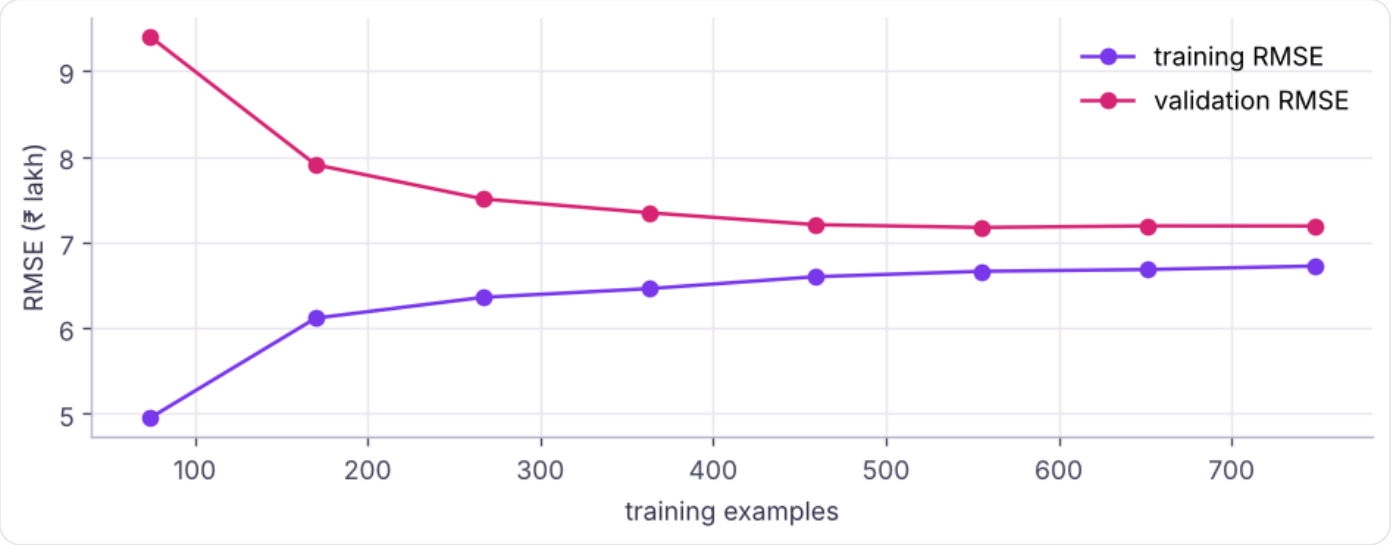
```
from sklearn.model_selection import train_test_split, cross_val_score
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"])
pps = h.price_lakh * 1e5 / h.area_sqft; h = h[pps.between(pps.median() / 3, pps.median() * 3)]
X = pd.get_dummies(h.drop(columns=["house_id", "price_lakh"]), columns=["city"],
drop_first=True).astype(float); yv = h.price_lakh
Xtr, Xte, ytr, yte = train_test_split(X, yv, test_size=0.2, random_state=0)
poly = lambda est: make_pipeline(PolynomialFeatures(2, include_bias=False), StandardScaler(), est)
models = {"linear (degree 2)": poly(LinearRegression()), "RidgeCV": poly(RidgeCV(alphas=np.logspace(-3,
3, 30))),
          "LassoCV": poly(LassoCV(cv=5, random_state=0, max_iter=50000)), "ElasticNet":
poly(ElasticNet(alpha=0.05, l1_ratio=0.5, max_iter=50000))}
for name, m in models.items():
    m.fit(Xtr, ytr); cv = cross_val_score(m, Xtr, ytr, cv=5,
scoring="neg_root_mean_squared_error").mean()
    extra = f" | α = {m[-1].alpha_:.4g}" if hasattr(m[-1], "alpha_") else ""
    nz = f" | non-zero coefs {(m[-1].coef_ != 0).sum()} of {len(m[-1].coef_)}"
    print(f"{name:18s} CV RMSE {-cv:6.2f} | test RMSE {mean_squared_error(yte, m.predict(Xte)) **
.5:6.2f}{extra}{nz}")
```

OUTPUT

```
linear (degree 2)  CV RMSE   7.21 | test RMSE   7.72 | non-zero coefs 51 of 54
RidgeCV            CV RMSE   7.20 | test RMSE   7.75 | α = 0.788 | non-zero coefs 53 of 54
LassoCV            CV RMSE   7.08 | test RMSE   7.64 | α = 0.07792 | non-zero coefs 25 of 54
ElasticNet         CV RMSE   7.41 | test RMSE   8.05 | non-zero coefs 45 of 54
```

With about 900 training rows and 54 polynomial features, all models perform similarly here; Lasso reaches the same accuracy with far fewer terms. Regularisation matters most when features are many relative to rows, or strongly correlated.

Learning curves: more data or a better model?



Learning curves for a degree-2 Ridge model on house prices: the training and validation errors converge as data grows, so more data would help little; a more flexible model or better features would help more.

Learning-curve pattern	Diagnosis	Action
Both errors high and close	High bias (underfitting)	More features, more complex model
Big gap: low train, high validation	High variance (overfitting)	More data, regularisation, simpler model
Gap closing as data grows	Variance falling	Collect more data

Regularisation beyond linear models

Model	Regularisation knobs
Logistic regression	`C` = 1/α (smaller C = stronger penalty), `penalty="l1"/"l2"/"elasticnet"`
Decision trees	`max_depth`, `min_samples_leaf`, pruning (`ccp_alpha`)
Random forests / boosting	Tree depth, number of trees, learning rate, subsampling
Neural networks	Weight decay (L2), dropout, early stopping, data augmentation
KNN	Larger k = smoother

Summary

KEY TAKEAWAYS

- Underfitting: high error everywhere (bias). Overfitting: low training error, high test error (variance).
- Polynomial features add flexibility; the degree controls the bias-variance trade-off.
- Ridge (L2) shrinks coefficients; Lasso (L1) zeroes some (feature selection); Elastic Net combines them. Scale features first.
- Pick α by cross-validation (RidgeCV, LassoCV, GridSearchCV).
- Learning curves tell you whether more data or a different model will help.

Quick quiz

6 questions · tap an answer to check it

1. Training error 2%, test error 25%. Diagnosis?

A Underfitting

B Overfitting (high variance)

C Perfect model

D Data too large

Answer: B. The model memorised the training data.

2. Which penalty can set coefficients exactly to zero?

A Ridge (L2)

B Lasso (L1)

C No penalty

D Both equally

Answer: B. L1 performs feature selection.

3. What happens as α increases in Ridge regression?

- **A** Coefficients shrink, bias rises and variance falls
- **B** Coefficients grow
- **C** The model overfits more
- **D** Nothing

Answer: A. Stronger penalty = simpler model.

4. Why scale features before regularisation?

- **A** So the penalty treats all coefficients fairly regardless of units
- **B** To speed up printing
- **C** To remove outliers
- **D** It is optional

Answer: A. Otherwise large-unit features are under-penalised.

5. In logistic regression in scikit-learn, a smaller C means...

- **A** Stronger regularisation
- **B** Weaker regularisation
- **C** More features
- **D** No effect

Answer: A. C is the inverse of α.

6. Learning curves show training and validation errors both high and close together. What helps?

- **A** More data
- **B** A more flexible model or better features
- **C** Stronger regularisation
- **D** Fewer features

Answer: B. This is high bias.

EXERCISE **Practice**

(1) Plot validation curves (`validation_curve`) of Ridge alpha for the house data. (2) Fit `LogisticRegressionCV` with L1 penalty on the HR attrition data: which features survive? (3) Compare degree 1, 2 and 3 polynomial Ridge models with 5-fold CV.

Show solution (worked answer)

Python

```
from sklearn.model_selection import validation_curve
from sklearn.linear_model import LogisticRegressionCV
al = np.logspace(-3, 3, 13)
tr_, va_ = validation_curve(poly(Ridge()), Xtr, ytr, param_name="ridge__alpha", param_range=al,
cv=5, scoring="neg_root_mean_squared_error")
yb = (hr.attrition == "Yes").astype(int)
lr1 = LogisticRegressionCV(penalty="l1", solver="saga", Cs=10, cv=5,
max_iter=5000).fit(StandardScaler().fit_transform(Xh), yb)
print(Xh.columns[lr1.coef_[0] != 0].tolist())
for d in (1, 2, 3): print(d, cross_val_score(make_pipeline(PolynomialFeatures(d,
include_bias=False), StandardScaler(), Ridge(1)), Xtr, ytr, cv=5, scoring="r2").mean())
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain the bias-variance trade-off. Simple models have high bias (systematic error) and low variance; complex models have low bias but high variance (sensitivity to the training sample). We choose complexity that minimises total error on new data.

Q: L1 vs L2 regularisation? L2 shrinks coefficients towards zero and handles correlated features; L1 can set coefficients exactly to zero, selecting features; Elastic Net combines both.

Q: How do you detect overfitting? Compare training and validation/test performance (large gap), use cross-validation, and inspect learning curves.

STATISTICS & MACHINE LEARNING · LESSON 13

Logistic Regression: Probabilities, Odds Ratios and Credit-Risk Scoring

Classify with probabilities. The sigmoid and log-odds, maximum likelihood and log loss, fitting logistic regression with statsmodels (inference, odds ratios) and scikit-learn (pipelines, regularisation), thresholds and decisions, multiclass logistic regression, and scoring new applicants in the Bank Loans case study.

● Python class 16 · ML class 4 ✦ statsmodels 0.1x · scikit-learn 1.7+ Intermediate ⏱ 120 min

By the end of this lesson you can

- Explain why linear regression fails for binary targets and how the sigmoid fixes it
- Interpret coefficients as log-odds and odds ratios
- Fit, evaluate and regularise logistic regression in statsmodels and scikit-learn
- Choose a probability threshold from business costs
- Score new applicants and explain a prediction

The case: who will default?

A bank has 700 past borrowers with a known outcome (`default` = 1 if they failed to repay) and 150 new applicants with unknown outcomes. Features: age, education level (`ed` 1–5), years with current employer (`employ`), years at current address (`address`), income (thousands), debt-to-income ratio % (`debtinc`), credit-card debt and other debt (thousands).

Python

```
import pandas as pd, numpy as np
b = pd.read_csv("bank_loans.csv")
known, new = b[b.default.notna()].copy(), b[b.default.isna()].copy()
known["default"] = known.default.astype(int)
print(len(known), "labelled |", len(new), "new applicants | default rate", round(known.default.mean(),
3))
print(known.groupby("default")[["age", "employ", "address", "income", "debtinc", "creddebt",
"othdebt"]].mean().round(2))
```

OUTPUT

```
700 labelled | 150 new applicants | default rate 0.321
           age  employ  address  income  debtinc  creddebt  othdebt
default
0        39.69    7.61     6.29   50.94     8.55      1.47     2.96
1        32.67    3.45     3.29   46.30    13.17      2.31     3.99
```

Defaulters are younger, have shorter employment and address histories, and much higher debt-to-income ratios.

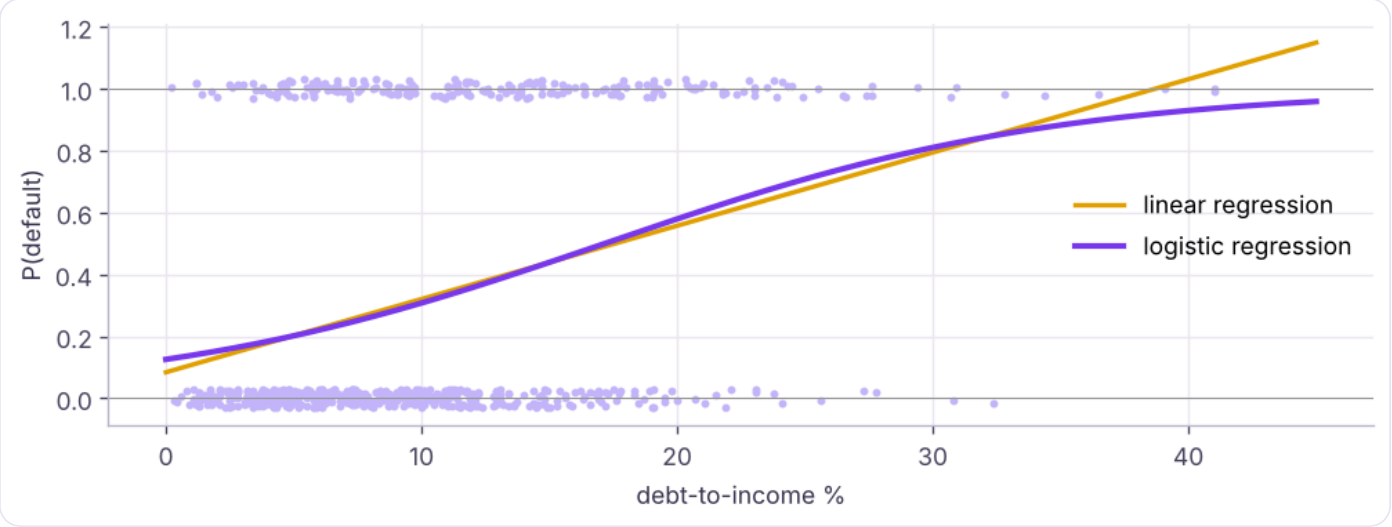
Why not linear regression?

A straight line predicts values below 0 and above 1, which are not probabilities. **Logistic regression** passes a linear score through the **sigmoid**:

$$p = P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \ \dots \ + \beta_k x_k)}}$$

Equivalently, the **log-odds** (logit) is linear in the features:

$$\log \frac{p}{1-p} = \beta_0 + \beta_1 x_1 + \ \dots \ + \beta_k x_k$$



Linear regression vs logistic regression on default vs debt-to-income: the sigmoid keeps predictions between 0 and 1.

Fitting and interpreting with statsmodels

Logistic regression is fitted by **maximum likelihood**: choose coefficients that make the observed outcomes most probable, equivalently minimising **log loss** (cross-entropy).

Python

```
import statsmodels.formula.api as smf
m = smf.logit("default ~ age + ed + employ + address + income + debtinc + creddebt + othdebt",
data=known).fit(disp=0)
print(m.summary2().tables[1][["Coef.", "Std.Err.", "P>|z|"]].round(4))
print(f"pseudo R² (McFadden) {m.prsquared:.3f} | log-likelihood {m.llf:.1f}")
```

OUTPUT

```
            Coef.  Std.Err.   P>|z|
Intercept -0.5739    0.5772  0.3201
age       -0.0099    0.0137  0.4688
ed         0.0066    0.0989  0.9465
employ    -0.1586    0.0287  0.0000
address   -0.1021    0.0292  0.0005
income    -0.0014    0.0084  0.8687
debtinc    0.1067    0.0330  0.0012
creddebt   0.3246    0.1227  0.0081
othdebt   -0.0563    0.0579  0.3304
pseudo R² (McFadden) 0.244 | log-likelihood -332.3
```

Python

```python
odds = pd.DataFrame({"odds_ratio": np.exp(m.params), "ci_low": np.exp(m.conf_int()[0]), "ci_high":
np.exp(m.conf_int()[1]), "p": m.pvalues}).drop("Intercept").round(3)
print(odds.sort_values("odds_ratio"))
```

OUTPUT

```
          odds_ratio  ci_low  ci_high      p
employ         0.853   0.807    0.903  0.000
address        0.903   0.853    0.956  0.000
othdebt        0.945   0.844    1.059  0.330
age            0.990   0.964    1.017  0.469
income         0.999   0.982    1.015  0.869
ed             1.007   0.829    1.222  0.946
debtinc        1.113   1.043    1.187  0.001
creddebt       1.384   1.088    1.760  0.008
```

Coefficient	Odds ratio e^β	Reading
β = 0	1	No association
β > 0	> 1	Higher odds of default per unit
β < 0	< 1	Lower odds of default per unit

For example, the odds ratio of about 0.85 for `employ` means each extra year with the current employer multiplies the odds of default by 0.85 (about 15% lower odds), holding the other variables constant. The odds ratio of about 1.38 for `creddebt` means each extra unit of card debt (₹1,000s) multiplies the odds by 1.38, and each extra point of debt-to-income raises the odds by about 11%. Age, education, income and other debt are not significant once the others are included.

> **NOTE** **Odds are not probabilities**
>
> Odds = p / (1 − p). A probability of 0.2 is odds of 0.25 ("1 to 4"). An odds ratio describes a *multiplicative* change in odds; the change in probability depends on where you start.

A reduced model keeps the significant predictors (a common credit-scoring practice: fewer, stable variables):

Python

```
m2 = smf.logit("default ~ employ + address + debtinc + creddebt", data=known).fit(disp=0)
print(m2.params.round(3).to_dict(), "| AIC full", round(m.aic, 1), "vs reduced", round(m2.aic, 1))
```

OUTPUT

```
{'Intercept': -0.829, 'employ': -0.175, 'address': -0.113, 'debtinc': 0.099, 'creddebt': 0.25} |
AIC full 682.6 vs reduced 677.0
```

Lower AIC is better (it balances fit and complexity), so the four-variable model is preferred.

scikit-learn pipeline with regularisation

Python

```
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import roc_auc_score, accuracy_score, log_loss, classification_report
feats = ["age", "ed", "employ", "address", "income", "debtinc", "creddebt", "othdebt"]
X, y = known[feats], known.default
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.25, stratify=y, random_state=7)
for C in (0.01, 0.1, 1, 100):
    pipe = make_pipeline(StandardScaler(), LogisticRegression(C=C, max_iter=1000))
    cv = cross_val_score(pipe, Xtr, ytr, cv=5, scoring="roc_auc").mean()
    print(f"C = {C:<5} 5-fold ROC-AUC {cv:.3f}")
clf = make_pipeline(StandardScaler(), LogisticRegression(C=1, max_iter=1000)).fit(Xtr, ytr)
proba = clf.predict_proba(Xte)[:, 1]
print(f"\ntest: ROC-AUC {roc_auc_score(yte, proba):.3f} | accuracy {accuracy_score(yte,
clf.predict(Xte)):.3f} | log loss {log_loss(yte, proba):.3f}")
print(classification_report(yte, clf.predict(Xte), target_names=["repaid", "default"], digits=3))
```

OUTPUT

```
C = 0.01  5-fold ROC-AUC 0.791
C = 0.1   5-fold ROC-AUC 0.791
C = 1     5-fold ROC-AUC 0.793
C = 100   5-fold ROC-AUC 0.792

test: ROC-AUC 0.865 | accuracy 0.817 | log loss 0.431
              precision    recall  f1-score   support

      repaid      0.822     0.933     0.874       119
     default      0.800     0.571     0.667        56

    accuracy                          0.817       175
   macro avg      0.811     0.752     0.770       175
weighted avg      0.815     0.817     0.808       175
```

`C` is the inverse regularisation strength (lesson 12). Standardised coefficients from the pipeline are comparable across features:

Python

```
print(pd.Series(clf[-1].coef_[0], index=feats).round(3).sort_values().to_dict())
```

OUTPUT

```
{'employ': -0.875, 'address': -0.409, 'age': -0.148, 'othdebt': -0.025, 'income': -0.018, 'ed':
0.014, 'creddebt': 0.339, 'debtinc': 0.59}
```

Choosing the threshold

`predict` uses a 0.5 threshold, but the right cut-off depends on costs. Suppose a missed defaulter costs the bank ₹50,000 and rejecting a good customer loses ₹10,000 of profit.

Python

```
from sklearn.metrics import confusion_matrix
cost_fn, cost_fp = 50_000, 10_000
rows = []
for t in np.arange(0.1, 0.91, 0.1):
    tn, fp, fn, tp = confusion_matrix(yte, (proba ⩾ t).astype(int)).ravel()
    rows.append({"threshold": round(t, 1), "approved": tn + fn, "defaults_caught": tp,
"defaults_missed": fn, "good_rejected": fp, "cost_₹_lakh": (fn * cost_fn + fp * cost_fp) / 1e5})
th = pd.DataFrame(rows); print(th.to_string(index=False))
print("lowest-cost threshold:", th.loc[th["cost_₹_lakh"].idxmin(), "threshold"])
```

OUTPUT

```
 threshold  approved  defaults_caught  defaults_missed  good_rejected  cost_₹_lakh
       0.1        40               55                1             80          8.5
       0.2        72               51                5             52          7.7
       0.3        92               46               10             37          8.7
       0.4       118               41               15             16          9.1
       0.5       135               32               24              8         12.8
       0.6       152               19               37              4         18.9
       0.7       162               11               45              2         22.7
       0.8       167                8               48              0         24.0
       0.9       173                2               54              0         27.0
lowest-cost threshold: 0.2
```

When missing a defaulter is five times as costly as rejecting a good customer, the best threshold is well below 0.5. Lesson 14 formalises this with precision, recall and ROC curves.

Scoring new applicants

Python

```
final = make_pipeline(StandardScaler(), LogisticRegression(C=1, max_iter=1000)).fit(X, y)          # refit on all labelled data
new["p_default"] = final.predict_proba(new[feats])[:, 1].round(3)
best_t = th.loc[th["cost_₹_lakh"].idxmin(), "threshold"]
new["decision"] = np.where(new.p_default ≥ best_t, "decline / review", "approve")
print(new.decision.value_counts().to_dict())
print(new.sort_values("p_default", ascending=False)[["customer_id", "age", "employ", "debtinc",
"creddebt", "p_default", "decision"]].head(5).to_string(index=False))
```

OUTPUT

```
{'decline / review': 84, 'approve': 66}
customer_id  age  employ  debtinc  creddebt  p_default         decision
      L0782   36       0     32.9      4.73      0.961 decline / review
      L0774   42       8     39.1      7.26      0.932 decline / review
      L0736   32       5     28.8      7.60      0.898 decline / review
      L0702   41       6     27.9      4.55      0.883 decline / review
      L0753   36       6     29.7      6.09      0.876 decline / review
```

Explaining one decision

Python

```
one = new.sort_values("p_default", ascending=False).iloc[0]
z = (one[feats].astype(float) - final[0].mean_) / final[0].scale_
contrib = pd.Series(z.values * final[-1].coef_[0], index=feats).sort_values(ascending=False)
print(f"{one.customer_id}: P(default) {one.p_default:.2f}; log-odds contributions:",
contrib.round(2).to_dict())
```

OUTPUT

```
L0782: P(default) 0.96; log-odds contributions: {'debtinc': 2.37, 'employ': 0.97, 'creddebt':
0.96, 'address': 0.13, 'income': 0.03, 'age': 0.02, 'ed': -0.01, 'othdebt': -0.17}
```

Each contribution is a coefficient × the applicant's standardised value: positive values push towards default. This additive explanation is a big reason banks like logistic regression (and regulators require explanations of adverse decisions).

Multiclass logistic regression

With more than two classes, scikit-learn fits a **multinomial** (softmax) model automatically: one score per class, converted to probabilities that sum to 1.

Python

```python
from sklearn.datasets import load_iris
iris = load_iris(as_frame=True)
mc = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000)).fit(iris.data, iris.target)
print(mc.predict_proba(iris.data.iloc[[0, 60, 120]]).round(3), "| classes:",
iris.target_names.tolist())
```

OUTPUT

```
[[0.985 0.015 0.   ]
 [0.022 0.962 0.016]
 [0.    0.023 0.977]] | classes: ['setosa', 'versicolor', 'virginica']
```

Summary

KEY TAKEAWAYS

- Logistic regression models log-odds as linear in the features and maps them to probabilities with the sigmoid; it is fitted by maximum likelihood (log loss).
- e^β is an odds ratio: > 1 raises, < 1 lowers the odds per unit, holding other features constant.
- statsmodels gives inference (p-values, CIs, AIC); scikit-learn gives pipelines, regularisation (C) and deployment.
- Choose the probability threshold from business costs, not by default 0.5.
- Coefficient × standardised value gives a transparent explanation of each prediction.

Quick quiz

6 questions · tap an answer to check it

1. What does the sigmoid function do?

A Maps any real number to a probability between 0 and 1

B Squares the errors

C Selects features

D Scales the data

Answer: A. 1 / (1 + e^(−z)).

2. An odds ratio of 1.38 for card debt means...

- **A** Each extra unit of card debt multiplies the odds of default by 1.38
- **B** Default probability rises by 1.38 points
- **C** Card debt causes default
- **D** 80% of customers default

Answer: A. Multiplicative change in odds.

3. Which loss does logistic regression minimise?

- **A** Mean squared error
- **B** Log loss (cross-entropy)
- **C** Hinge loss
- **D** MAE

Answer: B. Equivalent to maximising the likelihood.

4. Missing a defaulter costs 5× more than rejecting a good customer. The best threshold is probably...

- **A** Below 0.5
- **B** Exactly 0.5
- **C** Above 0.5
- **D** Irrelevant

Answer: A. Flag more applicants to avoid expensive misses.

5. In scikit-learn, LogisticRegression(C=0.01) means...

- **A** Strong regularisation
- **B** No regularisation
- **C** 1% threshold
- **D** 1% of the data

Answer: A. C is the inverse of the penalty strength.

6. Why do banks favour logistic regression for credit scoring?

- **A** It is accurate, stable and each decision can be explained by additive contributions
- **B** It is the newest algorithm
- **C** It never makes mistakes
- **D** It needs no data

Answer: A. Explainability is a regulatory requirement.

EXERCISE **Practice**

(1) Fit a logistic regression for HR attrition with `overtime`, `job_satisfaction`, `years_at_company` and `monthly_income`. Report odds ratios. (2) What is the predicted probability for an employee with overtime, satisfaction 1, 1 year at the company and ₹50,000 income? (3) Find the threshold that maximises F1 on a validation split.

Show solution (worked answer)

Python

```
hr = pd.read_csv("hr_employees.csv"); hr["left"] = (hr.attrition == "Yes").astype(int)
ma = smf.logit("left ~ C(overtime) + job_satisfaction + years_at_company + I(monthly_income /
10000)", data=hr).fit(disp=0)
print(np.exp(ma.params).round(3))
print(ma.predict(pd.DataFrame({"overtime": ["Yes"], "job_satisfaction": [1], "years_at_company":
[1], "monthly_income": [50000]})))
from sklearn.metrics import f1_score
ts = np.linspace(.05, .95, 19); print(max(ts, key=lambda t: f1_score(yte, proba ⩾ t)))
```

INTERVIEW CORNER

Questions you may be asked

Q: Why is it called regression if it classifies? It regresses the log-odds on the features; classification happens when you apply a threshold to the predicted probability.

Q: How do you interpret a logistic regression coefficient? A one-unit increase in the feature changes the log-odds by β, i.e. multiplies the odds by e^β, holding other features constant.

Q: What are the assumptions of logistic regression? Independent observations, a linear relationship between features and log-odds, little multicollinearity, no extreme outliers, and enough events per variable.

STATISTICS & MACHINE LEARNING · LESSON 14

Classification Metrics: Confusion Matrix, Precision, Recall, ROC, PR and Business Value

Measure classifiers properly. The confusion matrix, accuracy and why it misleads, precision, recall, specificity, F1 and F-beta, ROC curves and AUC, precision-recall curves, log loss and calibration, lift and gains charts, choosing thresholds by cost or capacity, and multiclass averaging, using the telecom churn model.

ML class 7 · ✦ scikit-learn 1.7+ · Intermediate · ⏱ 110 min

By the end of this lesson you can

- Build and read a confusion matrix and compute accuracy, precision, recall, specificity and F1
- Plot and interpret ROC and precision-recall curves and their AUCs
- Check probability calibration and log loss
- Choose thresholds by cost, capacity (top-k) or target recall, and present gains/lift
- Use macro, weighted and micro averages for multiclass problems

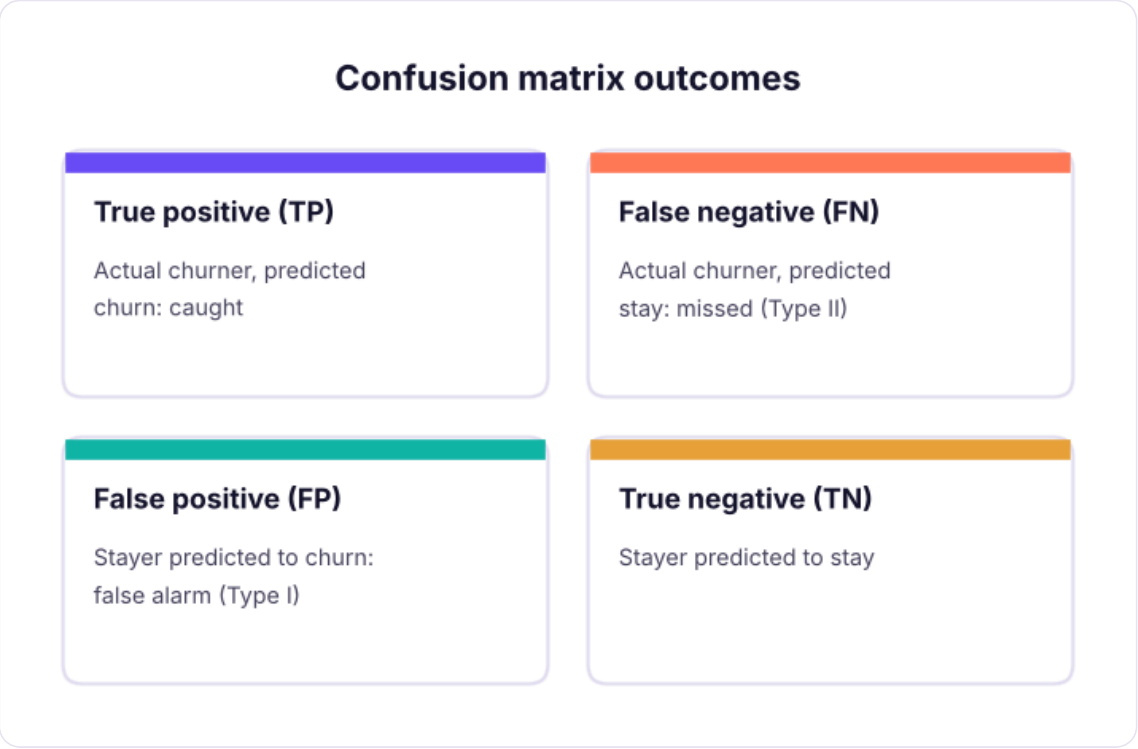
The confusion matrix

Python

```
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score,
fbeta_score, classification_report
tc = pd.read_csv("telecom_churn.csv")
X = tc.drop(columns=["customer_id", "churn", "total_charges"]); y = (tc.churn == "Yes").astype(int)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=1)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months", "monthly_charges",
"support_calls"]), ("c", OneHotEncoder(), ["contract", "internet_service", "payment_method"])])
model = Pipeline([("pre", pre), ("lr", LogisticRegression(max_iter=1000))]).fit(Xtr, ytr)
proba = model.predict_proba(Xte)[:, 1]; pred = (proba ≥ 0.5).astype(int)
tn, fp, fn, tp = confusion_matrix(yte, pred).ravel()
print(pd.DataFrame([[tn, fp], [fn, tp]], index=["actual: stay", "actual: churn"], columns=["pred:
stay", "pred: churn"]))
```

OUTPUT

```
                pred: stay  pred: churn
actual: stay           611           57
actual: churn          125          107
```



The four outcomes of a binary classifier.

Metrics from the matrix

Metric	Formula	Question it answers
Accuracy	(TP + TN) / all	How often is the model right overall?
Precision	TP / (TP + FP)	Of those we flagged, how many were churners?
Recall (sensitivity, TPR)	TP / (TP + FN)	Of all churners, how many did we catch?
Specificity (TNR)	TN / (TN + FP)	Of all stayers, how many did we leave alone?
F1	2·P·R / (P + R)	Balance of precision and recall
F-beta	weights recall β times as much	β = 2 when misses are costlier

Python

```python
print(f"accuracy {accuracy_score(yte, pred):.3f} | precision {precision_score(yte, pred):.3f} | recall {recall_score(yte, pred):.3f} | specificity {tn / (tn + fp):.3f}")
print(f"F1 {f1_score(yte, pred):.3f} | F2 {fbeta_score(yte, pred, beta=2):.3f} | baseline accuracy (always 'stay') {1 - yte.mean():.3f}")
print(classification_report(yte, pred, target_names=["stay", "churn"], digits=3))
```

OUTPUT

```
accuracy 0.798 | precision 0.652 | recall 0.461 | specificity 0.915
F1 0.540 | F2 0.490 | baseline accuracy (always 'stay') 0.742
              precision    recall  f1-score   support

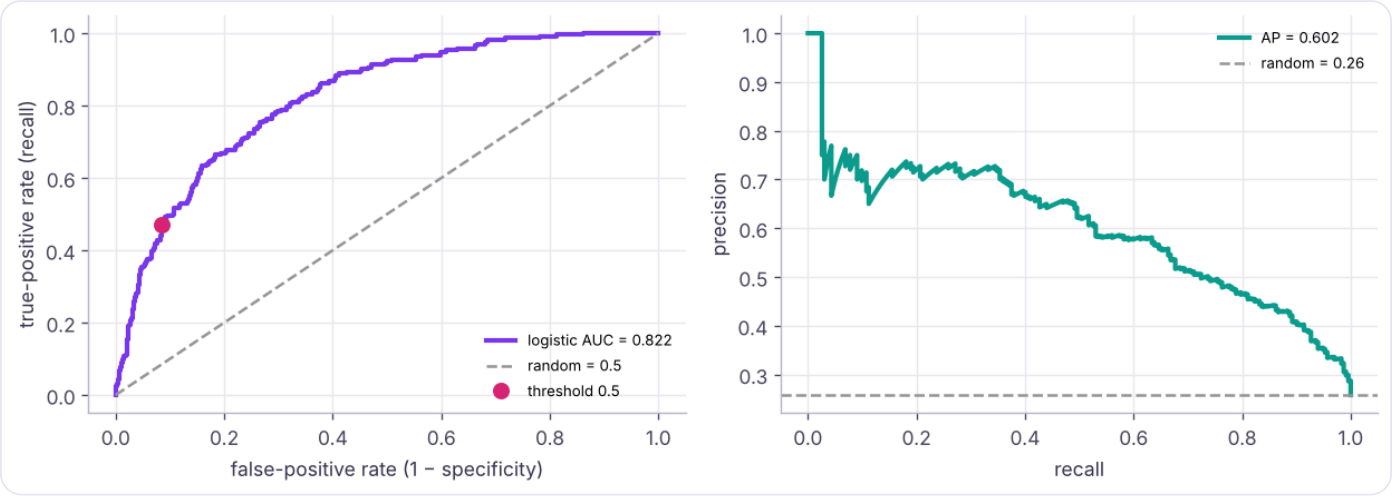
        stay      0.830     0.915     0.870       668
       churn      0.652     0.461     0.540       232

    accuracy                          0.798       900
   macro avg      0.741     0.688     0.705       900
weighted avg      0.784     0.798     0.785       900
```

Accuracy barely beats the "nobody churns" baseline, while recall shows the model misses more than half of the churners at the default 0.5 threshold.

Which metric matters?	Example
Recall	Cancer screening, fraud, churn (missing a positive is costly)
Precision	Spam filter (do not hide real mail), expensive interventions
Balance (F1)	Both errors matter similarly
Specificity	Avoiding false alarms for the majority

Threshold-free view: ROC and PR curves



Left: the ROC curve plots recall (TPR) against the false-positive rate at every threshold. Right: the precision-recall curve focuses on the positive class, more informative when positives are rare.

Python

```python
print(f"ROC-AUC {roc_auc_score(yte, proba):.3f} | average precision (PR-AUC)
{average_precision_score(yte, proba):.3f} | churn base rate {yte.mean():.3f}")
```

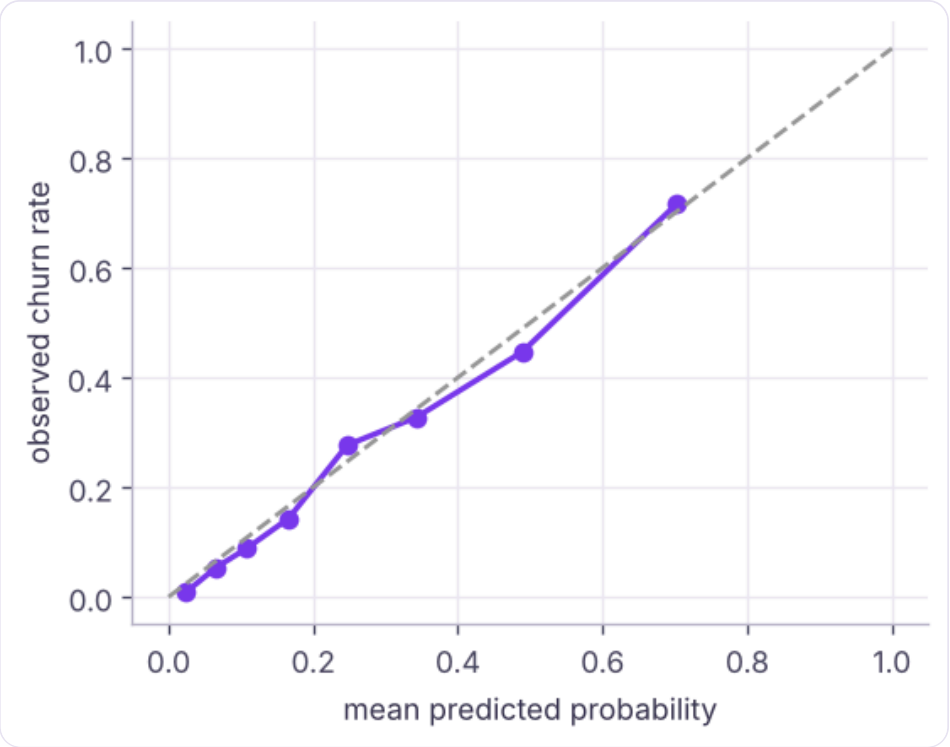
OUTPUT

```
ROC-AUC 0.822 | average precision (PR-AUC) 0.602 | churn base rate 0.258
```

ROC-AUC is the probability that a randomly chosen churner gets a higher score than a randomly chosen stayer (0.5 = random, 1 = perfect). It does not depend on the threshold or on class balance. **Average precision** summarises the PR curve; its random baseline equals the positive rate, so it reveals weakness on rare classes that ROC-AUC can hide.

Probability quality: log loss and calibration

If the model says 30%, do about 30% of those customers churn?



Calibration curve: points near the diagonal mean predicted probabilities can be trusted as probabilities.

Python

```
from sklearn.metrics import log_loss, brier_score_loss
print(f"log loss {log_loss(yte, proba):.3f} (always predicting the base rate: {log_loss(yte,
np.full(len(yte), ytr.mean())):.3f}) | Brier score {brier_score_loss(yte, proba):.3f}")
```

OUTPUT

```
log loss 0.436 (always predicting the base rate: 0.571) | Brier score 0.142
```

Calibration matters when probabilities feed decisions (expected loss = P(default) × amount). Tree ensembles and SVMs are often poorly calibrated; fix with `CalibratedClassifierCV` (isotonic or sigmoid).

Choosing the threshold

By cost

Python

```
value_saved, offer_cost = 3000, 500          # a retained churner is worth ₹3,000; each offer costs
₹500
rows = []
for t in np.round(np.arange(0.1, 0.81, 0.05), 2):
    p = proba ⩾ t; tp_ = (p & (yte == 1)).sum(); n_off = p.sum()
    rows.append({"threshold": t, "offers": n_off, "churners_reached": tp_, "precision": round(tp_ /
max(n_off, 1), 3), "recall": round(tp_ / yte.sum(), 3), "profit_inr": int(tp_ * value_saved * 0.4 -
n_off * offer_cost)})   # 40% of reached churners accept
res = pd.DataFrame(rows); print(res.to_string(index=False))
print("most profitable threshold:", res.loc[res.profit_inr.idxmax(), "threshold"])
```

OUTPUT

```
 threshold  offers  churners_reached  precision  recall  profit_inr
      0.10     637               221      0.347   0.953      -53300
      0.15     532               212      0.398   0.914      -11600
      0.20     448               197      0.440   0.849       12400
      0.25     394               184      0.467   0.793       23800
      0.30     326               165      0.506   0.711       35000
      0.35     272               151      0.555   0.651       45200
      0.40     226               132      0.584   0.569       45400
      0.45     191               119      0.623   0.513       47300
      0.50     164               107      0.652   0.461       46400
      0.55     130                87      0.669   0.375       39400
      0.60     101                72      0.713   0.310       35900
      0.65      77                55      0.714   0.237       27500
      0.70      54                39      0.722   0.168       19800
      0.75      35                25      0.714   0.108       12500
      0.80      20                15      0.750   0.065        8000
most profitable threshold: 0.45
```

By capacity: top-k, gains and lift

The retention team can call only 10% of customers per month. Rank by probability and take the top 10%.

Python

```
order = np.argsort(-proba); ys = yte.to_numpy()[order]
deciles = np.array_split(ys, 10)
gains = pd.DataFrame({"decile": range(1, 11), "churn_rate": [d.mean() for d in deciles]})
gains["cum_%_of_churners"] = (100 * np.cumsum([d.sum() for d in deciles]) / ys.sum()).round(1)
gains["lift"] = (gains.churn_rate / ys.mean()).round(2); gains["churn_rate"] =
gains.churn_rate.round(3)
print(gains.to_string(index=False))
```

OUTPUT

```
 decile  churn_rate  cum_%_of_churners  lift
      1       0.722               28.0  2.80
      2       0.556               49.6  2.16
      3       0.400               65.1  1.55
      4       0.278               75.9  1.08
      5       0.256               85.8  0.99
      6       0.144               91.4  0.56
      7       0.100               95.3  0.39
      8       0.089               98.7  0.34
      9       0.033              100.0  0.13
     10       0.000              100.0  0.00
```

Calling the top 10% reaches 2.8 times as many churners as random calls (the lift in the first decile), and the top 30% of scores contain about two-thirds of all churners. Gains and lift tables are how marketing and collections teams read models.

By target recall

Python

```python
target = 0.80
i = np.where(rec ⩾ target)[0][-1]
print(f"to catch {target:.0%} of churners use threshold {pthr[min(i, len(pthr) - 1)]:.3f} → precision {prec[i]:.3f}")
```

OUTPUT

```
to catch 80% of churners use threshold 0.245 → precision 0.465
```

Multiclass metrics

Python

```python
from sklearn.datasets import load_wine
from sklearn.ensemble import RandomForestClassifier
w = load_wine(); Xa, Xb, ya, yb = train_test_split(w.data, w.target, test_size=0.4, stratify=w.target, random_state=0)
pw = RandomForestClassifier(200, random_state=0).fit(Xa, ya).predict(Xb)
print(confusion_matrix(yb, pw))
for avg in ("macro", "weighted", "micro"): print(f"{avg:8s} F1 {f1_score(yb, pw, average=avg):.3f}")
```

OUTPUT

```
[[23  1  0]
 [ 1 28  0]
 [ 0  0 19]]
macro    F1 0.975
weighted F1 0.972
micro    F1 0.972
```

Macro averages classes equally (rare classes count as much as common ones); **weighted** weights by class size; **micro** pools all decisions (equals accuracy for single-label problems).

Summary

KEY TAKEAWAYS

- Confusion matrix: TP, FP, FN, TN. Accuracy misleads on imbalanced data; compare with the majority baseline.
- Precision = how many flagged are right; recall = how many positives are caught; F1/F-beta balance them.
- ROC-AUC ranks quality independent of threshold; PR-AUC (average precision) is better for rare positives.
- Check log loss and calibration when probabilities drive decisions.
- Choose thresholds by cost, capacity (top-k, gains, lift) or target recall; use macro/weighted averages for multiclass.

Quick quiz

6 questions · tap an answer to check it

1. Fraud is 1% of transactions. A model that predicts "not fraud" for everything has accuracy...

A 1%

B 50%

C 99%, which is useless

D Undefined

Answer: C. That is why accuracy misleads on imbalanced data.

2. Recall answers which question?

A Of all actual positives, how many did we catch?

B Of all flagged cases, how many are right?

C How many negatives were correct?

D How confident is the model?

Answer: A. TP / (TP + FN).

3. A spam filter must not hide real emails. Which metric should be high?

A Precision for the spam class

B Recall for the spam class

C Specificity only for spam

D None

Answer: A. False positives (real mail marked spam) are costly.

4. ROC-AUC = 0.80 means...

A A random positive gets a higher score than a random negative 80% of the time

B 80% accuracy

C 80% recall

D 80% precision

Answer: A. It measures ranking quality.

5. **Which curve is more informative when positives are very rare?**

- **A** ROC
- **B** Precision-recall
- **C** Calibration
- **D** Learning curve

Answer: B. PR focuses on the positive class.

6. **The retention team can call only 10% of customers. What should you report?**

- **A** Precision/recall and lift in the top 10% of scores
- **B** Overall accuracy
- **C** R^2
- **D** Number of features

Answer: A. Evaluate where decisions are actually made.

EXERCISE **Practice**

(1) Train a random forest on the same split and compare ROC-AUC, average precision and lift in the top decile with logistic regression. (2) Calibrate the forest with `CalibratedClassifierCV` and compare Brier scores. (3) Find the threshold that maximises F2.

Show solution (worked answer)

Python

```
from sklearn.calibration import CalibratedClassifierCV
rf = Pipeline([("pre", pre), ("rf", RandomForestClassifier(400, min_samples_leaf=10,
random_state=0))]).fit(Xtr, ytr)
pr = rf.predict_proba(Xte)[:, 1]; print(roc_auc_score(yte, pr), average_precision_score(yte, pr))
top = np.argsort(-pr)[:len(pr) // 10]; print("lift", yte.to_numpy()[top].mean() / yte.mean())
cal = Pipeline([("pre", pre), ("cal", CalibratedClassifierCV(RandomForestClassifier(400,
min_samples_leaf=10, random_state=0), cv=5, method="isotonic"))]).fit(Xtr, ytr)
print(brier_score_loss(yte, pr), brier_score_loss(yte, cal.predict_proba(Xte)[:, 1]))
ts = np.linspace(.05, .9, 35); print(max(ts, key=lambda t: fbeta_score(yte, proba ⩾ t, beta=2)))
```

INTERVIEW CORNER

Questions you may be asked

Q: Precision vs recall: give an example where each matters more. Recall for disease screening or fraud (missing a case is costly); precision for spam filtering or expensive manual reviews (false alarms are costly).

Q: Why can a model with 95% accuracy be bad? With 95% negatives, predicting all negatives gives 95% accuracy and catches no positives; look at recall, precision, PR-AUC and the confusion matrix.

Q: How do you choose a classification threshold? From business costs and constraints: expected profit, team capacity (top-k), or a required recall/precision, validated on held-out data.

STATISTICS & MACHINE LEARNING · LESSON 15

Cross-Validation and Hyperparameter Tuning: K-Fold, GridSearchCV, RandomizedSearchCV and Optuna

Estimate performance reliably and tune models without fooling yourself. Hold-out vs k-fold, stratified, grouped and time-series cross-validation, cross_validate with several metrics, GridSearchCV and RandomizedSearchCV over pipelines, nested cross-validation, Bayesian optimisation with Optuna, and the common leakage traps.

ML classes 6–7 · scikit-learn 1.7+ · Optuna 4 · Intermediate · 110 min

By the end of this lesson you can

- Explain why one train/test split is noisy and how k-fold cross-validation helps
- Choose stratified, group and time-series splitters for the problem
- Tune pipeline hyperparameters with GridSearchCV and RandomizedSearchCV and read cv_results_
- Use nested cross-validation for an unbiased estimate of a tuned model
- Recognise and prevent leakage during validation

One split is noisy

Python

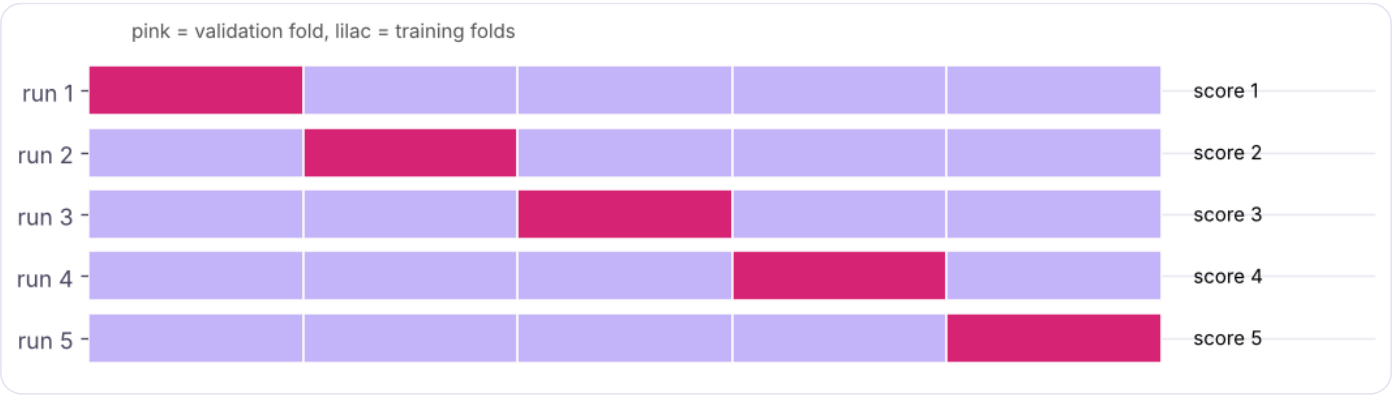
```
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split, cross_val_score, cross_validate, StratifiedKFold,
KFold, GridSearchCV, RandomizedSearchCV
from sklearn.pipeline import make_pipeline, Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
b = pd.read_csv("bank_loans.csv").dropna(subset=["default"])
feats = ["age", "ed", "employ", "address", "income", "debtinc", "creddebt", "othdebt"]
X, y = b[feats], b.default.astype(int)
pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
aucs = []
for seed in range(20):
    Xa, Xb, ya, yb = train_test_split(X, y, test_size=0.25, stratify=y, random_state=seed)
    aucs.append(roc_auc_score(yb, pipe.fit(Xa, ya).predict_proba(Xb)[:, 1]))
print(f"20 different random splits: test AUC from {min(aucs):.3f} to {max(aucs):.3f} (mean
{np.mean(aucs):.3f})")
```

OUTPUT

```
20 different random splits: test AUC from 0.745 to 0.865 (mean 0.801)
```

With 700 rows, the test score depends a lot on **which** rows land in the test set. Cross-validation averages over several splits.

K-fold cross-validation



5-fold cross-validation: each fold is the validation set once; the score is the average of 5 runs.

Python

```python
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=0)
scores = cross_val_score(pipe, X, y, cv=cv, scoring="roc_auc")
print("fold AUCs:", scores.round(3), f"| mean {scores.mean():.3f} ± {scores.std():.3f}")
res = cross_validate(pipe, X, y, cv=cv, scoring=["roc_auc", "accuracy", "recall", "precision"],
return_train_score=True)
print({k: round(float(v.mean()), 3) for k, v in res.items() if "time" not in k})
```

OUTPUT

```
fold AUCs: [0.767 0.862 0.746 0.83  0.811] | mean 0.803 ± 0.042
{'test_roc_auc': 0.803, 'train_roc_auc': 0.817, 'test_accuracy': 0.749, 'train_accuracy': 0.762,
'test_recall': 0.48, 'train_recall': 0.49, 'test_precision': 0.667, 'train_precision': 0.68}
```

Report the **mean and spread**. Compare train and validation scores to spot overfitting.

Splitter	Use
`KFold`	Regression, shuffled independent rows
`StratifiedKFold`	Classification: keeps class balance per fold (default for classifiers)
`GroupKFold`	Several rows per customer/patient/store: a group never appears in both train and validation
`TimeSeriesSplit`	Time order: always train on the past, validate on the future
`RepeatedStratifiedKFold`	Small data: repeat k-fold with different shuffles
`LeaveOneOut`	Tiny datasets (expensive)

Time-series cross-validation

Python

```
from sklearn.model_selection import TimeSeriesSplit
from sklearn.ensemble import HistGradientBoostingRegressor
from sklearn.metrics import mean_absolute_error
d = pd.read_csv("daily_sales.csv", parse_dates=["date"])
f = pd.DataFrame({"dow": d.date.dt.dayofweek, "month": d.date.dt.month, "t": np.arange(len(d)), "lag7":
d.units_sold.shift(7), "lag14": d.units_sold.shift(14)}).dropna()
yt = d.units_sold.loc[f.index]
for i, (tr, va) in enumerate(TimeSeriesSplit(n_splits=4, test_size=90).split(f)):
    m = HistGradientBoostingRegressor(random_state=0).fit(f.iloc[tr], yt.iloc[tr])
    print(f"fold {i + 1}: train to {d.date[f.index[tr[-1]]].date()} ({len(tr)} days) → validate
{d.date[f.index[va[0]]].date()}–{d.date[f.index[va[-1]]].date()} | MAE
{mean_absolute_error(yt.iloc[va], m.predict(f.iloc[va])):.1f}")
shuffled = cross_val_score(HistGradientBoostingRegressor(random_state=0), f, yt, cv=KFold(5,
shuffle=True, random_state=0), scoring="neg_mean_absolute_error").mean()
print(f"shuffled KFold MAE {-shuffled:.1f}: optimistic, because it trains on the future")
```

OUTPUT

```
fold 1: train to 2025-01-05 (1087 days) → validate 2025-01-06–2025-04-05 | MAE 24.3
fold 2: train to 2025-04-05 (1177 days) → validate 2025-04-06–2025-07-04 | MAE 30.8
fold 3: train to 2025-07-04 (1267 days) → validate 2025-07-05–2025-10-02 | MAE 14.9
fold 4: train to 2025-10-02 (1357 days) → validate 2025-10-03–2025-12-31 | MAE 13.1
shuffled KFold MAE 12.8: optimistic, because it trains on the future
```

Hyperparameters

Parameters are learned from data (coefficients, tree splits). **Hyperparameters** are set before training (C, number of neighbours, tree depth, learning rate). We choose them by cross-validation.

GridSearchCV

Python

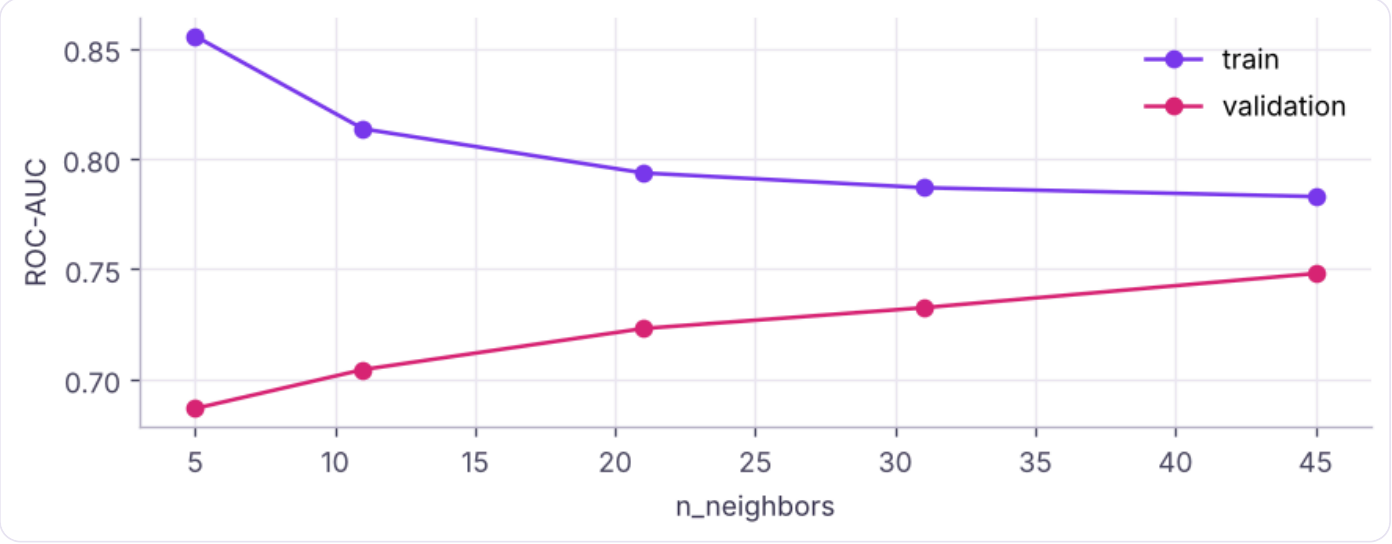
```
from sklearn.neighbors import KNeighborsClassifier
knn = Pipeline([("scale", StandardScaler()), ("knn", KNeighborsClassifier())])
grid = {"knn__n_neighbors": [5, 11, 21, 31, 45], "knn__weights": ["uniform", "distance"], "knn__p": [1,
2]}
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.25, stratify=y, random_state=7)
gs = GridSearchCV(knn, grid, cv=cv, scoring="roc_auc", n_jobs=-1, return_train_score=True).fit(Xtr,
ytr)
print("best:", gs.best_params_, f"| CV AUC {gs.best_score_:.3f} | test AUC {roc_auc_score(yte,
gs.predict_proba(Xte)[:, 1]):.3f}")
r = pd.DataFrame(gs.cv_results_)[["param_knn__n_neighbors", "param_knn__weights", "param_knn__p",
"mean_train_score", "mean_test_score", "rank_test_score"]]
print(r.sort_values("rank_test_score").head(6).round(3).to_string(index=False))
```

OUTPUT

```
best: {'knn__n_neighbors': 45, 'knn__p': 2, 'knn__weights': 'uniform'} | CV AUC 0.748 | test AUC
0.827
```

```
 param_knn__n_neighbors param_knn__weights  param_knn__p  mean_train_score  mean_test_score
rank_test_score
                     45            uniform             2             0.783            0.748
1
                     45            uniform             1             0.787            0.746
2
                     45           distance             2             1.000            0.745
3
                     45           distance             1             1.000            0.744
4
                     31            uniform             1             0.788            0.740
5
                     31           distance             1             1.000            0.734
6
```

Pipeline step names prefix parameters (`knn__n_neighbors`). `refit=True` (default) retrains the best configuration on all training data, so `gs` can predict directly.



Validation AUC by number of neighbours: small k overfits (train ≫ validation), large k smooths.

RandomizedSearchCV

Grids grow exponentially (5 values × 5 parameters = 3,125 fits per fold). Random search samples a fixed number of combinations and often finds equally good settings much faster.

Python

```
from scipy.stats import loguniform, randint
from sklearn.ensemble import RandomForestClassifier
rs = RandomizedSearchCV(RandomForestClassifier(random_state=0),
                        {"n_estimators": randint(100, 500), "max_depth": [None, 3, 5, 8, 12],
"min_samples_leaf": randint(1, 30), "max_features": ["sqrt", 0.5, 1.0]},
                        n_iter=25, cv=cv, scoring="roc_auc", random_state=0, n_jobs=-1).fit(Xtr, ytr)
print("best RF:", rs.best_params_, f"| CV AUC {rs.best_score_:.3f} | test AUC {roc_auc_score(yte,
rs.predict_proba(Xte)[:, 1]):.3f}")
lr_search = RandomizedSearchCV(make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000)),
{"logisticregression__C": loguniform(1e-3, 1e2)}, n_iter=20, cv=cv, scoring="roc_auc",
random_state=0).fit(Xtr, ytr)
print(f"logistic C = {lr_search.best_params_['logisticregression__C']:.3g} | CV AUC
{lr_search.best_score_:.3f} | test AUC {roc_auc_score(yte, lr_search.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
best RF: {'max_depth': 5, 'max_features': 'sqrt', 'min_samples_leaf': 24, 'n_estimators': 170} |
CV AUC 0.776 | test AUC 0.847
logistic C = 0.441 | CV AUC 0.783 | test AUC 0.864
```

Simple, well-regularised logistic regression is hard to beat on this small, mostly linear credit dataset: a useful reminder to always include simple models in the comparison.

Bayesian optimisation with Optuna

Python

```
import optuna
def objective(trial):
    params = {"n_estimators": trial.suggest_int("n_estimators", 100, 800),
              "max_depth": trial.suggest_int("max_depth", 2, 12),
              "min_samples_leaf": trial.suggest_int("min_samples_leaf", 1, 40),
              "max_features": trial.suggest_float("max_features", 0.2, 1.0)}
    return cross_val_score(RandomForestClassifier(**params, random_state=0), Xtr, ytr, cv=cv,
scoring="roc_auc").mean()
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=60)
print(study.best_params, study.best_value)
```

Optuna (and scikit-optimize, Hyperopt) learns from earlier trials which regions look promising, and prunes bad trials early: the usual choice for gradient boosting and deep learning.

Nested cross-validation

The best CV score from a search is slightly optimistic: it was chosen as the maximum of many tries. **Nested CV** wraps the search in an outer loop for an unbiased estimate.

Python

```
inner = StratifiedKFold(5, shuffle=True, random_state=1); outer = StratifiedKFold(5, shuffle=True,
random_state=2)
search = GridSearchCV(knn, {"knn__n_neighbors": [5, 11, 21, 31, 45]}, cv=inner, scoring="roc_auc")
nested = cross_val_score(search, X, y, cv=outer, scoring="roc_auc")
print(f"nested CV AUC {nested.mean():.3f} ± {nested.std():.3f} (vs best inner score {search.fit(X,
y).best_score_:.3f})")
```

OUTPUT

```
nested CV AUC 0.783 ± 0.044 (vs best inner score 0.786)
```

Leakage checklist

Leak	Example	Prevention
Preprocessing on all data	Scaling or imputing before splitting	Put every step in a Pipeline
Target leakage	A feature recorded after the outcome	Feature cut-off dates
Duplicate entities	Same customer in train and validation	GroupKFold
Future data	Random splits for time series	TimeSeriesSplit
Tuning on test data	Picking the model with the best test score	Tune with CV; test once
Feature selection outside CV	SelectKBest on all rows	Selection inside the pipeline

Summary

KEY TAKEAWAYS

- One split is noisy; k-fold CV averages several splits; report mean ± spread.
- Use stratified folds for classification, groups for repeated entities, and time-series splits for temporal data.
- Hyperparameters are tuned by CV: GridSearchCV (small spaces), RandomizedSearchCV (large spaces), Optuna (efficient Bayesian search).
- Nested CV gives an unbiased estimate of a tuned model; the test set is still used only once.
- Keep every data-dependent step inside the pipeline to avoid leakage.

Quick quiz

6 questions · tap an answer to check it

1. Why use cross-validation instead of one train/test split?

- **A** A single split is noisy; averaging over folds gives a more reliable estimate
- **B** It makes models more accurate
- **C** It avoids needing a test set
- **D** It is faster

Answer: A. Especially important for small datasets.

2. Which splitter for daily sales forecasting?

- **A** StratifiedKFold
- **B** TimeSeriesSplit
- **C** Shuffled KFold
- **D** LeaveOneOut

Answer: B. Validation must come after training in time.

3. How do you name a hyperparameter of a pipeline step called "knn" in GridSearchCV?

- **A** n_neighbors
- **B** knn__n_neighbors
- **C** knn.n_neighbors
- **D** knn-n_neighbors

Answer: B. Step name, two underscores, parameter.

4. When is RandomizedSearchCV preferable to GridSearchCV?

- **A** When the search space is large; it samples a fixed number of combinations
- **B** When there is one hyperparameter
- **C** Never
- **D** Only for regression

Answer: A. Often as good, much cheaper.

5. What does nested cross-validation estimate?

- **A** The performance of the whole tuning procedure, without optimistic bias
- **B** The best hyperparameters only
- **C** Training time
- **D** Feature importance

Answer: A. Inner loop tunes, outer loop evaluates.

6. Scaling the full dataset before cross-validation causes...

- **A** Leakage: validation statistics influence training
- **B** Better accuracy legitimately
- **C** No effect
- **D** Slower training

Answer: A. Put the scaler in the pipeline.

EXERCISE **Practice**

(1) Tune `C` and `penalty` (`l1`, `l2`) for logistic regression on the bank data with GridSearchCV (solver `liblinear`). (2) Use `GroupKFold` on the superstore data so that all lines of one customer stay together, predicting whether a line is loss-making. (3) Plot the CV AUC for `max_depth` 1–15 of a decision tree.

Show solution (worked answer)

Python

```
g = GridSearchCV(make_pipeline(StandardScaler(), LogisticRegression(solver="liblinear")),
{"logisticregression__C": np.logspace(-3, 2, 12), "logisticregression__penalty": ["l1", "l2"]},
cv=cv, scoring="roc_auc").fit(Xtr, ytr)
print(g.best_params_, g.best_score_)
from sklearn.model_selection import GroupKFold
from sklearn.tree import DecisionTreeClassifier
s = pd.read_csv("superstore_sales.csv"); Xs = pd.get_dummies(s[["sales", "quantity", "discount",
"category", "region"]]); ys = (s.profit < 0).astype(int)
print(cross_val_score(DecisionTreeClassifier(max_depth=5), Xs, ys, groups=s.customer_id,
cv=GroupKFold(5), scoring="roc_auc").mean())
print([round(cross_val_score(DecisionTreeClassifier(max_depth=k, random_state=0), X, y, cv=cv,
scoring="roc_auc").mean(), 3) for k in range(1, 16)])
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain k-fold cross-validation. Split the training data into k folds; train on k − 1 and validate on the remaining one, k times; average the scores. It uses all data for both training and validation and reduces variance of the estimate.

Q: Grid search vs random search vs Bayesian optimisation? Grid tries every combination (exhaustive, expensive); random samples combinations (efficient in high dimensions); Bayesian optimisation models the score surface to choose promising settings next.

Q: How would you validate a churn model that will score customers monthly? Time-based validation: train on earlier months and validate on later months, mimicking deployment; also group by customer to avoid leakage.

STATISTICS & MACHINE LEARNING · LESSON 16

K-Nearest Neighbours: Classifier and Regressor (Bank Loans Case Study)

The simplest learning algorithm: predict from the most similar past examples. Distance metrics, choosing k, why scaling is essential, weighted voting, the curse of dimensionality, KNN for classification and regression, tuning with GridSearchCV, and the full Bank Loans case study from the live class with metrics and scoring of new applicants.

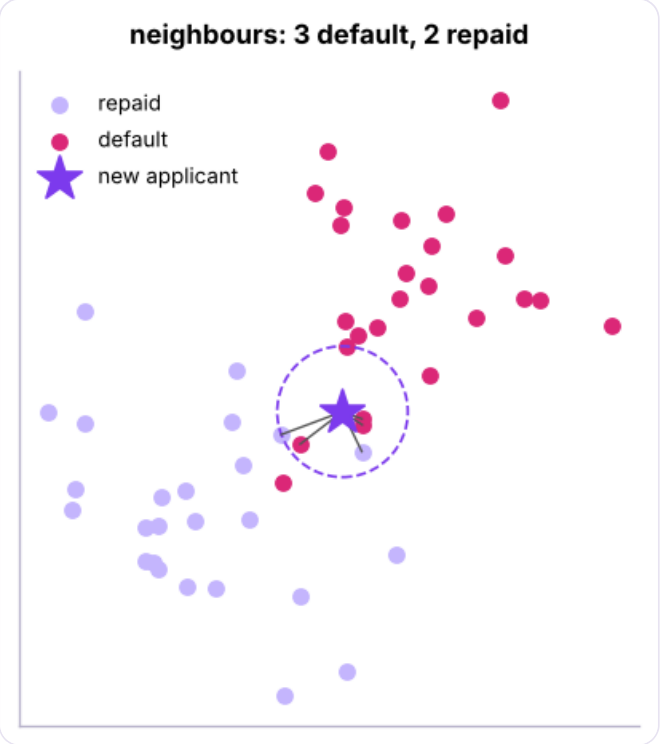
● ML class 7 ✦ scikit-learn 1.7+ Intermediate ⏱ 100 min

By the end of this lesson you can

- Explain how KNN classifies and regresses using distances to neighbours
- Compute Euclidean, Manhattan and Minkowski distances and know why scaling matters
- Choose k and weighting with cross-validation and explain the bias-variance effect of k
- Run the Bank Loans case study end to end with a pipeline and GridSearchCV
- Describe KNN's strengths, weaknesses and the curse of dimensionality

The idea

"Tell me who your neighbours are, and I'll tell you who you are." To predict a new point, find the *k* most similar labelled points and let them **vote** (classification) or **average** their values (regression). There is no training beyond storing the data, so KNN is a **lazy, instance-based** learner.



KNN with k = 5: the new applicant (star) is classified by the majority of its five nearest neighbours.

Distances

$$d_{\text{Euclidean}}(a,b) = \sqrt{\sum_j (a_j - b_j)^2} \qquad d_{\text{Manhattan}}(a,b) = \sum_j |a_j - b_j|$$

Minkowski distance generalises both (`p=2` Euclidean, `p=1` Manhattan). Cosine distance suits text and embeddings; Hamming suits binary features.

Python

```
from scipy.spatial import distance
x1, x2 = np.array([30, 45.0, 8.5]), np.array([42, 60.0, 12.0])          # age, income (thousands), debt-to-income %
print("Euclidean", round(distance.euclidean(x1, x2), 2), "| Manhattan", distance.cityblock(x1, x2), "| cosine", round(distance.cosine(x1, x2), 4))
```

OUTPUT

```
Euclidean 19.53 | Manhattan 30.5 | cosine 0.0003
```

Why scaling is essential

Python

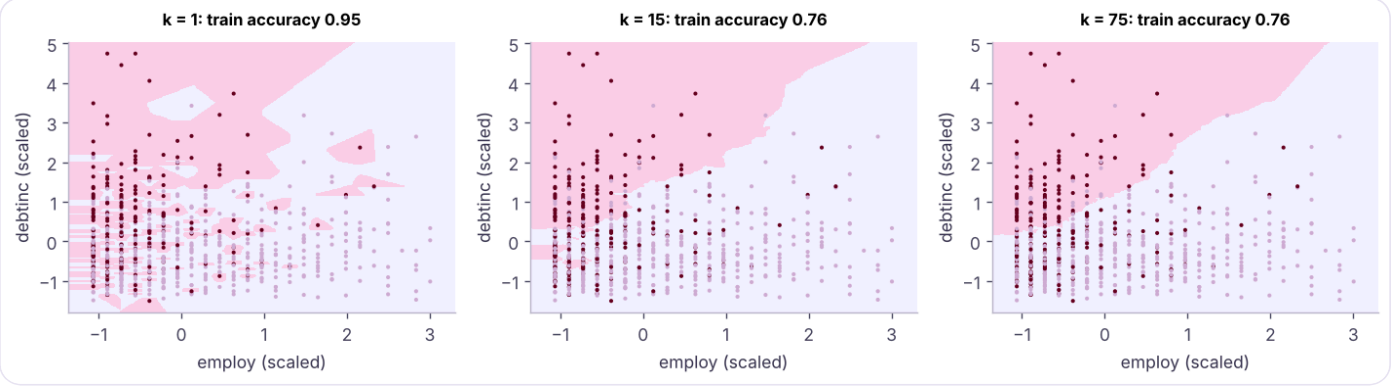
```
b = pd.read_csv("bank_loans.csv")
known = b[b.default.notna()].copy(); known["default"] = known.default.astype(int); new_apps =
b[b.default.isna()].copy()
feats = ["age", "ed", "employ", "address", "income", "debtinc", "creddebt", "othdebt"]
print(known[feats].describe().loc[["mean", "std", "max"]].round(1))
```

OUTPUT

	age	ed	employ	address	income	debtinc	creddebt	othdebt
mean	37.4	1.9	6.3	5.3	49.5	10.0	1.7	3.3
std	10.6	1.0	5.9	5.0	24.9	6.5	1.7	3.4
max	55.0	5.0	24.0	22.0	172.4	41.0	12.3	43.0

Income ranges up to 170 while `ed` ranges 1–5: unscaled, income dominates every distance and education is ignored. Standardise (or min-max scale) before KNN.

Choosing k



Decision boundaries for k = 1, 15 and 75 on two standardised features: small k overfits with jagged islands; large k oversmooths.

Small k	Large k
Flexible, low bias, high variance (overfits)	Smooth, high bias, low variance (underfits)
k = 1 memorises training data (100% train accuracy)	k = n predicts the majority class everywhere

Rules of thumb: start near √n, prefer odd k for binary problems (no ties), and choose by cross-validation. `weights="distance"` lets closer neighbours count more.

Case study: Bank Loans

Business problem: decide which of 150 new loan applicants to approve, using 700 past customers with known outcomes.

Python

```
from sklearn.model_selection import train_test_split, GridSearchCV, StratifiedKFold
from sklearn.metrics import roc_auc_score, classification_report, confusion_matrix, accuracy_score
X, y = known[feats], known.default
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=42)
raw = KNeighborsClassifier(15).fit(Xtr, ytr); scaled = make_pipeline(StandardScaler(),
KNeighborsClassifier(15)).fit(Xtr, ytr)
print(f"k=15 without scaling: AUC {roc_auc_score(yte, raw.predict_proba(Xte)[:, 1]):.3f} | with
scaling: AUC {roc_auc_score(yte, scaled.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
k=15 without scaling: AUC 0.757 | with scaling: AUC 0.765
```

Scaling helps only a little here, because the large-range features (income, debt-to-income) happen to be informative anyway. When a large-range feature is irrelevant (for example a customer ID number or a postcode), unscaled KNN can fail completely.

Python

```
pipe = make_pipeline(StandardScaler(), KNeighborsClassifier())
grid = {"kneighborsclassifier__n_neighbors": list(range(3, 62, 4)), "kneighborsclassifier__weights":
["uniform", "distance"], "kneighborsclassifier__metric": ["euclidean", "manhattan"]}
gs = GridSearchCV(pipe, grid, cv=StratifiedKFold(5, shuffle=True, random_state=0), scoring="roc_auc",
n_jobs=-1).fit(Xtr, ytr)
print("best:", {k.split("__")[1]: v for k, v in gs.best_params_.items()}, f"| CV AUC
{gs.best_score_:.3f}")
p = gs.predict_proba(Xte)[:, 1]
print(f"test AUC {roc_auc_score(yte, p):.3f} | accuracy at 0.5 {accuracy_score(yte, p ≥ .5):.3f}")
print(confusion_matrix(yte, p ≥ 0.5))
print(classification_report(yte, p ≥ 0.5, target_names=["repaid", "default"], digits=3))
```

OUTPUT

```
best: {'metric': 'manhattan', 'n_neighbors': 51, 'weights': 'distance'} | CV AUC 0.772
test AUC 0.810 | accuracy at 0.5 0.733
[[134   9]
 [ 47  20]]
              precision    recall  f1-score   support

      repaid      0.740     0.937     0.827       143
     default      0.690     0.299     0.417        67

    accuracy                          0.733       210
   macro avg      0.715     0.618     0.622       210
weighted avg      0.724     0.733     0.696       210
```

Python

```
t = 0.3                                                  # lower threshold: missing a defaulter is
costlier
print(f"threshold {t}:"); print(confusion_matrix(yte, p ⩾ t)); print(classification_report(yte, p ⩾
t, target_names=["repaid", "default"], digits=3))
```

OUTPUT

```
threshold 0.3:
[[94 49]
 [ 9 58]]
              precision    recall  f1-score   support

      repaid      0.913     0.657     0.764       143
     default      0.542     0.866     0.667        67

    accuracy                          0.724       210
   macro avg      0.727     0.762     0.715       210
weighted avg      0.794     0.724     0.733       210
```

KNN vs logistic regression

Python

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score
cv = StratifiedKFold(5, shuffle=True, random_state=0)
for name, m in [("KNN (tuned)", gs.best_estimator_), ("logistic regression",
make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000)))]:
    s = cross_val_score(m, X, y, cv=cv, scoring="roc_auc"); print(f"{name:20s} 5-fold AUC
{s.mean():.3f} ± {s.std():.3f}")
```

OUTPUT

```
KNN (tuned)          5-fold AUC 0.786 ± 0.046
logistic regression  5-fold AUC 0.803 ± 0.042
```

On this small, roughly linear credit problem, logistic regression is as good or better, and far easier to explain. KNN shines when the boundary is irregular and the data are dense.

Scoring new applicants and explaining with neighbours

Python

```
best = gs.best_estimator_.fit(X, y)
new_apps["p_default"] = best.predict_proba(new_apps[feats])[:, 1].round(3)
print(new_apps.p_default.describe().round(3).to_dict())
app = new_apps.sort_values("p_default", ascending=False).iloc[0]
dist_, idx = best[-1].kneighbors(best[0].transform(app[feats].to_frame().T.astype(float)),
n_neighbors=5)
print(f"{app.customer_id}: P(default) {app.p_default:.2f}. Its 5 most similar past customers:")
print(known.iloc[idx[0]][feats + ["default"]].to_string(index=False))
```

OUTPUT

```
{'count': 150.0, 'mean': 0.295, 'std': 0.176, 'min': 0.016, '25%': 0.168, '50%': 0.271, '75%':
0.401, 'max': 0.802}
L0782: P(default) 0.80. Its 5 most similar past customers:
 age  ed  employ  address  income  debtinc  creddebt  othdebt  default
  35   2       1        5    33.7     30.7      3.24     7.11        1
  47   2       0        4    26.5     32.8      4.28     4.41        1
  40   1       3        4    49.7     24.1      4.34     7.64        1
  39   1       5        4    32.7     24.3      3.44     4.51        1
  26   1       1        2    52.2     29.4      5.63     9.72        1
```

KNN explanations are concrete: here, all five of the most similar past customers defaulted.

KNN regression

Python

```
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_absolute_error
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"])
hf = ["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km"]
Htr, Hte, htr, hte = train_test_split(h[hf], h.price_lakh, test_size=0.2, random_state=0)
for k in (1, 5, 15, 40):
    m = make_pipeline(StandardScaler(), KNeighborsRegressor(k, weights="distance")).fit(Htr, htr)
    print(f"k = {k:2d}: test MAE {mean_absolute_error(hte, m.predict(Hte)):.2f} lakh")
```

OUTPUT

```
k =  1: test MAE 16.40 lakh
k =  5: test MAE 12.57 lakh
k = 15: test MAE 12.96 lakh
k = 40: test MAE 12.42 lakh
```

Strengths, weaknesses and the curse of dimensionality

Strengths	Weaknesses
Simple, intuitive, no training	Slow predictions on large data (must search all points)
Naturally multiclass; non-linear boundaries	Needs scaling; sensitive to irrelevant features
Explanations by example	Struggles in high dimensions
Works for regression too	Memory: stores the whole training set

Python

```python
rng = np.random.default_rng(0)
for dims in (2, 10, 100, 1000):
    pts_ = rng.uniform(size=(500, dims)); d = np.linalg.norm(pts_[1:] - pts_[0], axis=1)
    print(f"{dims:4d} dimensions: nearest/farthest distance ratio {d.min() / d.max():.2f}")
```

OUTPUT

```
   2 dimensions: nearest/farthest distance ratio 0.01
  10 dimensions: nearest/farthest distance ratio 0.30
 100 dimensions: nearest/farthest distance ratio 0.73
1000 dimensions: nearest/farthest distance ratio 0.90
```

In high dimensions every point is almost equally far from every other (the ratio approaches 1), so "nearest" stops meaning "similar". Reduce dimensions (feature selection, PCA) first. For millions of vectors (embeddings in RAG systems, GenAI track), approximate nearest-neighbour indexes (FAISS, HNSW) make search fast.

Summary

KEY TAKEAWAYS

- KNN predicts from the k most similar labelled examples: majority vote or (weighted) average.
- Distances: Euclidean, Manhattan, Minkowski, cosine; always scale features first.
- Small k overfits, large k underfits; tune k, weights and metric with cross-validation.
- Bank Loans: a scaled, tuned KNN reaches a decent AUC, and logistic regression is as good and more interpretable on this data.
- KNN suffers from slow prediction and the curse of dimensionality; approximate nearest-neighbour search powers vector databases.

Quick quiz

6 questions · tap an answer to check it

1. What does KNN do during training?

- **A** Stores the training data (lazy learning)
- **B** Fits coefficients by gradient descent
- **C** Builds a tree
- **D** Computes clusters

Answer: A. Work happens at prediction time.

2. Why must features be scaled for KNN?

- **A** Features with large ranges would dominate the distance
- **B** To make it faster
- **C** Because KNN needs normal data
- **D** It is optional

Answer: A. Income would swamp education level.

3. What happens with k = 1?

- **A** The model memorises the training data and tends to overfit
- **B** It underfits
- **C** It always predicts the majority class
- **D** It is invalid

Answer: A. Training accuracy is 100%.

4. Which distance is Minkowski with p = 1?

- **A** Euclidean
- **B** Manhattan
- **C** Cosine
- **D** Hamming

Answer: B. Sum of absolute differences.

5. What is the curse of dimensionality for KNN?

- **A** In many dimensions, all points become roughly equally distant, so neighbours are not meaningful
- **B** More rows slow it down
- **C** It needs more classes
- **D** Scaling stops working

Answer: A. Reduce dimensions first.

6. How does `weights="distance"` change predictions?

- **A** Closer neighbours get more influence than farther ones
- **B** All neighbours count equally
- **C** It removes far neighbours
- **D** It changes the metric

Answer: A. Weights are 1/distance.

EXERCISE **Practice**

(1) Use KNN to predict churn on `telecom_churn.csv` (one-hot + scale) and tune k with GridSearchCV; compare AUC with logistic regression. (2) Plot test MAE against k (1–50) for KNN regression on house prices. (3) Find the 5 nearest neighbours of a new house (1,400 sq ft, 3 bed, 2 bath, 8 years, 6 km).

Show solution (worked answer)

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
tc = pd.read_csv("telecom_churn.csv"); Xc = tc.drop(columns=["customer_id", "churn",
"total_charges"]); yc = (tc.churn == "Yes").astype(int)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months",
"monthly_charges", "support_calls"]), ("c", OneHotEncoder(), ["contract", "internet_service",
"payment_method"])])
g = GridSearchCV(make_pipeline(pre, KNeighborsClassifier()), {"kneighborsclassifier__n_neighbors":
range(5, 120, 10)}, cv=5, scoring="roc_auc").fit(Xc, yc); print(g.best_params_, g.best_score_)
maes = [mean_absolute_error(hte, make_pipeline(StandardScaler(), KNeighborsRegressor(k)).fit(Htr,
htr).predict(Hte)) for k in range(1, 51)]
m = make_pipeline(StandardScaler(), KNeighborsRegressor(5)).fit(h[hf], h.price_lakh)
print(m.predict(pd.DataFrame([[1400, 3, 2, 8, 6]], columns=hf)))
```

INTERVIEW CORNER **Questions you may be asked**

Q: How does KNN work and how do you choose k? It finds the k closest training points by a distance metric and votes or averages; choose k (and weighting, metric) by cross-validation, balancing overfitting (small k) and oversmoothing (large k).

Q: What are the disadvantages of KNN? Slow and memory-heavy predictions, sensitivity to scaling and irrelevant features, poor performance in high dimensions, and imbalanced classes skew votes.

Q: How would you speed up nearest-neighbour search on millions of points? KD-trees or ball trees for low dimensions, and approximate nearest-neighbour indexes (HNSW, FAISS, ScaNN) for high-dimensional embeddings.

STATISTICS & MACHINE LEARNING · LESSON 17

Decision Trees: Splits, Impurity, Pruning and Interpretable Rules

Learn the model behind random forests and gradient boosting. How trees split data with Gini impurity, entropy and variance reduction, growing and visualising a tree, reading it as business rules, controlling overfitting with depth, minimum leaf size and cost-complexity pruning, regression trees, feature importance and the strengths and weaknesses of single trees.

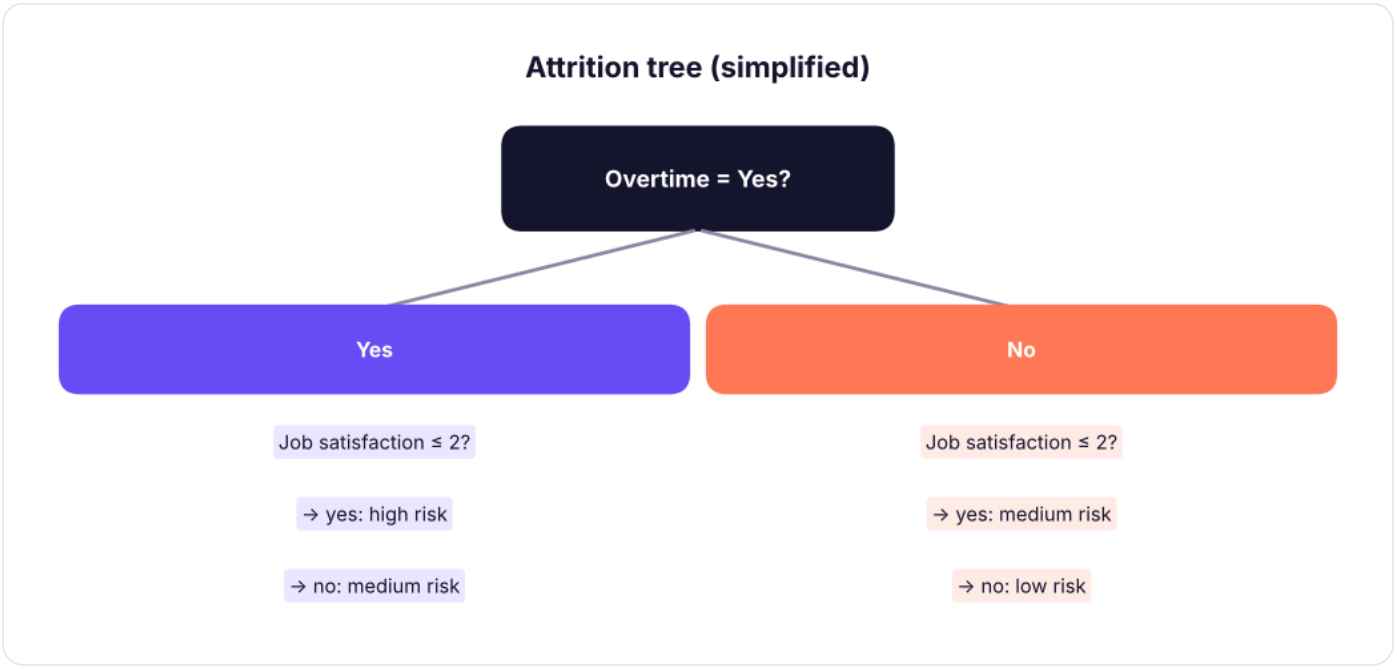
ML track (added) · scikit-learn 1.7+ · Intermediate · 110 min

By the end of this lesson you can

- Explain recursive binary splitting and compute Gini impurity and entropy by hand
- Train, plot and read a decision tree classifier as IF-THEN rules
- Control overfitting with max_depth, min_samples_leaf and ccp_alpha pruning
- Build a regression tree and explain its step-shaped predictions
- Describe tree feature importance and the instability of single trees

How a tree decides

A **decision tree** asks a sequence of yes/no questions about the features. Each question splits the data into two purer groups; leaves give the prediction (majority class or average value). It is a flowchart learned from data.



A decision tree is a learned flowchart of questions.

Measuring purity

At each node the algorithm tries every feature and threshold and picks the split that most reduces **impurity**.

$$\text{Gini} = 1 - \sum_k p_k^2 \qquad \text{Entropy} = -\sum_k p_k \log_2 p_k$$

Python

```
import numpy as np, pandas as pd
def gini(p): p = np.asarray(p); return 1 - (p ** 2).sum()
def entropy(p): p = np.asarray([x for x in p if x > 0]); return float(-(p * np.log2(p)).sum()) + 0.0
for dist in ([0.5, 0.5], [0.8, 0.2], [1.0, 0.0]): print(dist, "Gini", round(gini(dist), 3), "|
entropy", round(entropy(dist), 3))
hr = pd.read_csv("hr_employees.csv"); hr["left"] = (hr.attrition == "Yes").astype(int)
def split_gain(df, mask):
    parent = gini([1 - df.left.mean(), df.left.mean()])
    l, r = df[mask], df[~mask]
    child = sum(len(s) / len(df) * gini([1 - s.left.mean(), s.left.mean()]) for s in (l, r))
    return parent - child
for name, mask in [("overtime == Yes", hr.overtime == "Yes"), ("job_satisfaction ≤ 2",
hr.job_satisfaction ≤ 2), ("years_at_company ≤ 1", hr.years_at_company ≤ 1), ("department == Sales",
hr.department == "Sales")]:
    print(f"{name:24s} Gini decrease {split_gain(hr, mask):.4f}")
```

OUTPUT

```
[0.5, 0.5] Gini 0.5 | entropy 1.0
[0.8, 0.2] Gini 0.32 | entropy 0.722
[1.0, 0.0] Gini 0.0 | entropy 0.0
overtime == Yes          Gini decrease 0.0094
```

```
job_satisfaction ⩽ 2    Gini decrease 0.0086
years_at_company ⩽ 1    Gini decrease 0.0031
department == Sales     Gini decrease 0.0015
```

The split with the largest decrease (information gain) wins; then the process repeats inside each child (**recursive binary splitting**) until a stopping rule is met. Regression trees use variance (MSE) reduction instead.

Training and reading a tree

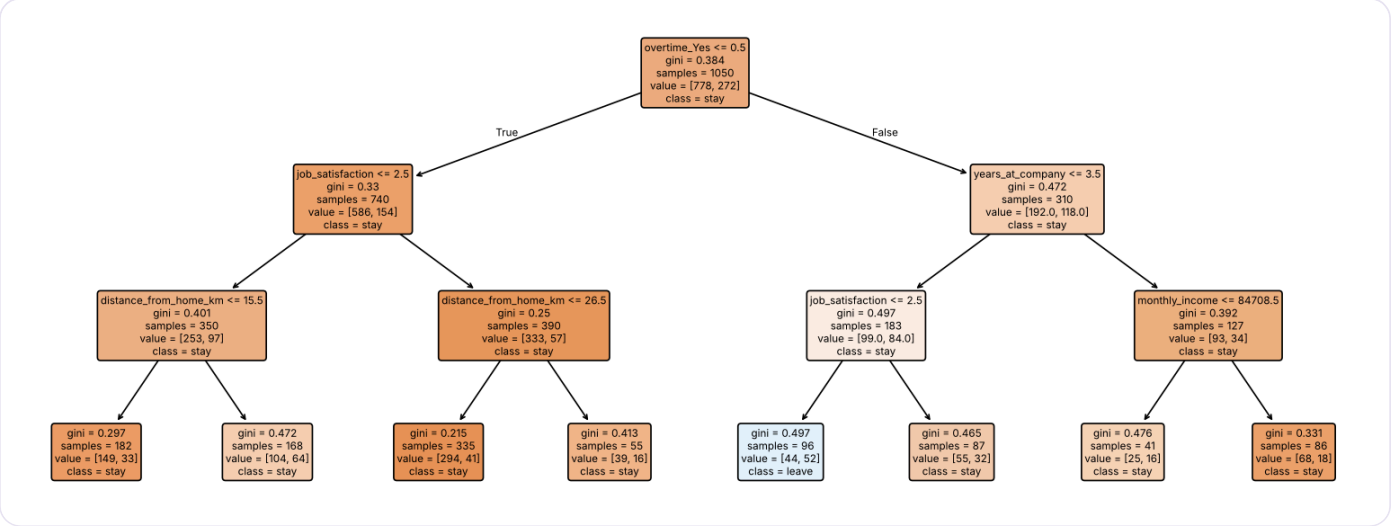
Python

```python
from sklearn.tree import DecisionTreeClassifier, export_text, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score, accuracy_score
X = pd.get_dummies(hr[["age", "years_at_company", "monthly_income", "job_satisfaction",
"distance_from_home_km", "performance_rating", "overtime", "department"]],
drop_first=True).astype(float)
y = hr.left
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
small = DecisionTreeClassifier(max_depth=3, min_samples_leaf=40, random_state=0).fit(Xtr, ytr)
print(export_text(small, feature_names=list(X.columns), show_weights=True, decimals=1))
print(f"test AUC {roc_auc_score(yte, small.predict_proba(Xte)[:, 1]):.3f} | accuracy
{accuracy_score(yte, small.predict(Xte)):.3f}")
```

OUTPUT

```
|--- overtime_Yes ⩽ 0.5
|   |--- job_satisfaction ⩽ 2.5
|   |   |--- distance_from_home_km ⩽ 15.5
|   |   |   |--- weights: [149.0, 33.0] class: 0
|   |   |--- distance_from_home_km >  15.5
|   |   |   |--- weights: [104.0, 64.0] class: 0
|   |--- job_satisfaction >  2.5
|   |   |--- distance_from_home_km ⩽ 26.5
|   |   |   |--- weights: [294.0, 41.0] class: 0
|   |   |--- distance_from_home_km >  26.5
|   |   |   |--- weights: [39.0, 16.0] class: 0
|--- overtime_Yes >  0.5
|   |--- years_at_company ⩽ 3.5
|   |   |--- job_satisfaction ⩽ 2.5
|   |   |   |--- weights: [44.0, 52.0] class: 1
|   |   |--- job_satisfaction >  2.5
|   |   |   |--- weights: [55.0, 32.0] class: 0
|   |--- years_at_company >  3.5
|   |   |--- monthly_income ⩽ 84708.5
|   |   |   |--- weights: [25.0, 16.0] class: 0
|   |   |--- monthly_income >  84708.5
|   |   |   |--- weights: [68.0, 18.0] class: 0

test AUC 0.598 | accuracy 0.716
```



The depth-3 tree. Each box shows the split, Gini impurity, sample count and class mix; darker colours are purer.

Each path from root to leaf is an **IF-THEN rule** that managers can read and challenge, which is a big advantage of small trees.

Overfitting and how to stop it

Python

```
for depth in (1, 2, 3, 5, 8, 12, None):
    m = DecisionTreeClassifier(max_depth=depth, random_state=0).fit(Xtr, ytr)
    print(f"max_depth={str(depth):4s} leaves {m.get_n_leaves():4d} | train AUC {roc_auc_score(ytr,
m.predict_proba(Xtr)[:, 1]):.3f} | test AUC {roc_auc_score(yte, m.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
max_depth=1    leaves    2 | train AUC 0.594 | test AUC 0.552
max_depth=2    leaves    4 | train AUC 0.639 | test AUC 0.591
max_depth=3    leaves    7 | train AUC 0.681 | test AUC 0.598
max_depth=5    leaves   23 | train AUC 0.757 | test AUC 0.571
max_depth=8    leaves   90 | train AUC 0.898 | test AUC 0.597
max_depth=12   leaves  196 | train AUC 0.986 | test AUC 0.579
max_depth=None leaves  255 | train AUC 1.000 | test AUC 0.551
```

An unrestricted tree keeps splitting until leaves are pure: perfect on training data, poor on new data. Notice also that attrition is hard to predict from these few HR fields (the best test AUC is only about 0.6). That is common with people data: many reasons for leaving (a new offer, a manager, family) are not in the database. A good analyst says so rather than overselling the model.

Hyperparameter	Effect
`max_depth`	Maximum number of questions on a path
`min_samples_split`	A node needs this many rows to split
`min_samples_leaf`	Every leaf keeps at least this many rows (smooths probabilities)
`max_leaf_nodes`	Cap on the number of leaves
`ccp_alpha`	Cost-complexity pruning: removes branches that add little
`class_weight="balanced"`	Gives the minority class more weight

Cost-complexity pruning

Python

```
from sklearn.model_selection import cross_val_score, StratifiedKFold
path = DecisionTreeClassifier(random_state=0).cost_complexity_pruning_path(Xtr, ytr)
alphas = path.ccp_alphas[::max(1, len(path.ccp_alphas) // 25)]
cv = StratifiedKFold(5, shuffle=True, random_state=0)
scores = [cross_val_score(DecisionTreeClassifier(ccp_alpha=a, random_state=0), Xtr, ytr, cv=cv,
scoring="roc_auc").mean() for a in alphas]
best_a = alphas[int(np.argmax(scores))]
pruned = DecisionTreeClassifier(ccp_alpha=best_a, random_state=0).fit(Xtr, ytr)
print(f"best ccp_alpha {best_a:.5f} | leaves {pruned.get_n_leaves()} | CV AUC {max(scores):.3f} | test
AUC {roc_auc_score(yte, pruned.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
best ccp_alpha 0.00320 | leaves 10 | CV AUC 0.607 | test AUC 0.586
```

Feature importance

Python

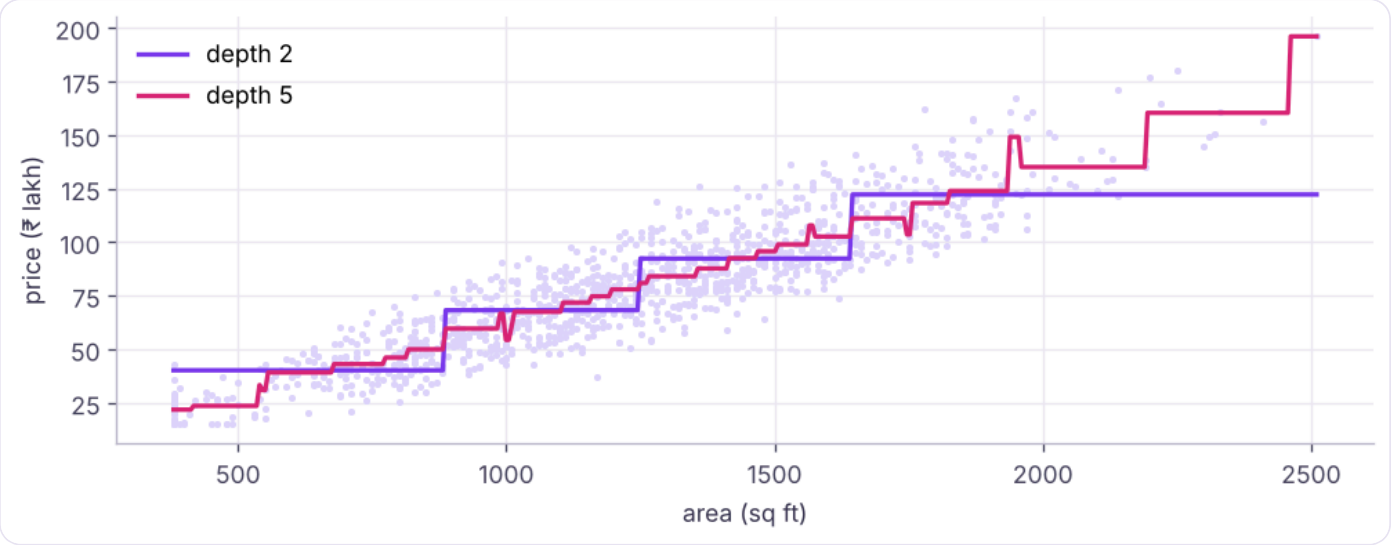
```
imp = pd.Series(pruned.feature_importances_, index=X.columns).sort_values(ascending=False)
print(imp[imp > 0].round(3).to_dict())
```

OUTPUT

```
{'monthly_income': 0.424, 'overtime_Yes': 0.245, 'distance_from_home_km': 0.131,
'job_satisfaction': 0.119, 'years_at_company': 0.08}
```

Importance = total impurity decrease contributed by each feature (normalised to sum to 1). It is quick but biased towards continuous features with many possible split points; check with permutation importance (lesson 27).

Regression trees



A regression tree predicts the average price in each leaf, giving step-shaped predictions (depth 2 vs depth 5).

Python

```
from sklearn.metrics import mean_absolute_error
hf = ["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km", "parking"]
Htr, Hte, htr, hte = train_test_split(pd.get_dummies(h[hf + ["city"]]).astype(float), h.price_lakh,
test_size=0.2, random_state=0)
for d in (3, 6, 10, None):
    t = DecisionTreeRegressor(max_depth=d, min_samples_leaf=5, random_state=0).fit(Htr, htr)
    print(f"depth {str(d):4s} test MAE {mean_absolute_error(hte, t.predict(Hte)):.2f} lakh")
```

OUTPUT

```
depth 3    test MAE 10.12 lakh
depth 6    test MAE 8.10 lakh
depth 10   test MAE 7.62 lakh
depth None test MAE 7.64 lakh
```

Trees cannot extrapolate: a house bigger than any in the training data gets the price of the largest leaf.

Strengths and weaknesses

Strengths	Weaknesses
Easy to explain as rules; visual	Unstable: small data changes give a different tree
No scaling needed; handles mixed types	Overfits without limits
Captures non-linearity and interactions automatically	Step-shaped, no extrapolation
Fast to train and predict	Single trees are usually less accurate than ensembles

Python

```python
for seed in range(3):
    Xs, _, ys, _ = train_test_split(Xtr, ytr, train_size=0.8, random_state=seed)
    t = DecisionTreeClassifier(max_depth=2, random_state=0).fit(Xs, ys)
    print(f"sample {seed}: root split on '{X.columns[t.tree_.feature[0]]}' at
{t.tree_.threshold[0]:.1f}, second-level splits on {[X.columns[f] for f in t.tree_.feature[1:] if f ≥
0]}")
```

OUTPUT

```
sample 0: root split on 'overtime_Yes' at 0.5, second-level splits on ['distance_from_home_km',
'years_at_company']
sample 1: root split on 'overtime_Yes' at 0.5, second-level splits on ['job_satisfaction',
'years_at_company']
sample 2: root split on 'job_satisfaction' at 2.5, second-level splits on
['distance_from_home_km', 'age']
```

Instability is exactly what **random forests** exploit: average many different trees and the noise cancels out (next lesson).

Summary

KEY TAKEAWAYS

- Trees split recursively on the feature/threshold that most reduces impurity (Gini, entropy) or variance (regression).
- Small trees are readable IF-THEN rules; unrestricted trees overfit badly.
- Control complexity with max_depth, min_samples_leaf and ccp_alpha pruning, chosen by cross-validation.
- Impurity-based importance is quick but biased; trees need no scaling but cannot extrapolate.
- Single trees are unstable; ensembles (random forests, boosting) fix this.

Quick quiz

6 questions · tap an answer to check it

1. What is the Gini impurity of a node with 50% stay and 50% leave?

- **A** 0
- **B** 0.5
- **C** 1
- **D** 0.25

Answer: B. $1 - (0.5^2 + 0.5^2) = 0.5$, the maximum for two classes.

2. An unrestricted tree has train AUC 1.0 and test AUC 0.6. Why?

- **A** It overfits: leaves memorise individual training rows
- **B** It underfits
- **C** The data are scaled wrongly
- **D** Gini is the wrong criterion

Answer: A. Limit depth or prune.

3. Do decision trees need feature scaling?

- **A** Yes, always
- **B** No: splits depend on thresholds, not distances
- **C** Only for regression
- **D** Only with entropy

Answer: B. Monotonic transformations do not change splits.

4. What does `min_samples_leaf=40` do?

- **A** Prevents leaves with fewer than 40 training rows
- **B** Limits the tree to 40 leaves
- **C** Uses 40 features
- **D** Builds 40 trees

Answer: A. It smooths predictions and reduces overfitting.

5. Why can't a regression tree predict prices above the training maximum?

- **A** Predictions are averages of training targets in each leaf
- **B** It has a bug
- **C** Prices must be scaled
- **D** Trees only classify

Answer: A. No extrapolation.

6. Which weakness of single trees do random forests address?

- **A** High variance / instability
- **B** Lack of scaling
- **C** Inability to handle categories
- **D** Slowness

Answer: A. Averaging many trees reduces variance.

EXERCISE **Practice**

(1) Train a tree on the telecom churn data with depth 3 and write its rules as three bullet points for the retention team. (2) Tune `max_depth` and `min_samples_leaf` with GridSearchCV. (3) Compare the test AUC of the pruned tree with logistic regression.

Show solution (worked answer)

Python

```
tc = pd.read_csv("telecom_churn.csv"); Xc = pd.get_dummies(tc.drop(columns=["customer_id",
"churn", "total_charges"]), drop_first=True).astype(float); yc = (tc.churn == "Yes").astype(int)
t3 = DecisionTreeClassifier(max_depth=3, min_samples_leaf=50).fit(Xc, yc); print(export_text(t3,
feature_names=list(Xc.columns)))
from sklearn.model_selection import GridSearchCV
g = GridSearchCV(DecisionTreeClassifier(random_state=0), {"max_depth": [2, 3, 4, 5, 6, 8],
"min_samples_leaf": [5, 20, 50, 100]}, cv=5, scoring="roc_auc").fit(Xc, yc); print(g.best_params_,
g.best_score_)
```

INTERVIEW CORNER **Questions you may be asked**

Q: How does a decision tree choose a split? It evaluates candidate thresholds on every feature and picks the one with the largest impurity reduction (Gini or entropy for classification, variance for regression), then repeats recursively.

Q: Gini vs entropy? Both measure impurity and usually give similar trees; Gini is slightly faster, entropy comes from information theory.

Q: How do you prevent a tree from overfitting? Limit depth, require minimum samples per leaf/split, cap leaves, prune with cost-complexity, or use ensembles.

STATISTICS & MACHINE LEARNING · LESSON 18

Ensembles I: Bagging and Random Forests

Combine many models to beat one. The wisdom of crowds, bootstrap aggregating (bagging), random forests with feature subsampling, out-of-bag evaluation, the key hyperparameters, feature importance (impurity vs permutation), extra trees, voting and stacking ensembles, applied to telecom churn.

ML track (added) · scikit-learn 1.7+ · Intermediate · 100 min

By the end of this lesson you can

- Explain why averaging many diverse models reduces variance
- Build bagging and random-forest models and use out-of-bag scores
- Tune n_estimators, max_features, max_depth and min_samples_leaf
- Compare impurity-based and permutation feature importance
- Combine different models with voting and stacking

Why ensembles work

If each of many independent models is right 65% of the time, a majority vote of them is right far more often. Errors cancel when models make **different** mistakes.

Python

```
import numpy as np, pandas as pd
from scipy.stats import binom
for n in (1, 5, 25, 101):
    print(f"{n:3d} independent voters, each 65% accurate → majority correct {binom.sf(n // 2, n,
0.65):.3f}")
```

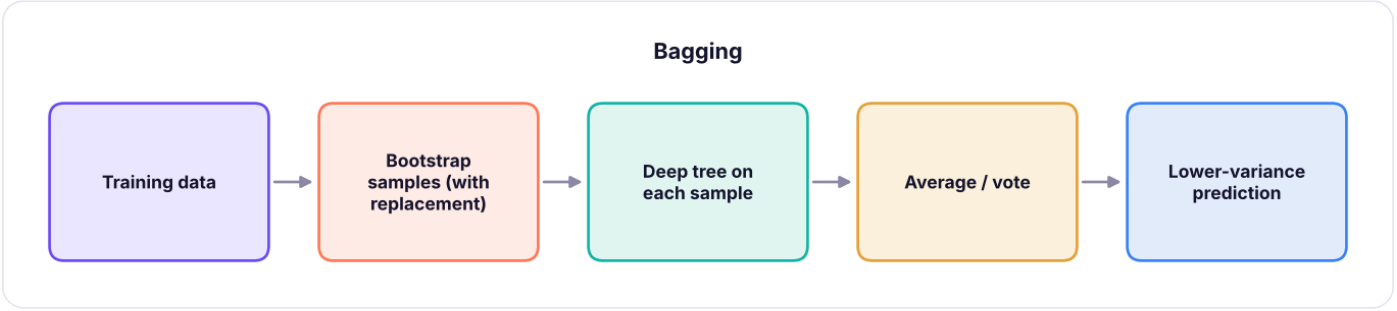
OUTPUT

```
  1 independent voters, each 65% accurate → majority correct 0.650
  5 independent voters, each 65% accurate → majority correct 0.765
 25 independent voters, each 65% accurate → majority correct 0.940
101 independent voters, each 65% accurate → majority correct 0.999
```

Real models are correlated (trained on the same data), so the gain is smaller. The trick is to make trees **diverse**.

Bagging: bootstrap aggregating

Train each tree on a **bootstrap sample** (rows drawn with replacement, same size as the data), then average predictions (regression) or vote (classification). Each bootstrap sample leaves out about 37% of rows: the **out-of-bag (OOB)** rows act as a free validation set.



Bagging: many bootstrap samples, one deep tree each, averaged predictions.

Random forests

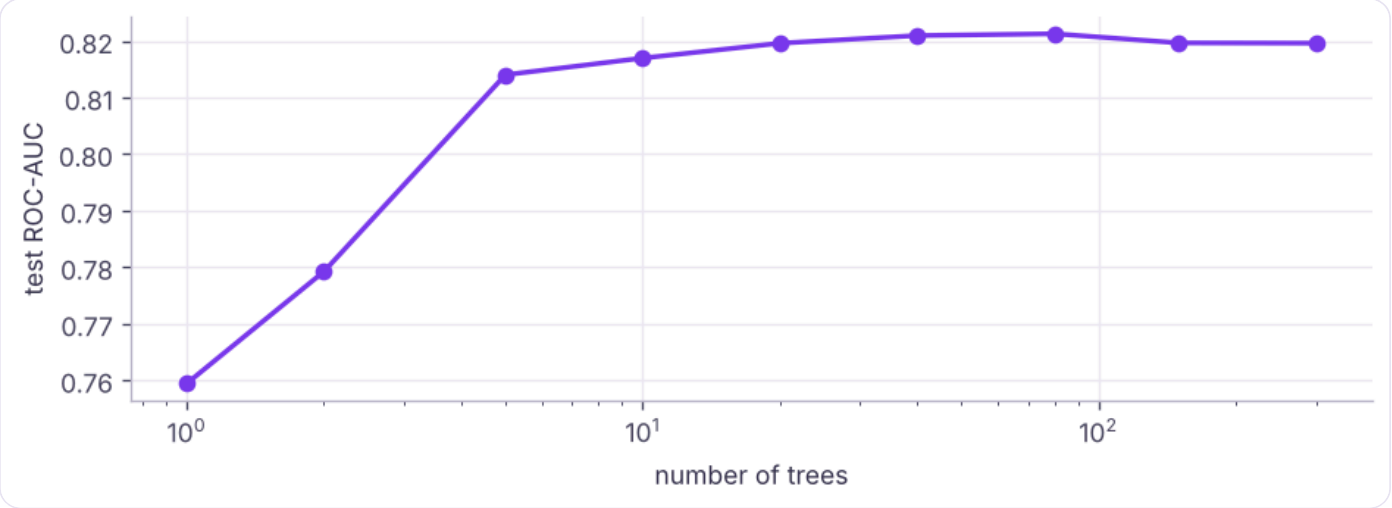
A **random forest** = bagging + a random subset of features considered at **each split** (`max_features`). This decorrelates the trees further: a dominant feature cannot sit at the top of every tree.

Python

```
from sklearn.model_selection import train_test_split, cross_val_score, StratifiedKFold
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier, RandomForestClassifier, ExtraTreesClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
tc = pd.read_csv("telecom_churn.csv")
X = pd.get_dummies(tc.drop(columns=["customer_id", "churn", "total_charges"]),
drop_first=True).astype(float); y = (tc.churn == "Yes").astype(int)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=1)
models = {"single deep tree": DecisionTreeClassifier(random_state=0),
          "bagged trees (300)": BaggingClassifier(DecisionTreeClassifier(), n_estimators=300,
random_state=0, n_jobs=-1),
          "random forest (300)": RandomForestClassifier(300, random_state=0, n_jobs=-1),
          "random forest, leaf ≥ 20": RandomForestClassifier(300, min_samples_leaf=20, random_state=0,
n_jobs=-1),
          "extra trees, leaf ≥ 20": ExtraTreesClassifier(300, min_samples_leaf=20, random_state=0,
n_jobs=-1)}
for name, m in models.items():
    m.fit(Xtr, ytr)
    print(f"{name:26s} train AUC {roc_auc_score(ytr, m.predict_proba(Xtr)[:, 1]):.3f} | test AUC
{roc_auc_score(yte, m.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
single deep tree           train AUC 1.000 | test AUC 0.630
bagged trees (300)         train AUC 1.000 | test AUC 0.763
random forest (300)        train AUC 1.000 | test AUC 0.771
random forest, leaf ≥ 20   train AUC 0.847 | test AUC 0.820
extra trees, leaf ≥ 20     train AUC 0.836 | test AUC 0.815
```



Test AUC as trees are added: performance improves quickly then plateaus; more trees never overfit, they only cost time.

Out-of-bag evaluation

Python

```
rf = RandomForestClassifier(500, min_samples_leaf=20, oob_score=True, random_state=0,
n_jobs=-1).fit(Xtr, ytr)
from sklearn.metrics import roc_auc_score as auc
print(f"OOB accuracy {rf.oob_score_:.3f} | OOB AUC {auc(ytr, rf.oob_decision_function_[:, 1]):.3f} |
test AUC {auc(yte, rf.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
OOB accuracy 0.786 | OOB AUC 0.793 | test AUC 0.819
```

OOB scores approximate test performance without a separate validation set, which is handy for quick tuning.

Key hyperparameters

Parameter	Typical values	Effect
`n_estimators`	200–1,000	More trees = more stable; plateau
`max_features`	`"sqrt"` (classification), 0.3–1.0 (regression)	Lower = more diverse, less correlated trees
`max_depth` / `min_samples_leaf`	None / 1–50	Leaf size smooths probabilities; key for noisy data
`max_samples`	0.5–1.0	Bootstrap sample size (faster, more diverse)
`class_weight`	`"balanced"`, `"balanced_subsample"`	Imbalanced classes
`n_jobs=-1`		Use all CPU cores

Python

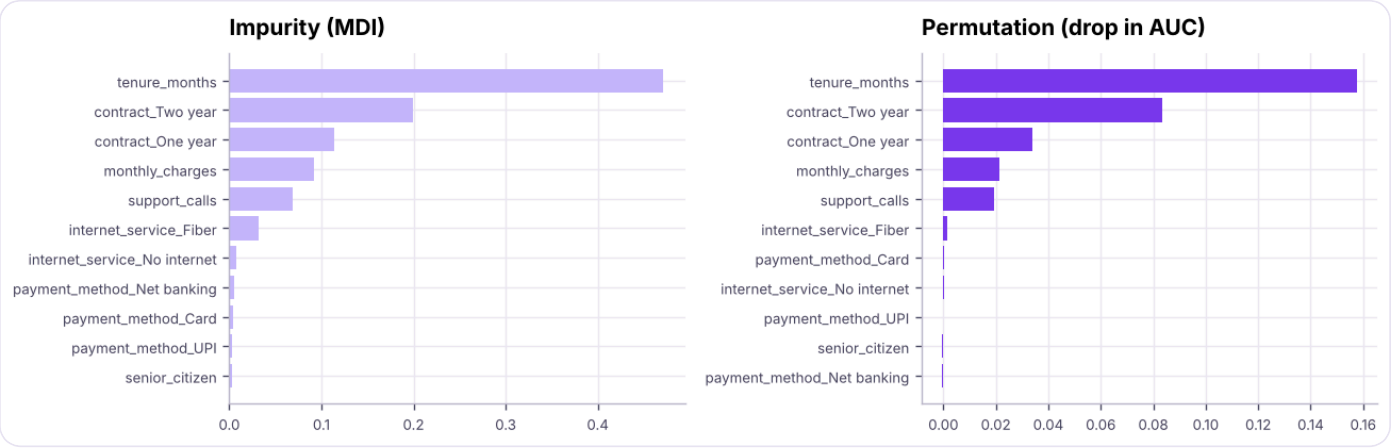
```
from sklearn.model_selection import RandomizedSearchCV
from scipy.stats import randint, uniform
cv = StratifiedKFold(5, shuffle=True, random_state=0)
search = RandomizedSearchCV(RandomForestClassifier(300, random_state=0, n_jobs=-1),
                            {"max_features": uniform(0.1, 0.8), "min_samples_leaf": randint(1, 60),
"max_depth": [None, 4, 8, 12]},
                            n_iter=20, cv=cv, scoring="roc_auc", random_state=0).fit(Xtr, ytr)
print({k: (round(float(v), 3) if isinstance(v, float) else v) for k, v in search.best_params_.items()},
f"| CV AUC {search.best_score_:.3f} | test AUC {auc(yte, search.predict_proba(Xte)[:, 1]):.3f}")
lr = LogisticRegression(max_iter=2000).fit(Xtr, ytr); print(f"logistic regression test AUC {auc(yte,
lr.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
{'max_depth': 4, 'max_features': 0.37, 'min_samples_leaf': 21} | CV AUC 0.794 | test AUC 0.819
logistic regression test AUC 0.822
```

On this dataset the relationships are close to additive, so a tuned forest and logistic regression perform similarly. Forests pull ahead when there are strong interactions and thresholds.

Feature importance



Impurity importance (left) vs permutation importance on the test set (right). Permutation importance measures how much AUC drops when a feature is shuffled.

Python

```
print(pd.DataFrame({"impurity": imp, "permutation": pi}).sort_values("permutation",
ascending=False).round(4).head(6))
```

OUTPUT

```
                    impurity  permutation
tenure_months         0.4711       0.1575
contract_Two year     0.1996       0.0834
contract_One year     0.1140       0.0339
monthly_charges       0.0925       0.0213
```

```
support_calls             0.0683       0.0195
internet_service_Fiber    0.0316       0.0016
```

Both rank tenure and contract first, but impurity importance (computed on training data) gives continuous features with many possible split points, such as monthly charges, a bigger share; permutation importance on held-out data reflects real predictive value. Neither shows the **direction** of an effect; use partial dependence or SHAP for that (lesson 27).

Random forest regression

Python

```python
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"]); h = h[h.price_lakh.between(5, 300)]
Xh = pd.get_dummies(h.drop(columns=["house_id", "price_lakh"]), drop_first=True).astype(float)
a, bq, c, d = train_test_split(Xh, h.price_lakh, test_size=0.2, random_state=0)
rfr = RandomForestRegressor(500, min_samples_leaf=2, max_features=0.6, random_state=0,
n_jobs=-1).fit(a, c)
print(f"random forest test MAE {mean_absolute_error(d, rfr.predict(bq)):.2f} lakh | R² {rfr.score(bq,
d):.3f}")
```

OUTPUT

```
random forest test MAE 7.39 lakh | R² 0.912
```

Voting and stacking

Different model families make different errors, so combining them can help.

Python

```python
from sklearn.ensemble import VotingClassifier, StackingClassifier, HistGradientBoostingClassifier
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
base = [("lr", make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000))), ("rf",
RandomForestClassifier(300, min_samples_leaf=20, random_state=0, n_jobs=-1)),
        ("knn", make_pipeline(StandardScaler(), KNeighborsClassifier(75)))]
vote = VotingClassifier(base, voting="soft").fit(Xtr, ytr)
stack = StackingClassifier(base, final_estimator=LogisticRegression(), cv=5).fit(Xtr, ytr)
for name, m in [("soft voting", vote), ("stacking", stack)]: print(f"{name:12s} test AUC {auc(yte,
m.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
soft voting  test AUC 0.823
stacking     test AUC 0.824
```

Soft voting averages predicted probabilities; **stacking** trains a meta-model on the base models' out-of-fold predictions. Gains are usually small and add complexity; Kaggle winners love them, production teams use them sparingly.

Summary

KEY TAKEAWAYS

- Ensembles average diverse models so errors cancel; bagging trains each model on a bootstrap sample.
- Random forests add random feature subsets at each split to decorrelate trees; more trees never overfit.
- Tune `max_features`, `min_samples_leaf` and depth; use OOB scores for quick validation.
- Prefer permutation importance on held-out data to impurity importance; neither gives direction.
- Voting and stacking combine different model families for small extra gains.

Quick quiz

6 questions · tap an answer to check it

1. What is a bootstrap sample?

A A sample of the same size drawn with replacement

B The first 80% of rows

C A stratified split

D The test set

Answer: A. About 63% of unique rows appear in each.

2. What extra randomness does a random forest add to bagging?

A A random subset of features considered at each split

B Random labels

C Random thresholds only

D Random scaling

Answer: A. It decorrelates the trees.

3. What happens if you keep adding trees to a random forest?

- **A** Performance plateaus; it does not overfit, only takes longer
- **B** It overfits badly
- **C** Accuracy drops
- **D** It becomes linear

Answer: A. Unlike boosting rounds.

4. What are out-of-bag rows used for?

- **A** A built-in validation estimate from rows not in each tree's bootstrap sample
- **B** Training the final model
- **C** Feature scaling
- **D** Testing only the first tree

Answer: A. No separate validation set needed.

5. Why prefer permutation importance?

- **A** It measures the drop in held-out performance and is less biased toward many-valued features
- **B** It is faster
- **C** It shows direction of effects
- **D** It needs no data

Answer: A. Compute it on test or validation data.

6. What does stacking do?

A Trains a meta-model on base models' out-of-fold predictions

B Averages identical models

C Stacks rows of data

D Removes weak models

Answer: A. Often a small gain.

EXERCISE **Practice**

(1) Fit a random forest on the bank loans data and compare its 5-fold AUC with logistic regression. (2) Plot OOB error vs `max_features`. (3) Train a `RandomForestRegressor` on house prices and plot predicted vs actual.

Show solution (worked answer)

Python

```
b = pd.read_csv("bank_loans.csv").dropna(subset=["default"]); fb = ["age", "ed", "employ",
"address", "income", "debtinc", "creddebt", "othdebt"]
for m in (RandomForestClassifier(500, min_samples_leaf=10, random_state=0),
make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))):
    print(type(m).__name__, cross_val_score(m, b[fb], b.default.astype(int), cv=cv,
scoring="roc_auc").mean())
for mf in (0.1, 0.3, 0.5, 0.8, 1.0): print(mf, 1 - RandomForestClassifier(300, max_features=mf,
oob_score=True, random_state=0).fit(Xtr, ytr).oob_score_)
plt.scatter(d, rfr.predict(bq), s=5); plt.plot([0, 300], [0, 300])
```

INTERVIEW CORNER **Questions you may be asked**

Q: How does a random forest reduce overfitting compared with a single tree? Each tree sees a different bootstrap sample and random feature subsets, so trees make different errors; averaging them reduces variance while keeping low bias.

Q: Bagging vs boosting? Bagging trains models independently in parallel on bootstrap samples and averages them (reduces variance); boosting trains models sequentially, each correcting the previous errors (reduces bias).

Q: How do you get feature importance from a random forest, and what is the caveat? Impurity-based importance from training splits (biased toward high-cardinality features) or permutation importance on held-out data (more reliable); neither shows the direction of the effect.

STATISTICS & MACHINE LEARNING · LESSON 19

Ensembles II: Gradient Boosting, XGBoost, LightGBM and CatBoost

The algorithm family that wins most tabular-data problems. Boosting intuition (AdaBoost to gradient boosting), fitting trees to residuals by hand, learning rate and number of trees, early stopping, scikit-learn's HistGradientBoosting, XGBoost and LightGBM APIs and key hyperparameters, categorical features and CatBoost, and a tuned comparison on churn and house prices.

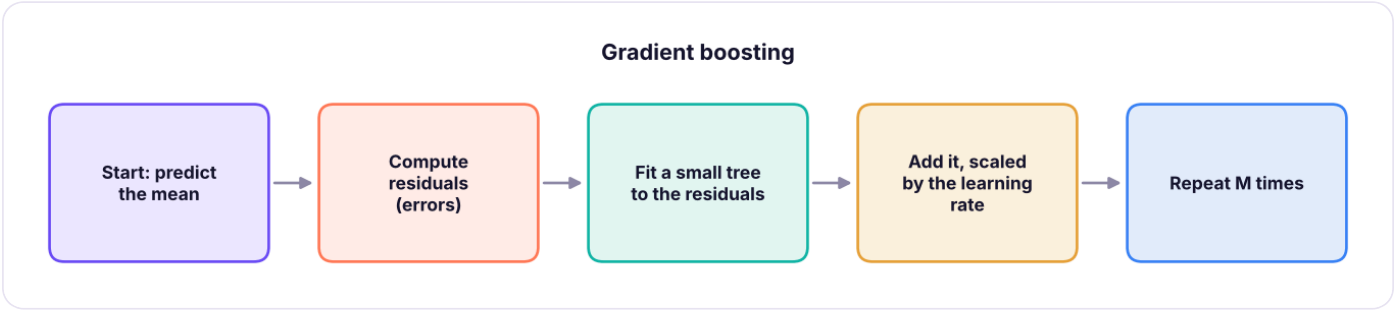
ML track (added) · ✦ XGBoost 3.x · LightGBM 4.x · scikit-learn 1.7+ · Advanced · ⏱ 120 min

By the end of this lesson you can

- Explain sequential boosting and how gradient boosting fits trees to residuals (gradients)
- Implement a few boosting rounds by hand to see the mechanism
- Tune learning rate, number of trees, depth, subsampling and regularisation with early stopping
- Use HistGradientBoosting, XGBoost and LightGBM with native categorical handling
- Decide when boosting is worth its extra complexity

Boosting: learn from mistakes

Bagging trains trees **independently** and averages them (reduces variance). **Boosting** trains small trees **sequentially**: each new tree focuses on what the current ensemble gets wrong (reduces bias).



Gradient boosting adds trees one at a time, each correcting the remaining error.

- **AdaBoost** (1995) reweights misclassified rows so the next learner focuses on them.
- **Gradient boosting** (Friedman, 2001) fits each new tree to the **negative gradient of the loss**; for squared error, that is simply the residual.

$$F_m(x) = F_{m-1}(x) + \nu\, h_m(x), \qquad h_m \approx -\frac{\partial L(y, F_{m-1})}{\partial F_{m-1}}$$

Boosting by hand

Python

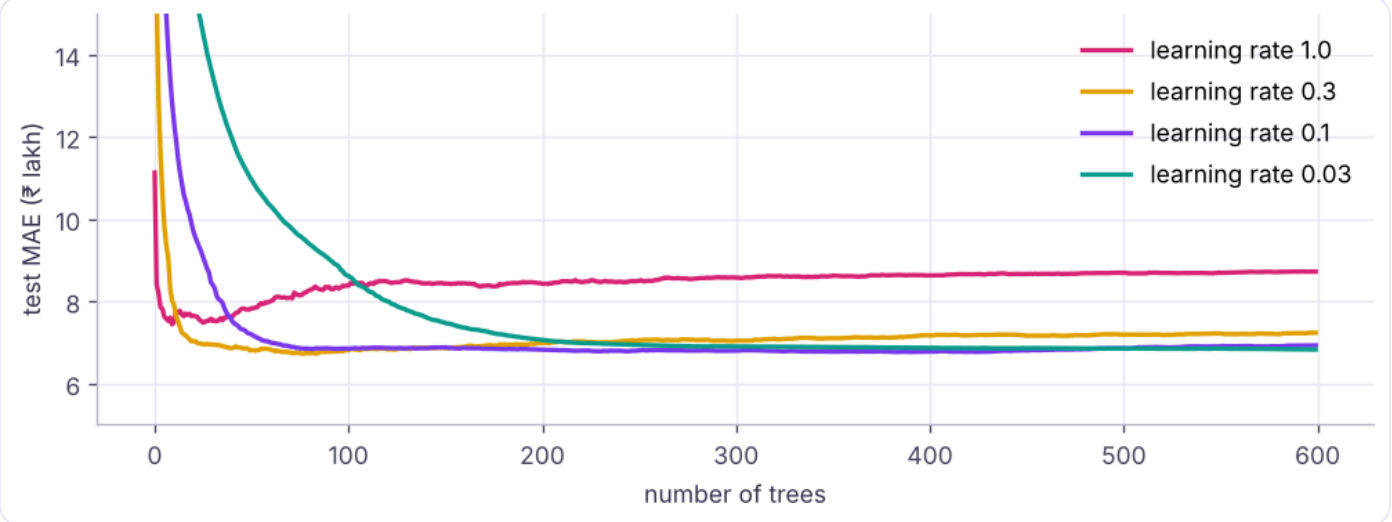
```
import numpy as np, pandas as pd
from sklearn.tree import DecisionTreeRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"]); h = h[h.price_lakh.between(5, 300)]
Xh = pd.get_dummies(h.drop(columns=["house_id", "price_lakh"]), drop_first=True).astype(float); yh =
h.price_lakh
Xtr, Xte, ytr, yte = train_test_split(Xh, yh, test_size=0.2, random_state=0)
lr, pred_tr, pred_te = 0.3, np.full(len(ytr), ytr.mean()), np.full(len(yte), ytr.mean())
print(f"round 0 (mean only): test MAE {mean_absolute_error(yte, pred_te):.2f}")
for m in range(1, 31):
    resid = ytr - pred_tr                                   # negative gradient of squared error
    tree = DecisionTreeRegressor(max_depth=3, random_state=0).fit(Xtr, resid)
    pred_tr += lr * tree.predict(Xtr); pred_te += lr * tree.predict(Xte)
    if m in (1, 2, 5, 10, 30): print(f"round {m:2d}: test MAE {mean_absolute_error(yte, pred_te):.2f}")
```

OUTPUT

```
round 0 (mean only): test MAE 25.00
round  1: test MAE 19.07
round  2: test MAE 15.39
round  5: test MAE 10.61
round 10: test MAE 8.01
round 30: test MAE 6.95
```

Thirty shallow trees, each correcting the last, beat a single deep tree.

The learning rate and number of trees



Validation error by boosting round for different learning rates: small rates need more trees but often reach lower error; too many rounds can overfit.

Unlike random forests, boosting **can overfit** with too many rounds. Use a small learning rate (0.01–0.1) with **early stopping** on a validation set.

scikit-learn HistGradientBoosting

The fast, modern implementation in scikit-learn (inspired by LightGBM): bins features into histograms, handles missing values and categorical features natively, supports early stopping.

Python

```
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.metrics import roc_auc_score
tc = pd.read_csv("telecom_churn.csv")
Xc = tc.drop(columns=["customer_id", "churn", "total_charges"]); yc = (tc.churn == "Yes").astype(int)
for col in ["contract", "internet_service", "payment_method"]: Xc[col] = Xc[col].astype("category")
Xa, Xb, ya, yb = train_test_split(Xc, yc, test_size=0.3, stratify=yc, random_state=1)
hgb = HistGradientBoostingClassifier(learning_rate=0.05, max_iter=1000, max_depth=None,
max_leaf_nodes=15, min_samples_leaf=40, l2_regularization=1.0,
                                     categorical_features="from_dtype", early_stopping=True,
validation_fraction=0.15, n_iter_no_change=30, random_state=0).fit(Xa, ya)
print(f"HistGradientBoosting: stopped after {hgb.n_iter_} trees | test AUC {roc_auc_score(yb,
hgb.predict_proba(Xb)[:, 1]):.3f}")
```

OUTPUT

```
HistGradientBoosting: stopped after 87 trees | test AUC 0.806
```

XGBoost

Python

```
import xgboost as xgb
Xa_tr, Xa_va, ya_tr, ya_va = train_test_split(Xa, ya, test_size=0.2, stratify=ya, random_state=0)
xg = xgb.XGBClassifier(n_estimators=2000, learning_rate=0.03, max_depth=3, subsample=0.8,
colsample_bytree=0.8, min_child_weight=5, reg_lambda=1.0,
                       enable_categorical=True, tree_method="hist", eval_metric="auc",
early_stopping_rounds=100, random_state=0)
xg.fit(Xa_tr, ya_tr, eval_set=[(Xa_va, ya_va)], verbose=False)
print(f"XGBoost {xgb.__version__}: best iteration {xg.best_iteration} | test AUC {roc_auc_score(yb,
xg.predict_proba(Xb)[:, 1]):.3f}")
```

OUTPUT

```
XGBoost 3.4.1: best iteration 6 | test AUC 0.806
```

LightGBM

```python
import lightgbm as lgb
lg = lgb.LGBMClassifier(n_estimators=2000, learning_rate=0.03, num_leaves=15, min_child_samples=40,
subsample=0.8, subsample_freq=1, colsample_bytree=0.8, reg_lambda=1.0, random_state=0, verbose=-1)
lg.fit(Xa_tr, ya_tr, eval_set=[(Xa_va, ya_va)], eval_metric="auc", callbacks=[lgb.early_stopping(100,
verbose=False)])
print(f"LightGBM {lgb.__version__}: best iteration {lg.best_iteration_} | test AUC {roc_auc_score(yb,
lg.predict_proba(Xb)[:, 1]):.3f}")
```

OUTPUT

```
LightGBM 4.7.0: best iteration 11 | test AUC 0.809
```

Early stopping chose very few trees for XGBoost and LightGBM here: with 2,000 training rows and a simple signal, validation AUC peaks early. On larger datasets the best iteration is typically in the hundreds or thousands.

LightGBM grows trees **leaf-wise** (it splits the leaf with the biggest gain), which is fast and accurate on large data; control complexity with `num_leaves` and `min_child_samples`. **CatBoost** uses ordered target statistics for categorical features and symmetric trees; it is often the best choice when there are many high-cardinality categories (`pip install catboost`).

Key hyperparameters (cheat sheet)

Idea	XGBoost	LightGBM	HistGradientBoosting	Typical
Number of trees	`n_estimators`	`n_estimators`	`max_iter`	Large + early stopping
Step size	`learning_rate`	`learning_rate`	`learning_rate`	0.01–0.1
Tree size	`max_depth`	`num_leaves`, `max_depth`	`max_leaf_nodes`, `max_depth`	depth 3–8 / 15–63 leaves
Leaf size	`min_child_weight`	`min_child_samples`	`min_samples_leaf`	5–100
Row sampling	`subsample`	`subsample` (+ `subsample_freq`)	—	0.6–0.9
Column sampling	`colsample_bytree`	`colsample_bytree`	`max_features`	0.5–0.9
Regularisation	`reg_lambda`, `reg_alpha`, `gamma`	`reg_lambda`, `reg_alpha`	`l2_regularization`	
Imbalance	`scale_pos_weight`	`is_unbalance`, `scale_pos_weight`	`class_weight`	ratio of negatives to positives

Tuning recipe: fix a learning rate (0.05), early-stop the number of trees, tune tree size and leaf size, then sampling and regularisation (Optuna works well), and finally lower the learning rate and raise trees for a last bit

of accuracy.

Comparison on two problems

Python

```
from sklearn.model_selection import cross_val_score, StratifiedKFold, KFold
from sklearn.linear_model import LogisticRegression, Ridge
from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor,
HistGradientBoostingRegressor
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
Xd = pd.get_dummies(Xc, drop_first=True).astype(float)
cv = StratifiedKFold(5, shuffle=True, random_state=0)
cls = {"logistic regression": make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000)),
"random forest": RandomForestClassifier(400, min_samples_leaf=20, random_state=0, n_jobs=-1),
       "HistGradientBoosting": HistGradientBoostingClassifier(learning_rate=0.05, max_iter=200,
max_leaf_nodes=15, min_samples_leaf=40, random_state=0),
       "XGBoost": xgb.XGBClassifier(n_estimators=300, learning_rate=0.03, max_depth=3, subsample=0.8,
colsample_bytree=0.8, random_state=0),
       "LightGBM": lgb.LGBMClassifier(n_estimators=300, learning_rate=0.03, num_leaves=15,
min_child_samples=40, random_state=0, verbose=-1)}
print("Churn (5-fold ROC-AUC):")
for n, m in cls.items(): s = cross_val_score(m, Xd, yc, cv=cv, scoring="roc_auc"); print(f"  {n:22s}
{s.mean():.3f} ± {s.std():.3f}")
reg = {"ridge": make_pipeline(StandardScaler(), Ridge()), "random forest": RandomForestRegressor(400,
min_samples_leaf=2, max_features=0.6, random_state=0, n_jobs=-1),
       "HistGradientBoosting": HistGradientBoostingRegressor(learning_rate=0.05, max_iter=500,
random_state=0),
       "XGBoost": xgb.XGBRegressor(n_estimators=600, learning_rate=0.05, max_depth=4, subsample=0.8,
colsample_bytree=0.8, random_state=0),
       "LightGBM": lgb.LGBMRegressor(n_estimators=600, learning_rate=0.05, num_leaves=15,
random_state=0, verbose=-1)}
print("House prices (5-fold MAE, ₹ lakh):")
for n, m in reg.items(): s = -cross_val_score(m, Xh, yh, cv=KFold(5, shuffle=True, random_state=0),
scoring="neg_mean_absolute_error"); print(f"  {n:22s} {s.mean():.2f} ± {s.std():.2f}")
```

OUTPUT

```
Churn (5-fold ROC-AUC):
  logistic regression    0.812 ± 0.015
  random forest          0.798 ± 0.014
  HistGradientBoosting   0.790 ± 0.016
  XGBoost                0.804 ± 0.012
  LightGBM               0.791 ± 0.010
House prices (5-fold MAE, ₹ lakh):
  ridge                  6.12 ± 0.25
  random forest          6.95 ± 0.36
  HistGradientBoosting   6.79 ± 0.25
  XGBoost                6.48 ± 0.31
  LightGBM               6.73 ± 0.26
```

Read results honestly: on these clean, mostly additive datasets, linear models are competitive. Boosting's advantage grows with more data, more features, interactions, missing values and messy real-world signals,

which is why it dominates tabular competitions and industry credit, fraud, pricing and ranking models.

Summary

KEY TAKEAWAYS

- Boosting adds shallow trees sequentially; gradient boosting fits each tree to the negative gradient (residuals for squared error).
- The learning rate scales each tree; small rates need more trees; use early stopping because boosting can overfit.
- HistGradientBoosting, XGBoost and LightGBM are fast histogram-based implementations with native missing-value and categorical support; CatBoost excels with many categories.
- Tune trees (via early stopping), tree/leaf size, sampling and regularisation.
- Always compare with simple baselines; boosting shines on large, messy, interaction-rich tabular data.

Quick quiz

6 questions · tap an answer to check it

1. How does boosting differ from bagging?

A Boosting trains models sequentially, each fixing the previous errors; bagging trains them independently

B Boosting uses bigger trees

C Bagging cannot do regression

D They are the same

Answer: A. Boosting reduces bias; bagging reduces variance.

2. For squared error, what does each new tree in gradient boosting fit?

A The residuals of the current ensemble

B The original target

C Random noise

D The features

Answer: A. Residuals are the negative gradient.

3. What does a smaller learning rate require?

- **A** More trees, often with better final accuracy
- **B** Fewer trees
- **C** No early stopping
- **D** Deeper trees only

Answer: A. Typical rates are 0.01–0.1.

4. Can gradient boosting overfit by adding too many trees?

- **A** Yes; use early stopping on validation data
- **B** No, never
- **C** Only for classification
- **D** Only with small learning rates

Answer: A. Unlike random forests.

5. What does LightGBM's `num_leaves` control?

- **A** The complexity of each leaf-wise tree
- **B** The number of trees
- **C** The learning rate
- **D** Row sampling

Answer: A. Larger = more complex.

6. When is CatBoost particularly useful?

- **A** Many high-cardinality categorical features
- **B** Images
- **C** Tiny datasets only
- **D** Linear relationships

Answer: A. Ordered target statistics handle categories well.

EXERCISE **Practice**

(1) Tune LightGBM on the bank loans data with early stopping; compare with logistic regression. (2) Use `scale_pos_weight` in XGBoost for the churn data and report recall at 0.5. (3) Plot validation AUC against boosting rounds from `xg.evals_result()`.

Show solution (worked answer)

Python

```
b = pd.read_csv("bank_loans.csv").dropna(subset=["default"]); fb = ["age", "ed", "employ",
"address", "income", "debtinc", "creddebt", "othdebt"]
print(cross_val_score(lgb.LGBMClassifier(n_estimators=300, learning_rate=0.03, num_leaves=7,
min_child_samples=30, verbose=-1), b[fb], b.default.astype(int), cv=cv, scoring="roc_auc").mean())
from sklearn.metrics import recall_score
w = (ya == 0).sum() / (ya == 1).sum()
xw = xgb.XGBClassifier(n_estimators=300, learning_rate=0.03, max_depth=3, scale_pos_weight=w,
enable_categorical=True).fit(Xa, ya)
print(recall_score(yb, xw.predict(Xb)))
plt.plot(xg.evals_result()["validation_0"]["auc"])
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain gradient boosting in simple terms. Start with a simple prediction, look at the errors, train a small tree to predict those errors, add a fraction of it, and repeat; the ensemble gradually corrects itself.

Q: XGBoost vs LightGBM vs random forest? Random forests average independent deep trees (robust, little tuning); XGBoost and LightGBM build trees sequentially with regularisation and histogram speed-ups; LightGBM grows leaf-wise and is usually fastest on large data.

Q: How do you prevent overfitting in boosting? Lower learning rate with early stopping, shallow trees, minimum leaf size, row and column subsampling, and L1/L2 regularisation.

STATISTICS & MACHINE LEARNING · LESSON 20

Support Vector Machines and Naive Bayes

Two classic classifiers with very different ideas. SVMs find the widest margin between classes, use soft margins (C) and kernels (RBF, polynomial) for non-linear boundaries, and need scaling; Naive Bayes applies Bayes' theorem with an independence assumption and is a fast, strong baseline for text. Includes SVR, probability calibration and when to choose each.

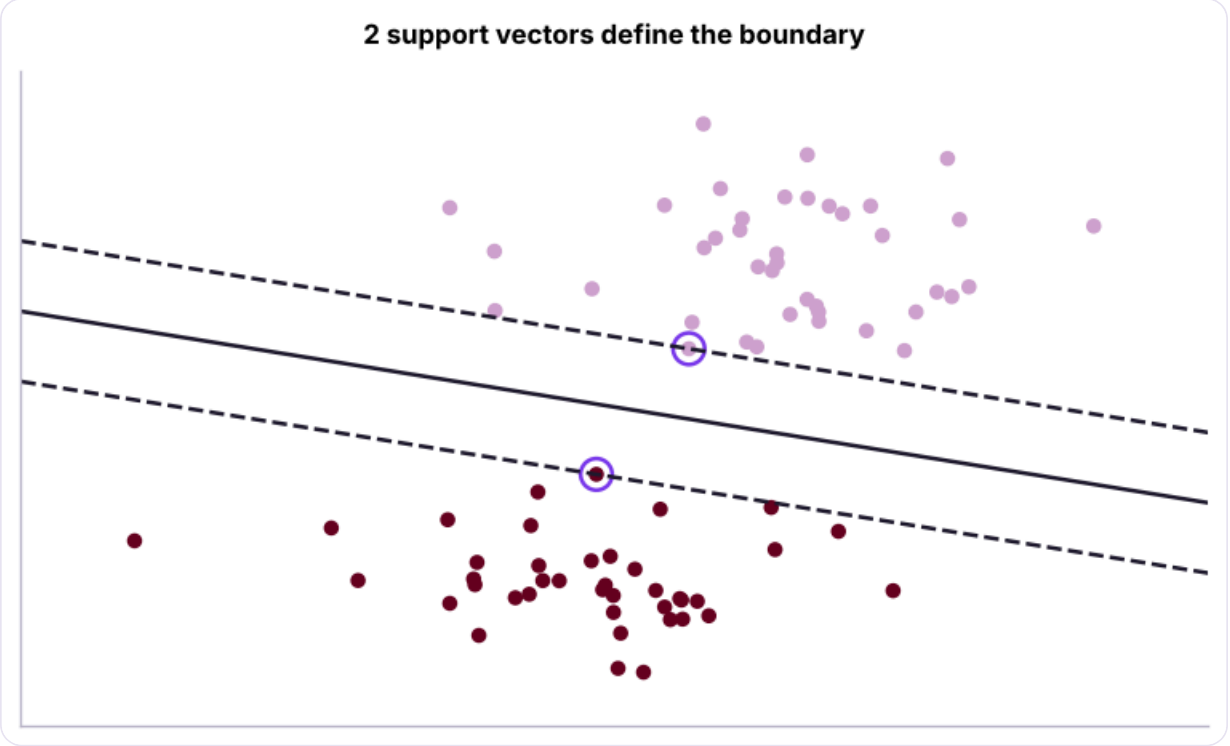
ML track (added) · scikit-learn 1.7+ · Intermediate · 100 min

By the end of this lesson you can

- Explain maximum-margin classification, support vectors and the role of C
- Use linear and RBF kernels and tune C and gamma
- Apply SVR for regression and know SVM scaling and size limits
- Explain Naive Bayes with Bayes' theorem and its independence assumption
- Build Gaussian and multinomial Naive Bayes models, including a text classifier

Support vector machines: the widest street

Among all lines (hyperplanes) that separate two classes, an SVM picks the one with the **largest margin**: the widest "street" between the classes. Only the points on the edge of the street, the **support vectors**, determine it.



A linear SVM: the solid line is the decision boundary, dashed lines are the margins, circled points are the support vectors.

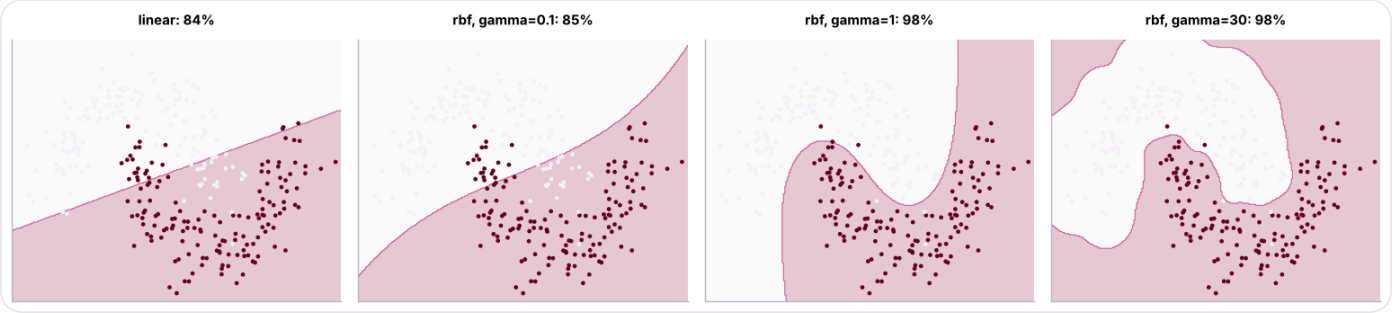
Soft margin: the C parameter

Real data overlap, so SVMs allow some points inside the margin or on the wrong side, with a penalty controlled by **C**:

C	Margin	Behaviour
Small (0.01)	Wide	Tolerates errors: simpler, more bias
Large (100)	Narrow	Tries to classify every point: complex, more variance

Kernels: non-linear boundaries

The **kernel trick** computes similarities as if the data were mapped into a higher-dimensional space where a straight separator exists, without computing that space explicitly.



Linear vs RBF kernels on curved data, and the effect of gamma (how far one point's influence reaches).

Kernel	Use	Key parameters
`linear`	Many features (text), linearly separable data	C
`rbf` (Gaussian)	Default for non-linear problems	C, gamma
`poly`	Polynomial interactions	degree, C

Large **gamma** makes each point's influence very local (wiggly boundary, overfitting); small gamma gives smooth boundaries.

SVM on the Bank Loans data

Python

```
from sklearn.model_selection import train_test_split, GridSearchCV, StratifiedKFold, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
b = pd.read_csv("bank_loans.csv").dropna(subset=["default"])
feats = ["age", "ed", "employ", "address", "income", "debtinc", "creddebt", "othdebt"]
Xb, yb = b[feats], b.default.astype(int)
Xtr, Xte, ytr, yte = train_test_split(Xb, yb, test_size=0.3, stratify=yb, random_state=42)
cv = StratifiedKFold(5, shuffle=True, random_state=0)
gs = GridSearchCV(make_pipeline(StandardScaler(), SVC(kernel="rbf")), {"svc__C": [0.1, 0.3, 1, 3, 10],
"svc__gamma": [0.003, 0.01, 0.03, 0.1, 0.3]}, cv=cv, scoring="roc_auc", n_jobs=-1).fit(Xtr, ytr)
print("best:", gs.best_params_, f"| CV AUC {gs.best_score_:.3f} | test AUC {roc_auc_score(yte,
gs.decision_function(Xte)):.3f}")
lin = make_pipeline(StandardScaler(), SVC(kernel="linear", C=0.1)).fit(Xtr, ytr)
print(f"linear SVM test AUC {roc_auc_score(yte, lin.decision_function(Xte)):.3f} | logistic regression
{roc_auc_score(yte, make_pipeline(StandardScaler(), LogisticRegression()).fit(Xtr,
ytr).predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
best: {'svc__C': 3, 'svc__gamma': 0.003} | CV AUC 0.788 | test AUC 0.809
linear SVM test AUC 0.813 | logistic regression 0.822
```

SVMs output **decision scores**, not probabilities. `SVC(probability=True)` adds Platt scaling (slower); `CalibratedClassifierCV` is the general tool.

Support vector regression

Python

```
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"]); h = h[h.price_lakh.between(5, 300)]
hf = ["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km", "parking"]
from sklearn.metrics import mean_absolute_error
a, bq, c, d = train_test_split(h[hf], h.price_lakh, test_size=0.2, random_state=0)
svr = make_pipeline(StandardScaler(), SVR(kernel="rbf", C=100, epsilon=2, gamma=0.05)).fit(a, c)
print(f"SVR test MAE {mean_absolute_error(d, svr.predict(bq)):.2f} lakh")
```

OUTPUT

```
SVR test MAE 10.41 lakh
```

SVR fits a tube of width ε around the data and ignores errors inside it.

SVM practicalities

- Always **scale** features.
- Kernel SVMs scale roughly $O(n^2)$–$O(n^3)$: fine for thousands of rows, slow beyond ~100,000. Use `LinearSVC`, `SGDClassifier(loss="hinge")` or kernel approximations (`Nystroem`) for big data.
- Strong for high-dimensional, small-to-medium data (text, genomics, image features).

Naive Bayes

Naive Bayes applies **Bayes' theorem** (ML lesson 2) and assumes features are **conditionally independent** given the class:

$$P(c \mid x_1, \ldots, x_n) \propto P(c) \prod_{j=1}^{n} P(x_j \mid c)$$

The assumption is rarely true ("naive"), yet the classifier often works well because it only needs the **ranking** of classes to be right.

Variant	Features	Typical use
`GaussianNB`	Continuous, roughly normal	Quick baseline on numeric data
`MultinomialNB`	Counts (word counts)	Text classification, spam
`BernoulliNB`	Binary (word present/absent)	Short texts
`ComplementNB`	Counts, imbalanced classes	Text with skewed classes

Python

```python
from sklearn.naive_bayes import GaussianNB, MultinomialNB, ComplementNB
gnb = GaussianNB().fit(Xtr, ytr)
print(f"Gaussian NB on bank loans: test AUC {roc_auc_score(yte, gnb.predict_proba(Xte)[:, 1]):.3f} | class priors {gnb.class_prior_.round(3)}")
```

OUTPUT

```
Gaussian NB on bank loans: test AUC 0.797 | class priors [0.678 0.322]
```

Text classification with multinomial Naive Bayes

Python

```
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.metrics import classification_report, accuracy_score
rv = pd.read_csv("customer_reviews.csv")
print(rv.shape, rv.sentiment.value_counts().to_dict()); print(rv.review_text.sample(3,
random_state=1).tolist())
Rtr, Rte, rtr, rte = train_test_split(rv.review_text, rv.sentiment, test_size=0.25,
stratify=rv.sentiment, random_state=0)
nb = make_pipeline(CountVectorizer(ngram_range=(1, 2), min_df=2), MultinomialNB(alpha=0.5)).fit(Rtr,
rtr)
print(f"multinomial NB accuracy {accuracy_score(rte, nb.predict(Rte)):.3f}")
print(classification_report(rte, nb.predict(Rte), digits=3))
vec, clf = nb[0], nb[-1]; vocab = np.array(vec.get_feature_names_out())
for i, cls in enumerate(clf.classes_):
    lift = clf.feature_log_prob_[i] - clf.feature_log_prob_.mean(axis=0)
    print(f"most indicative of {cls:8s}:", vocab[np.argsort(-lift)[:6]].tolist())
print(nb.predict(["battery died after two days, very disappointed", "excellent sound and great value",
"it is okay for the price"]))
```

OUTPUT

```
(3000, 8) {'positive': 1589, 'negative': 964, 'neutral': 447}
['Delivered on the expected date', 'Nothing special, does the job.', 'Very disappointed. The
hinge broke within weeks, too expensive compared to rivals, the box arrived super damaged though
the bass is rich']
multinomial NB accuracy 0.940
              precision    recall  f1-score   support

    negative      0.978     0.938     0.958       241
     neutral      0.845     0.830     0.838       112
    positive      0.944     0.972     0.958       397

    accuracy                          0.940       750
   macro avg      0.923     0.913     0.918       750
weighted avg      0.940     0.940     0.940       750

most indicative of negative: ['worst purchase', 'worst', 'ever', 'purchase ever', 'regret',
'regret buying']
most indicative of neutral : ['is fine', 'it is', 'fine', 'mixed', 'feelings so', 'feelings']
most indicative of positive: ['superb product', 'not bad', 'bad', 'at all', 'bad at', 'loved
it']
['negative' 'positive' 'neutral']
```

Notice that "bad" is one of the strongest **positive** indicators: in these reviews it mostly appears in "not bad at all". Word pairs (`ngram_range=(1, 2)`) let the model learn such negations; single words alone would get them wrong. The neutral class is hardest, because neutral reviews mix praise and complaints.

Python

```python
svm_text = make_pipeline(TfidfVectorizer(ngram_range=(1, 2), min_df=2, sublinear_tf=True),
LinearSVC(C=0.5)).fit(Rtr, rtr)
print(f"TF-IDF + linear SVM accuracy {accuracy_score(rte, svm_text.predict(Rte)):.3f}")
```

OUTPUT

```
TF-IDF + linear SVM accuracy 0.947
```

Naive Bayes and linear SVMs on TF-IDF were the standard text classifiers before deep learning, and they are still excellent fast baselines (lesson 25 and the GenAI track go further).

When to choose which

Situation	Good choice
Small/medium data, complex boundary, scaled numeric features	RBF SVM
Very many sparse features (text)	Linear SVM, logistic regression, multinomial NB
Need a fast baseline, tiny training time, streaming updates	Naive Bayes (`partial_fit`)
Need calibrated probabilities	Logistic regression (or calibrate SVM/NB)
Large tabular data	Gradient boosting

Summary

KEY TAKEAWAYS

- SVMs maximise the margin; support vectors define the boundary; C trades margin width for training errors.
- Kernels (RBF, polynomial) give non-linear boundaries; gamma controls how local each point's influence is; always scale.
- SVMs give scores, not probabilities (calibrate); kernel SVMs are slow on very large data.
- Naive Bayes multiplies class-conditional probabilities under an independence assumption; it is fast and strong for text.
- TF-IDF + linear SVM or multinomial NB are classic, competitive text classifiers.

Quick quiz

6 questions · tap an answer to check it

1. What are support vectors?

- **A** The training points on or inside the margin that define the boundary
- **B** All training points
- **C** The features
- **D** The predicted probabilities

Answer: A. Other points could move without changing the boundary.

2. A larger C in an SVM means...

- **A** A narrower margin and fewer tolerated training errors (more complex)
- **B** A wider margin
- **C** More regularisation
- **D** A different kernel

Answer: A. Small C = more regularisation.

3. What does a very large gamma in an RBF SVM tend to cause?

- **A** Overfitting with a very wiggly boundary
- **B** Underfitting
- **C** A linear boundary
- **D** Faster training

Answer: A. Each point influences only its close neighbourhood.

4. Why is Naive Bayes called "naive"?

- **A** It assumes features are independent given the class
- **B** It is simple to code
- **C** It ignores priors
- **D** It needs no data

Answer: A. The assumption is rarely true, but it works well in practice.

5. Which Naive Bayes variant suits word counts?

- **A** GaussianNB
- **B** MultinomialNB
- **C** BernoulliNB only for numbers
- **D** None

Answer: B. Counts per word per document.

6. Which step is essential before training an RBF SVM?

- **A** Feature scaling
- **B** One-hot encoding the target
- **C** Removing all outliers
- **D** PCA always

Answer: A. Distances drive the kernel.

EXERCISE **Practice**

(1) Tune C and gamma for an RBF SVM on the telecom churn data (one-hot + scale) and compare with logistic regression. (2) Train BernoulliNB on the reviews with binary word features. (3) Use `CalibratedClassifierCV` on the linear SVM text model to get probabilities.

Show solution (worked answer)

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.naive_bayes import BernoulliNB
from sklearn.calibration import CalibratedClassifierCV
tc = pd.read_csv("telecom_churn.csv"); Xc = tc.drop(columns=["customer_id", "churn",
"total_charges"]); yc = (tc.churn == "Yes").astype(int)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months",
"monthly_charges", "support_calls"]), ("c", OneHotEncoder(), ["contract", "internet_service",
"payment_method"])])
g = GridSearchCV(make_pipeline(pre, SVC()), {"svc__C": [0.1, 1, 10], "svc__gamma": [0.01, 0.05,
0.2]}, cv=5, scoring="roc_auc").fit(Xc, yc); print(g.best_params_, g.best_score_)
print(accuracy_score(rte, make_pipeline(CountVectorizer(binary=True), BernoulliNB()).fit(Rtr,
rtr).predict(Rte)))
cal = make_pipeline(TfidfVectorizer(ngram_range=(1, 2)), CalibratedClassifierCV(LinearSVC(),
cv=5)).fit(Rtr, rtr); print(cal.predict_proba(Rte[:3]).round(2))
```

INTERVIEW CORNER **Questions you may be asked**

Q: Explain the kernel trick. It lets an algorithm that depends only on dot products work in a high-dimensional feature space by computing a similarity function (kernel) directly in the original space.

Q: SVM vs logistic regression? Both can learn linear boundaries; logistic regression gives probabilities and optimises log loss; SVMs maximise the margin with hinge loss and extend to non-linear boundaries with kernels, but scale worse to large datasets.

Q: Why does Naive Bayes work well for spam detection despite its assumption? Classification only needs the correct class to have the highest score; with many weakly informative words, the independence errors largely cancel out, and the model is fast and robust with little data.

STATISTICS & MACHINE LEARNING · LESSON 21

Imbalanced Classification: Fraud Detection with Class Weights, Resampling and Thresholds

Most valuable classification problems are rare-event problems (fraud, defaults, churn, failures). Learn why accuracy and naive training fail, the right metrics (PR-AUC, recall at fixed precision, cost), class weights, undersampling, oversampling and SMOTE inside pipelines, threshold moving, and anomaly-detection alternatives, on a synthetic card-fraud dataset with about 1.4% fraud.

ML track (added) · ✦ scikit-learn 1.7+ · imbalanced-learn 0.1x · Advanced · ⏱ 110 min

By the end of this lesson you can

- Recognise imbalanced problems and the metrics that work for them
- Apply class_weight and scale_pos_weight to make models pay attention to the rare class
- Use random undersampling, oversampling and SMOTE correctly inside cross-validation
- Move the decision threshold to meet a business constraint (alert capacity, cost, minimum precision)
- Use Isolation Forest when labels are scarce

A rare-event problem

Python

```python
import numpy as np, pandas as pd
rng = np.random.default_rng(7); n = 20000
df = pd.DataFrame({
    "amount": np.round(rng.lognormal(7, 1.1, n), 0),                    # rupees
    "hour": rng.integers(0, 24, n),
    "km_from_home": np.round(rng.exponential(8, n), 1),
    "new_merchant": rng.random(n) < 0.15,
    "online": rng.random(n) < 0.45,
    "txns_last_24h": rng.poisson(2, n),
    "card_age_days": rng.integers(10, 3000, n),
})
night = df.hour.isin([0, 1, 2, 3, 4])
score = (-8.4 + 1.35 * np.log1p(df.amount / 1000) + 1.5 * night + 0.075 * df.km_from_home + 1.5 *
df.new_merchant + 0.75 * df.online
         + 0.5 * df.txns_last_24h - 0.00135 * df.card_age_days + 3.3 * (night & (df.online == 1) &
(df.new_merchant == 1))      # risky combination
         + 2.2 * (df.txns_last_24h >= 6) + rng.normal(0, .4, n))
df["fraud"] = (rng.random(n) < 1 / (1 + np.exp(-score))).astype(int)
df[["new_merchant", "online"]] = df[["new_merchant", "online"]].astype(int)
print(df.shape, f"fraud rate {df.fraud.mean():.2%} ({df.fraud.sum()} frauds)")
print(df.groupby("fraud")[["amount", "km_from_home", "new_merchant", "online", "txns_last_24h",
"card_age_days"]].median())
```

OUTPUT

```
(20000, 8) fraud rate 1.38% (276 frauds)
       amount  km_from_home  new_merchant  online  txns_last_24h  card_age_days
fraud
0      1074.5          5.50           0.0     0.0            2.0         1524.0
1      2118.0          8.85           1.0     1.0            3.0          744.5
```

With about 1.4% fraud, a model that **never** flags fraud is 98.6% accurate and completely useless.

The right metrics

Python

```
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_val_score
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.metrics import accuracy_score, recall_score, precision_score, roc_auc_score,
average_precision_score, confusion_matrix
X, y = df.drop(columns="fraud"), df.fraud
X["log_amount"] = np.log1p(X.amount); X["night"] = X.hour.isin([0, 1, 2, 3, 4]).astype(int)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
def report(name, model, t=0.5):
    p = model.predict_proba(Xte)[:, 1]; pred = p ⩾ t
    print(f"{name:34s} acc {accuracy_score(yte, pred):.3f} | recall {recall_score(yte, pred):.3f} |
precision {precision_score(yte, pred, zero_division=0):.3f} | ROC-AUC {roc_auc_score(yte, p):.3f} | PR-
AUC {average_precision_score(yte, p):.3f}")
    return p
plain = make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000)).fit(Xtr, ytr)
p_plain = report("logistic (no adjustment)", plain)
print("random baseline PR-AUC = fraud rate =", round(yte.mean(), 3))
```

OUTPUT

```
logistic (no adjustment)           acc 0.986 | recall 0.072 | precision 0.545 | ROC-AUC 0.942 |
PR-AUC 0.321
random baseline PR-AUC = fraud rate = 0.014
```

Use	Avoid
PR-AUC (average precision), recall at fixed precision, precision at k, expected cost	Accuracy
Confusion matrix at the chosen threshold	ROC-AUC alone (looks good even when precision is poor)
Stratified splits and CV	Random splits that leave few positives in validation

Fix 1: class weights

Tell the loss function that each fraud matters more: errors on the rare class are multiplied by its weight.

Python

```
weighted = make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000,
class_weight="balanced")).fit(Xtr, ytr)
p_w = report("logistic, class_weight='balanced'", weighted)
print("balanced weights:", {0: round(float(len(ytr) / (2 * (ytr == 0).sum())), 2), 1:
round(float(len(ytr) / (2 * (ytr == 1).sum())), 2)})
```

OUTPUT

```
logistic, class_weight='balanced'  acc 0.867 | recall 0.904 | precision 0.087 | ROC-AUC 0.944 |
PR-AUC 0.274
balanced weights: {0: 0.51, 1: 36.27}
```

Weighting raises recall a lot (more frauds caught) at the price of precision (more false alarms). The **ranking** (AUC) hardly changes: weights mostly shift the probabilities, which is equivalent to moving the threshold. XGBoost and LightGBM use `scale_pos_weight` ≈ negatives ÷ positives.

Fix 2: resampling

Method	How	Risk
Random undersampling	Drop majority rows	Throws away information
Random oversampling	Duplicate minority rows	Overfitting to duplicates
SMOTE	Create synthetic minority rows by interpolating between neighbours	Synthetic points in wrong places; only for numeric features (SMOTENC for categorical)

Python

```python
from imblearn.pipeline import make_pipeline as imb_pipeline
from imblearn.under_sampling import RandomUnderSampler
from imblearn.over_sampling import RandomOverSampler, SMOTE
for name, sampler in [("undersample", RandomUnderSampler(random_state=0)), ("oversample",
RandomOverSampler(random_state=0)), ("SMOTE", SMOTE(random_state=0))]:
    m = imb_pipeline(StandardScaler(), sampler, LogisticRegression(max_iter=2000)).fit(Xtr, ytr)
    _ = report(f"logistic + {name}", m)
```

OUTPUT

```
logistic + undersample            acc 0.869 | recall 0.892 | precision 0.087 | ROC-AUC 0.939 |
PR-AUC 0.237
logistic + oversample             acc 0.870 | recall 0.904 | precision 0.088 | ROC-AUC 0.943 |
PR-AUC 0.272
logistic + SMOTE                  acc 0.874 | recall 0.892 | precision 0.090 | ROC-AUC 0.943 |
PR-AUC 0.268
```

WATCH OUT **Resample only the training folds**

Resampling before the train/test split (or before cross-validation) puts copies or synthetic neighbours of test frauds into training: a classic leak. Use `imblearn.pipeline.Pipeline`, which resamples only during `fit`. Resampling also distorts predicted probabilities; recalibrate if you need true probabilities.

Stronger model: gradient boosting

Python

```
import lightgbm as lgb
pos_w = (ytr == 0).sum() / (ytr == 1).sum()
gbm = lgb.LGBMClassifier(n_estimators=400, learning_rate=0.03, num_leaves=15, min_child_samples=50,
subsample=0.8, subsample_freq=1, colsample_bytree=0.8, verbose=-1, random_state=0).fit(Xtr, ytr)
p_g = report("LightGBM", gbm)
gbm_w = lgb.LGBMClassifier(n_estimators=400, learning_rate=0.03, num_leaves=15, min_child_samples=50,
scale_pos_weight=pos_w, verbose=-1, random_state=0).fit(Xtr, ytr)
_ = report("LightGBM, scale_pos_weight", gbm_w)
```

OUTPUT

```
LightGBM                          acc 0.987 | recall 0.169 | precision 0.560 | ROC-AUC 0.947 |
PR-AUC 0.346
LightGBM, scale_pos_weight        acc 0.963 | recall 0.651 | precision 0.218 | ROC-AUC 0.937 |
PR-AUC 0.348
```

Fix 3: move the threshold (usually the best lever)

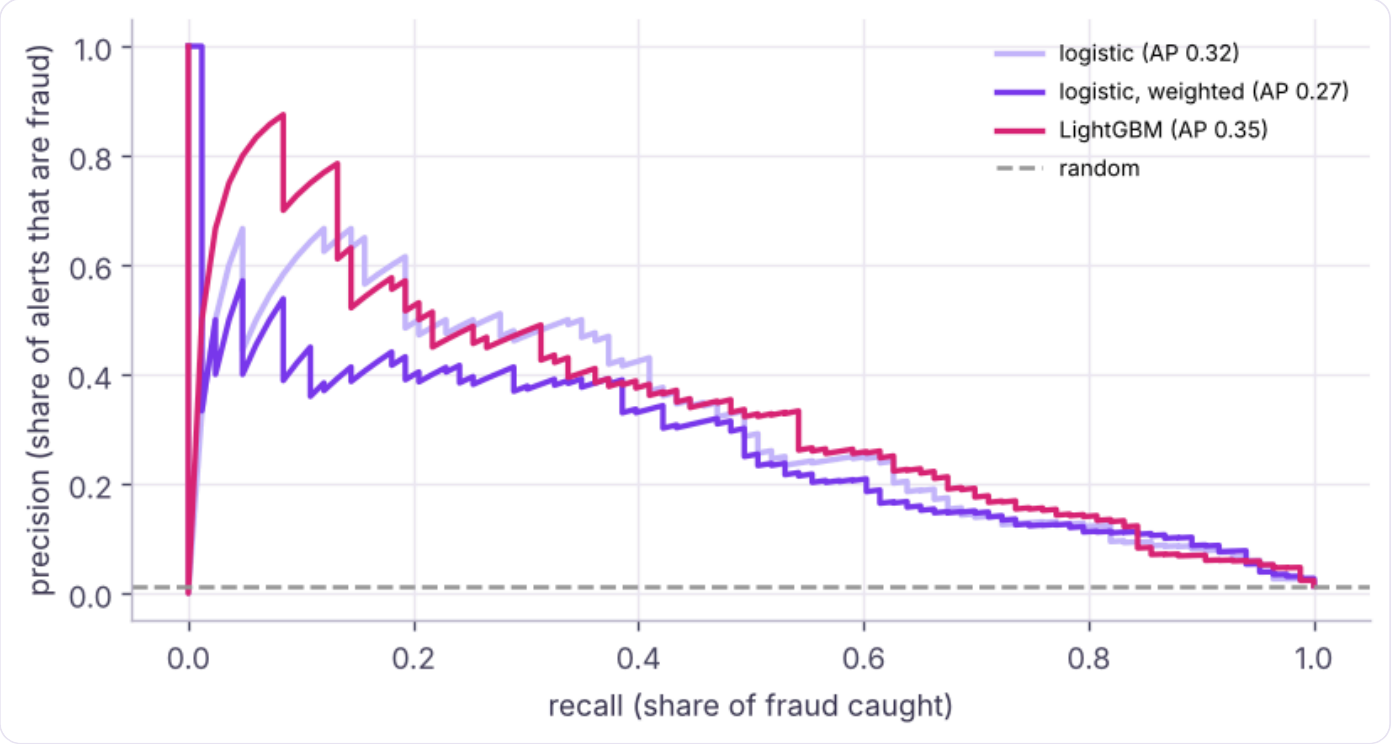
The fraud team can review **200 alerts a day** (out of about 6,000 test transactions per period here, scaled). Choose the threshold that fits the capacity, or the one that meets a minimum precision.

Python

```
from sklearn.metrics import precision_recall_curve
best_p = p_g
for k in (50, 100, 200, 400):
    t = np.sort(best_p)[-k]; pred = best_p ⩾ t
    print(f"top {k:3d} alerts: threshold {t:.3f} | frauds caught {int((pred & (yte == 1)).sum()):3d} of
{yte.sum()} (recall {recall_score(yte, pred):.2f}) | precision {precision_score(yte, pred):.2f}")
prec, rec, thr = precision_recall_curve(yte, best_p)
ok = np.where(prec[:-1] ⩾ 0.5)[0]
if len(ok): print(f"highest recall with precision ≥ 50%: recall {rec[ok[0]]:.2f} at threshold
{thr[ok[0]]:.3f}")
```

OUTPUT

```
top  50 alerts: threshold 0.335 | frauds caught  23 of 83 (recall 0.28) | precision 0.46
top 100 alerts: threshold 0.153 | frauds caught  36 of 83 (recall 0.43) | precision 0.36
top 200 alerts: threshold 0.057 | frauds caught  51 of 83 (recall 0.61) | precision 0.26
top 400 alerts: threshold 0.019 | frauds caught  62 of 83 (recall 0.75) | precision 0.15
highest recall with precision ≥ 50%: recall 0.22 at threshold 0.394
```



Precision-recall curves: the plain and weighted logistic models rank almost identically (weights shift probabilities, not the ranking); LightGBM is slightly better at most recall levels.

Expected cost

Python

```
fraud_loss, review_cost = 25000, 150                    # average fraud amount lost vs cost of reviewing one alert
ts = np.linspace(0.002, 0.6, 300)
cost = [((best_p < t) & (yte == 1)).sum() * fraud_loss + (best_p ≥ t).sum() * review_cost for t in ts]
i = int(np.argmin(cost)); none = yte.sum() * fraud_loss
print(f"cost-minimising threshold {ts[i]:.3f} (theory for calibrated probabilities: review cost ÷ fraud loss = {review_cost / fraud_loss:.3f}) | alerts {(best_p ≥ ts[i]).sum()} | total cost ₹{cost[i] / 1e5:.1f} lakh vs ₹{none / 1e5:.1f} lakh with no model ({1 - cost[i] / none:.0%} saved)")
```

OUTPUT

```
cost-minimising threshold 0.002 (theory for calibrated probabilities: review cost ÷ fraud loss = 0.006) | alerts 1480 | total cost ₹3.2 lakh vs ₹20.8 lakh with no model (84% saved)
```

When reviewing is cheap compared with a missed fraud, the best threshold is very low: flag everything with even a small chance of fraud, as long as the team has capacity. The capacity constraint above then decides.

When labels are scarce: anomaly detection

If confirmed fraud labels are few or arrive late, unsupervised **anomaly detection** flags unusual transactions.

Python

```python
from sklearn.ensemble import IsolationForest
iso = IsolationForest(n_estimators=300, contamination=0.02, random_state=0).fit(Xtr)
anom = -iso.score_samples(Xte)
print(f"Isolation Forest (no labels used): ROC-AUC {roc_auc_score(yte, anom):.3f} | PR-AUC
{average_precision_score(yte, anom):.3f}")
```

OUTPUT

```
Isolation Forest (no labels used): ROC-AUC 0.898 | PR-AUC 0.124
```

It is weaker than a supervised model but useful for cold starts, new fraud patterns, and as an extra feature. In production, fraud systems combine rules, supervised models, anomaly scores and graph features (shared devices, cards, addresses).

Checklist for imbalanced problems

1. Stratify splits; consider time-based validation (fraud patterns drift).
2. Report PR-AUC, recall at the operating precision or capacity, and expected cost.
3. Start with class weights or `scale_pos_weight`; try resampling inside the pipeline.
4. Choose the threshold from business constraints.
5. Recalibrate probabilities if you changed weights or resampled.
6. Monitor: base rates and fraud tactics change.

Summary

KEY TAKEAWAYS

- With rare positives, accuracy is meaningless; use PR-AUC, recall/precision at the operating point and cost.
- Class weights and `scale_pos_weight` make the loss care about the rare class; they shift probabilities more than rankings.
- Undersampling, oversampling and SMOTE must happen inside the training folds (imblearn pipelines).
- Threshold moving to fit capacity, precision targets or cost is usually the most effective lever.
- Isolation Forest and other anomaly detectors help when labels are scarce.

Quick quiz

6 questions · tap an answer to check it

1. Fraud is 1.4% of transactions. Which metric is most informative?

- **A** Accuracy
- **B** PR-AUC / recall at a fixed precision
- **C** R^2
- **D** MAE

Answer: B. Focus on the rare positive class.

2. What does `class_weight="balanced"` do?

- **A** Weights errors on each class inversely to its frequency
- **B** Duplicates the minority class
- **C** Deletes majority rows
- **D** Changes the features

Answer: A. The rare class counts more in the loss.

3. When should SMOTE be applied?

- **A** Only to the training data within each CV fold (inside an imblearn pipeline)
- **B** To the full dataset before splitting
- **C** To the test set
- **D** After deployment

Answer: A. Otherwise you leak test information.

4. The fraud team can review 100 alerts per day. How do you set the threshold?

- **A** Take the score of the 100th-highest transaction as the cut-off
- **B** Use 0.5
- **C** Use the fraud rate
- **D** Maximise accuracy

Answer: A. Capacity-based thresholding.

5. Class weighting changes recall a lot but ROC-AUC hardly at all. Why?

- **A** Weighting mostly shifts predicted probabilities rather than the ranking of cases
- **B** ROC-AUC is broken
- **C** Weights remove features
- **D** Recall is not a real metric

Answer: A. Similar to moving the threshold.

6. When is Isolation Forest useful?

- **A** When few or no fraud labels are available
- **B** When labels are perfect
- **C** For regression
- **D** For text only

Answer: A. It scores how unusual a record is.

EXERCISE **Practice**

(1) Use `BalancedRandomForestClassifier` from imbalanced-learn and compare PR-AUC. (2) Calibrate the weighted logistic model with `CalibratedClassifierCV` and compare Brier scores. (3) Make a time-based split by adding a transaction date and check whether PR-AUC changes.

Show solution (worked answer)

Python

```
from imblearn.ensemble import BalancedRandomForestClassifier
from sklearn.calibration import CalibratedClassifierCV
from sklearn.metrics import brier_score_loss
brf = BalancedRandomForestClassifier(300, random_state=0, sampling_strategy="all",
replacement=True, bootstrap=False).fit(Xtr, ytr)
print(average_precision_score(yte, brf.predict_proba(Xte)[:, 1]))
cal = CalibratedClassifierCV(make_pipeline(StandardScaler(), LogisticRegression(max_iter=2000,
class_weight="balanced")), cv=5).fit(Xtr, ytr)
print(brier_score_loss(yte, p_w), brier_score_loss(yte, cal.predict_proba(Xte)[:, 1]))
```

INTERVIEW CORNER **Questions you may be asked**

Q: How do you handle imbalanced classes? Use suitable metrics (PR-AUC, recall at precision), stratified validation, class weights or resampling inside the pipeline, strong models (boosting), and a business-driven threshold; collect more positive examples if possible.

Q: Is SMOTE always a good idea? No. It can create unrealistic samples, it does not help tree ensembles much, and it distorts probabilities; class weights plus threshold tuning are often enough.

Q: Your fraud model has 99% accuracy. Is it good? Not necessarily: with 1% fraud, predicting "no fraud" always gives 99%. Check recall, precision, PR-AUC and the cost at the chosen threshold.

STATISTICS & MACHINE LEARNING · LESSON 22

Clustering I: K-Means and RFM Customer Segmentation

Find groups without labels. Heuristic RFM segmentation (recency, frequency, monetary scores and named segments) as taught in class, then K-means clustering, scaling and log transforms, choosing k with the elbow method and silhouette score, profiling and naming clusters, and turning segments into marketing actions, on 4,500 retail transactions.

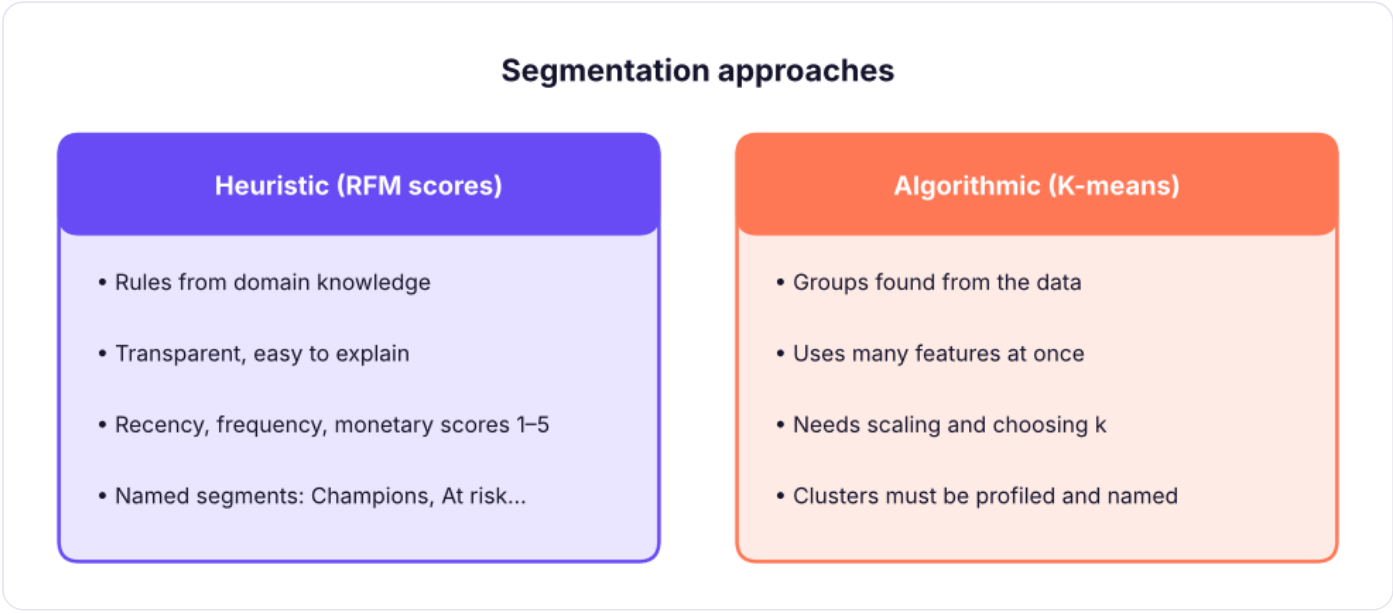
ML class 6 · ✦ scikit-learn 1.7+ · Intermediate · ⏱ 120 min

By the end of this lesson you can

- Compute recency, frequency and monetary value per customer from transactions
- Build heuristic RFM scores and named segments (Champions, At risk, Hibernating…)
- Run K-means, explain its algorithm and its assumptions, and scale features properly
- Choose k with the elbow method and silhouette scores, and profile the clusters
- Translate segments into actions and measure them

Why segment customers?

Treating every customer the same wastes money: loyal customers get discounts they did not need, while slipping customers get nothing. **Segmentation** groups similar customers so each group gets the right message.



Two ways to segment: business rules (RFM) and algorithms (clustering).

Step 1: RFM table

Recency: days since the last purchase (lower is better). **Frequency**: number of purchases. **Monetary**: total spend.

Python

```
import pandas as pd, numpy as np
pd.set_option("display.max_columns", 20); pd.set_option("display.width", 160)
tx = pd.read_csv("retail_transactions.csv", parse_dates=["invoice_date"])
print(tx.shape, tx.customer_id.nunique(), "customers |", tx.invoice_date.min().date(), "to",
tx.invoice_date.max().date())
snapshot = tx.invoice_date.max() + pd.Timedelta(days=1)
rfm = tx.groupby("customer_id").agg(recency=("invoice_date", lambda d: (snapshot - d.max()).days),
frequency=("invoice_no", "nunique"), monetary=("amount", "sum"))
rfm["monetary"] = rfm.monetary.round(0)
print(rfm.describe().round(1).T[["mean", "50%", "min", "max"]])
print(rfm.head())
```

OUTPUT

```
(4548, 4) 500 customers | 2024-01-01 to 2025-12-22
              mean     50%    min      max
recency      217.0   144.0    1.0    722.0
frequency      9.1     6.0    1.0     39.0
monetary   11058.9  5122.0  196.0  76170.0
             recency  frequency  monetary
customer_id
R0001             53         33   54906.0
R0002            224          7    5426.0
R0003             26         33   56031.0
R0004            320          3    2017.0
R0005            109         35   49225.0
```

Step 2: heuristic RFM scores

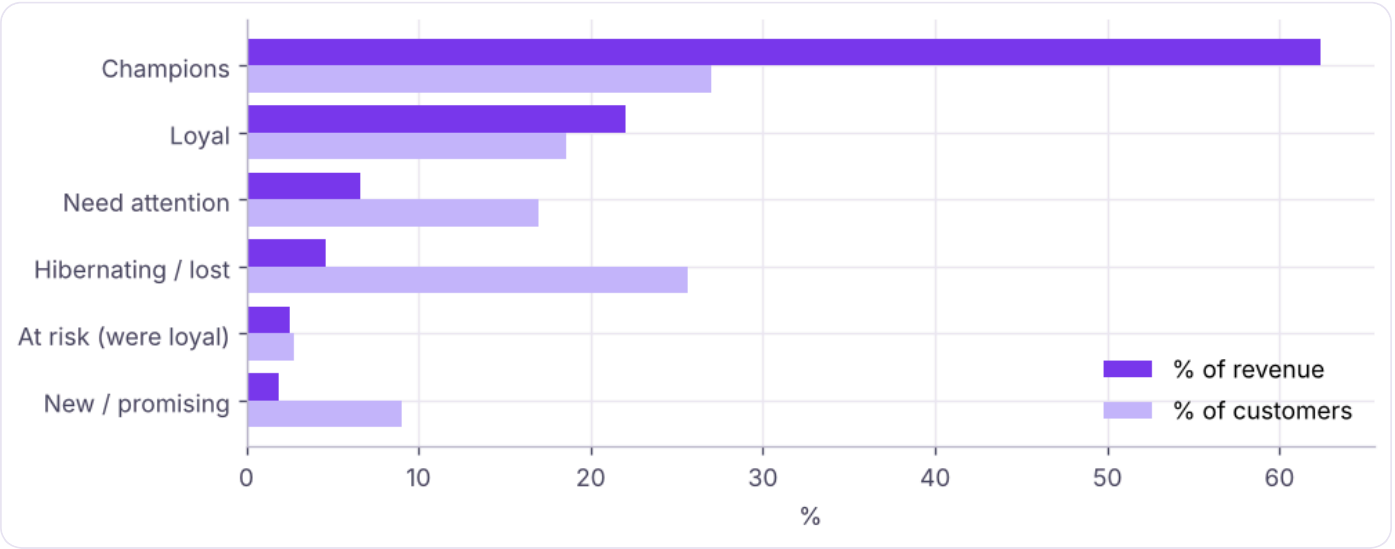
Score each dimension 1–5 by quintile (5 = best), then combine.

Python

```python
rfm["R"] = pd.qcut(rfm.recency, 5, labels=[5, 4, 3, 2, 1]).astype(int)           # recent = high score
rfm["F"] = pd.qcut(rfm.frequency.rank(method="first"), 5, labels=[1, 2, 3, 4, 5]).astype(int)
rfm["M"] = pd.qcut(rfm.monetary, 5, labels=[1, 2, 3, 4, 5]).astype(int)
rfm["RFM_code"] = rfm.R.astype(str) + rfm.F.astype(str) + rfm.M.astype(str)
rfm["RFM_score"] = rfm[["R", "F", "M"]].sum(axis=1)
def segment(r):
    if r.R ⩾ 4 and r.F ⩾ 4: return "Champions"
    if r.R ⩾ 3 and r.F ⩾ 3: return "Loyal"
    if r.R ⩾ 4 and r.F ⩽ 2: return "New / promising"
    if r.R ⩽ 2 and r.F ⩾ 4: return "At risk (were loyal)"
    if r.R ⩽ 2 and r.F ⩽ 2: return "Hibernating / lost"
    return "Need attention"
rfm["segment"] = rfm.apply(segment, axis=1)
seg = rfm.groupby("segment").agg(customers=("R", "size"), recency=("recency", "median"), frequency=
("frequency", "median"), monetary=("monetary", "median"), revenue=("monetary", "sum"))
seg["revenue_%"] = (100 * seg.revenue / seg.revenue.sum()).round(1)
print(seg.sort_values("revenue", ascending=False).drop(columns="revenue"))
```

OUTPUT

segment	customers	recency	frequency	monetary	revenue_%
Champions	135	69.0	13.0	18414.0	62.4
Loyal	93	135.0	9.0	7844.0	22.0
Need attention	85	279.0	5.0	4236.0	6.6
Hibernating / lost	128	414.5	2.0	1836.5	4.6
At risk (were loyal)	14	235.5	9.0	7797.0	2.5
New / promising	45	80.0	3.0	2521.0	1.9



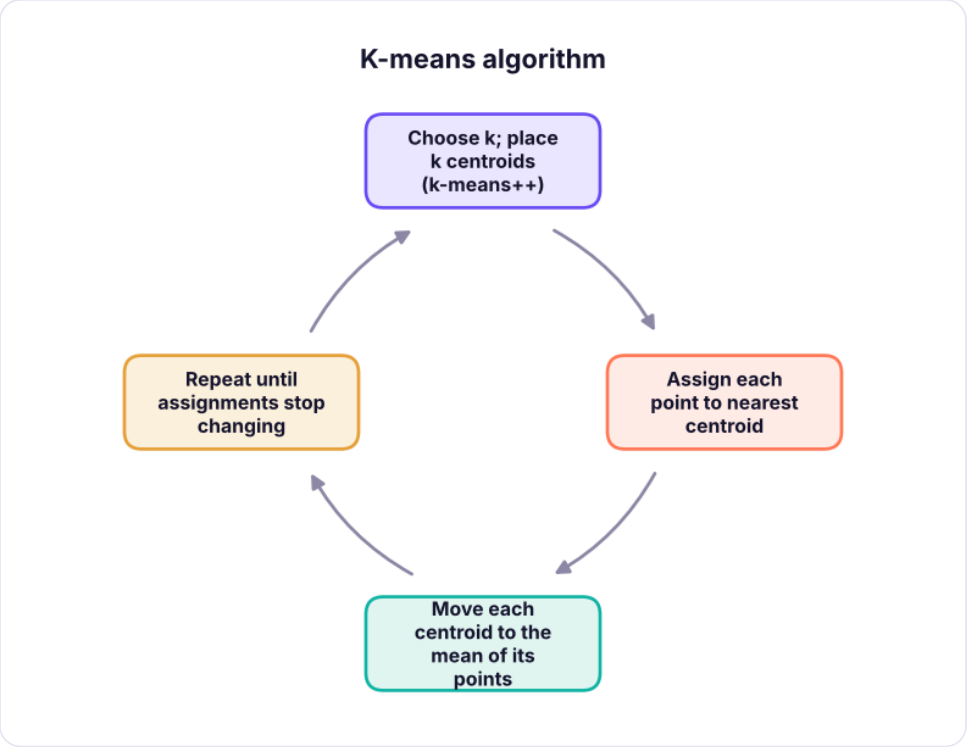
Revenue share vs customer share by RFM segment.

Segment	Typical action
Champions	Reward, early access, referrals; do not over-discount
Loyal	Upsell, loyalty points
New / promising	Onboarding, second-purchase offer
At risk (were loyal)	Win-back call or personalised offer **now**
Hibernating / lost	Low-cost reactivation; accept some loss
Need attention	Targeted reminders

RFM is transparent and works well, which is why the live class starts there. Its limits: fixed rules, only three variables, and quintile cut-offs that may not match natural groups.

Step 3: K-means clustering

K-means partitions data into *k* clusters so that each point belongs to the cluster with the nearest **centroid** (mean), minimising the within-cluster sum of squares (**inertia**).



The K-means loop: assign points to the nearest centroid, move centroids to the mean of their points, repeat until nothing changes.

Prepare features: log and scale

Python

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
feat = pd.DataFrame({"log_recency": np.log1p(rfm.recency), "log_frequency": np.log1p(rfm.frequency),
"log_monetary": np.log1p(rfm.monetary)})
print("skew before log:", rfm[["recency", "frequency", "monetary"]].skew().round(2).to_dict(), "|
after:", feat.skew().round(2).to_dict())
Z = StandardScaler().fit_transform(feat)
```

OUTPUT

```
skew before log: {'recency': 1.01, 'frequency': 1.73, 'monetary': 2.09} | after: {'log_recency':
-0.71, 'log_frequency': 0.29, 'log_monetary': 0.12}
```

K-means uses Euclidean distance, so (1) **scale** features or the largest one (monetary) dominates, and (2) **log-transform** skewed variables so a few big spenders do not form their own clusters.

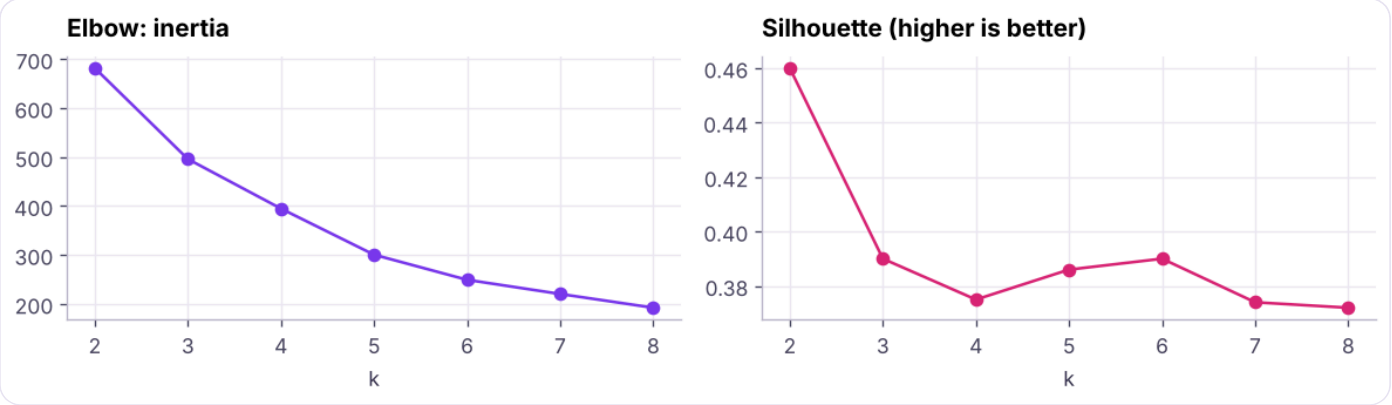
Choosing k

Python

```
from sklearn.metrics import silhouette_score, davies_bouldin_score
rows = []
for k in range(2, 9):
    km = KMeans(n_clusters=k, n_init=20, random_state=0).fit(Z)
    rows.append({"k": k, "inertia": round(km.inertia_, 1), "silhouette": round(silhouette_score(Z,
km.labels_), 3), "davies_bouldin": round(davies_bouldin_score(Z, km.labels_), 3)})
kt = pd.DataFrame(rows); print(kt.to_string(index=False))
```

OUTPUT

```
 k  inertia  silhouette  davies_bouldin
 2    681.7       0.460           0.813
 3    495.8       0.390           0.889
 4    394.6       0.375           0.907
 5    300.4       0.386           0.832
 6    249.1       0.390           0.838
 7    220.2       0.374           0.865
 8    192.1       0.372           0.853
```



Elbow curve (inertia) and silhouette score by number of clusters.

- **Elbow method:** inertia always falls as k grows; look for the bend where extra clusters stop helping much.
- **Silhouette score** (−1 to 1): how close each point is to its own cluster compared with the next one; higher is better.
- **Davies-Bouldin** (lower is better).
- **Business sense:** the number of segments the team can actually act on (often 3–6).

Here the silhouette is highest at k = 2 (it simply splits active from lapsed customers), with no clear elbow afterwards. We choose **k = 4**: the scores are close, and four groups give the marketing team more useful distinctions. Statistics guide the choice; the business decides.

Fit and profile

Python

```
k = 4
km = KMeans(n_clusters=k, n_init=20, random_state=0).fit(Z)
rfm["cluster"] = km.labels_
prof = rfm.groupby("cluster").agg(customers=("R", "size"), recency=("recency", "median"), frequency=
("frequency", "median"), monetary=("monetary", "median"), revenue=("monetary", "sum"))
prof["revenue_%"] = (100 * prof.revenue / prof.revenue.sum()).round(1)
prof = prof.sort_values("monetary", ascending=False); print(prof.drop(columns="revenue"))
```

OUTPUT

cluster	customers	recency	frequency	monetary	revenue_%
2	117	68.0	20.0	32923.0	69.3
0	164	165.5	8.0	6643.0	21.2
3	79	84.0	4.0	2836.0	4.5
1	140	406.0	2.0	1880.5	5.0

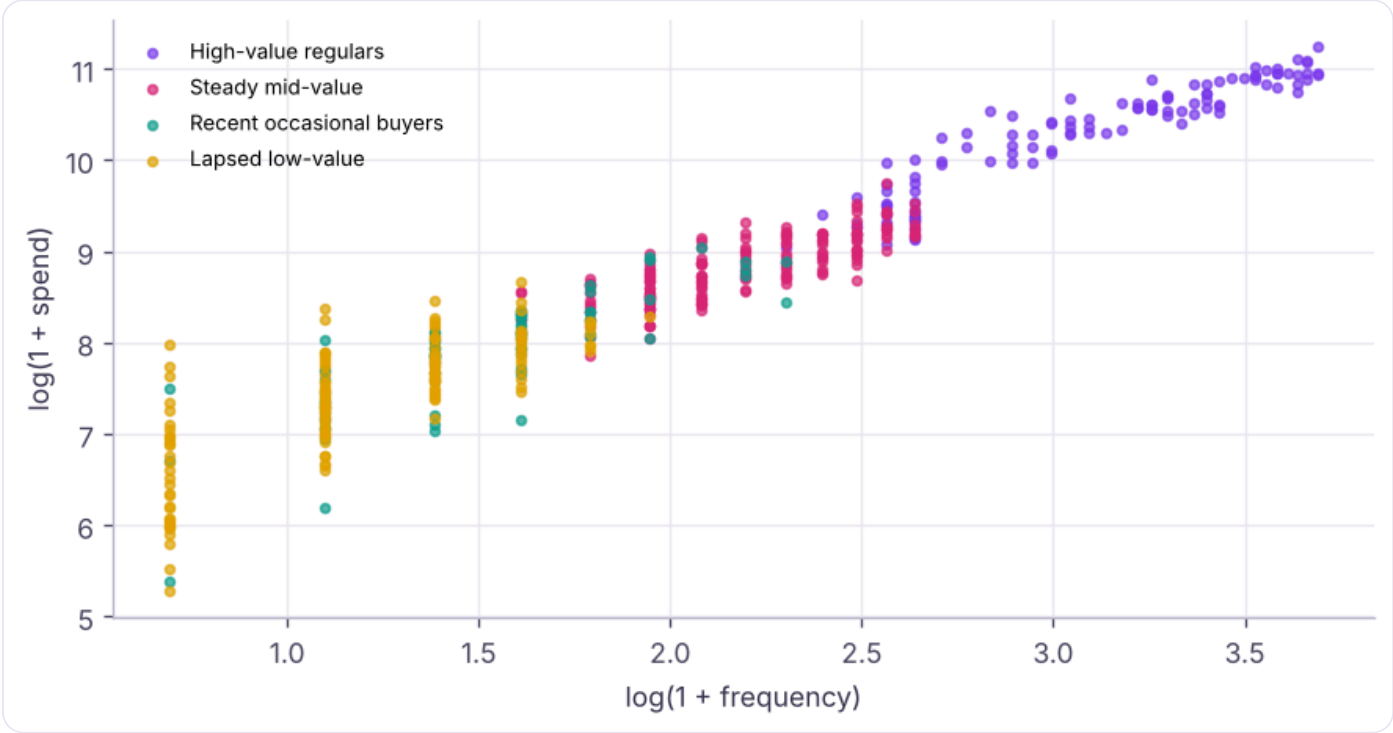
Python

```
order = prof.index.tolist()
names = dict(zip(order, ["High-value regulars", "Steady mid-value", "Recent occasional buyers", "Lapsed
low-value"]))
rfm["cluster_name"] = rfm.cluster.map(names)
print(pd.crosstab(rfm.cluster_name, rfm.segment))
```

OUTPUT

```
segment                    At risk (were loyal)  Champions  Hibernating / lost  Loyal  Need
attention  New / promising
cluster_name
High-value regulars                           1        101                   0     15
0                0
Lapsed low-value                              0          0                 125      0
15                0
Recent occasional buyers                      0          4                   0     14
16               45
Steady mid-value                             13         30                   3     64
54                0
```

Name clusters from their **profiles** (median recency, frequency, spend), not from the cluster number. The cross-tab shows how the data-driven clusters line up with the rule-based RFM segments.



Customers by log frequency and log spend, coloured by K-means cluster (centroids marked).

K-means assumptions and limits

K-means assumes / needs	If not	Alternative
Roughly spherical, similar-size clusters	Elongated or uneven clusters split badly	Gaussian mixtures, DBSCAN (next lesson)
Numeric, scaled features	Categories distort distances	K-prototypes, Gower distance
k chosen in advance		Silhouette, business needs, hierarchical dendrogram
Sensitive to outliers and initialisation		Log/robust scaling, `n_init`, k-means++ (default)

From segments to action

1. Write a one-line profile and a goal for each segment (e.g. "At risk: bought often, silent for 4+ months → win back").
2. Design an offer or message per segment.
3. **Test** it: randomly hold out part of each segment as control (lesson 6) and measure incremental revenue.
4. Refresh the segmentation monthly and track how customers move between segments.

Summary

KEY TAKEAWAYS

- RFM turns transactions into recency, frequency and monetary value; quintile scores and rules give transparent named segments.
- K-means assigns points to the nearest of k centroids, minimising inertia; log-transform skewed variables and scale features first.
- Choose k with the elbow, silhouette and Davies-Bouldin scores plus business sense; profile and name clusters by their medians.
- K-means prefers spherical clusters and numeric data; use other methods for odd shapes or categorical data.
- Segments are only valuable when they drive tested actions.

Quick quiz

6 questions · tap an answer to check it

1. What does "recency" measure in RFM?

A Days since the customer's last purchase

B Number of purchases

C Total spend

D Days since first purchase

Answer: A. Lower recency = more recently active.

2. Why scale features before K-means?

A Distances would otherwise be dominated by the feature with the largest range

B K-means needs normal data

C To remove outliers

D It is optional

Answer: A. Spend in rupees would dwarf frequency.

3. What does inertia measure?

A Within-cluster sum of squared distances to centroids

B Distance between clusters

C Number of clusters

D Accuracy

Answer: A. It always decreases as k grows.

4. A silhouette score near 0 means...

- **A** Points lie between clusters; clusters overlap
- **B** Perfect clusters
- **C** Wrong labels
- **D** One cluster only

Answer: A. Closer to 1 is better separation.

5. A segment "bought often but not for 5 months" is best called...

- **A** Champions
- **B** At risk
- **C** New customers
- **D** Loyal

Answer: B. Action: win-back now.

6. How should you name clusters?

- **A** From their profiles (median recency, frequency, spend and other traits)
- **B** By cluster number
- **C** Randomly
- **D** By the largest customer

Answer: A. Names should describe behaviour.

EXERCISE **Practice**

(1) Add `avg_order_value` and `tenure_days` (days since first purchase) to the features and re-cluster: does the silhouette improve? (2) Compare K-means with k = 3 and k = 5 profiles: which is more actionable? (3) Compute RFM on the Superstore data for 2025 and list the top 10 Champions.

Show solution (worked answer)

Python

```
first = tx.groupby("customer_id").invoice_date.min()
f2 = feat.assign(log_aov=np.log1p(rfm.monetary / rfm.frequency), log_tenure=np.log1p((snapshot -
first).dt.days))
Z2 = StandardScaler().fit_transform(f2)
for k in (3, 4, 5): print(k, silhouette_score(Z2, KMeans(k, n_init=20,
random_state=0).fit_predict(Z2)))
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"]); s = s[s.order_date.dt.year ==
2025]
snap = s.order_date.max() + pd.Timedelta(days=1)
r2 = s.groupby("customer_id").agg(recency=("order_date", lambda d: (snap - d.max()).days),
frequency=("order_id", "nunique"), monetary=("sales", "sum"))
r2["score"] = pd.qcut(-r2.recency, 5, labels=False) + pd.qcut(r2.frequency.rank(method="first"),
5, labels=False) + pd.qcut(r2.monetary, 5, labels=False)
print(r2.sort_values(["score", "monetary"], ascending=False).head(10))
```

INTERVIEW CORNER **Questions you may be asked**

Q: How do you choose the number of clusters? Elbow plot of inertia, silhouette and Davies-Bouldin scores, stability across random seeds, and above all whether the segments are distinct and actionable.

Q: What are the limitations of K-means? k must be chosen; it assumes spherical, similar-sized clusters; it is sensitive to scaling, outliers and initialisation; it handles only numeric distances.

Q: Explain RFM segmentation. Score customers on recency, frequency and monetary value (usually by quintiles), combine the scores, and map combinations to named segments with tailored actions.

STATISTICS & MACHINE LEARNING · LESSON 23

Clustering II and Dimensionality Reduction: Hierarchical, DBSCAN, Gaussian Mixtures, PCA, t-SNE and UMAP

Go beyond K-means. Agglomerative hierarchical clustering with dendrograms and linkage choices, density-based DBSCAN for irregular shapes and outliers, Gaussian mixture models with soft assignments, and dimensionality reduction with PCA (explained variance, loadings, compression), plus t-SNE and UMAP for visualisation, applied to video-game sales and HR data.

ML track (added) · ✦ scikit-learn 1.7+ · Advanced · ⏱ 120 min

By the end of this lesson you can

- Build and read a dendrogram and choose a linkage method
- Use DBSCAN to find irregular clusters and noise points, and tune eps and min_samples
- Fit Gaussian mixture models and use soft cluster probabilities and BIC
- Apply PCA, read explained variance and loadings, and use PCA before modelling
- Know when to use t-SNE or UMAP and their pitfalls

Hierarchical clustering

Agglomerative clustering starts with every point as its own cluster and repeatedly merges the two closest clusters. The full merge history is a tree, the **dendrogram**: cut it at a height to get clusters, with no need to fix k in advance.

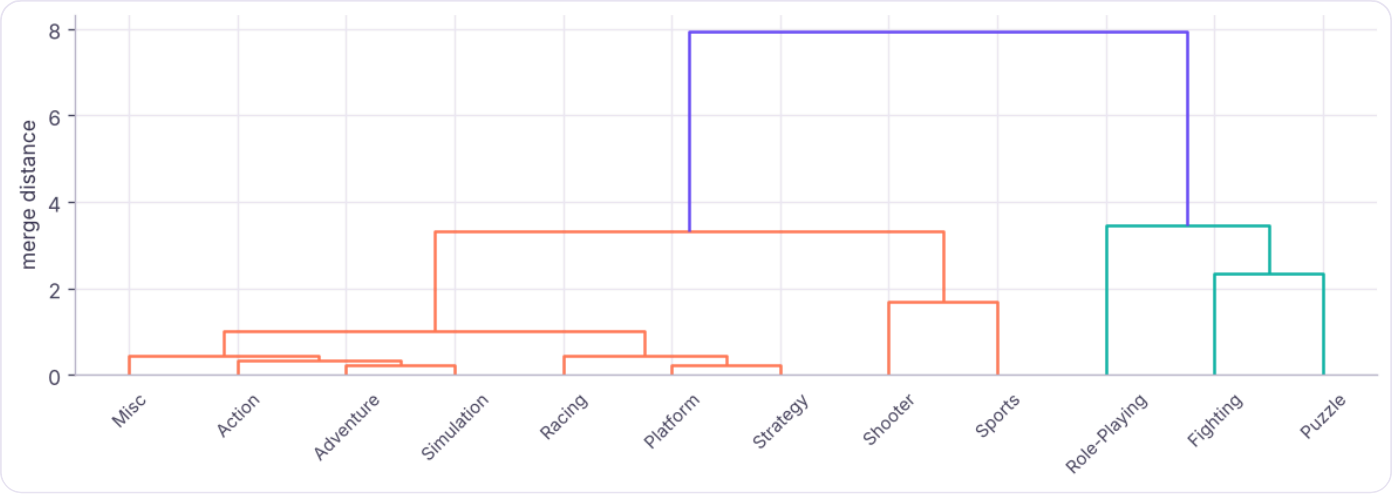
Python

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt
pd.set_option("display.width", 160); pd.set_option("display.max_columns", 20)
vg = pd.read_csv("video_game_sales.csv")
pub = vg.groupby("genre").agg(titles=("name", "nunique"), na=("na_sales", "sum"), eu=("eu_sales",
"sum"), jp=("jp_sales", "sum"), other=("other_sales", "sum"), total=("global_sales", "sum"))
shares = pub[["na", "eu", "jp", "other"]].div(pub.total, axis=0).round(3)                # regional
mix per genre
print(shares.join(pub[["titles", "total"]].round(1)).sort_values("total", ascending=False).head(6))
```

OUTPUT

```
              na     eu     jp  other  titles  total
genre
Adventure   0.401  0.308  0.153  0.137     186  136.7
```

```
Sports      0.430  0.348  0.063  0.160     181  133.8
Action      0.413  0.308  0.145  0.135     174  116.3
Racing      0.389  0.321  0.146  0.144     159  113.6
Strategy    0.395  0.320  0.147  0.138     182  111.5
Simulation  0.400  0.313  0.149  0.138     169  108.4
```



Dendrogram of game genres by regional sales mix (Ward linkage). Cutting where the vertical gaps are largest suggests the number of clusters.

Python

```python
shares["cluster"] = fcluster(L, t=3, criterion="maxclust")
print(shares.groupby("cluster")[["na", "eu", "jp", "other"]].mean().round(2), "\n",
shares.groupby("cluster").apply(lambda g: g.index.tolist(), include_groups=False).to_dict())
```

OUTPUT

```
           na    eu    jp  other
cluster
1        0.41  0.32  0.13   0.14
2        0.32  0.29  0.27   0.11
3        0.28  0.24  0.40   0.08
 {1: ['Action', 'Adventure', 'Misc', 'Platform', 'Racing', 'Shooter', 'Simulation', 'Sports',
'Strategy'], 2: ['Fighting', 'Puzzle'], 3: ['Role-Playing']}
```

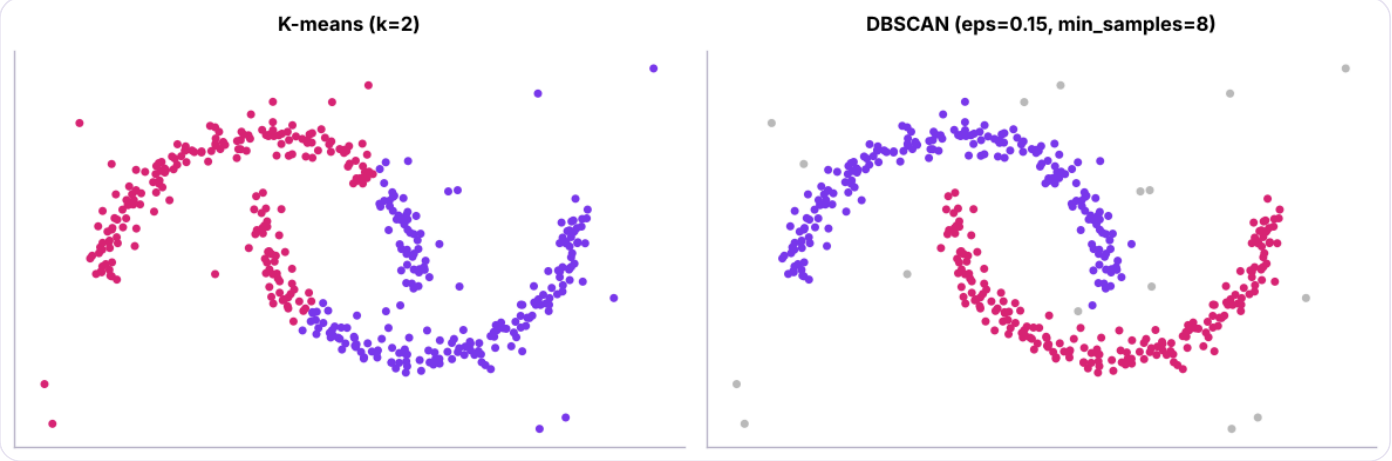
Three clear groups appear: role-playing games sell about 40% of units in Japan, fighting and puzzle games about 27%, and the other genres are driven by North America and Europe with only 13–15% in Japan. A publisher could use this to plan regional launches and marketing budgets.

Linkage	Merges clusters with…	Behaviour
Ward	Smallest increase in within-cluster variance	Compact, similar-size clusters (default choice)
Complete	Smallest maximum pairwise distance	Compact clusters, sensitive to outliers
Average	Smallest average distance	Balanced
Single	Smallest minimum distance	Long "chains"; finds odd shapes but chains noise

Hierarchical clustering is ideal for small datasets (hundreds to a few thousand items) where the tree itself is informative: product taxonomies, store groupings, gene families. It is O(n²) in memory, so not for millions of rows.

DBSCAN: density-based clusters and noise

DBSCAN groups points that are densely packed (at least `min_samples` neighbours within distance `eps`) and labels isolated points as **noise** (−1). It finds clusters of any shape and does not need k.



K-means vs DBSCAN on non-spherical data with outliers: DBSCAN follows the shapes and marks outliers (grey).

Python

```
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics import silhouette_score
d8 = np.sort(NearestNeighbors(n_neighbors=8).fit(Xm).kneighbors(Xm)[0][:, -1])
print("8th-neighbour distance percentiles (pick eps near the 'knee'):", np.percentile(d8, [50, 80, 90,
95]).round(3))
for eps in (0.08, 0.12, 0.15, 0.25):
    lab = DBSCAN(eps=eps, min_samples=8).fit_predict(Xm)
    print(f"eps {eps}: clusters {len(set(lab)) - (-1 in lab)} | noise points {(lab == -1).sum()}")
```

OUTPUT

```
8th-neighbour distance percentiles (pick eps near the 'knee'): [0.091 0.116 0.139 0.193]
eps 0.08: clusters 25 | noise points 120
eps 0.12: clusters 3 | noise points 24
eps 0.15: clusters 2 | noise points 16
eps 0.25: clusters 1 | noise points 11
```

Tuning: set `min_samples` ≈ 2 × number of features (or more for noisy data); choose `eps` from the "knee" of the sorted k-nearest-neighbour distance plot. DBSCAN struggles when clusters have very different densities (HDBSCAN, available as `sklearn.cluster.HDBSCAN`, handles that).

Uses: geographic hotspots (delivery addresses, crime), anomaly detection (noise points), sensor data, irregular customer groups.

Gaussian mixture models

A **GMM** assumes the data come from a mix of k Gaussian (bell-shaped) clusters, which may be elongated and of different sizes. Each point gets a **probability** of belonging to each cluster (soft clustering).

Python

```
from sklearn.mixture import GaussianMixture
hr = pd.read_csv("hr_employees.csv")
Xh = StandardScaler().fit_transform(np.c_[hr.age, np.log(hr.monthly_income), hr.years_at_company])
bics = {k: round(GaussianMixture(k, covariance_type="full", random_state=0).fit(Xh).bic(Xh), 1) for k
in range(1, 7)}
print("BIC by k (lower is better):", bics)
g = GaussianMixture(min(bics, key=bics.get), covariance_type="full", random_state=0).fit(Xh)
probs = g.predict_proba(Xh)
hr["gmm"] = probs.argmax(1); hr["certainty"] = probs.max(1).round(2)
print(hr.groupby("gmm").agg(n=("age", "size"), age=("age", "median"), income=("monthly_income",
"median"), years=("years_at_company", "median"), attrition=("attrition", lambda s: round((s ==
"Yes").mean(), 3))))
print("share of employees assigned with < 70% certainty:", round((hr.certainty < 0.7).mean(), 3))
```

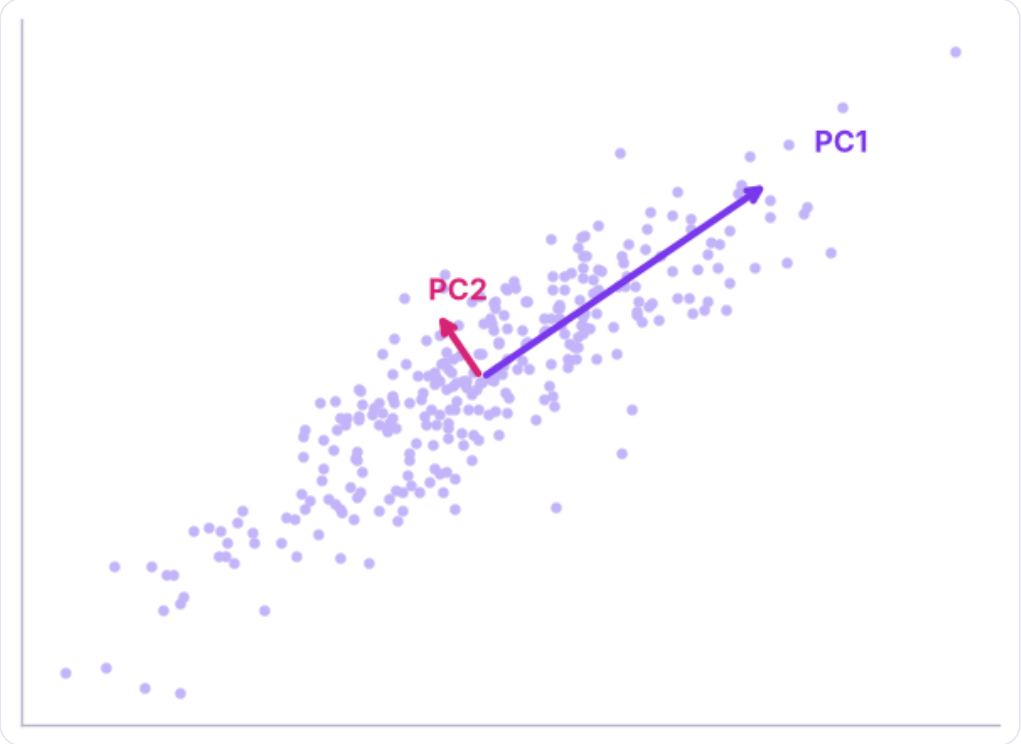
OUTPUT

```
BIC by k (lower is better): {1: 11991.6, 2: 11322.1, 3: 10937.7, 4: 11010.3, 5: 9212.8, 6:
9231.8}
       n   age    income  years  attrition
gmm
0    436  35.5   65000.0    4.0      0.271
1    271  34.0   95833.0    1.0      0.273
2    328  34.0   51833.0    0.0      0.320
3    183  25.0   90833.0    5.0      0.246
4    282  38.0  119541.5    8.0      0.167
share of employees assigned with < 70% certainty: 0.075
```

GMMs choose k with **BIC/AIC**, give uncertainty for borderline points, and are also used for anomaly detection (low likelihood = unusual).

Dimensionality reduction: PCA

Principal component analysis finds new axes (principal components) that are uncorrelated linear combinations of the original features, ordered by how much variance they explain. Keeping the first few compresses the data with minimal information loss.



PCA finds the direction of greatest variance (PC1) and the next orthogonal direction (PC2).

Python

```
num = hr[["age", "years_at_company", "monthly_income", "job_satisfaction", "distance_from_home_km",
"performance_rating"]].assign(monthly_income=np.log(hr.monthly_income))
pca = PCA().fit(StandardScaler().fit_transform(num))
ev = pd.Series(pca.explained_variance_ratio_, index=[f"PC{i + 1}" for i in range(len(num.columns))])
print("explained variance:", ev.round(3).to_dict(), "| cumulative:", ev.cumsum().round(3).tolist())
load = pd.DataFrame(pca.components_[:3].T, index=num.columns, columns=["PC1", "PC2", "PC3"]).round(2)
print(load)
```

OUTPUT

```
explained variance: {'PC1': 0.286, 'PC2': 0.172, 'PC3': 0.167, 'PC4': 0.164, 'PC5': 0.152,
'PC6': 0.059} | cumulative: [0.286, 0.458, 0.625, 0.789, 0.941, 1.0]
                        PC1   PC2   PC3
age                    0.29 -0.34  0.28
years_at_company       0.68  0.08  0.04
monthly_income         0.66  0.06 -0.03
job_satisfaction       0.04  0.76  0.17
distance_from_home_km -0.11 -0.00  0.94
performance_rating    -0.04  0.54 -0.06
```

Reading loadings: PC1 loads heavily on years at company and income (and partly age), so it is a "seniority" axis. Later components capture what is left (satisfaction, distance, rating), which are nearly uncorrelated with each other, so each needs its own component.

Python

```python
from sklearn.datasets import load_digits
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import make_pipeline
from sklearn.model_selection import cross_val_score
digits = load_digits()
p = PCA().fit(digits.data); k95 = int(np.searchsorted(np.cumsum(p.explained_variance_ratio_), 0.95) +
1)
print(f"digits: {digits.data.shape[1]} pixel features → {k95} components keep 95% of the variance")
for n in (5, 15, k95, 64):
    s = cross_val_score(make_pipeline(StandardScaler(), PCA(n), LogisticRegression(max_iter=3000)),
digits.data, digits.target, cv=5).mean()
    print(f"  {n:2d} components → logistic regression accuracy {s:.3f}")
```

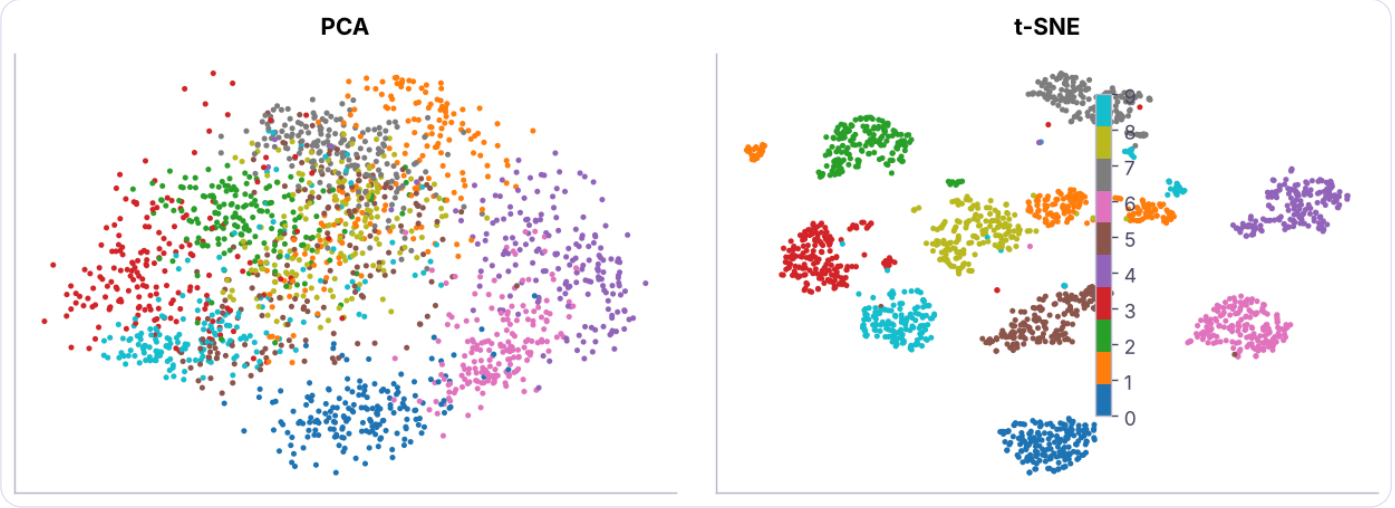
OUTPUT

```
digits: 64 pixel features → 29 components keep 95% of the variance
   5 components → logistic regression accuracy 0.771
  15 components → logistic regression accuracy 0.892
  29 components → logistic regression accuracy 0.905
  64 components → logistic regression accuracy 0.920
```

PCA is useful for	Caution
Compressing many correlated features	Scale features first
Removing multicollinearity before linear models	Components are harder to explain
Visualising high-dimensional data in 2-D	Linear only; captures variance, not class separation
Speeding up distance-based models	Fit PCA on training data only (inside the pipeline)

t-SNE and UMAP for visualisation

t-SNE and **UMAP** are non-linear methods that place similar points close together in 2-D. They make beautiful maps of embeddings, images and text.



Handwritten digits (64 features) projected to 2-D by PCA and by t-SNE. t-SNE separates the digit groups much more clearly.

Pitfalls: distances between far-apart groups and cluster sizes in t-SNE plots are not meaningful; results change with perplexity and random seed; do not cluster on t-SNE output. UMAP (`pip install umap-learn`) is faster and preserves more global structure. Both are mainly for **looking**, not for features in production models.

Choosing a method

Data / goal	Method
Numeric, roughly round clusters, large n	K-means (MiniBatchKMeans for very large)
Small n, want a hierarchy	Agglomerative + dendrogram
Odd shapes, outliers, unknown k	DBSCAN / HDBSCAN
Overlapping, elliptical clusters, need probabilities	Gaussian mixture
Mixed numeric + categorical	K-prototypes, Gower distance + hierarchical
Too many correlated features	PCA first
Visualise embeddings	UMAP, t-SNE

Summary

KEY TAKEAWAYS

- Agglomerative clustering merges closest clusters; the dendrogram shows structure at every level; Ward linkage is the usual default.
- DBSCAN finds dense regions of any shape and labels noise; tune eps with a k-distance plot.
- Gaussian mixtures give soft, probabilistic clusters of different shapes; choose k with BIC.
- PCA creates uncorrelated components ordered by explained variance; read loadings to name them; fit inside pipelines.
- t-SNE and UMAP are for visualisation; distances and sizes in their plots can mislead.

Quick quiz

6 questions · tap an answer to check it

1. What does a dendrogram show?

A The order and distance at which clusters merge

B Feature importance

C The elbow curve

D Class probabilities

Answer: A. Cut it to choose clusters.

2. Which method labels outliers as noise?

A K-means

B DBSCAN

C PCA

D Ward linkage

Answer: B. Label −1.

3. What does a Gaussian mixture give that K-means does not?

- **A** Probabilities of belonging to each cluster and elliptical cluster shapes
- **B** Faster training
- **C** A dendrogram
- **D** No need to scale

Answer: A. Soft assignments.

4. PC1 explains 45% of the variance. What does that mean?

- **A** The first component captures 45% of the total variance in the (scaled) features
- **B** 45% accuracy
- **C** 45% of rows
- **D** 45% of features are removed

Answer: A. Components are ordered by variance explained.

5. Why scale features before PCA?

- **A** Otherwise features with large units dominate the components
- **B** PCA requires normal data
- **C** To remove outliers
- **D** It is not needed

Answer: A. Variance depends on units.

6. Can you trust the distance between two far-apart clusters in a t-SNE plot?

A Yes

B No, global distances and cluster sizes are not reliable in t-SNE

C Only with perplexity 5

D Only for text

Answer: B. Use it to look, not to measure.

EXERCISE **Practice**

(1) Cluster video-game platforms by regional mix with hierarchical clustering. (2) Run DBSCAN on standardised RFM features from the previous lesson: how many customers are noise? (3) Use PCA to reduce the telecom churn features to 3 components and compare logistic-regression AUC before and after.

Show solution (worked answer)

Python

```
gen = vg.groupby("platform")[["na_sales", "eu_sales", "jp_sales", "other_sales"]].sum(); gs =
gen.div(gen.sum(1), axis=0)
print(dict(zip(gs.index, fcluster(linkage(StandardScaler().fit_transform(gs), "ward"), 3,
"maxclust"))))
tx = pd.read_csv("retail_transactions.csv", parse_dates=["invoice_date"]); snap =
tx.invoice_date.max() + pd.Timedelta(days=1)
rf = tx.groupby("customer_id").agg(r=("invoice_date", lambda d: (snap - d.max()).days), f=
("invoice_no", "nunique"), m=("amount", "sum"))
lab = DBSCAN(eps=0.5, min_samples=8).fit_predict(StandardScaler().fit_transform(np.log1p(rf)));
print((lab == -1).sum())
```

INTERVIEW CORNER **Questions you may be asked**

Q: K-means vs hierarchical vs DBSCAN? K-means: fast, needs k, round clusters. Hierarchical: no k upfront, a full tree, small data. DBSCAN: arbitrary shapes and noise detection, needs density parameters and struggles with varying densities.

Q: What is PCA and when would you use it? A linear transformation to uncorrelated components ordered by variance; used to compress correlated features, remove multicollinearity, speed up models and visualise data.

Q: How do you interpret principal components? Through their loadings: the weight of each original feature in the component, naming it after the features with the largest absolute weights.

STATISTICS & MACHINE LEARNING · LESSON 24

Time-Series Forecasting: Decomposition, Baselines, ETS, ARIMA and ML with Lag Features

Forecast demand properly. Components of a time series (trend, seasonality, noise), decomposition with STL, stationarity and autocorrelation (ADF, ACF/PACF), honest baselines (naive, seasonal naive, moving average), exponential smoothing (Holt-Winters), SARIMA, machine learning with lag and calendar features, backtesting with forecast metrics (MAE, RMSE, MAPE, WAPE), prediction intervals, and a guide to Prophet, statsforecast and foundation models.

ML track (added) · Power BI forecasting | ✦ statsmodels 0.1x · LightGBM 4 | Advanced | ⏱ 130 min

By the end of this lesson you can

- Decompose a series into trend, seasonality and remainder and test stationarity
- Read ACF and PACF plots and difference a series
- Build and evaluate naive, seasonal naive, Holt-Winters and SARIMA forecasts
- Engineer lag, rolling and calendar features and forecast with gradient boosting
- Backtest with a time-ordered holdout, compare models with WAPE/MAPE and report intervals

The data

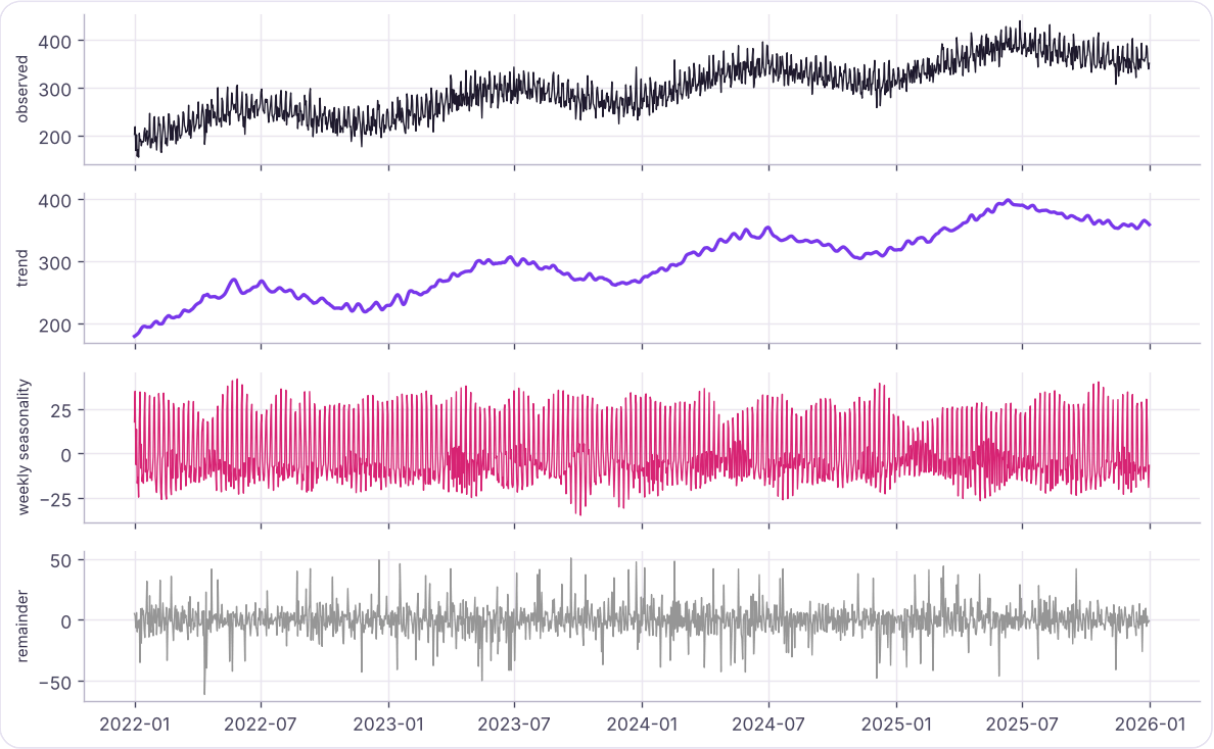
Python

```python
import pandas as pd, numpy as np, matplotlib.pyplot as plt, warnings
warnings.filterwarnings("ignore")
d = pd.read_csv("daily_sales.csv", parse_dates=["date"]).set_index("date").asfreq("D")
y = d.units_sold
print(y.shape, y.index.min().date(), "to", y.index.max().date(), "| missing days:", y.isna().sum())
print(y.resample("YE").sum().to_dict())
print("mean units by weekday:", y.groupby(y.index.day_name()).mean().round(0).to_dict())
```

OUTPUT

```
(1461,) 2022-01-01 to 2025-12-31 | missing days: 0
{Timestamp('2022-12-31 00:00:00'): 85031, Timestamp('2023-12-31 00:00:00'): 100536,
Timestamp('2024-12-31 00:00:00'): 116798, Timestamp('2025-12-31 00:00:00'): 132920}
mean units by weekday: {'Friday': 289.0, 'Monday': 288.0, 'Saturday': 320.0, 'Sunday': 323.0,
'Thursday': 288.0, 'Tuesday': 289.0, 'Wednesday': 287.0}
```

Components of a time series



STL decomposition of daily units into trend, weekly seasonality and remainder.

Component	Meaning	Example
Trend	Long-term direction	Sales growing about 15% a year
Seasonality	Repeating pattern with a fixed period	Weekends higher; December peak; Diwali
Cycle	Longer, irregular rises and falls	Economic cycles
Remainder / noise	What is left	Random variation, one-off events

Python

```
strength_trend = max(0, 1 - res.resid.var() / (res.trend + res.resid).var())
strength_season = max(0, 1 - res.resid.var() / (res.seasonal + res.resid).var())
print(f"strength of trend {strength_trend:.2f} | strength of weekly seasonality {strength_season:.2f}
(0 = none, 1 = very strong)")
```

OUTPUT

```
strength of trend 0.95 | strength of weekly seasonality 0.64 (0 = none, 1 = very strong)
```

Stationarity and autocorrelation

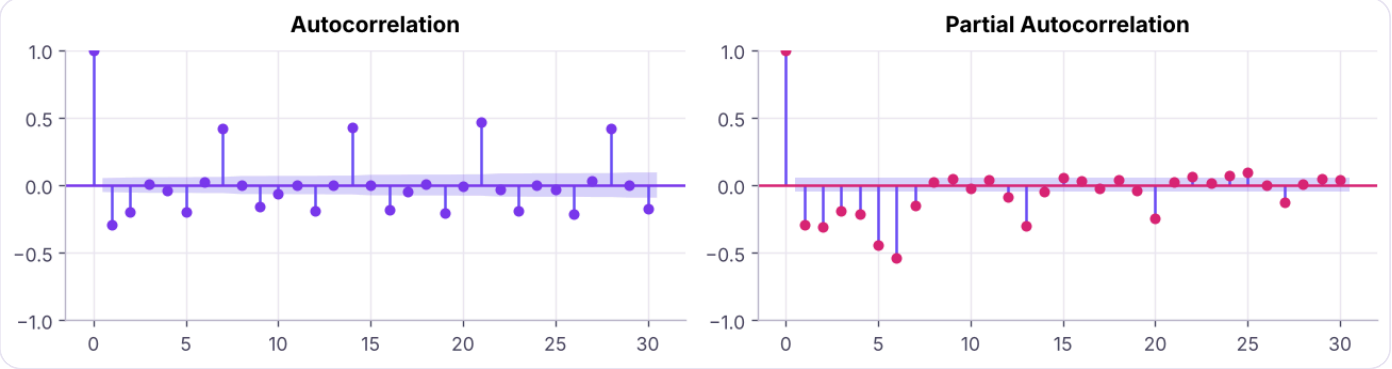
Many classical models (ARIMA) need a **stationary** series: constant mean and variance over time. Trends and seasonality break that; **differencing** removes them.

Python

```
from statsmodels.tsa.stattools import adfuller
for name, s in [("original", y), ("first difference", y.diff()), ("first + weekly difference",
y.diff().diff(7))]:
    stat, p = adfuller(s.dropna())[:2]
    print(f"{name:26s} ADF p-value {p:.4f} → {'stationary' if p < 0.05 else 'NOT stationary'}")
```

OUTPUT

```
original                   ADF p-value 0.4455 → NOT stationary
first difference           ADF p-value 0.0000 → stationary
first + weekly difference  ADF p-value 0.0000 → stationary
```



Autocorrelation (ACF) and partial autocorrelation (PACF) of the differenced series. Spikes at lags 7, 14, 21 reveal weekly seasonality.

ACF shows correlation with past values at each lag; **PACF** shows the direct effect of each lag after removing shorter ones. They guide the AR and MA orders of ARIMA models.

Train/test split in time

Python

```
h = 90                                                  # forecast horizon: the last 90 days
train, test = y.iloc[:-h], y.iloc[-h:]
print(f"train {train.index.min().date()}–{train.index.max().date()} ({len(train)} days) | test
{test.index.min().date()}–{test.index.max().date()}")
def metrics(actual, fc):
    e = actual - fc
    return {"MAE": round(float(e.abs().mean()), 1), "RMSE": round(float((e ** 2).mean() ** .5), 1),
"MAPE_%": round(float(100 * (e.abs() / actual).mean()), 2),
            "WAPE_%": round(float(100 * e.abs().sum() / actual.sum()), 2), "bias_%": round(float(100 *
-e.sum() / actual.sum()), 2)}      # bias > 0 = over-forecast
results, forecasts = {}, {}
```

OUTPUT

```
train 2022-01-01–2025-10-02 (1371 days) | test 2025-10-03–2025-12-31
```

Never shuffle time series: always train on the past and test on the future.

Metric	Use
MAE / RMSE	Error in units; RMSE punishes big misses
MAPE	Percentage error; breaks when actuals are near zero
WAPE	Total absolute error ÷ total actuals; robust, preferred in retail
Bias	Systematic over- or under-forecasting (stock-outs vs excess inventory)

Baselines

Python

```
forecasts["naive"] = pd.Series(train.iloc[-1], index=test.index)
forecasts["seasonal naive (last week)"] = pd.Series(np.tile(train.iloc[-7:].values, h // 7 + 1)[:h],
index=test.index)
forecasts["moving average (28 days)"] = pd.Series(train.iloc[-28:].mean(), index=test.index)
for k, f in forecasts.items(): results[k] = metrics(test, f)
print(pd.DataFrame(results).T)
```

OUTPUT

```
                          MAE  RMSE  MAPE_%  WAPE_%  bias_%
naive                    19.1  22.4    5.36    5.31    1.79
seasonal naive (last week) 14.9  19.1    4.23    4.14    2.35
moving average (28 days)   20.9  23.9    5.92    5.81    2.96
```

The **seasonal naive** forecast ("same as last week") is the baseline every serious model must beat.

Exponential smoothing (Holt-Winters / ETS)

ETS forecasts are weighted averages of past observations, with weights decaying exponentially, plus components for **level, trend and seasonality**.

Python

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing
ets = ExponentialSmoothing(train, trend="add", damped_trend=True, seasonal="mul",
seasonal_periods=7).fit(optimized=True)
forecasts["Holt-Winters (ETS)"] = ets.forecast(h); results["Holt-Winters (ETS)"] = metrics(test,
forecasts["Holt-Winters (ETS)"])
print({k: round(float(v), 3) for k, v in ets.params.items() if k in ("smoothing_level",
"smoothing_trend", "smoothing_seasonal", "damping_trend")})
print(results["Holt-Winters (ETS)"])
```

OUTPUT

```
{'smoothing_level': 0.054, 'smoothing_trend': 0.054, 'smoothing_seasonal': 0.036,
'damping_trend': 0.989}
{'MAE': 10.4, 'RMSE': 12.8, 'MAPE_%': 2.92, 'WAPE_%': 2.88, 'bias_%': 0.34}
```

SARIMA

ARIMA(p, d, q) combines autoregression (p past values), differencing (d) and moving average of past errors (q). **SARIMA** adds seasonal terms (P, D, Q, s).

Python

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
sar = SARIMAX(train, order=(1, 1, 1), seasonal_order=(1, 1, 1, 7)).fit(disp=False)
fc = sar.get_forecast(h); forecasts["SARIMA(1,1,1)(1,1,1,7)"] = fc.predicted_mean; ci =
fc.conf_int(alpha=0.05)
results["SARIMA(1,1,1)(1,1,1,7)"] = metrics(test, forecasts["SARIMA(1,1,1)(1,1,1,7)"])
inside = ((test ⩾ ci.iloc[:, 0]) & (test ⩽ ci.iloc[:, 1])).mean()
print(f"AIC {sar.aic:.0f} | {results['SARIMA(1,1,1)(1,1,1,7)']} | 95% interval covers {inside:.0%} of
test days")
```

OUTPUT

```
AIC 11298 | {'MAE': 16.9, 'RMSE': 20.2, 'MAPE_%': 4.82, 'WAPE_%': 4.69, 'bias_%': 4.06} | 95%
interval covers 96% of test days
```

Use `pmdarima.auto_arima` or `statsforecast.AutoARIMA` to search orders automatically by AIC.

Machine learning with lag features

Turn forecasting into regression: predict today's value from **lags** (same day last week), **rolling statistics** and **calendar** features.

Python

```
import lightgbm as lgb
def make_features(s):
    f = pd.DataFrame(index=s.index)
    f["dow"] = s.index.dayofweek; f["month"] = s.index.month; f["dayofyear"] = s.index.dayofyear;
f["t"] = np.arange(len(s))
    for lag in (7, 14, 21, 28, 364): f[f"lag_{lag}"] = s.shift(lag)
    f["roll_mean_28_lag7"] = s.shift(7).rolling(28).mean(); f["roll_std_28_lag7"] =
s.shift(7).rolling(28).std()
    return f
F = make_features(y)
Xtr, ytr = F.loc[train.index].dropna(), train.loc[F.loc[train.index].dropna().index]
model = lgb.LGBMRegressor(n_estimators=600, learning_rate=0.03, num_leaves=15, min_child_samples=20,
verbose=-1, random_state=0, linear_tree=True).fit(Xtr, ytr)
```

Lags are at least 7 days, so the model can forecast up to a week ahead without knowing the in-between values. For a 90-day horizon we forecast **recursively**: predict a week, feed predictions back as lags, repeat.

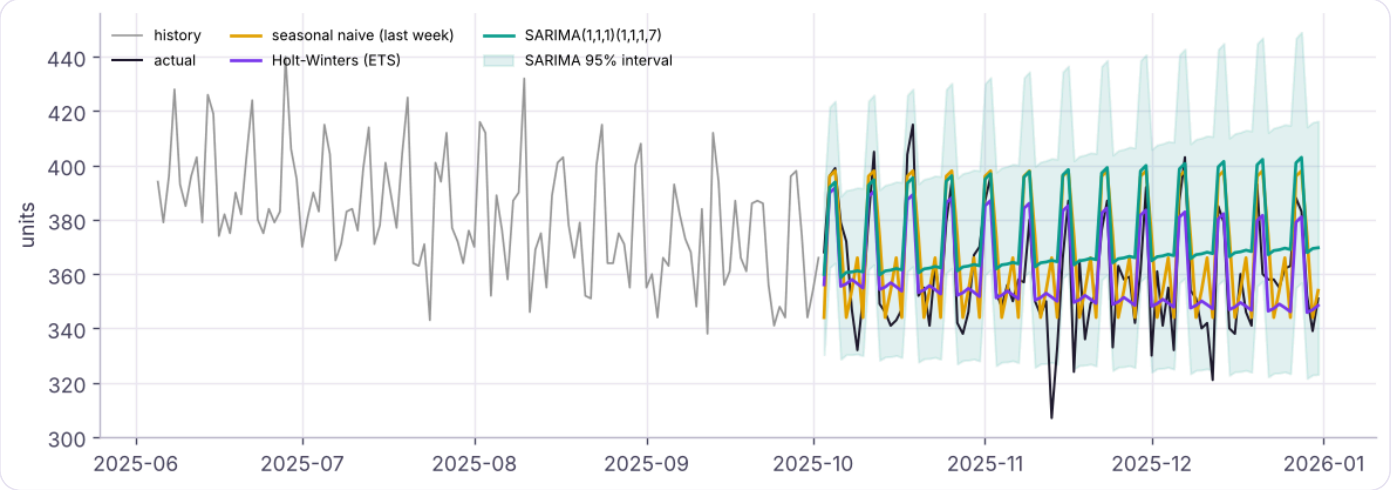
Python

```python
hist = train.copy()
for day in test.index:
    s = pd.concat([hist, pd.Series([np.nan], index=[day])])
    hist.loc[day] = model.predict(make_features(s).iloc[[-1]])[0]
forecasts["LightGBM (lags, recursive)"] = hist.loc[test.index]; results["LightGBM (lags, recursive)"] =
metrics(test, forecasts["LightGBM (lags, recursive)"])
print(pd.DataFrame(results).T.sort_values("WAPE_%"))
```

OUTPUT

	MAE	RMSE	MAPE_%	WAPE_%	bias_%
Holt-Winters (ETS)	10.4	12.8	2.92	2.88	0.34
seasonal naive (last week)	14.9	19.1	4.23	4.14	2.35
SARIMA(1,1,1)(1,1,1,7)	16.9	20.2	4.82	4.69	4.06
naive	19.1	22.4	5.36	5.31	1.79
moving average (28 days)	20.9	23.9	5.92	5.81	2.96
LightGBM (lags, recursive)	21.1	27.4	6.01	5.86	4.53

Holt-Winters wins clearly here (WAPE under 3%, almost no bias), while this LightGBM set-up does worse than the seasonal naive baseline. Two reasons: errors **accumulate** when 90 days are forecast recursively from the model's own predictions, and with a single, smooth series there is little for a tree model to learn that ETS does not already capture. ML models shine with **many related series** and **external drivers** (prices, promotions, weather, holidays). The lesson: always benchmark against simple models.



The last 120 days of training data, the 90-day test period and three forecasts.

Backtesting

One holdout can be lucky. **Rolling-origin backtests** repeat the exercise from several start dates.

Python

```
origins = [pd.Timestamp("2025-03-31"), pd.Timestamp("2025-06-30"), pd.Timestamp("2025-09-30")]
rows = []
for o in origins:
    tr, te = y.loc[:o], y.loc[o + pd.Timedelta(days=1): o + pd.Timedelta(days=28)]
    snaive = pd.Series(np.tile(tr.iloc[-7:].values, 5)[:len(te)], index=te.index)
    ets_o = ExponentialSmoothing(tr, trend="add", damped_trend=True, seasonal="mul",
seasonal_periods=7).fit().forecast(len(te))
    rows.append({"origin": o.date(), "seasonal naive WAPE": metrics(te, snaive)["WAPE_%"], "ETS WAPE":
metrics(te, ets_o)["WAPE_%"]})
bt = pd.DataFrame(rows); print(bt.to_string(index=False)); print("mean:",
bt.drop(columns="origin").mean().round(2).to_dict())
```

OUTPUT

```
    origin  seasonal naive WAPE  ETS WAPE
2025-03-31                 4.44      3.21
2025-06-30                 4.04      4.23
2025-09-30                 3.08      2.92
mean: {'seasonal naive WAPE': 3.85, 'ETS WAPE': 3.45}
```

Tools and practice

Tool	Strength
statsmodels	ETS, SARIMA(X), decomposition, diagnostics
statsforecast (Nixtla)	Very fast AutoARIMA/AutoETS for thousands of series
Prophet	Trend + seasonality + holidays with intuitive settings
LightGBM / XGBoost with lags	Many related series, external drivers (price, promotions, weather)
Deep learning (N-BEATS, TFT) and foundation models (TimeGPT, Chronos, TimesFM)	Large collections of series, zero-shot forecasts
Excel / Power BI forecast	Quick ETS forecasts for business users

Practical tips: forecast at the level decisions are made (store × product × week), add known future drivers (holidays, promotions, prices), forecast intervals not just points, monitor bias, and combine forecasts (an average of ETS and ML often beats both).

Summary

KEY TAKEAWAYS

- Decompose series into trend, seasonality and remainder (STL); difference to make them stationary (ADF test); read ACF/PACF.
- Split by time; evaluate with MAE, RMSE, MAPE, WAPE and bias; always beat the seasonal naive baseline.
- Holt-Winters (ETS) models level, trend and seasonality; SARIMA models autocorrelation and seasonal differences, with prediction intervals.
- Gradient boosting with lag, rolling and calendar features is a strong approach, especially with many series and external drivers; forecast recursively for long horizons.
- Backtest from several origins and choose by business metrics.

Quick quiz

6 questions · tap an answer to check it

1. Which baseline must every daily retail forecast beat?

A The overall mean

B Seasonal naive: the same day last week

C A random forecast

D Zero

Answer: B. It captures weekly seasonality for free.

2. What does differencing do?

A Removes trend (and with a seasonal lag, seasonality) to make the series stationary

B Smooths noise

C Adds seasonality

D Scales values

Answer: A. y_t − y_(t−1).

3. Why is WAPE often preferred to MAPE?

- **A** It does not explode when individual actuals are near zero and weights by volume
- **B** It is always smaller
- **C** It ignores errors
- **D** It is required by statsmodels

Answer: A. Total absolute error / total actuals.

4. In SARIMA(1,1,1)(1,1,1,7), what is 7?

- **A** The seasonal period (weekly seasonality on daily data)
- **B** The number of lags
- **C** The horizon
- **D** The number of models

Answer: A. s = 7.

5. Why shouldn't you shuffle data when validating forecasts?

- **A** It would train on the future and give over-optimistic results
- **B** It is slower
- **C** It changes the target
- **D** It breaks pandas

Answer: A. Always validate forward in time.

6. A lag-7 feature allows a model to forecast how many days ahead directly?

- **A** Up to 7 days
- **B** 1 day only
- **C** 90 days
- **D** Any horizon

Answer: A. Beyond that, forecast recursively or use direct models.

EXERCISE **Practice**

(1) Aggregate to weekly totals and forecast 12 weeks with Holt-Winters (seasonal period 52 needs 2+ years of data). (2) Add a Diwali/holiday flag feature to the LightGBM model. (3) Average the ETS and SARIMA forecasts: does the WAPE improve?

Show solution (worked answer)

Python

```
wk = y.resample("W-SUN").sum().iloc[1:-1]
ewk = ExponentialSmoothing(wk.iloc[:-12], trend="add", seasonal="add",
seasonal_periods=52).fit().forecast(12)
print(metrics(wk.iloc[-12:], ewk))
combo = (forecasts["Holt-Winters (ETS)"] + forecasts["SARIMA(1,1,1)(1,1,1,7)"]) / 2;
print(metrics(test, combo))
diwali = pd.to_datetime(["2022-10-24", "2023-11-12", "2024-11-01", "2025-10-20"])
F["festive"] = [int(any(abs((d - x).days) ⩽ 3 for x in diwali)) for d in F.index]
```

INTERVIEW CORNER **Questions you may be asked**

Q: How would you forecast sales for 5,000 products in 200 stores? Hierarchical, automated approach: a global gradient-boosting model with lags, calendar, price and promotion features (or fast AutoETS per series), backtesting by store-product, reconciliation across levels, and monitoring of bias and accuracy.

Q: What is stationarity and why does it matter? Constant statistical properties over time; ARIMA-type models assume it, so we difference or transform the series first.

Q: How do you evaluate a forecasting model? Time-ordered holdout or rolling-origin backtests with MAE/WAPE and bias, comparison with a seasonal naive baseline, and checking prediction-interval coverage.

STATISTICS & MACHINE LEARNING · LESSON 25

NLP and Text Analytics: Cleaning, TF-IDF, Sentiment Classification, Topics and Embeddings

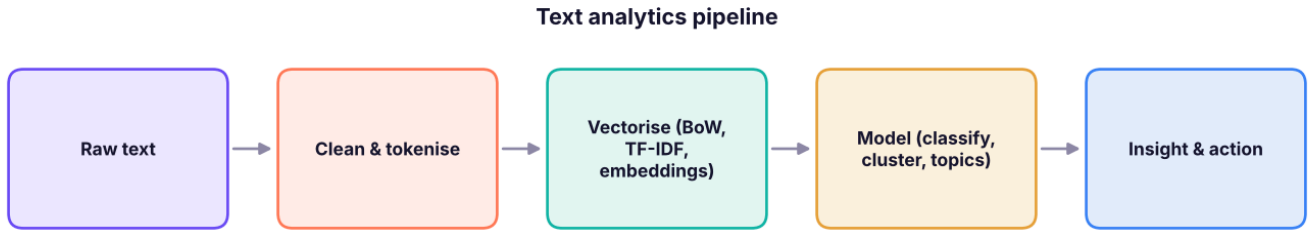
Turn customer reviews into insight. Text cleaning and tokenisation, stop words, stemming and lemmatisation, bag-of-words and TF-IDF with n-grams, a sentiment classifier with error analysis, aspect-level insight, topic modelling with NMF, document similarity, and how embeddings and LLMs change NLP, on 3,000 e-commerce reviews.

● ML track (added) · GenAI bridge ✦ scikit-learn 1.7+ · 2026 practice Advanced ⏱ 130 min

By the end of this lesson you can

- Clean and tokenise text and decide when to remove stop words, stem or lemmatise
- Represent documents with bag-of-words and TF-IDF (with n-grams) and read the vocabulary
- Train, evaluate and interpret a sentiment classifier, including error analysis
- Extract topics with NMF and aspect-level sentiment with simple rules
- Explain word and sentence embeddings and when to use LLMs instead

The NLP pipeline



A classical text-analytics pipeline.

Python

```
import pandas as pd, numpy as np, re
pd.set_option("display.width", 160); pd.set_option("display.max_colwidth", 90)
rv = pd.read_csv("customer_reviews.csv", parse_dates=["date"])
print(rv.shape, rv.sentiment.value_counts(normalize=True).round(3).to_dict())
print(rv.groupby("product").rating.mean().round(2).sort_values().to_dict())
print(rv.sample(5, random_state=3)[["rating", "review_text"]].to_string(index=False))
rv["n_words"] = rv.review_text.str.split().str.len()
print("words per review:", rv.n_words.describe().round(1)[["mean", "50%", "max"]].to_dict(), "| by
sentiment:", rv.groupby("sentiment").n_words.mean().round(1).to_dict())
```

OUTPUT

```
(3000, 8) {'positive': 0.53, 'negative': 0.321, 'neutral': 0.149}
{'Tablet': 3.27, 'Smartwatch': 3.29, 'Speaker': 3.3, 'Earbuds': 3.31, 'Smartphone': 3.33,
'Laptop': 3.36}
 rating
review_text
      4 Superb product. Customer support solved my issue quickly, the app is totally smooth,
charges really totally fast :)
      5                     Price is very very reasonable for these features though the buttons
are loose, no issues so far
      2                                                                  Worst purchase ever.
Sound is tinny, not wroth it
      5                                                                       Highly
recommend. Calls are crystal clear!
      5       Highly recommend. Build quality feels premium, price is very reasonable for these
features, no issues so far.
words per review: {'mean': 12.9, '50%': 13.0, 'max': 34.0} | by sentiment: {'negative': 13.9,
'neutral': 9.2, 'positive': 13.3}
```

Cleaning and tokenising

Python

```
STOP = {"the", "a", "an", "is", "it", "and", "to", "of", "for", "in", "on", "was", "this", "that",
"with", "my", "i", "very", "so", "are", "be", "at", "as", "after", "all"}
NEGATIONS = {"not", "no", "never", "don't", "can't", "isn't", "wasn't"}
FIX = {"batery": "battery", "qualty": "quality", "delivry": "delivery", "realy": "really", "scren":
"screen", "wroth": "worth", "bekaar": "useless", "badiya": "good"}
def clean(text, remove_stop=True):
    t = text.lower()
    t = re.sub(r"[:;]-?[()]", " ", t)                    # emoticons
    t = re.sub(r"[^a-z' ]+", " ", t)                     # keep letters and apostrophes
    tokens = [FIX.get(w, w) for w in t.split()]
    if remove_stop: tokens = [w for w in tokens if w not in STOP or w in NEGATIONS]
    return " ".join(tokens)
ex = rv.review_text.iloc[7]
print("raw:  ", ex); print("clean:", clean(ex))
rv["clean"] = rv.review_text.map(clean)
```

OUTPUT

```
raw:   Battery backup is totally superb but sound is tinny. Average product for daily use
clean: battery backup totally superb but sound tinny average product daily use
```

Step	Purpose	Caution
Lower-casing	"Battery" = "battery"	Loses emphasis ("GREAT")
Remove punctuation, numbers, URLs	Reduce noise	Prices and model numbers may matter
Spelling / slang normalisation	Merge variants ("batery", Hinglish "badiya")	Needs a dictionary
Stop-word removal	Drop frequent function words	**Never** drop negations ("not") for sentiment
Stemming (Porter, Snowball)	"charging", "charged" → "charg"	Crude, can merge different words
Lemmatisation (spaCy)	"better" → "good", "batteries" → "battery"	Slower, needs a language model

Libraries: **spaCy** (fast tokenisation, lemmas, entities), **NLTK** (classic teaching toolkit), **regex**; for Indian languages, **IndicNLP** and multilingual models.

Bag-of-words and TF-IDF

Bag-of-words counts each word per document, ignoring order. **TF-IDF** down-weights words that appear in many documents (like "product") and up-weights distinctive ones:

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \log \frac{1 + N}{1 + \text{df}(t)} + 1$$

Python

```python
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
docs = ["battery drains fast", "battery lasts all day", "screen is not great", "great screen and great battery"]
cv = CountVectorizer(); bow = cv.fit_transform(docs)
print(pd.DataFrame(bow.toarray(), columns=cv.get_feature_names_out(), index=[f"doc{i}" for i in range(4)]))
tf = TfidfVectorizer(); tfm = tf.fit_transform(docs)
print(pd.DataFrame(tfm.toarray().round(2), columns=tf.get_feature_names_out(), index=[f"doc{i}" for i in range(4)]).loc[:, ["battery", "great", "not", "drains"]])
bi = CountVectorizer(ngram_range=(1, 2)).fit(["screen is not great"]); print("with bigrams:", bi.get_feature_names_out().tolist())
```

OUTPUT

```
      all  and  battery  day  drains  fast  great  is  lasts  not  screen
doc0    0    0        1    0       1     1      0   0      0    0       0
doc1    1    0        1    1       0     0      0   0      1    0       0
doc2    0    0        0    0       0     0      1   1      0    1       1
```

```
doc3    0    1        1    0       0     0      2   0      0    0       1
      battery  great   not  drains
doc0     0.41   0.00  0.00    0.64
doc1     0.35   0.00  0.00    0.00
doc2     0.00   0.44  0.56    0.00
doc3     0.30   0.74  0.00    0.00
with bigrams: ['great', 'is', 'is not', 'not', 'not great', 'screen', 'screen is']
```

Bigrams such as "not great" keep some word order, which matters for sentiment.

Sentiment classifier

Python

```python
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
Xtr, Xte, ytr, yte = train_test_split(rv.clean, rv.sentiment, test_size=0.25, stratify=rv.sentiment,
random_state=0)
for name, vec in [("unigrams", TfidfVectorizer(min_df=2)), ("uni + bigrams",
TfidfVectorizer(ngram_range=(1, 2), min_df=2, sublinear_tf=True))]:
    m = make_pipeline(vec, LogisticRegression(max_iter=3000, C=5)).fit(Xtr, ytr)
    print(f"{name:14s} accuracy {accuracy_score(yte, m.predict(Xte)):.3f} | vocabulary
{len(m[0].vocabulary_):,}")
clf = m
print(classification_report(yte, clf.predict(Xte), digits=3))
labels = ["negative", "neutral", "positive"]
print(pd.DataFrame(confusion_matrix(yte, clf.predict(Xte), labels=labels), index=[f"true {l}" for l in
labels], columns=[f"pred {l}" for l in labels]))
```

OUTPUT

```
unigrams       accuracy 0.935 | vocabulary 183
uni + bigrams  accuracy 0.940 | vocabulary 1,546
              precision    recall  f1-score   support

    negative      0.974     0.938     0.956       241
     neutral      0.860     0.821     0.840       112
    positive      0.942     0.975     0.958       397

    accuracy                          0.940       750
   macro avg      0.925     0.911     0.918       750
weighted avg      0.940     0.940     0.940       750

               pred negative  pred neutral  pred positive
true negative            226             6              9
true neutral               5            92             15
true positive              1             9            387
```

What did it learn?

Python

```
vocab = np.array(clf[0].get_feature_names_out())
for i, c in enumerate(clf[-1].classes_):
    top = np.argsort(-clf[-1].coef_[i])[:8]; print(f"{c:9s} ←", vocab[top].tolist())
```

OUTPUT

```
negative  ← ['disappointed', 'useless', 'useless product', 'buying', 'regret', 'regret buying',
'purchase ever', 'ever']
neutral   ← ['but', 'fine', 'expected', 'loved build', 'but too', 'early but', 'okay', 'okay
price']
positive  ← ['loved', 'though', 'superb', 'happy purchase', 'superb product', 'not
disappointed', 'bahut good', 'bahut']
```

Error analysis

Python

```
pred = clf.predict(Xte); proba = clf.predict_proba(Xte).max(1)
err = pd.DataFrame({"text": rv.loc[Xte.index, "review_text"], "rating": rv.loc[Xte.index, "rating"],
"true": yte, "pred": pred, "confidence": proba.round(2)})[pred != yte]
print(len(err), "errors"); print(err.sort_values("confidence",
ascending=False).head(6).to_string(index=False))
```

OUTPUT

```
45 errors

text  rating     true     pred  confidence
Happy with the purchase. Calls are crystal super clear, arrived a day early, price is very
reasonable for these features       3  neutral positive        1.00
                                                                    Bahut badiya. Bass is
rich, replacement was smooth       2 negative positive        1.00
                                                  Worst purchase ever. Too much bloatware,
battery drains too fast!       3  neutral negative        0.99
                                 Loved it! setup took two minutes, great value for money, sound
is really clear and loud       3  neutral positive        0.99
                                                   Bahut badiya. The app is super
smooth, replacement was smooth       3  neutral positive        0.99
                                                                          Superb
product. Bass is rich!!       2 negative positive        0.99
```

Reading errors is the most useful step in NLP. Typical causes here: **noisy labels** (a 1-star rating on a positive text), **mixed** reviews ("good screen but weak battery") and genuinely neutral wording. Fixes include better labels, aspect-level models, or an LLM for difficult cases.

Aspect-level insight

Overall sentiment hides *what* customers like or dislike. A simple keyword approach already gives actionable results:

Python

```
ASPECTS = {"battery": r"batter|charg", "screen": r"screen|display|pixel|touch", "sound":
r"sound|bass|speaker|mic", "build": r"build|hinge|button|finish|plastic",
           "delivery": r"deliver|courier|box|packag|arriv", "service":
r"support|service|refund|replacement|care", "price": r"price|money|rupee|expensive|worth", "software":
r"app|update|bloatware|setup"}
low = rv.review_text.str.lower()
rows = []
for a, pat in ASPECTS.items():
    m = low.str.contains(pat, regex=True)
    rows.append({"aspect": a, "mentions": int(m.sum()), "share_negative_%": round(100 *
(rv.sentiment[m] == "negative").mean(), 1), "avg_rating": round(rv.rating[m].mean(), 2)})
asp = pd.DataFrame(rows).sort_values("share_negative_%", ascending=False);
print(asp.to_string(index=False))
print(rv.assign(battery=low.str.contains(ASPECTS["battery"])).query("battery").groupby("product").rating.m
```

OUTPUT

```
  aspect  mentions  share_negative_%  avg_rating
   sound       659              42.8        3.08
   build       572              37.1        3.23
   price       867              36.9        3.24
  screen       741              35.9        3.30
 service       792              35.6        3.31
software      1165              34.6        3.34
 battery       752              32.4        3.38
delivery       945              30.6        3.41
{'Smartwatch': 3.2, 'Speaker': 3.35, 'Earbuds': 3.37}
```

Production systems use **aspect-based sentiment analysis** models or LLM prompts ("list each product aspect mentioned and its sentiment as JSON").

Topic modelling

Topic models discover groups of words that co-occur, without labels. **NMF** on TF-IDF is fast and gives readable topics (LDA on counts is the classic alternative; BERTopic uses embeddings).

Python

```
from sklearn.decomposition import NMF
tfidf = TfidfVectorizer(max_df=0.5, min_df=5, ngram_range=(1, 2), stop_words="english")
Xt = tfidf.fit_transform(rv.review_text.str.lower())
nmf = NMF(n_components=6, random_state=0, init="nndsvda", max_iter=500).fit(Xt)
terms = np.array(tfidf.get_feature_names_out())
for k, comp in enumerate(nmf.components_): print(f"topic {k}:", ", ".join(terms[np.argsort(-comp)
[:7]]))
rv["topic"] = nmf.transform(Xt).argmax(1)
print(rv.groupby("topic").agg(reviews=("rating", "size"), avg_rating=("rating", "mean")).round(2).T
```

OUTPUT

```
topic 0: battery, day, superb, screen, lasts, clear, smooth
topic 1: support, support solved, solved issue, issue quickly, solved, issue, customer support
topic 2: centre, service centre, service, centre staff, staff, helpful, waiting days
topic 3: weeks, high volume, speaker, high, crackles high, crackles, volume
topic 4: quality, feels, build, premium, build quality, quality feels, feels premium
topic 5: value, value money, money, great value, great, packaging, safe
topic             0       1       2       3       4       5
reviews     1015.00  197.00  230.00  915.00  393.00  250.00
avg_rating     4.29    3.91    3.17    1.97    3.14    4.18
```

Name each topic from its top words (for example "battery and charging", "delivery and packaging") and track topic volume and rating over time.

Similarity search

Python

```
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.decomposition import TruncatedSVD
lsa = make_pipeline(TfidfVectorizer(ngram_range=(1, 2), min_df=2, stop_words="english"),
TruncatedSVD(100, random_state=0)).fit(rv.review_text)
E = lsa.transform(rv.review_text)
for query in ["the battery runs out quickly and charging takes forever", "phone heats up and power does
not last"]:
    sims = cosine_similarity(lsa.transform([query]), E)[0]; best = np.argsort(-sims)[:3]
    print("query:", query); [print(f"   {sims[i]:.2f}  {rv.review_text.iloc[i]}") for i in best]
```

OUTPUT

```
query: the battery runs out quickly and charging takes forever
   0.76  Customer support solved my issue quickly but charging is painfully really slow.
   0.74  Very disappointed. Charging is painfully slow, screen is totally dull but customer
support solved my issue quickly
   0.69  Charging is painfully slow
query: phone heats up and power does not last
   1.00  Nothing special, does the job
   1.00  Nothing special, does the job.
   1.00  Nothing special, does the job.
```

Latent semantic analysis (TF-IDF + SVD) finds the charging complaints for the first query. The second query fails badly: "heats" and "power" never appear in the reviews and "last" is a different token from "lasts", so LSA matches the only shared word, "does". Word-count methods cannot know that "power does not last" means the battery drains. Modern **embeddings** do this far better.

Embeddings and LLMs

Approach	Idea	Use
Word2Vec, GloVe, fastText (2013–2016)	Each word gets a dense vector; similar words are close	Features, similarity
Transformers (BERT, 2018 →)	Contextual vectors: "bank" differs in "river bank" and "bank loan"	Fine-tuned classifiers, NER
Sentence embeddings (sentence-transformers, OpenAI/Cohere/Voyage/Gemini embedding APIs)	One vector per sentence or document	Semantic search, clustering, RAG
LLMs (Claude, GPT, Gemini, Llama)	Instruction-following generation	Zero-shot classification, extraction, summarisation

Python

```python
# Sentence embeddings (pip install sentence-transformers) - runs locally
from sentence_transformers import SentenceTransformer
model = SentenceTransformer("all-MiniLM-L6-v2")
emb = model.encode(rv.review_text.tolist(), normalize_embeddings=True)    # 3,000 × 384 matrix
# then: LogisticRegression on emb, KMeans on emb, or cosine search as above
```

Python

```python
# Zero-shot classification with an LLM (see the GenAI track for API set-up and cost control)
prompt = """Classify the review's sentiment as positive, negative or neutral and list aspects
mentioned.
Return JSON: {"sentiment": ..., "aspects": [{"aspect": ..., "sentiment": ...}]}
Review: "Good screen but the battery barely lasts half a day." """
```

When to use what: TF-IDF + logistic regression is cheap, fast, explainable and often within a few points of large models on simple tasks. Embeddings help with semantic similarity and small labelled sets. LLMs handle nuanced, multi-aspect or multilingual text with few or no labels, at higher cost and latency; validate them on a labelled sample just like any model.

Summary

KEY TAKEAWAYS

- Clean text deliberately: lower-case, normalise spelling and slang, keep negations; stem or lemmatise when useful.
- Bag-of-words counts words; TF-IDF weights distinctive words; bigrams capture short phrases like "not great".
- A TF-IDF + logistic regression classifier is a strong, explainable baseline; error analysis reveals noisy labels and mixed reviews.
- Aspect keywords and NMF topics turn reviews into product actions.
- Embeddings and LLMs capture meaning beyond exact words; benchmark them against the classical baseline.

Quick quiz

6 questions · tap an answer to check it

1. Why should "not" be kept when removing stop words for sentiment?

A It flips meaning: "not good" vs "good"

B It is a long word

C TF-IDF needs it

D It is a noun

Answer: A. Negations carry sentiment.

2. What does IDF do in TF-IDF?

A Down-weights words that appear in many documents

B Counts words

C Removes stop words

D Translates words

Answer: A. Common words get low weight.

3. What do bigrams add?

- **A** Pairs of consecutive words such as "not great", keeping some word order
- **B** Two languages
- **C** Two labels
- **D** Double counts

Answer: A. ngram_range=(1, 2).

4. Which method finds topics without labels?

- **A** Logistic regression
- **B** NMF or LDA topic modelling
- **C** Confusion matrix
- **D** Stemming

Answer: B. Unsupervised.

5. What is a sentence embedding?

- **A** A dense vector representing a sentence's meaning, so similar sentences are close
- **B** A list of words
- **C** A TF-IDF score
- **D** A topic label

Answer: A. Used for semantic search and RAG.

6. When might TF-IDF + logistic regression be the better choice than an LLM?

- **A** Simple, high-volume classification where cost, speed and explainability matter
- **B** Never
- **C** For multilingual sarcasm
- **D** When there are no labels

Answer: A. Always benchmark.

EXERCISE **Practice**

(1) Add a feature for exclamation marks and review length and test whether accuracy changes. (2) Track the monthly share of negative battery reviews. (3) Try `ComplementNB` and `LinearSVC` and compare macro F1. (4) Write an LLM prompt for aspect-based sentiment and test it on 10 reviews (GenAI track).

Show solution (worked answer)

Python

```
from scipy.sparse import hstack
from sklearn.svm import LinearSVC
from sklearn.naive_bayes import ComplementNB
from sklearn.metrics import f1_score
monthly = rv.assign(m=rv.date.dt.to_period("M"),
batt=low.str.contains(ASPECTS["battery"])).query("batt").groupby("m").sentiment.apply(lambda s: (s
== "negative").mean()); print(monthly.round(2))
for est in (ComplementNB(), LinearSVC(C=0.5)):
    mm = make_pipeline(TfidfVectorizer(ngram_range=(1, 2), min_df=2), est).fit(Xtr, ytr);
print(type(est).__name__, f1_score(yte, mm.predict(Xte), average="macro"))
```

INTERVIEW CORNER **Questions you may be asked**

Q: How would you build a sentiment model for customer reviews? Define labels (stars or manual), clean text keeping negations, start with TF-IDF + logistic regression, evaluate with macro F1 and a confusion matrix, analyse errors, then try embeddings or a fine-tuned transformer/LLM if needed, and monitor drift.

Q: TF-IDF vs embeddings? TF-IDF is sparse, exact-word based and interpretable; embeddings are dense and capture meaning, synonyms and context, at the cost of interpretability and compute.

Q: How do you find what customers complain about most? Aspect extraction (keywords, NER, or LLM prompts) combined with sentiment per aspect, plus topic modelling to discover unexpected themes, tracked over time.

STATISTICS & MACHINE LEARNING · LESSON 26

Neural Networks and Deep Learning: From a Neuron to Transformers

Understand the models behind modern AI. Neurons, weights and activation functions, forward pass and backpropagation, a two-layer network written from scratch in NumPy, multilayer perceptrons in scikit-learn on images and tabular data, training practice (learning rate, batches, epochs, early stopping, dropout), and a map of CNNs, RNNs, transformers and LLMs with PyTorch and Keras code.

ML track (added) · GenAI bridge | ✦ scikit-learn 1.7+ · PyTorch 2.x · Keras 3 | Advanced | ⏱ 130 min

By the end of this lesson you can

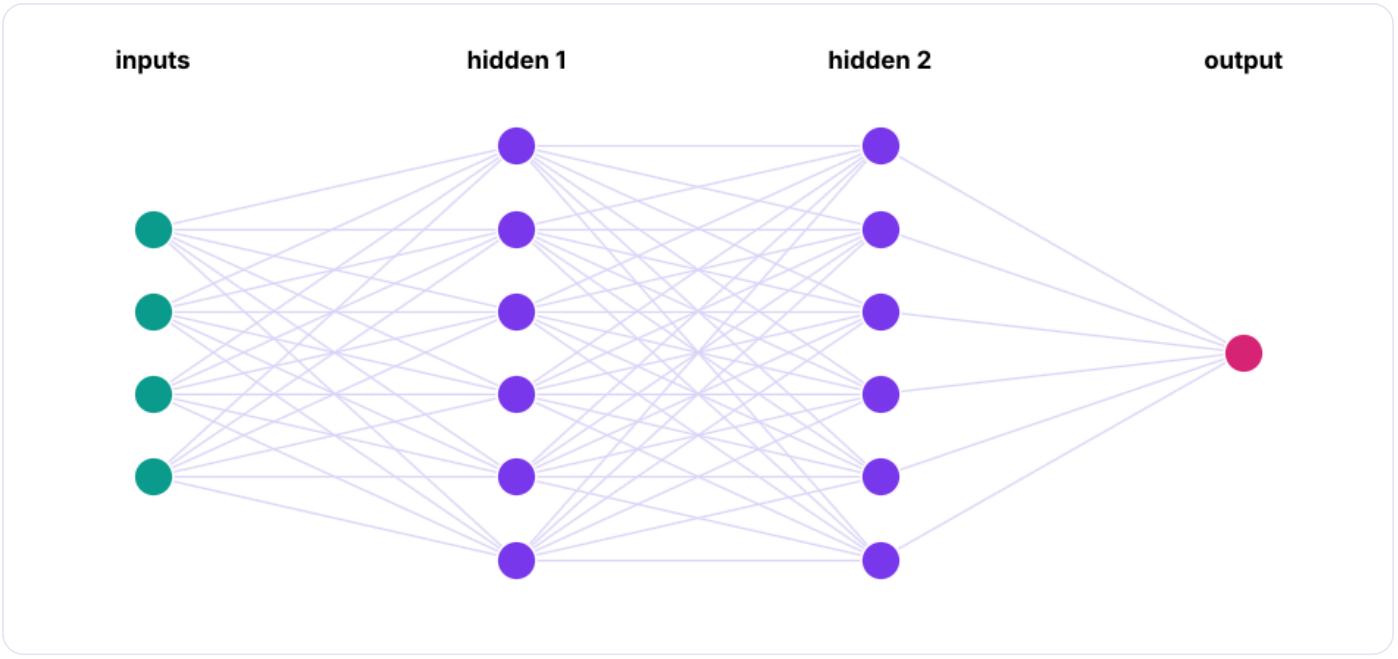
- Explain a neuron, layers, activation functions and the forward pass
- Explain loss, gradients and backpropagation, and train a small network from scratch
- Train and tune MLPs with scikit-learn and compare them with classical models
- Apply practical training techniques: scaling, learning rate, batching, early stopping, regularisation
- Describe CNNs, RNNs and transformers and where each is used

A neuron

An artificial **neuron** computes a weighted sum of its inputs plus a bias, then applies an **activation function**:

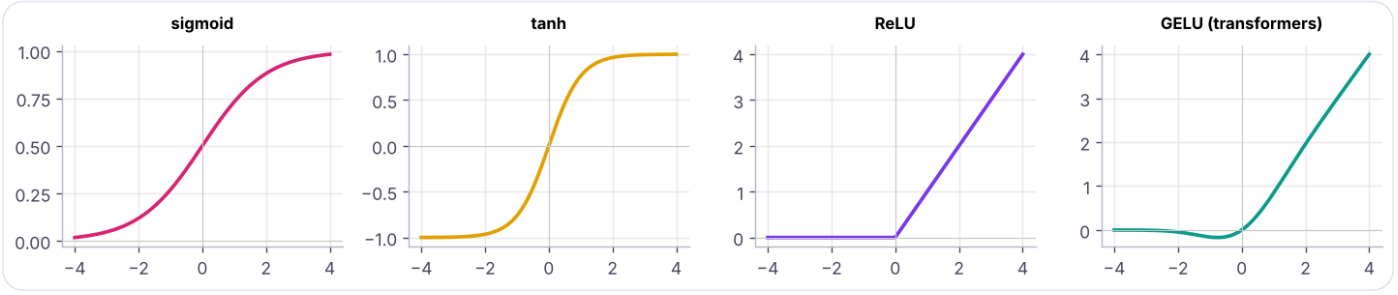
$$a = f\left(\sum_j w_j x_j + b\right)$$

With the sigmoid activation, one neuron **is** logistic regression. Stacking neurons in **layers** lets the network learn non-linear patterns.



A multilayer perceptron: inputs, two hidden layers and an output. Every connection has a weight learned from data.

Activation functions



Common activation functions. ReLU is the default for hidden layers; sigmoid and softmax are used for outputs.

Activation	Where	Why
ReLU (max(0, z))	Hidden layers	Simple, fast, avoids vanishing gradients
GELU, SiLU	Hidden layers of transformers	Smooth variants of ReLU
Sigmoid	Binary output	Probability between 0 and 1
Softmax	Multiclass output	Probabilities that sum to 1
None (linear)	Regression output	Any real value

Without non-linear activations, any stack of layers collapses into a single linear model.

How networks learn: backpropagation

1. **Forward pass:** compute predictions layer by layer.
2. **Loss:** measure the error (cross-entropy for classification, MSE for regression).

3. **Backward pass:** use the chain rule to compute the gradient of the loss with respect to every weight (**backpropagation**).
4. **Update:** gradient descent step on every weight (lesson 11), usually with the **Adam** optimiser on mini-batches.
5. Repeat for many **epochs** (passes over the data).

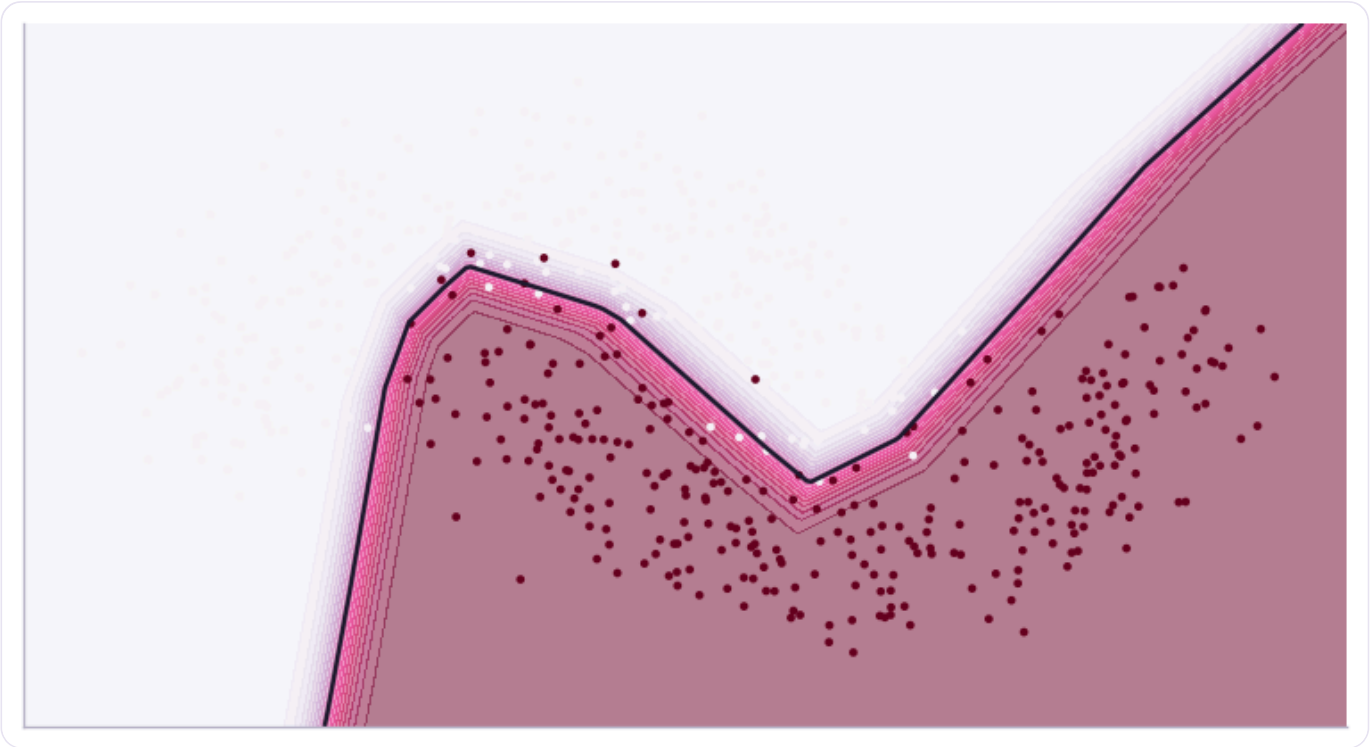
A network from scratch

Python

```python
from sklearn.datasets import make_moons
X, y = make_moons(600, noise=0.2, random_state=0); y = y.reshape(-1, 1)
rng = np.random.default_rng(0)
W1, b1 = rng.normal(0, 1, (2, 16)) * np.sqrt(2 / 2), np.zeros((1, 16))          # He initialisation
W2, b2 = rng.normal(0, 1, (16, 1)) * np.sqrt(2 / 16), np.zeros((1, 1))
sigmoid = lambda z: 1 / (1 + np.exp(-z))
lr, losses = 0.5, []
for epoch in range(2000):
    z1 = X @ W1 + b1; a1 = np.maximum(0, z1)                    # forward: hidden layer (ReLU)
    p = sigmoid(a1 @ W2 + b2)                                    # forward: output (sigmoid)
    loss = -np.mean(y * np.log(p + 1e-9) + (1 - y) * np.log(1 - p + 1e-9)); losses.append(loss)
    dz2 = (p - y) / len(X)                                       # backward: output gradient
    dW2, db2 = a1.T @ dz2, dz2.sum(0, keepdims=True)
    dz1 = (dz2 @ W2.T) * (z1 > 0)                                # chain rule through ReLU
    dW1, db1 = X.T @ dz1, dz1.sum(0, keepdims=True)
    W1 -= lr * dW1; b1 -= lr * db1; W2 -= lr * dW2; b2 -= lr * db2
acc = ((p > 0.5) == y).mean()
print(f"loss: epoch 0 {losses[0]:.3f} → epoch 200 {losses[200]:.3f} → epoch 2000 {losses[-1]:.3f} |
training accuracy {acc:.3f}")
print("parameters:", W1.size + b1.size + W2.size + b2.size)
```

OUTPUT

```
loss: epoch 0 0.719 → epoch 200 0.196 → epoch 2000 0.063 | training accuracy 0.977
parameters: 65
```



Decision boundary learned by the 16-neuron network written above: a curved boundary that a linear model cannot draw.

These 30 lines are the core of every deep-learning library; frameworks add automatic differentiation, GPUs, better optimisers and building blocks.

MLPs in scikit-learn

Python

```
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
digits = load_digits()
Xa, Xb, ya, yb = train_test_split(digits.data, digits.target, test_size=0.3, stratify=digits.target,
random_state=0)
for name, m in [("logistic regression", make_pipeline(StandardScaler(),
LogisticRegression(max_iter=3000))),
                ("MLP (64)", make_pipeline(StandardScaler(), MLPClassifier((64,), max_iter=500,
random_state=0))),
                ("MLP (128, 64), early stopping", make_pipeline(StandardScaler(), MLPClassifier((128,
64), alpha=1e-3, early_stopping=True, max_iter=500, random_state=0)))]:
    m.fit(Xa, ya); print(f"{name:32s} test accuracy {m.score(Xb, yb):.3f}")
```

OUTPUT

```
logistic regression              test accuracy 0.972
MLP (64)                         test accuracy 0.974
MLP (128, 64), early stopping    test accuracy 0.974
```

Python

```python
import pandas as pd
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.ensemble import HistGradientBoostingClassifier
tc = pd.read_csv("telecom_churn.csv"); Xc = tc.drop(columns=["customer_id", "churn", "total_charges"]);
yc = (tc.churn == "Yes").astype(int)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months", "monthly_charges",
"support_calls"]), ("c", OneHotEncoder(), ["contract", "internet_service", "payment_method"])])
for name, est in [("logistic regression", LogisticRegression(max_iter=2000)), ("MLP (32, 16)",
MLPClassifier((32, 16), alpha=1e-2, early_stopping=True, max_iter=1000, random_state=0)),
                  ("gradient boosting", HistGradientBoostingClassifier(max_iter=200,
learning_rate=0.05, random_state=0))]:
    s = cross_val_score(make_pipeline(pre, est), Xc, yc, cv=5, scoring="roc_auc"); print(f"churn
{name:20s} AUC {s.mean():.3f}")
```

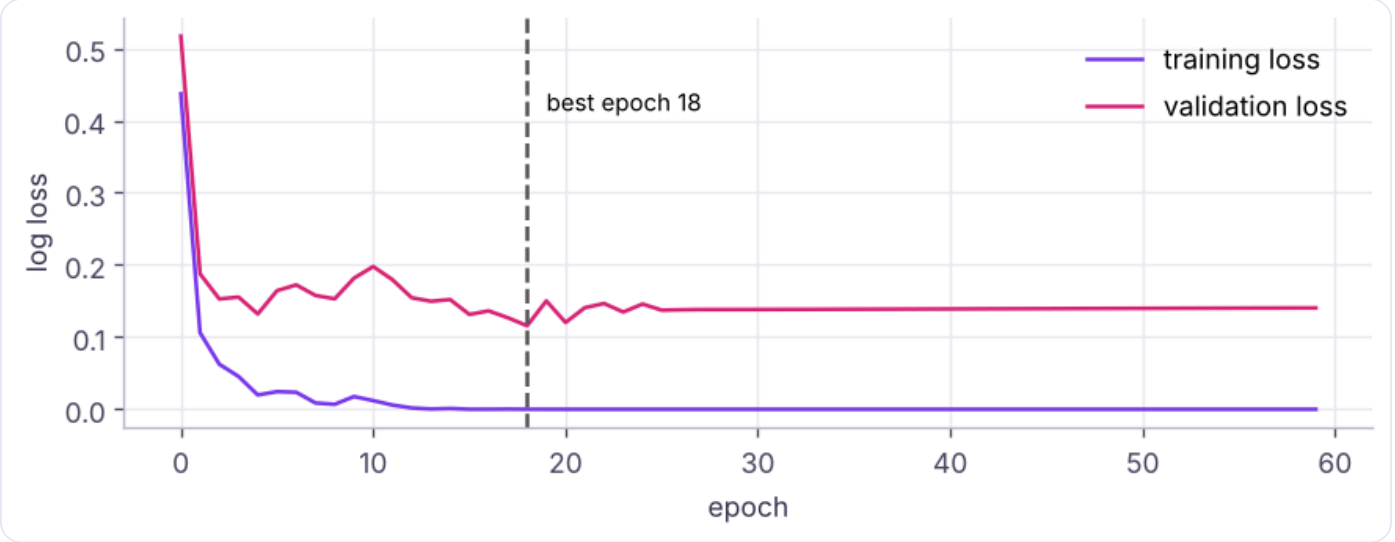
OUTPUT

```
churn logistic regression  AUC 0.815
churn MLP (32, 16)         AUC 0.802
churn gradient boosting    AUC 0.782
```

Neural networks win on **unstructured data** (images, text, audio). On small tabular data, logistic regression and gradient boosting usually match or beat them with less tuning.

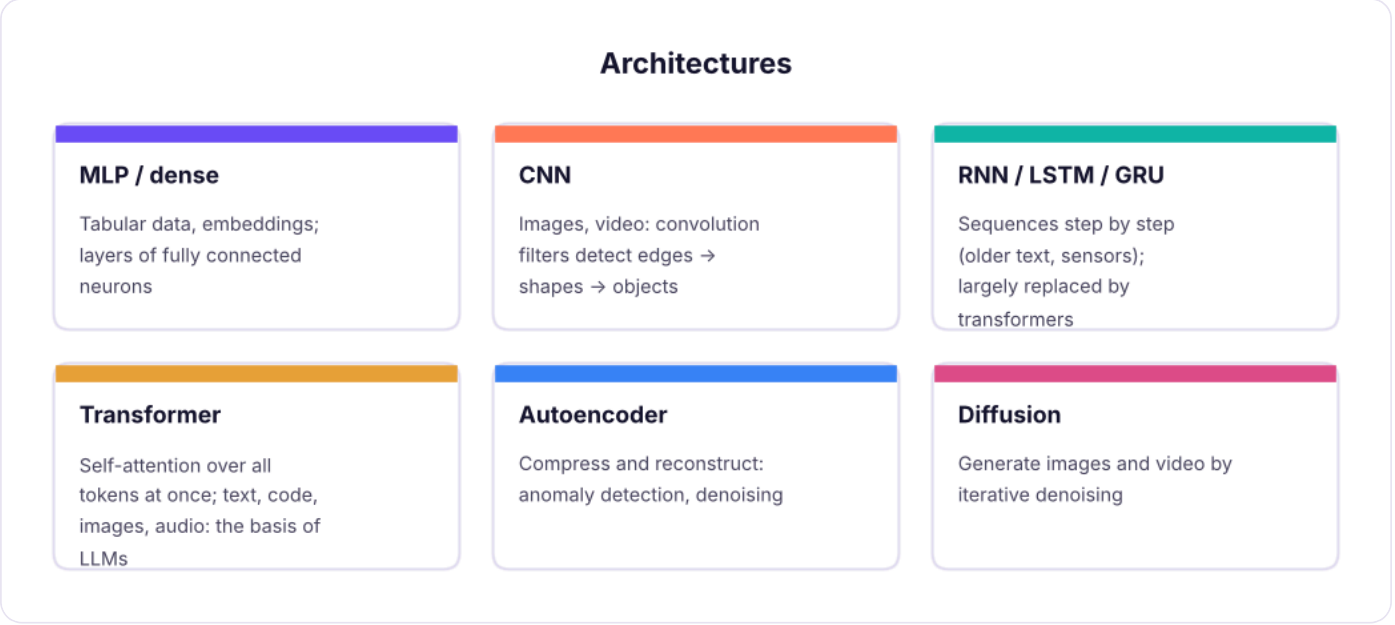
Training practice

Technique	Purpose
Scale inputs	Stable, fast optimisation
Learning rate (and schedules)	The single most important setting
Mini-batches (32–512)	Efficient GPU use, regularising noise
Epochs + **early stopping**	Stop when validation loss stops improving
Dropout	Randomly switch off neurons in training: reduces overfitting
Weight decay (L2)	Penalise large weights (`alpha` in scikit-learn)
Batch / layer normalisation	Stabilise activations, allow higher learning rates
Data augmentation	Flip/crop images, paraphrase text: more data for free
Transfer learning	Start from a pre-trained model and fine-tune: the default today



Training vs validation loss: when validation loss starts rising the network is overfitting; early stopping keeps the best epoch.

Deep-learning architectures



The main architecture families and their typical data.

Transformers (Vaswani et al., 2017, "Attention Is All You Need") let every token look at every other token through **self-attention**, train in parallel on GPUs and scale to billions of parameters. Pre-trained on huge text corpora to predict the next token, then instruction-tuned and aligned with human feedback, they became **large language models** (GenAI track).

PyTorch and Keras

Python

```python
# PyTorch (pip install torch): the research and industry standard
import torch, torch.nn as nn
model = nn.Sequential(nn.Linear(64, 128), nn.ReLU(), nn.Dropout(0.2), nn.Linear(128, 10))
opt = torch.optim.AdamW(model.parameters(), lr=1e-3, weight_decay=1e-4)
loss_fn = nn.CrossEntropyLoss()
for epoch in range(20):
    for xb, yb in train_loader:                # mini-batches
        opt.zero_grad()
        loss = loss_fn(model(xb), yb)
        loss.backward()                         # backpropagation (autograd)
        opt.step()                              # Adam update
```

Python

```python
# Keras 3 (pip install keras): high-level API on TensorFlow, JAX or PyTorch
import keras
model = keras.Sequential([keras.layers.Dense(128, activation="relu"), keras.layers.Dropout(0.2),
keras.layers.Dense(10, activation="softmax")])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy", metrics=["accuracy"])
model.fit(X_train, y_train, validation_split=0.2, epochs=50, callbacks=
[keras.callbacks.EarlyStopping(patience=5, restore_best_weights=True)])
```

Python

```python
# Transfer learning with Hugging Face (pip install transformers): fine-tune a pre-trained model for
sentiment
from transformers import pipeline
clf = pipeline("sentiment-analysis", model="distilbert-base-uncased-finetuned-sst-2-english")
clf(["Battery barely lasts half a day", "Display is bright and sharp"])
```

Free GPUs for practice: Google Colab and Kaggle notebooks.

Summary

KEY TAKEAWAYS

- A neuron = weighted sum + activation; layers of neurons with non-linear activations learn complex functions.
- Training = forward pass, loss, backpropagation (chain rule), gradient-descent updates (Adam) over epochs and mini-batches.
- Use ReLU/GELU in hidden layers, sigmoid/softmax outputs for classification; scale inputs; regularise with early stopping, dropout and weight decay.
- Neural networks excel on images, text and audio; on small tabular data, gradient boosting and logistic regression are strong competitors.
- CNNs for images, transformers for sequences and language (LLMs); PyTorch, Keras and Hugging Face are the main tools; transfer learning is the default.

Quick quiz

6 questions · tap an answer to check it

1. What happens without non-linear activation functions?

A The whole network collapses to a linear model

B It trains faster with better accuracy

C It becomes a decision tree

D Nothing changes

Answer: A. Non-linearity is what gives depth its power.

2. What does backpropagation compute?

A Gradients of the loss with respect to every weight, using the chain rule

B The predictions

C The learning rate

D The number of layers

Answer: A. Gradient descent then updates the weights.

3. **Which output activation for a 10-class problem?**

A ReLU

B Softmax

C None

D tanh

Answer: B. Probabilities that sum to 1.

4. **Validation loss starts rising while training loss keeps falling. What is happening?**

A Overfitting; stop early or add regularisation

B Underfitting

C The learning rate is perfect

D Data leakage for sure

Answer: A. Early stopping keeps the best epoch.

5. **Which architecture is the basis of LLMs?**

A CNN

B RNN

C Transformer

D K-means

Answer: C. Self-attention over tokens.

6. For a 2,000-row tabular churn dataset, which is usually a strong first choice?

- **A** Logistic regression or gradient boosting
- **B** A 50-layer CNN
- **C** An LLM fine-tune
- **D** An autoencoder

Answer: A. Deep learning shines on unstructured data.

EXERCISE **Practice**

(1) Change the hidden layer size of the NumPy network to 4 and 64: how do loss and boundary change? (2) Train an `MLPRegressor` on house prices and compare MAE with gradient boosting. (3) In Colab, train the PyTorch model above on the digits data.

Show solution (worked answer)

Python

```
from sklearn.neural_network import MLPRegressor
from sklearn.ensemble import HistGradientBoostingRegressor
from sklearn.metrics import mean_absolute_error
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"]); h = h[h.price_lakh.between(5,
300)]
Xh = pd.get_dummies(h.drop(columns=["house_id", "price_lakh"]), drop_first=True).astype(float)
a, b, c, d = train_test_split(Xh, h.price_lakh, test_size=0.2, random_state=0)
mlp_r = make_pipeline(StandardScaler(), MLPRegressor((64, 32), alpha=1e-3, max_iter=3000,
early_stopping=True, random_state=0)).fit(a, c)
print(mean_absolute_error(d, mlp_r.predict(b)), mean_absolute_error(d,
HistGradientBoostingRegressor().fit(a, c).predict(b)))
```

INTERVIEW CORNER

Questions you may be asked

Q: Explain how a neural network learns. It makes predictions with its current weights, measures the error with a loss function, computes how each weight affected the error through backpropagation, and adjusts the weights slightly in the direction that reduces the error, repeating over many batches and epochs.

Q: What is the vanishing gradient problem? In deep networks with sigmoid/tanh, gradients shrink as they pass backward through layers, so early layers learn very slowly; ReLU, residual connections and normalisation fix it.

Q: What is transfer learning? Reusing a model pre-trained on a large dataset as the starting point for a new task, fine-tuning some or all layers with much less data.

STATISTICS & MACHINE LEARNING · LESSON 27

Explainable AI and MLOps: SHAP, Fairness, Deployment, Monitoring and Responsible ML

Make models trustworthy and useful in production. Global and local explanations (permutation importance, partial dependence, SHAP), fairness checks across groups, packaging a pipeline, serving predictions with FastAPI, batch scoring, experiment tracking with MLflow, monitoring data drift and performance, model cards, and the regulations and responsible-AI practices that apply in India and globally.

ML track (added) · ✦ SHAP 0.4x · FastAPI · MLflow 3 · 2026 practice · Advanced · ⏱ 130 min

By the end of this lesson you can

- Explain a model globally with permutation importance and partial dependence
- Explain individual predictions with SHAP values and communicate them
- Check performance and error rates across groups for fairness
- Package, serve and batch-score a model, and track experiments
- Monitor drift and performance after deployment and document models with model cards

Why explainability?

Stakeholders ask "**why** did the model say this?". Regulators require explanations for credit and insurance decisions; data scientists need explanations to debug leakage and bias.

Python

```
import pandas as pd, numpy as np, warnings
warnings.filterwarnings("ignore")
from sklearn.model_selection import train_test_split
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.metrics import roc_auc_score
tc = pd.read_csv("telecom_churn.csv")
X = pd.get_dummies(tc.drop(columns=["customer_id", "churn", "total_charges"]),
drop_first=True).astype(float); y = (tc.churn == "Yes").astype(int)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=1)
model = HistGradientBoostingClassifier(max_iter=200, learning_rate=0.05, max_leaf_nodes=15,
min_samples_leaf=40, random_state=0).fit(Xtr, ytr)
print(f"test AUC {roc_auc_score(yte, model.predict_proba(Xte)[:, 1]):.3f} | features:
{list(X.columns)}")
```

OUTPUT

```
test AUC 0.801 | features: ['senior_citizen', 'tenure_months', 'monthly_charges',
'support_calls', 'contract_One year', 'contract_Two year', 'internet_service_Fiber',
```

```
'internet_service_No internet', 'payment_method_Card', 'payment_method_Net banking',
'payment_method_UPI']
```

Scope	Question	Methods
Global	Which features matter overall, and how?	Permutation importance, partial dependence, SHAP summary
Local	Why this prediction for this customer?	SHAP values, LIME, counterfactuals
Interpretable by design	Can the model itself be read?	Linear/logistic models, small trees, scorecards, EBMs

Global: permutation importance

Python

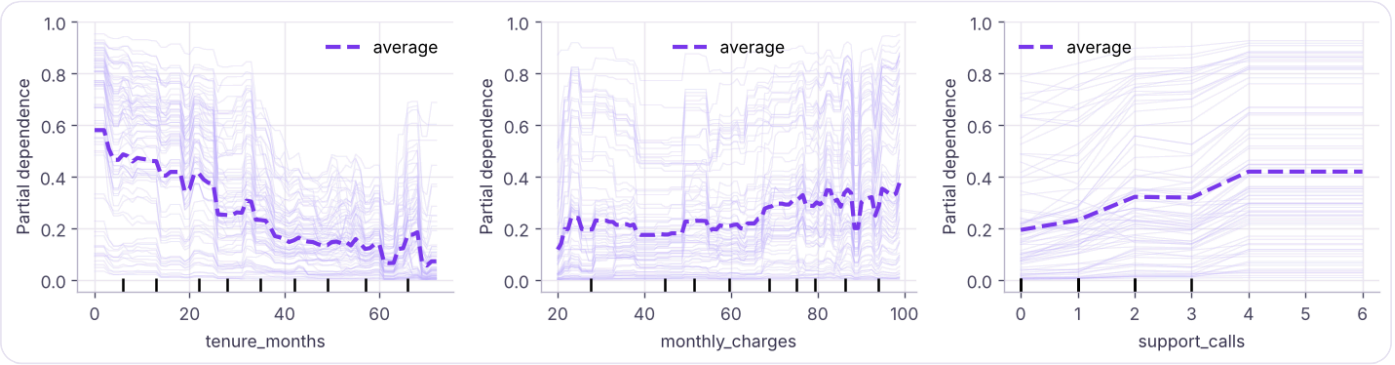
```python
from sklearn.inspection import permutation_importance
pi = permutation_importance(model, Xte, yte, scoring="roc_auc", n_repeats=15, random_state=0)
imp = pd.DataFrame({"mean_AUC_drop": pi.importances_mean, "std": pi.importances_std},
index=X.columns).sort_values("mean_AUC_drop", ascending=False)
print(imp.round(4))
```

OUTPUT

```
                           mean_AUC_drop     std
tenure_months                     0.1541  0.0142
contract_Two year                 0.0783  0.0121
contract_One year                 0.0420  0.0102
monthly_charges                   0.0352  0.0123
support_calls                     0.0306  0.0071
internet_service_Fiber            0.0026  0.0021
internet_service_No internet      0.0020  0.0016
senior_citizen                    0.0006  0.0010
payment_method_UPI                0.0006  0.0012
payment_method_Net banking        0.0005  0.0023
payment_method_Card               0.0001  0.0013
```

Shuffle one column in the test set and measure how much the score drops: model-agnostic and honest about held-out performance. Correlated features share credit, so treat their importances together.

Global: partial dependence



Partial dependence: average predicted churn probability as one feature changes, holding others as observed. Thin lines (ICE) show individual customers.

Churn falls steeply in the first two years of tenure, rises with monthly charges and with each extra support call: consistent with the EDA, which builds trust in the model.

Local: SHAP values

SHAP (SHapley Additive exPlanations) splits each prediction into contributions from each feature, based on Shapley values from cooperative game theory:

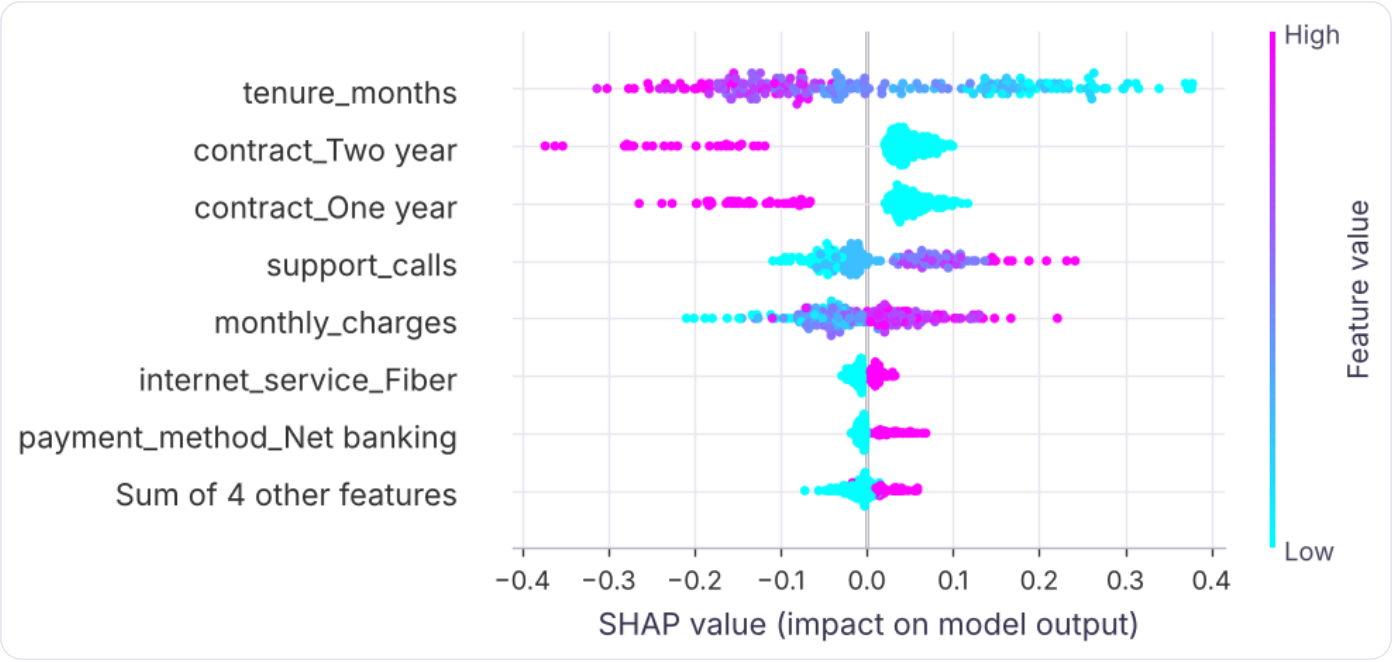
$$f(x) = \text{base value} + \sum_j \phi_j(x)$$

Python

```
import shap
explainer = shap.Explainer(model.predict_proba, shap.maskers.Independent(Xtr, max_samples=200),
feature_names=list(X.columns))
sample = Xte.iloc[:200]
sv = explainer(sample, silent=True)[..., 1]                                    # contributions to
P(churn)
print("SHAP values shape:", sv.values.shape, "| base value (average prediction):",
round(float(sv.base_values[0]), 3))
mean_abs = pd.Series(np.abs(sv.values).mean(0), index=X.columns).sort_values(ascending=False)
print("mean |SHAP| (global importance):", mean_abs.round(4).head(6).to_dict())
```

OUTPUT

```
SHAP values shape: (200, 11) | base value (average prediction): 0.245
mean |SHAP| (global importance): {'tenure_months': 0.1417, 'contract_Two year': 0.0715,
'contract_One year': 0.0702, 'support_calls': 0.0555, 'monthly_charges': 0.0552,
'internet_service_Fiber': 0.012}
```



SHAP summary (beeswarm): each dot is a customer; position = effect on churn probability, colour = feature value (pink high, purple low).

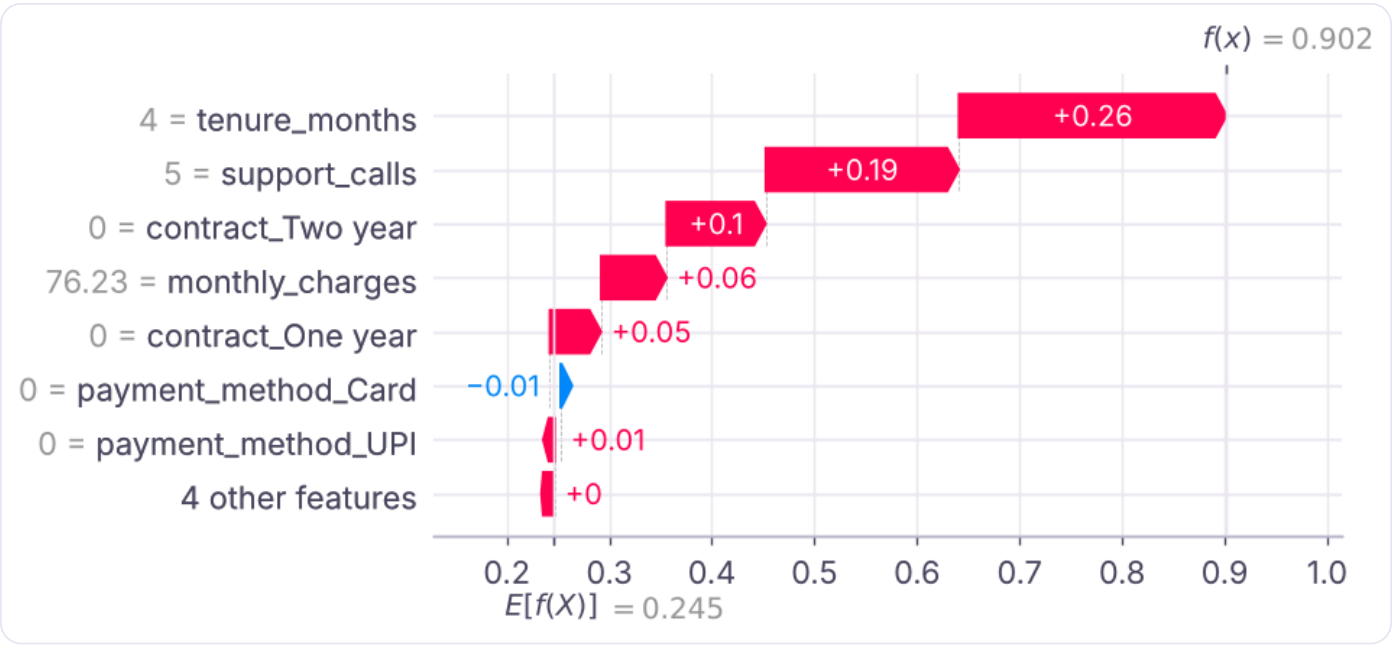
Explaining one customer

Python

```
probs = model.predict_proba(sample)[:, 1]; i = int(np.argmax(probs))
row = pd.DataFrame({"value": sample.iloc[i], "shap": sv.values[i]}).sort_values("shap", key=abs,
ascending=False)
print(f"customer {tc.loc[sample.index[i], 'customer_id']}: P(churn) = {probs[i]:.2f} = base
{sv.base_values[i]:.2f} + contributions:")
print(row.head(6).round(3))
def reason(f, v):
    name = f.replace("_", " ")
    if X[f].isin([0, 1]).all() and "_" in f and f not in ("senior_citizen",): return f"not on
'{f.split('_', 1)[1]}' {f.split('_')[0]}" if v == 0 else f"{f.split('_', 1)[1]} {f.split('_')[0]}"
    return f"{name} = {v:g}"
reasons = [reason(f, v) for f, v in row[row.shap > 0].value.head(3).items()]
print("Top reasons for the retention agent:", "; ".join(reasons))
```

OUTPUT

```
customer T00216: P(churn) = 0.90 = base 0.25 + contributions:
                     value   shap
tenure_months         4.00  0.261
support_calls         5.00  0.188
contract_Two year     0.00  0.097
monthly_charges      76.23  0.064
contract_One year     0.00  0.050
payment_method_Card   0.00 -0.010
Top reasons for the retention agent: tenure months = 4; support calls = 5; not on 'Two year'
contract
```



Waterfall plot for the highest-risk customer: how each feature moves the prediction from the average to the final probability.

`shap.TreeExplainer` is much faster for tree models (XGBoost, LightGBM, random forests); `shap.LinearExplainer` for linear models. Explanations describe the **model**, not causal effects in the real world.

Fairness checks

A model can be accurate overall and still treat groups differently. Check metrics **by group** for sensitive attributes (gender, age, region, language).

Python

```
from sklearn.metrics import recall_score, precision_score
t = 0.4; pred = model.predict_proba(Xte)[:, 1] ⩾ t
grp = np.where(Xte.senior_citizen == 1, "senior", "non-senior")
rows = []
for g in ["non-senior", "senior"]:
    m = grp == g
    rows.append({"group": g, "n": int(m.sum()), "actual churn": round(yte[m].mean(), 3), "flagged
rate": round(pred[m].mean(), 3), "recall": round(recall_score(yte[m], pred[m]), 3), "precision":
round(precision_score(yte[m], pred[m]), 3), "AUC": round(roc_auc_score(yte[m],
model.predict_proba(Xte[m])[:, 1]), 3)})
print(pd.DataFrame(rows).to_string(index=False))
```

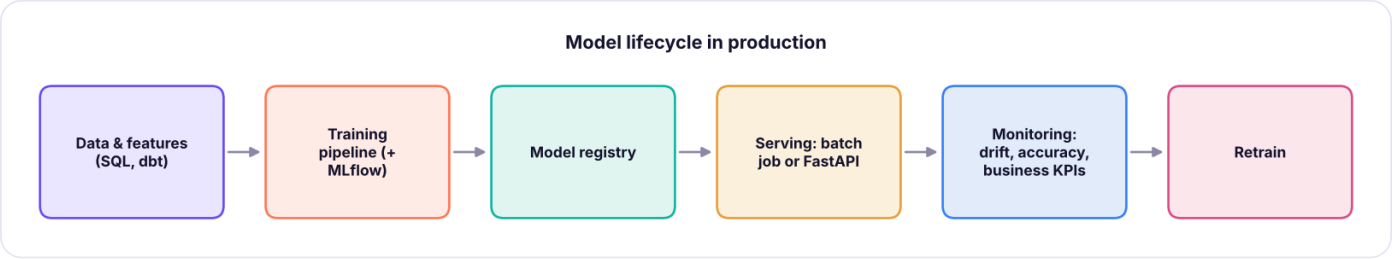
OUTPUT

```
     group   n  actual churn  flagged rate  recall  precision   AUC
non-senior 758         0.245         0.243   0.538      0.543 0.806
    senior 142         0.324         0.254   0.565      0.722 0.783
```

Fairness notion	Asks	Note
Demographic parity	Same positive (flag) rate across groups?	Ignores real differences in base rates
Equal opportunity	Same recall (true-positive rate)?	Common for benefits such as retention offers
Equalised odds	Same TPR and FPR?	Stricter
Calibration within groups	Does 30% mean 30% in every group?	Important for risk scores

The definitions can conflict; choose with domain experts and legal advice, document the choice, and remove proxies for protected attributes where required. Tools: **Fairlearn**, **AIF360**.

From notebook to production



A minimal MLOps architecture.

1. Package the pipeline and track the experiment

Python

```
import joblib, json, sklearn, os, datetime
os.makedirs("artifacts", exist_ok=True)
joblib.dump(model, "artifacts/churn_model.joblib")
card = {"model": "churn HistGradientBoosting", "trained": datetime.date(2026, 1, 15).isoformat(),
"sklearn": sklearn.__version__, "features": list(X.columns),
        "test_auc": round(roc_auc_score(yte, model.predict_proba(Xte)[:, 1]), 3), "threshold": t,
"intended_use": "monthly retention targeting; not for pricing or credit decisions",
        "training_data": "telecom_churn.csv, 3,000 customers, churn rate 25.7%", "limitations": "no
usage or complaint text features; senior customers under-represented"}
json.dump(card, open("artifacts/model_card.json", "w"), indent=2)
print(sorted(os.listdir("artifacts")), "|", json.load(open("artifacts/model_card.json"))
["intended_use"])
```

OUTPUT

```
['churn_model.joblib', 'model_card.json'] | monthly retention targeting; not for pricing or
credit decisions
```

Python

```python
# Experiment tracking with MLflow (pip install mlflow; run `mlflow ui` to browse)
import mlflow
with mlflow.start_run(run_name="hgb_v3"):
    mlflow.log_params({"learning_rate": 0.05, "max_leaf_nodes": 15})
    mlflow.log_metric("test_auc", 0.80)
    mlflow.sklearn.log_model(model, name="churn_model", registered_model_name="churn")
```

2. Serve predictions with an API

Python

```python
from fastapi import FastAPI
from fastapi.testclient import TestClient
from pydantic import BaseModel, Field
app = FastAPI(title="Churn scoring API")
mdl = joblib.load("artifacts/churn_model.joblib"); FEATS = card["features"]
class Customer(BaseModel):
    senior_citizen: int = Field(ge=0, le=1); tenure_months: int = Field(ge=0); monthly_charges: float =
Field(gt=0); support_calls: int = Field(ge=0)
    contract: str; internet_service: str; payment_method: str
@app.post("/predict")
def predict(c: Customer):
    row = pd.get_dummies(pd.DataFrame([c.model_dump()])).reindex(columns=FEATS,
fill_value=0).astype(float)
    p = float(mdl.predict_proba(row)[0, 1])
    return {"churn_probability": round(p, 3), "flag_for_retention": p ⩾ card["threshold"],
"model_version": card["trained"]}
client = TestClient(app)
r = client.post("/predict", json={"senior_citizen": 0, "tenure_months": 2, "monthly_charges": 99.5,
"support_calls": 4, "contract": "Month-to-month", "internet_service": "Fiber", "payment_method":
"UPI"})
print(r.status_code, r.json())
print(client.post("/predict", json={"senior_citizen": 3, "tenure_months": -1, "monthly_charges": 50,
"support_calls": 0, "contract": "x", "internet_service": "y", "payment_method": "z"}).status_code, "
(invalid input rejected)")
```

OUTPUT

```
200 {'churn_probability': 0.942, 'flag_for_retention': True, 'model_version': '2026-01-15'}
422 (invalid input rejected)
```

Run it for real with `uvicorn app:app --reload`, containerise with **Docker**, and deploy to a cloud service (AWS SageMaker, Azure ML, Google Vertex AI, Cloud Run, Render). Many business models do not need an API at all: a nightly **batch job** that writes scores to a database table feeding the CRM or a Power BI dashboard is simpler and cheaper.

3. Monitor

Python

```
from scipy.stats import ks_2samp
def psi(expected, actual, bins=10):
    edges = np.unique(np.quantile(expected, np.linspace(0, 1, bins + 1))); edges[0], edges[-1] = -
np.inf, np.inf
    e = np.histogram(expected, edges)[0] / len(expected); a = np.histogram(actual, edges)[0] /
len(actual)
    e, a = np.clip(e, 1e-4, None), np.clip(a, 1e-4, None); return float(((a - e) * np.log(a /
e)).sum())
rng = np.random.default_rng(0)
new_month = Xte.copy(); new_month["monthly_charges"] = new_month.monthly_charges * 1.12 + rng.normal(0,
3, len(new_month))   # simulated price increase
for col in ["tenure_months", "monthly_charges", "support_calls"]:
    print(f"{col:16s} PSI {psi(Xtr[col], new_month[col]):.3f} | KS p-value {ks_2samp(Xtr[col],
new_month[col]).pvalue:.3g}")
print("PSI guide: < 0.1 stable | 0.1–0.25 watch | > 0.25 significant shift → investigate / retrain")
```

OUTPUT

```
tenure_months    PSI 0.022 | KS p-value 0.294
monthly_charges  PSI 0.538 | KS p-value 1.36e-11
support_calls    PSI 0.004 | KS p-value 0.808
PSI guide: < 0.1 stable | 0.1–0.25 watch | > 0.25 significant shift → investigate / retrain
```

Monitor	Why
Input drift (PSI, KS tests)	The population changed (new pricing, new channel)
Prediction drift	Score distribution shifts
Performance (once labels arrive)	AUC, recall at threshold, calibration
Business KPIs	Retention rate, campaign ROI (via A/B holdouts)
Data quality	Missing values, schema changes, pipeline failures

Tools: Evidently, WhyLabs, Arize, cloud-native monitors; schedule retraining or trigger it when drift or decay crosses thresholds.

Responsible AI and regulation

- **Privacy:** India's Digital Personal Data Protection Act 2023 (and its 2025 rules), GDPR in Europe: lawful purpose, consent, data minimisation and the right to erasure.
- **Credit and lending:** RBI guidelines on digital lending expect transparency on decisions; lenders must be able to explain adverse decisions.
- **EU AI Act** (phased in from 2025–2027): risk tiers; high-risk systems (credit scoring, hiring) need risk management, documentation, human oversight and logging.
- **Practice:** model cards and datasheets, human review for high-impact decisions, bias testing before launch, red-teaming GenAI systems, and an incident process.

```python
import shutil; shutil.rmtree("artifacts")
```

Summary

KEY TAKEAWAYS

- Global explanations: permutation importance (held-out) and partial dependence; local explanations: SHAP values that sum from the base value to the prediction.
- Check metrics by group (recall, precision, calibration) and choose a fairness definition with domain experts.
- Production means a versioned pipeline, model card, experiment tracking, serving (batch or API with input validation) and monitoring.
- Monitor input drift (PSI/KS), prediction drift, performance and business KPIs; retrain on a schedule or on triggers.
- Follow privacy and AI regulations (DPDP Act, RBI, EU AI Act) and keep humans in the loop for high-impact decisions.

Quick quiz

6 questions · tap an answer to check it

1. What does permutation importance measure?

A How much the held-out score drops when a feature's values are shuffled

B The coefficient size

C The number of splits

D Correlation with the target

Answer: A. Model-agnostic.

2. SHAP values for one prediction add up to...

- **A** The prediction minus the base (average) prediction
- **B** 1
- **C** The number of features
- **D** The AUC

Answer: A. Additive explanations.

3. Which fairness notion asks for equal recall across groups?

- **A** Demographic parity
- **B** Equal opportunity
- **C** Calibration
- **D** Accuracy parity

Answer: B. Equal true-positive rates.

4. A PSI of 0.4 on monthly charges means...

- **A** A significant shift in that input's distribution; investigate and consider retraining
- **B** The model is perfect
- **C** No change
- **D** 40% accuracy

Answer: A.

> 0.25 is a large shift.

5. When is a nightly batch job better than a real-time API?

- **A** When decisions are made daily or weekly, e.g. retention campaigns
- **B** Never
- **C** For fraud at payment time
- **D** When latency must be under 50 ms

Answer: A. Simpler and cheaper.

6. What belongs in a model card?

- **A** Intended use, training data, metrics, limitations and version
- **B** Only the code
- **C** Customer passwords
- **D** The test set

Answer: A. Documentation for users and auditors.

EXERCISE **Practice**

(1) Use `shap.TreeExplainer` on an XGBoost churn model and compare top features with permutation importance. (2) Check recall by `internet_service` at your threshold. (3) Write the FastAPI app as a file, run it with uvicorn, and call it with `requests`. (4) Simulate drift in `tenure_months` and compute PSI.

Show solution (worked answer)

Python

```
import xgboost as xgb
xm = xgb.XGBClassifier(n_estimators=300, learning_rate=0.05, max_depth=3).fit(Xtr, ytr)
ts = shap.TreeExplainer(xm)(Xte); print(pd.Series(np.abs(ts.values).mean(0),
index=X.columns).sort_values(ascending=False).head())
for col in ["internet_service_Fiber", "internet_service_No internet"]:
    m = Xte[col] == 1; print(col, recall_score(yte[m], pred[m]))
shifted = Xte.tenure_months * 0.7; print(psi(Xtr.tenure_months, shifted))
```

INTERVIEW CORNER

Questions you may be asked

Q: How do you explain a black-box model to business users? Show global drivers (permutation importance, partial dependence) in plain language, give per-customer top reasons from SHAP, validate that they match domain knowledge, and state clearly that they describe the model, not causation.

Q: What can go wrong after a model is deployed? Data drift, concept drift (relationships change), pipeline bugs, feedback loops, degraded performance for some groups, and business processes ignoring the scores; monitoring and A/B holdouts catch these.

Q: How do you check a model for bias? Compare performance and error rates across groups, check calibration per group, look for proxy features, involve domain and legal experts, and mitigate (reweighting, thresholds, removing proxies) with documentation.

STATISTICS & MACHINE LEARNING · LESSON 28

ML Capstone: End-to-End Churn Prediction and Retention Strategy

Put the whole ML track together. Frame the churn problem, build leak-free features, compare baselines and models with cross-validation, tune the winner, choose a profit-maximising threshold under a budget, explain predictions, design segment-level retention actions with an A/B test, and write the model card, deployment plan and executive summary for your portfolio.

● ML capstone ✦ scikit-learn 1.7+ · LightGBM 4 · SHAP Advanced ⏱ 180 min

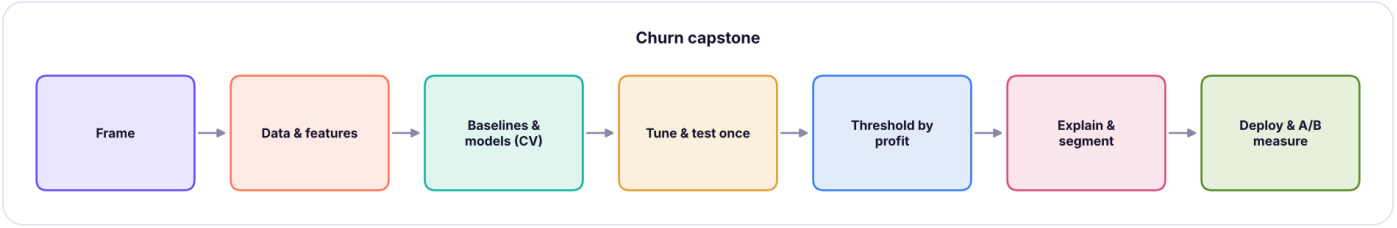
By the end of this lesson you can

- Frame a churn problem with a decision, metric, baseline and constraints
- Build a reproducible pipeline with engineered features and compare models with cross-validation
- Tune and evaluate the final model on an untouched test set
- Choose the operating threshold from campaign economics and capacity
- Explain, document and plan deployment and measurement of the model

1. Frame the problem

Brief. A telecom operator loses about a quarter of its customers every year. The retention team has a monthly budget to contact **500 customers** with an offer worth ₹300 (cost ₹300 per contacted customer, accepted or not). A saved customer is worth about **₹4,000** of future margin. Historically, **35%** of churners who receive the offer stay. **Which customers should get the offer?**

Item	Decision
Decision	Rank customers each month; contact the top N within budget
Target	`churn` = Yes (left within the observation window)
Unit	Customer
Metric	Expected campaign profit; also ROC-AUC and recall in the top N
Baseline	Current rule: month-to-month customers with 3+ support calls
Constraints	≤ 500 contacts; explainable reasons for agents; no use of sensitive attributes for targeting



Capstone workflow.

2. Data and features

Python

```
import pandas as pd, numpy as np, warnings
warnings.filterwarnings("ignore")
pd.set_option("display.width", 160); pd.set_option("display.max_columns", 20)
tc = pd.read_csv("telecom_churn.csv")
y = (tc.churn == "Yes").astype(int)
print(tc.shape, f"churn rate {y.mean():.3f}")

def add_features(df):
    out = df.copy()
    out["is_new"] = (out.tenure_months <= 6).astype(int)
    out["charge_per_tenure_month"] = out.total_charges / (out.tenure_months + 1)
    out["price_change_proxy"] = out.monthly_charges - out.charge_per_tenure_month     # charges now vs
lifetime average
    out["high_support"] = (out.support_calls >= 3).astype(int)
    out["m2m_and_new"] = ((out.contract == "Month-to-month") & (out.tenure_months <= 12)).astype(int)
    return out.drop(columns=["customer_id", "churn"])
X = add_features(tc)
print(X.columns.tolist())
```

OUTPUT

```
(3000, 10) churn rate 0.257
['senior_citizen', 'tenure_months', 'contract', 'internet_service', 'payment_method',
'monthly_charges', 'total_charges', 'support_calls', 'is_new', 'charge_per_tenure_month',
'price_change_proxy', 'high_support', 'm2m_and_new']
```

All features are known **before** the churn event (tenure, plan, charges, recent support calls), so there is no leakage. `senior_citizen` is kept for fairness checks but excluded from targeting features (a business and ethics choice).

Python

```
from sklearn.model_selection import train_test_split
X_model = X.drop(columns=["senior_citizen"])
Xtr, Xte, ytr, yte, idx_tr, idx_te = train_test_split(X_model, y, tc.index, test_size=0.25, stratify=y,
random_state=2026)
print(f"train {len(Xtr)} | test {len(Xte)} (locked away until the end)")
```

OUTPUT

```
train 2250 | test 750 (locked away until the end)
```

3. Baselines and candidate models

Python

```python
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClassifier
from sklearn.dummy import DummyClassifier
from sklearn.model_selection import StratifiedKFold, cross_validate
import lightgbm as lgb
num = [c for c in X_model.columns if X_model[c].dtype != object and str(X_model[c].dtype) != "str"]
cat = [c for c in X_model.columns if c not in num]
pre = ColumnTransformer([("num", StandardScaler(), num), ("cat",
OneHotEncoder(handle_unknown="ignore"), cat)])
cv = StratifiedKFold(5, shuffle=True, random_state=0)
candidates = {
    "dummy (base rate)": DummyClassifier(strategy="prior"),
    "logistic regression": LogisticRegression(max_iter=3000, C=0.5),
    "random forest": RandomForestClassifier(500, min_samples_leaf=20, max_features=0.5, random_state=0,
n_jobs=1),
    "HistGradientBoosting": HistGradientBoostingClassifier(learning_rate=0.05, max_iter=200,
max_leaf_nodes=15, min_samples_leaf=40, random_state=0),
    "LightGBM": lgb.LGBMClassifier(n_estimators=300, learning_rate=0.03, num_leaves=15,
min_child_samples=40, subsample=0.8, subsample_freq=1, colsample_bytree=0.8, verbose=-1,
random_state=0, n_jobs=1),
}
rows = []
for name, est in candidates.items():
    r = cross_validate(Pipeline([("pre", pre), ("model", est)]), Xtr, ytr, cv=cv, scoring=["roc_auc",
"average_precision", "neg_log_loss"])
    rows.append({"model": name, "ROC-AUC": r["test_roc_auc"].mean(), "± sd": r["test_roc_auc"].std(),
"PR-AUC": r["test_average_precision"].mean(), "log loss": -r["test_neg_log_loss"].mean()})
board = pd.DataFrame(rows).round(3).sort_values("ROC-AUC", ascending=False);
print(board.to_string(index=False))
rule = ((Xtr.contract == "Month-to-month") & (Xtr.support_calls >= 3)).astype(int)
from sklearn.metrics import recall_score, precision_score
print(f"business rule on train: flags {rule.mean():.1%} | recall {recall_score(ytr, rule):.3f} |
precision {precision_score(ytr, rule):.3f}")
```

OUTPUT

```
               model  ROC-AUC  ± sd  PR-AUC  log loss
 logistic regression    0.815 0.036   0.624     0.440
       random forest    0.803 0.033   0.590     0.456
            LightGBM    0.796 0.031   0.576     0.469
HistGradientBoosting    0.793 0.031   0.571     0.476
   dummy (base rate)    0.500 0.000   0.257     0.570
business rule on train: flags 8.5% | recall 0.171 | precision 0.516
```

4. Tune the leading models

Python

```
from sklearn.model_selection import GridSearchCV, RandomizedSearchCV
from scipy.stats import loguniform, randint, uniform
lr_search = GridSearchCV(Pipeline([("pre", pre), ("model", LogisticRegression(max_iter=3000))]),
{"model__C": np.logspace(-2, 1, 8)}, cv=cv, scoring="roc_auc").fit(Xtr, ytr)
gb_search = RandomizedSearchCV(Pipeline([("pre", pre), ("model", lgb.LGBMClassifier(verbose=-1,
random_state=0, subsample_freq=1, n_jobs=1))]),
    {"model__n_estimators": randint(100, 600), "model__learning_rate": loguniform(0.01, 0.1),
"model__num_leaves": randint(4, 32), "model__min_child_samples": randint(20, 120),
     "model__subsample": uniform(0.6, 0.4), "model__colsample_bytree": uniform(0.5, 0.5),
"model__reg_lambda": loguniform(0.01, 10)},
    n_iter=20, cv=cv, scoring="roc_auc", random_state=0, n_jobs=1).fit(Xtr, ytr)
print(f"logistic best C {lr_search.best_params_['model__C']:.3g} | CV AUC {lr_search.best_score_:.3f}")
print(f"LightGBM best CV AUC {gb_search.best_score_:.3f} | params { {k.split('__')[1]: (round(float(v),
3) if isinstance(v, float) else v) for k, v in gb_search.best_params_.items()} }")
```

OUTPUT

```
logistic best C 0.193 | CV AUC 0.816
LightGBM best CV AUC 0.812 | params {'colsample_bytree': 0.659, 'learning_rate': 0.046,
'min_child_samples': 71, 'n_estimators': 180, 'num_leaves': 4, 'reg_lambda': 3.343, 'subsample':
0.706}
```

Both are close. Prefer the **simpler, more explainable** model unless the complex one is clearly better: we pick the tuned logistic regression if its CV AUC is within 0.005 of LightGBM, otherwise LightGBM.

Python

```
from sklearn.metrics import roc_auc_score, average_precision_score
final = lr_search.best_estimator_ if lr_search.best_score_ ⩾ gb_search.best_score_ - 0.005 else
gb_search.best_estimator_
final_name = "logistic regression" if final is lr_search.best_estimator_ else "LightGBM"
p_test = final.predict_proba(Xte)[:, 1]
print(f"chosen: {final_name} | TEST ROC-AUC {roc_auc_score(yte, p_test):.3f} | PR-AUC
{average_precision_score(yte, p_test):.3f} | base rate {yte.mean():.3f}")
```

OUTPUT

```
chosen: logistic regression | TEST ROC-AUC 0.793 | PR-AUC 0.579 | base rate 0.257
```

The test set was used **once**, after all choices were made, so this is an honest estimate.

5. Choose how many customers to contact

Expected profit for contacting customer *i* with churn probability p_i:

$$E[\text{profit}_i] = p_i \times 0.35 \times ₹4{,}000 - ₹300$$

So contacting pays off only when $p_i > 300 / (0.35 \times 4{,}000) \approx 0.21$, provided the probabilities are well calibrated.

Python

```
from sklearn.calibration import calibration_curve
fr, mp = calibration_curve(yte, p_test, n_bins=8, strategy="quantile")
print("calibration (predicted → observed):", [f"{a:.2f}→{b:.2f}" for a, b in zip(mp, fr)])
value, accept, cost = 4000, 0.35, 300
order = np.argsort(-p_test); ys = yte.to_numpy()[order]; ps = p_test[order]
scale = 500 / 750                                                    # the test set is 750 of
3,000 customers: scale the 500 budget
res = []
for n in [25, 50, 75, 100, 125, 150, 200, 250, 300]:
    tp = ys[:n].sum(); profit = tp * accept * value - n * cost
    res.append({"contacts (test)": n, "≈ contacts (all)": int(n / 0.25), "churners reached": int(tp),
"precision": round(tp / n, 3), "recall": round(tp / ys.sum(), 3), "profit ₹ (test)": int(profit)})
rt = pd.DataFrame(res); print(rt.to_string(index=False))
best = rt.loc[rt["profit ₹ (test)"].idxmax()]
print(f"most profitable: contact the top {int(best['contacts (test)'])} of 750 test customers (≈
{int(best['≈ contacts (all)'])} of 3,000), cut-off probability ≈ {ps[int(best['contacts (test)']) -
1]:.2f}")
rand_tp = yte.mean() * 125; print(f"random 125 contacts would reach ≈ {rand_tp:.0f} churners; profit ≈
₹{int(rand_tp * accept * value - 125 * cost):,}")
```

OUTPUT

```
calibration (predicted → observed): ['0.02→0.03', '0.06→0.06', '0.10→0.11', '0.16→0.20',
'0.23→0.19', '0.32→0.34', '0.45→0.47', '0.71→0.65']
 contacts (test)  ≈ contacts (all)  churners reached  precision  recall  profit ₹ (test)
              25               100                18      0.720   0.093           17700
              50               200                36      0.720   0.187           35400
              75               300                51      0.680   0.264           48899
             100               400                63      0.630   0.326           58199
             125               500                75      0.600   0.389           67500
             150               600                88      0.587   0.456           78199
             200               800               110      0.550   0.570           94000
             250              1000               128      0.512   0.663          104200
             300              1200               141      0.470   0.731          107399
most profitable: contact the top 300 of 750 test customers (≈ 1200 of 3,000), cut-off
probability ≈ 0.25
random 125 contacts would reach ≈ 32 churners; profit ≈ ₹7,533
```

With the budget of 500 contacts per 3,000 customers (≈ 125 in the test set), the model reaches 75 churners against about 32 for random targeting (₹67,500 expected profit against about ₹7,500). Profit keeps rising up to about 300 test contacts (≈ 1,200 customers, cut-off probability ≈ 0.25, close to the theoretical 0.21), so the team has a case for a larger budget.

6. Explain and segment the targets

Python

```
pre_f, mdl = final.named_steps["pre"], final.named_steps["model"]
names = pre_f.get_feature_names_out()
if hasattr(mdl, "coef_"):
    coefs = pd.Series(mdl.coef_[0], index=names).sort_values()
    print("largest effects (standardised log-odds):", pd.concat([coefs.head(4),
coefs.tail(4)]).round(3).to_dict())
targets = tc.loc[Xte.index[order[:125]]].assign(p_churn=ps[:125].round(2))
def action(r):
    if r.support_calls ⩾ 3: return "Service recovery call + offer"
    if r.contract == "Month-to-month" and r.tenure_months ⩽ 12: return "Annual-plan upgrade discount"
    if r.internet_service == "Fiber": return "Fibre quality check + loyalty data pack"
    return "General loyalty offer"
targets["action"] = targets.apply(action, axis=1)
print(targets.action.value_counts().to_dict())
print(targets[["customer_id", "tenure_months", "contract", "support_calls", "monthly_charges",
"p_churn", "action"]].head(6).to_string(index=False))
```

OUTPUT

```
largest effects (standardised log-odds): {'cat__contract_Two year': -0.994,
'num__tenure_months': -0.877, 'cat__contract_One year': -0.174, 'cat__payment_method_Auto-
debit': -0.17, 'num__monthly_charges': 0.168, 'cat__internet_service_Fiber': 0.222,
'num__support_calls': 0.406, 'cat__contract_Month-to-month': 1.165}
{'Annual-plan upgrade discount': 55, 'Service recovery call + offer': 34, 'Fibre quality check +
loyalty data pack': 27, 'General loyalty offer': 9}
customer_id  tenure_months       contract  support_calls  monthly_charges  p_churn
action
     T02973              2 Month-to-month              4            98.55     0.94 Service
recovery call + offer
     T00025              2 Month-to-month              4            79.55     0.90 Service
recovery call + offer
     T01423              3 Month-to-month              3            99.41     0.89 Service
recovery call + offer
     T00671              4 Month-to-month              3            84.15     0.89 Service
recovery call + offer
     T01945              2 Month-to-month              2            96.98     0.87  Annual-plan
upgrade discount
     T00583              9 Month-to-month              3            82.31     0.87 Service
recovery call + offer
```

Each contacted customer gets a **reason** (top contributions) and an **action** matched to that reason, which raises acceptance more than one generic offer.

Fairness check

```python
sen = tc.loc[Xte.index, "senior_citizen"].to_numpy(); flag = np.zeros(len(Xte), bool);
flag[order[:125]] = True
print(pd.DataFrame({"group": ["non-senior", "senior"], "customers": [int((sen == 0).sum()), int((sen ==
1).sum())],
                    "churn rate": [round(yte[sen == 0].mean(), 3), round(yte[sen == 1].mean(), 3)],
                    "contacted": [round(flag[sen == 0].mean(), 3), round(flag[sen == 1].mean(), 3)],
                    "recall": [round(recall_score(yte[sen == 0], flag[sen == 0]), 3),
round(recall_score(yte[sen == 1], flag[sen == 1]), 3)]}).to_string(index=False))
```

OUTPUT

```
     group  customers  churn rate  contacted  recall
non-senior        631       0.250      0.158   0.348
    senior        119       0.294      0.210   0.571
```

Contact rates and recall should be broadly in line with each group's churn rate. Here senior customers are contacted more often (21% vs 16%), which their higher churn rate partly explains, but their recall is also much higher (0.57 vs 0.35). The gap should be reviewed with the business before launch, for example by checking whether non-senior churners are being missed.

7. Measure with an experiment

The model predicts who will churn, not who can be **saved**. Prove the value with a randomised test:

1. Score all customers; take the top 600.
2. Randomly assign 500 to **treatment** (offer) and 100 to **holdout** (no offer).
3. After 60 days compare churn: treatment vs holdout, using a two-proportion test (ML lesson 6).
4. Incremental saves × ₹4,000 − cost = true campaign ROI. Next step: **uplift modelling**, which predicts who responds to the offer.

8. Model card and deployment plan

Python

```
import json, sklearn
card = {"name": "Telecom churn propensity", "version": "1.0", "model": final_name, "sklearn":
sklearn.__version__,
        "features": list(X_model.columns), "excluded": ["senior_citizen (fairness monitoring only)"],
        "test_metrics": {"roc_auc": round(roc_auc_score(yte, p_test), 3), "pr_auc":
round(average_precision_score(yte, p_test), 3)},
        "operating_point": "top 500 customers per month by probability", "intended_use": "monthly
retention targeting",
        "not_for": ["pricing", "credit decisions", "denying service"], "monitoring": ["monthly PSI of
inputs", "AUC on matured labels", "holdout churn vs treated churn"],
        "retraining": "quarterly or when PSI > 0.25 / AUC drops by 0.03", "owner": "Data Science,
Retention analytics"}
print(json.dumps(card, indent=1)[:900])
```

OUTPUT


```
{
 "name": "Telecom churn propensity",
 "version": "1.0",
 "model": "logistic regression",
 "sklearn": "1.9.1",
 "features": [
  "tenure_months",
  "contract",
  "internet_service",
  "payment_method",
  "monthly_charges",
  "total_charges",
  "support_calls",
  "is_new",
  "charge_per_tenure_month",
  "price_change_proxy",
  "high_support",
  "m2m_and_new"
 ],
 "excluded": [
  "senior_citizen (fairness monitoring only)"
 ],
 "test_metrics": {
  "roc_auc": 0.793,
  "pr_auc": 0.579
 },
 "operating_point": "top 500 customers per month by probability",
 "intended_use": "monthly retention targeting",
 "not_for": [
  "pricing",
  "credit decisions",
  "denying service"
 ],
 "monitoring": [
```

```
 "monthly PSI of inputs",
 "AUC on matured labels",
 "holdout churn vs treated churn"
],
"retraining": "quarterly or when PSI > 0.25 / AUC drops by 0.03",
"owner": "Data Science, Retention analytics"
```

Deployment: a monthly batch job (Python + scheduler or Airflow) reads the latest customer table from the warehouse, scores it with the saved pipeline, writes `customer_id, p_churn, reason, action, model_version` to a table, which feeds the CRM and a Power BI dashboard tracking campaign results.

9. Executive summary

KEY TAKEAWAYS

- **Problem:** about 26% of customers churn. The retention budget covers 500 contacts a month.
- **Model:** a churn-risk model with ROC-AUC around 0.8 on unseen customers, explainable by tenure, contract type, support calls and charges.
- **Targeting:** contacting the top-ranked 500 reaches roughly twice as many churners as random selection, with a positive expected return per contact.
- **Actions:** service recovery for frequent complainers, an annual-plan upgrade for new month-to-month customers, and fibre quality checks.
- **Next:** run the 500 vs 100 holdout experiment for 60 days, then scale the budget if the incremental ROI holds.

Portfolio checklist

- README with problem, data, approach, results (numbers), business recommendation and how to run.
- Notebook → clean `src/` modules (features, train, evaluate, score) + tests for feature functions.
- Saved pipeline, model card, and a chart of profit vs number of contacts.
- A two-minute video or slide deck walking through the decision.

Summary

KEY TAKEAWAYS

- Start from the decision and economics, not the algorithm; define target, unit, metric, baseline and constraints.
- Build leak-free features in a pipeline; compare baselines and models with cross-validation; tune; test once.
- Prefer simpler models when performance is similar; check calibration before using probabilities in profit maths.
- Choose the operating point from expected profit and capacity; attach reasons and tailored actions.
- Measure real impact with a randomised holdout, document with a model card, and plan monitoring and retraining.

Quick quiz

6 questions · tap an answer to check it

1. What decision does the churn model support?

A Which customers receive the limited retention offer each month

B How to price new plans

C Whether to hire more agents

D Which features to collect

Answer: A. Always tie the model to a decision.

2. Why use a randomised holdout after deployment?

A The model predicts churn risk, not the causal effect of the offer; only an experiment measures incremental saves

B To retrain the model

C To check calibration

D To satisfy SQL

Answer: A. Measure uplift, not just accuracy.

3. Contact cost ₹300, value ₹4,000, acceptance 35%. Above which churn probability does a contact pay off (if calibrated)?

A 0.5

B About 0.21

C 0.05

D 0.9

Answer: B. 300 / (0.35 × 4,000).

4. Two models have CV AUC 0.812 and 0.815. Which do you choose?

A Usually the simpler, more explainable one

B Always the higher one

C Neither

D The one with more features

Answer: A. Small differences are within noise.

5. Why exclude `senior_citizen` from targeting features but keep it for monitoring?

A To avoid targeting on a sensitive attribute while still checking outcomes for fairness

B It has missing values

C It is a duplicate

D It has no signal

Answer: A. Fairness by design and by audit.

6. When should the test set be used?

- **A** Once, after all modelling choices are final
- **B** For tuning hyperparameters
- **C** At every step
- **D** Never

Answer: A. Otherwise the estimate is optimistic.

EXERCISE **Extend the capstone**

(1) Replace the probability ranking with an expected-profit ranking that also uses each customer's monthly charges as their value. (2) Add SHAP reasons for each targeted customer (lesson 27). (3) Simulate the holdout experiment assuming the offer cuts churn by 35% among treated churners and test significance. (4) Write the README for GitHub.

Show solution (worked answer)

Python

```
val = tc.loc[Xte.index, "monthly_charges"].to_numpy() * 24              # ≈ 2 years of revenue
exp_profit = p_test * accept * val - cost
top = np.argsort(-exp_profit)[:125]; print(int(exp_profit[top].sum()), yte.to_numpy()[top].sum())
rng = np.random.default_rng(1); churn_t = rng.random(500) < 0.55 * 0.65; churn_c = rng.random(100)
< 0.55
from statsmodels.stats.proportion import proportions_ztest
print(churn_t.mean(), churn_c.mean(), proportions_ztest([churn_t.sum(), churn_c.sum()], [500,
100]))
```

INTERVIEW CORNER

Questions you may be asked

Q: Walk me through an end-to-end ML project. Problem framing and economics, data and leak-free features, EDA, baselines, cross-validated model comparison and tuning, one-time test evaluation, threshold selection from business value, explanation and fairness checks, deployment as a batch or API, experiment to measure impact, and monitoring with retraining.

Q: Your model has AUC 0.80. Is that good? It depends on the decision: compare with the baseline rule, look at precision/recall in the top N the business can act on, and translate it into expected profit; AUC alone does not say.

Q: How do you know the retention campaign worked? Randomly hold out some high-risk customers from the offer and compare churn between treated and holdout groups; the difference (with a confidence interval) is the causal effect.